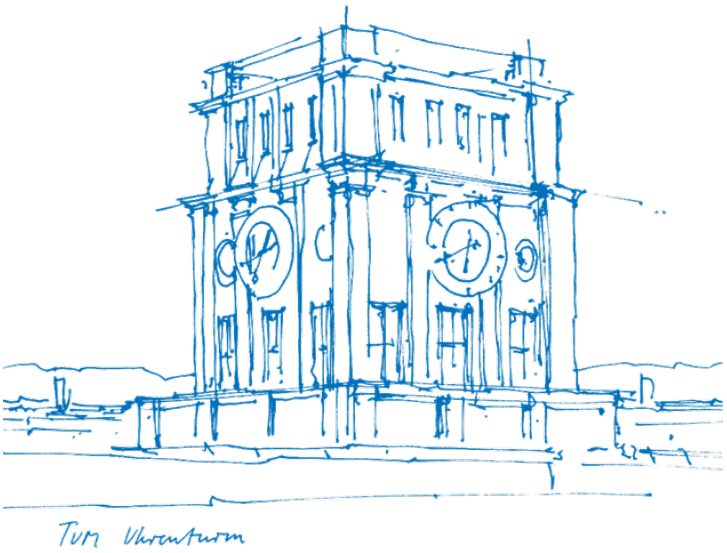
Chair of Aerodynamics and Fluid Mechanics
Department of Aerospace and Geodesy
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# Assessment of RANS Wall Functions for Predicting Aerothermal Heating in Hypersonic Cold-Wall Flows

**Michael Roman Gnadlinger**



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**Michael Roman Gnadlinger**

Thesis for the attainment of the academic degree

**Master of Science (M.Sc.)**

at the Department of Aerospace and Geodesy of the Technical University of Munich.

**Examiner:**
Prof. Dr.-Ing. Christian Stemmer

**Supervisor:**
Dr. Friedrich Ulrich

**Submitted:**
Munich, October 19, 2026

I hereby declare that this thesis is entirely the result of my own work except where otherwise indicated. I have only used the resources given in the list of references.

Munich, October 19, 2026 Michael Roman Gnadlinger

# Abstract

Aerodynamic heating poses a major challenge in hypersonic flight, making thermal design a central consideration in vehicle development. This requires accurate prediction of both the heating imposed by the flow and the resulting temperatures within the vehicle structure. Coupled simulations of the flow and structural heat conduction throughout the flight trajectory are computationally demanding, limiting the speed of design iterations. Wall functions offer a way to reduce this cost by relaxing the near-wall mesh requirements of Reynolds-averaged Navier–Stokes simulations. This work investigates existing analytical wall functions, implements them within a computational framework, and systematically assesses their accuracy in predicting aerothermal heat-flux. By isolating individual sources of error, the study identifies promising wall-function closures but reveals that their performance is substantially constrained by the underlying turbulence models and conventional meshing practices. Both retain assumptions that are shown to not reliably extend to hypersonic flows over strongly cooled walls. A novel mesh-resolution criterion addresses the latter limitation and challenges the longstanding assumption that wall functions inherently struggle in the buffer layer. The turbulence-model assessment further identifies the assumption of a constant turbulent Prandtl number as a likely source of discrepancies in predicted heat-flux and thermal boundary layer profiles. These findings demonstrate that reliable application of analytical wall functions requires joint assessment of closure accuracy, mesh resolution, and turbulence modelling, and identify priorities for improving the accuracy and computational efficiency of hypersonic aerothermal predictions.
Furthermore, proceeding through this work, various important corrections were discovered and successfully submitted for correction, most notably a correction to the momentum thickness definition in Schlichting's Boundary Layer Theory [1], a typo in the reported viscosity law constant, hindering reproducibility, in Malik and Anderson's laminar boundary layer paper [2] and an error in the implementation of the compressible Spalart–Allmaras turbulence model in OpenFOAM. For completeness, the accepted proofs and arguments supporting these and other corrections are in the appendices E and F.

# Acknowledgements

I would like to thank Prof. Dr.-Ing. Christian Stemmer for taking the time and serving as the examiner of this thesis. I am particularly grateful to my supervisor, Dr. Friedrich Ulrich, for his guidance throughout this work and for introducing me to Hypersonica in the first place. I also thank Dr. Maximilian Maigler of Hypersonica for introducing me to the topic of wall-function modelling and inspiring this investigation.
I am deeply grateful to Hypersonica GmbH for the opportunity to conduct this research and for providing access to its high-performance computing resources. These resources enabled the extensive simulation campaign that forms the basis of the findings and conclusions presented in this thesis.

# AI Transparency Statement

As a non-native English speaker, artificial intelligence (AI) in the form of large language models (LLMs) were used in the written part of this thesis work to assist with grammar, sentence structure, and overall clarity and readability of the text. AI tools were also employed to help with the code implementation and debugging of the wall functions (Chapter 5), the required extensions to the OpenFOAM/HiSA simulation framework (Section 3.1) and in the provision of various post-processing scripts. The AI tools were used following the Springer Nature guidelines for responsible AI use in research publishing[1], where any code or written text, was reviewed and verified by the author to ensure accuracy, correctness and accountability.

[1] https://www.nature.com/nature-portfolio/editorial-policies/ai

# Data Availability Statement

All simulation files underlying the results presented in this thesis are available in the GitHub repository at `https://github.com/Gnadi27/MasterThesis`. The repository also contains the custom libraries used to extend the OpenFOAM/HiSA simulation framework, together with the implementations of the investigated wall functions.
The data presented in this thesis are not subject to company confidentiality or export restrictions. For further information, please contact the author.

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# Nomenclature

The following tables collect recurring quantities and notation. Reference states, heat-flux signs and formulation-specific parameters are defined where used. A dash denotes a dimensionless quantity. Symbols with more than one meaning are distinguished by their context and units.

| Symbol | Meaning | SI unit |
|---|---|---|
| **Latin symbols** | | |
| $a$ | Speed of sound | $\mathrm{m\,s^{-1}}$ |
| $B_q$ | Dimensionless wall heat-transfer rate; negative for wall cooling | – |
| $c_p$ | Specific heat at constant pressure | $\mathrm{J\,kg^{-1}\,K^{-1}}$ |
| $C_f$ | Skin-friction coefficient | – |
| $C_h,\ St$ | Stanton number, with the stated reference normalization | – |
| $h$ | Specific enthalpy | $\mathrm{J\,kg^{-1}}$ |
| $k$ | Turbulent kinetic energy per unit mass | $\mathrm{m^2\,s^{-2}}$ |
| $M,\ M_\tau$ | Mach number and friction Mach number $M_\tau = u_\tau/a_w$ | – |
| $p$ | Static pressure | $\mathrm{Pa}$ |
| $Pr,\ Pr_t$ | Molecular and turbulent Prandtl numbers | – |
| $q_w$ | Wall heat-flux; the sign convention is specified locally | $\mathrm{W\,m^{-2}}$ |
| $r$ | Recovery factor | – |
| $R$ | Specific gas constant | $\mathrm{J\,kg^{-1}\,K^{-1}}$ |
| $Re_x$ | Reynolds number based on streamwise distance | – |
| $Re_\theta$ | Momentum-thickness Reynolds number using freestream or edge viscosity | – |
| $Re_{\delta 2}$ | Momentum-thickness Reynolds number using wall viscosity | – |
| $Re_\tau$ | Friction Reynolds number based on the stated boundary-layer thickness | – |
| $Re_{\mathrm{cell1}}$ | First-cell acoustic Reynolds number, $\rho_w a_w y_1/\mu_w$ | – |
| $T$ | Temperature; $T_w$ is the wall temperature and $T_r$ the recovery temperature | $\mathrm{K}$ |
| $u,\ U$ | Streamwise or wall-tangential velocity; locally stated averaging | $\mathrm{m\,s^{-1}}$ |
| $u_\tau$ | Friction velocity, $\sqrt{\lvert\tau_w\rvert/\rho_w}$ | $\mathrm{m\,s^{-1}}$ |
| $x,\ y$ | Streamwise and wall-normal coordinates | $\mathrm{m}$ |
| **Greek symbols and mesh measures** | | |
| $\gamma$ | Ratio of specific heats | – |
| $\delta$ | Boundary-layer thickness, with the stated edge criterion | $\mathrm{m}$ |
| $\Delta x$ | Streamwise mesh spacing | $\mathrm{m}$ |
| $\Delta y_1$ | Full thickness of the first wall-adjacent cell | $\mathrm{m}$ |
| $h$ | Full cell-centre separation $y_R - y_L$ in the internal-face stencil analysis | $\mathrm{m}$ |
| $\epsilon$ | Relative tolerance on the checked gradient or coefficient-interpolation error | – |
| $\theta$ | Momentum thickness | $\mathrm{m}$ |
| $\kappa,\ \lambda$ | Thermal conductivity | $\mathrm{W\,m^{-1}\,K^{-1}}$ |
| $\mu$ | Dynamic viscosity | $\mathrm{Pa\,s}$ |
| $\nu$ | Kinematic viscosity | $\mathrm{m^2\,s^{-1}}$ |
| $\rho$ | Density | $\mathrm{kg\,m^{-3}}$ |
| $\tau_w$ | Wall shear stress | $\mathrm{Pa}$ |
| $\omega$ | Specific turbulence-dissipation rate in the SST model | $\mathrm{s^{-1}}$ |

For the wall-normal cells used here, the first-cell-centre distance is $y_1 = \delta_n = \Delta y_1/2$. It is distinct from the wall-function sampling height $y_s$, the internal-face position $y_f$ and the cell-centre separation $h$. Source figures that use $z$ as the wall-normal coordinate are identified in their captions.

| Notation | Meaning |
|---|---|
| **Subscripts** | |
| $w$ | Wall value |
| $\infty$ | Freestream value |
| $e,\ \delta$ | Boundary-layer edge value |
| $r$ | Recovery state |
| $\mathrm{lam},\ t,\ T,\ \mathrm{eff}$ | Laminar, turbulent ($t$ or $T$) and effective quantities; $t$ may also denotes a tangential component where stated |
| $1,\ 2$ | Cell or sampling indices; pre-shock and post-shock states in shock relations |
| $s,\ m$ | Wall-function sampling location and Griffin matching location, respectively |
| $f$ | Internal face; finest-grid value in mesh comparisons |
| $L,\ R$ | Neighbouring cell centres in the face stencil |
| $i,\ 99$ | Inlet condition and 99% boundary-layer edge criterion, respectively |
| $\mathrm{ref}$ | Specified reference value |
| $\mathrm{DNS},\ \mathrm{RANS},\ \mathrm{SA}$ | DNS, RANS or Spalart–Allmaras reference |
| $\mathrm{WR},\ \mathrm{WF}$ | Wall-resolved or wall-function result |
| $\mathrm{std},\ \mathrm{target}$ | Standard evaluation or prescribed target value |
| $\mathrm{TL},\ \mathrm{VD}$ | Trettel–Larsson or van Driest transformed quantity |
| $\mathrm{grad},\ \mathrm{interp}$ | Gradient reconstruction or coefficient interpolation |
| $\mathrm{poly},\ \max$ | Polynomial fit or maximum value |
| **Superscripts, accents and operators** | |
| $+$ | Wall units: lengths scaled by $\nu_w/u_\tau$, velocities by $u_\tau$ |
| $y^*$ | Semi-local wall coordinate |
| $^{\mathrm{OF}}$ | Native OpenFOAM output, with its stated sign convention |
| $\overline{(\cdot)}$ | Reynolds mean of a flow quantity |
| $\widetilde{\nu}$ | Spalart–Allmaras working variable |
| $\boldsymbol{u},\ \boldsymbol{n}$ | Bold symbols denote vectors |
| $',\ '',\ '''$ | Derivatives with respect to physical wall-normal distance in the stencil analysis |
| $D_f$ | Discrete internal-face gradient operator |
| $\nabla,\ \partial$ | Spatial gradient and partial differentiation |
| $\partial/\partial n_{\mathrm{out}}$ | Derivative along the outward unit normal of the fluid domain |
| $(\cdot)\vert_w$ | Evaluation at the wall |
| $\lvert\cdot\rvert,\ \mathcal{O}(\cdot)$ | Absolute value and asymptotic order, respectively |

# 1 Introduction

Reliable prediction of aerodynamic heating is essential for the thermal design of hypersonic vehicles. Assessing aerothermal heating along a whole flight trajectory makes computational requirement more severe, especially when fast, iterative engineering is asked for. Wall functions promise a way to reduce the computational effort by reducing the near-wall resolution required by Reynolds-averaged Navier–Stokes (RANS) simulations by inferring wall shear stress and heat-flux from the flow samples away from the wall.

A wall-function CFD simulation depends on the wall-function's closure relations, the RANS field supplying its input and the discretization of the remaining boundary layer. Comparing the resulting heat-flux with a reference therefore cannot, by itself, identify the source of a discrepancy. This thesis assesses these influences through successive diagnostic comparisons.

## 1.1 Research objectives and approach

The objective is to assess the accuracy and numerical requirements of analytical wall functions for hypersonic cold-wall RANS calculations. Three questions guide the investigation:

1. How accurately can existing wall-functions solve wall heat flux and shear stress for hypersonic cold-wall cases?
2. What error sources exist and drive the accuracy of these wall-functions?
3. How can these error sources be isolated and reduced?

The investigation begins with verification and validation of the used OpenFOAM/HiSA numerical framework using canonical shock and boundary layer test cases to assess its capabilities and limitations. Wall-resolved RANS calculations are then compared with direct numerical simulation (DNS) references to assess turbulence and heat-transfer modelling, supported by mesh-convergence studies that establish the required near-wall resolution. Selected analytical wall functions [3, 4] are first tested on prescribed boundary layer profiles to assess their closure relation predictions without feedback from a full CFD flow solver. They are then evaluated in coupled RANS calculations, comparing conventional meshes with meshes adapted to the local boundary layer profiles. Together, these comparisons examine how turbulence modelling, wall-function relations and mesh resolution affect wall shear stress and heat flux, and identify requirements for reliable predictions in the investigated cold-wall flows.

The contribution is a joint assessment of turbulence modelling, wall-function closure and mesh resolution, including a novel boundary layer-informed spacing criterion for internal-face viscous gradients and transport-coefficient interpolation, required for wall-functions in hypersonic cold-wall problems.

## 1.2 Scope and limitations

The main wall-function assessment is restricted to attached, fully turbulent zero-pressure-gradient (ZPG) flat-plate boundary layers. This setting retains the cold-wall thermal-profile challenge while limiting additional influences associated with pressure gradients, separation and complex geometry. Substantial discrepancies encountered even on these ZPG flat-plate canonical cases motivate examining their origins before drawing conclusions about more complex applications, leaving the assessment of complex geometries to the outlook.

The benchmarks use thermally and calorically perfect gases with prescribed wall temperatures or adiabatic wall conditions. Hypersonic cold-wall cases are the principal focus, while adiabatic and lower-Mach comparisons provide context for interpreting their behaviour. The case selection and reference-data limitations are detailed in Section 6.1. Conclusions remain conditional on these cases and the investigated implementation. Pressure gradients, separation, shock–boundary-layer interactions and complex geometries require further wall-function assessment. Conjugate heat transfer motivates the application, but coupled structural thermal response is not investigated here. The separate solver assessment includes additional canonical flows to examine the underlying numerical framework.

Chapter 2 establishes the physical and modelling background. Chapter 3 describes the numerical framework. Chapter 4 presents its verification and validation using canonical test cases. Chapter 5 introduces the wall-function formulations and their implementation. Chapter 6 defines the evaluation methodology with the DNS references used; Chapter 7 presents the diagnostic comparisons of the isolated errors. Chapter 8 draws the conclusions and identifies priorities for further work.

# 2 Physical and modelling background

This chapter reviews compressible turbulent boundary layers, RANS turbulence models and wall-function fundamentals as the basis for assessing hypersonic cold-wall heating predictions. The discussion focuses on the physical mechanisms and modelling assumptions relevant to this thesis, with further details available in the cited literature.

## 2.1 Hypersonic compressible turbulent boundary layers and wall heat-transfer

The boundary layer is referred to as the region of flow near a viscous wall, where the bulk flow needs to decelerate down to the no-slip condition at the wall. Whole books have been dedicated to the boundary layer, i.e. Schlichting's "Boundary-Layer Theory" [1]. Hence this section is kept to a minimum only highlighting certain aspects of the turbulent hypersonic cold-wall boundary layer, which are relevant for the later parts of this work.

### 2.1.1 Difference in boundary layer profiles

Figures 2.1 and 2.2 show the difference in the Reynolds averaged velocity and temperature boundary layer profiles from DNS calculations of a transonic cold-wall (C09) a hypersonic near-adiabatic-wall (M6Tw076) and hypersonic cold-wall (M6Tw025) flow. The non-dimensional wall coordinate is $y^+ = yu_\tau/\nu_w$, where $y$ is the physical wall-normal distance, $u_\tau = \sqrt{\tau_w/\rho_w}$ is the friction velocity and $\nu_w = \mu_w/\rho_w$ is the kinematic viscosity at the wall. Each profile is normalized by its own freestream velocity or temperature. M6Tw076 and M6Tw025 are the near-adiabatic and strongly cooled hypersonic cases of Zhang et al. [5]. C09 is the cooled transonic case at $M_\infty = 0.8$ of Iwatani et al. [6], sampled in the fully turbulent region.

While the velocity profiles of these three cases show a similar trend, the temperature boundary layer profile of the hypersonic cold-wall M6Tw025 profile has a pronounced temperature peak inside of the boundary layer. The near-adiabatic M6Tw076 case has a nearly flat near-wall temperature profile. It retains a small nonzero wall gradient and a weak interior maximum, followed by a gradual decrease towards the freestream. C09 also has a small interior maximum (Figure 2.2 inset), but with much less kinetic energy in the freestream that can be converted to internal energy when the flow is being slowed to the wall, this peak is much less severe. Thus, the strength and location of the thermal maximum distinguish these profiles. The cases differ in Reynolds number and wall thermal condition, so this comparison does not isolate the effect of Mach number or compressibility effects.

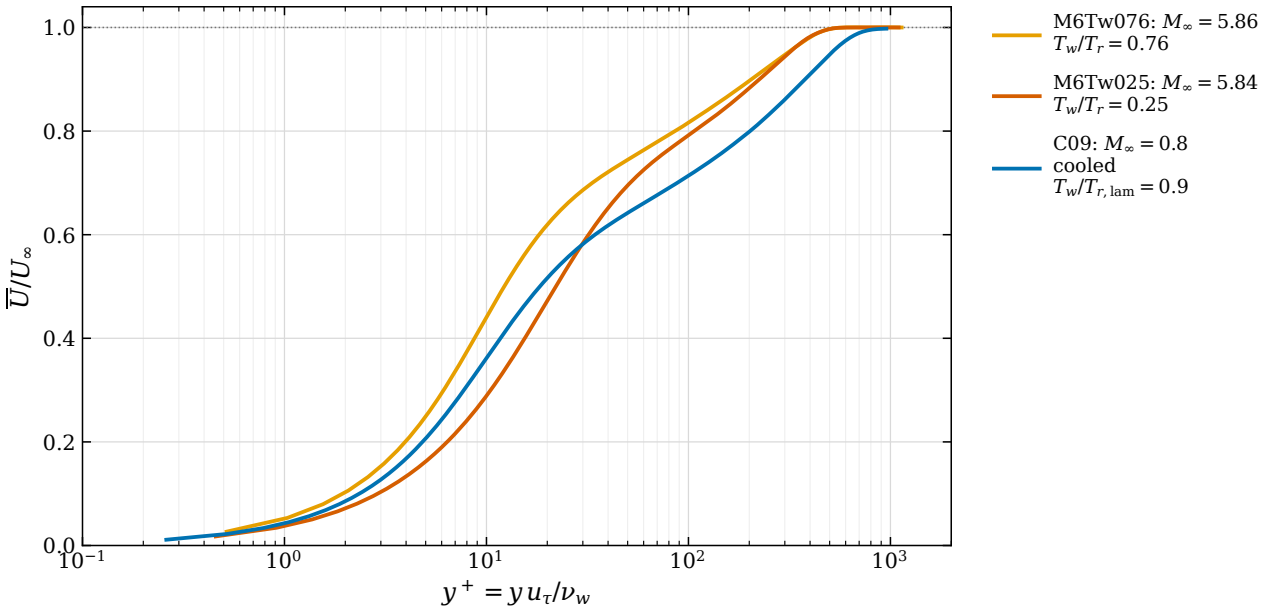



**Figure 2.1** Freestream-normalized Reynolds-mean velocity profiles for the hypersonic M6Tw076 and M6Tw025 cases [5] and the fully turbulent subsonic C09 case [6]. All profiles use their own conventional wall units on the logarithmic $y^+$ axis.

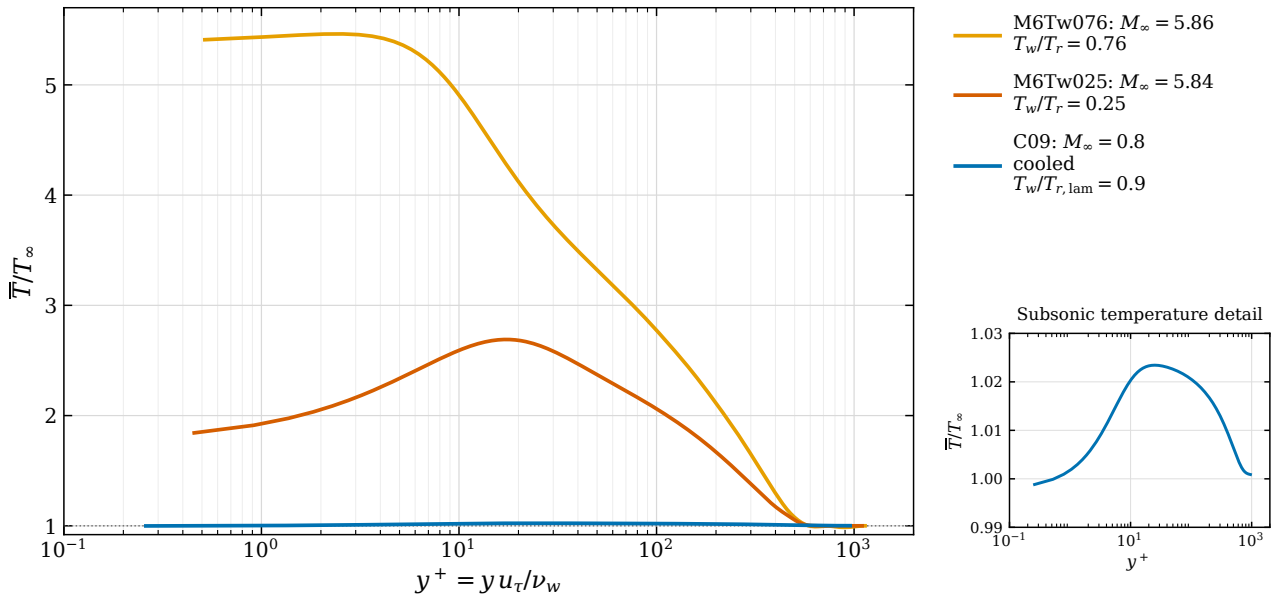



**Figure 2.2** Freestream-normalized Reynolds-mean temperature profiles for the DNS cases in Figure 2.1. The hypersonic cold-wall case M6Tw025 has a pronounced interior temperature peak, $\overline{T}_{\max}/T_\infty \approx 2.69$ at $y^+ \approx 17.5$, followed by a decline towards the freestream. The near-adiabatic M6Tw076 case has only a weak near-wall maximum. The inset resolves the much smaller C09 peak, $\overline{T}_{\max}/T_\infty \approx 1.023$.

### 2.1.2 Compressibility and cold-wall effects and Morkovin's hypothesis

Throughout this thesis the friction Mach number

$$M_\tau = \frac{u_\tau}{a_w}, \qquad u_\tau = \sqrt{\frac{\tau_w}{\rho_w}}, \tag{2.1}$$

where $a_w$ is the speed of sound at the wall, $\tau_w$ the wall shear stress and $\rho_w$ the density at the wall, is being used. The friction Mach number is discussed by Smits and Dussauge [7] and is used here as an indicator for intrinsic compressibility effects, following Hasan et al. [8].

To indicate the strength of the cold-wall effects in the used DNS references either the dimensionless wall heat-transfer rate

$$B_q = \frac{q_w}{\rho_w c_p u_\tau T_w}, \qquad u_\tau = \sqrt{\frac{\tau_w}{\rho_w}}, \tag{2.2}$$

with the wall heat-flux $q_w$, wall shear stress $\tau_w$ and heat capacity at constant pressure $c_p$ is used. Or, as a simpler a-priori assessment, the ratio of isothermal wall temperature to adiabatic recovery temperature $\frac{T_w}{T_r}$, e.g. [5].

A detailed discussion of compressibility effects is beyond the scope of this work, the reader is referred to Pecnik and Hasan [9]. Morkovin's hypothesis [10] provides a physical rationale for extending such models to compressible flow: when intrinsic compressibility effects are sufficiently small, much of the change in turbulence can be accounted for through mean-property variations [11]. The reference DNS studies used for assessing the turbulence models (see Section 6.1) provide substantial support for this approximate similarity under the investigated conditions. Furthermore, the classical strong Reynolds analogy (SRA) introduces additional assumptions concerning temperature fluctuations [12]. Modified forms, including HSRA, accommodate departures from the classical relations and show approximate agreement over much of the outer boundary layer. These findings support assessing standard SA and SST formulations as compressible-flow models, but do not establish the accuracy of their momentum and thermal closures, particularly near strongly cooled walls.

### 2.1.3 Conclusion and implications to thesis work

The pronounced thermal-profile differences in the investigated hypersonic cold-wall case have important implications for wall-function mesh design (Section 7.4.2). The linear-$y^+$ profiles in Figures 7.25 and 7.26 relate these profile shapes to finite wall-normal spacing, because $y^+$ is proportional to physical wall distance within each case. For wall-resolved calculations, the relevant issue is variation of the temperature gradient across the wall-to-first-cell distance: a steep but linear profile would still give an exact secant approximation to the wall derivative. Section 6.2 establishes the first-cell acoustic-Reynolds-number guideline through mesh-convergence studies for the investigated cases.

## 2.2 RANS turbulence modelling

This section introduces the two RANS turbulence models used in this thesis, $k$–$\omega$ SST and Spalart–Allmaras, and discusses their assumptions, limitations and previous assessments. The discussion focuses on aspects relevant to the present work. A comprehensive account of turbulence modelling and its rich historical development is beyond its scope. For a broader introduction, the reader is referred to David Wilcox's *Turbulence Modeling for CFD* [13], which served as a valuable reference throughout this work.

### 2.2.1 Menter's $k$–$\omega$ SST turbulence model

The $k$–$\omega$ shear stress transport (SST) model is widely used in engineering RANS simulations. The OpenFOAM implementation follows the formulation of Menter et al. [14], with updated coefficients from Menter et al. [15]. Its central feature is the blending of the Wilcox $k$–$\omega$ model near the wall with a transformed standard high-Reynolds-number $k$-$\varepsilon$ formulation away from the wall [16].

The validation presented in the principal SST development papers [16, 14, 15] does not include a dedicated supersonic-freestream case. Menter's original study [16] includes a transonic hump case, assessed through the surface-pressure distribution. Wilcox states that his $k$–$\omega$ model can predict shock-separated flows across all Mach numbers without compressibility corrections [13, p. 128 and Section 5.4.7]. However,

the compressible flat-plate comparisons in that discussion assess velocity profiles, leaving heat-transfer accuracy untested by those comparisons.

Modifications of SST for compressible flows have since been proposed [17, 18, 19] and warrant further investigation. Nevertheless, SST remains widely used for hypersonic aerothermal predictions. The comparison by Hoste et al. [20] includes NASA and DLR solvers, Ansys Fluent and STAR-CCM+ and highlights the strong reliance on the non-corrected turbulence models for aerothermal heating. Their hypersonic cone–flare results show substantial limitations in heat-transfer prediction, motivating careful assessment of the model for the flows considered here.

Unless otherwise stated, the inflow turbulent-to-molecular viscosity ratio is set to $\frac{\mu_T}{\mu} = 0.001$, in accordance with the NASA Wind-US recommendation.[1] Preliminary sensitivity checks for the wall-resolved analysis in Section 7.2 found no significant changes in predicted heat-flux or boundary layer profiles when the freestream turbulence intensity $I$ was varied. These checks are not documented in detail here and do not constitute a systematic sensitivity study.

#### The Eddy viscosity concept

The $k$–$\omega$ SST model solves two transport equations: one for the turbulent kinetic energy $k$ and one for the specific dissipation rate $\omega$. Together, these quantities determine the eddy viscosity. In the basic $k$–$\omega$ formulation, the kinematic eddy viscosity is

$$\nu_T = \frac{k}{\omega} = \sqrt{k}\, l_\nu, \qquad l_\nu = \frac{\sqrt{k}}{\omega}. \tag{2.3}$$

Here, $\sqrt{k}$ is a turbulent velocity scale and $l_\nu$ the corresponding viscosity length scale. The SST model modifies this relation through a shear-stress limiter.

Linear eddy-viscosity models use the Boussinesq hypothesis to relate the anisotropic part of the Reynolds-stress tensor to the mean strain rate through a scalar eddy viscosity. This assumption restricts the normal-stress anisotropy that the model can represent. The limitation is particularly relevant where differences in turbulent normal stresses drive secondary flows, as in three-dimensional corners.

It also applies to zero-pressure-gradient flat-plate boundary layers. Under the locally parallel approximation, the Boussinesq hypothesis predicts equal turbulent normal stresses. By contrast, the flat-plate DNS of Zhang et al. used later in this work exhibit unequal normal stresses [5, Figure 13(a,c)]. This discrepancy illustrates a limitation in the representation of the Reynolds-stress tensor with its consequences for mean velocity and wall shear must be assessed separately.

Frahnert and Dallmann [21] question the physical justification of the eddy-viscosity concept and discuss shortcomings of two-equation models even in flat-plate boundary layers. The $k$–$\omega$ formulation likewise relies on modelling assumptions and dimensional reasoning [13, Section 4.3.1]. These limitations motivate assessing the resulting profiles and wall quantities directly, as done in Section 7.2.

### 2.2.2 Spalart–Allmaras turbulence model

The Spalart–Allmaras (SA) model is a widely used one-equation linear eddy-viscosity model that transports a modified turbulent kinematic viscosity. Like SST, it uses the Boussinesq hypothesis and is therefore subject to the limitations of the linear eddy-viscosity assumption discussed above.

The native OpenFOAM implementation cites Spalart and Allmaras [22]. During this work, comparison with the density-conservative formulation given by Allmaras, Johnson and Spalart [23] revealed a discrepancy in its diffusion term: placing density inside the diffusive flux introduces an additional density-gradient contribution for variable-density flow. The correction described in Section 3.5.3 restores the intended continuum

[1] `https://www.grc.nasa.gov/WWW/winddocs/user/keywords/turbulence.html`

diffusion structure. For the investigated ZPG flat-plates, it produced no significant change in heat-flux or boundary layer profiles. Nevertheless, all SA results reported in this thesis use the corrected implementation. The discrepancy and proposed correction were also reported to the OpenFOAM developers.[2]

### 2.2.3 Previous assessments of RANS models for hypersonic boundary layers

The widespread use of SST and SA has prompted numerous assessments. The following selection provides context for their performance in attached hypersonic boundary layers, with particular attention to the distinction between skin-friction and heat-transfer accuracy. It is not a comprehensive review. Studies using the same ZPG flat-plate benchmarks as this thesis are discussed alongside the present results in Section 7.2.4.

Bardina et al. [24] assessed turbulence models for ten flows: seven relatively simple free-shear and ZPG boundary layer flows, and three more complex flows involving separation. Their assessment included grid-refinement studies and sensitivity to initial and boundary conditions. SST performed best overall, followed by SA, although the ranking depended on the case. For the Mach-5 adiabatic-wall ZPG flat-plate, SA gave the closest overall agreement with empirical skin-friction and velocity-profile correlations. The unmodified models performed poorly for the compressible mixing layer, while compressibility corrections improved the predictions, but the authors identified a need for further improvement in free-shear flows. They also noted difficulties in recovering flows downstream of reattachment. These findings illustrate the dependence of model performance on the flow and validation quantity. In particular, agreement for an adiabatic flat-plate does not establish heat-transfer accuracy under wall cooling.

Roy and Blottner [25] considered a Mach-8 cold-wall ZPG flat-plate, for which SA and the Menter $k$–$\omega$ model agreed well with the van Driest II skin-friction relation. This comparison tests consistency with an empirical relation and provides no direct assessment of wall heat-flux. In their Mach-20 re-entry comparison at an altitude of $24.4\,\mathrm{km}$, SA overpredicted the turbulent wall heat-flux by approximately $15\%$, while the Menter model predicted it to within $40\%$ of the measurements. However, the flight measurements were taken on the leeward side at an angle of attack of $0.14^\circ$, whereas the simulations assumed axisymmetric flow at $0^\circ$. This difference limits how confidently the heating discrepancies can be attributed to turbulence modelling alone.

The later review by Roy and Blottner [26] covers experimental validation of RANS models for two-dimensional and axisymmetric hypersonic flows within the calorically perfect-gas regime. Its recommended database spans seven geometries, including flat-plates and cylinders. The review considers 18 one- and two-equation models integrated to the wall, with detailed assessments of six extensively studied models. Across the reviewed cases, surface pressure is generally predicted more successfully than heat-transfer. This conclusion includes shock-interaction flows and should not be read as a finding restricted to attached flat-plates.

#### Van Driest II as an assessment reference

Van Driest II theory [27] provides a skin-friction reference in some of the turbulence-model assessments discussed in Section 2.2, and in the primary paper of the Li et al. wall-function [3], as discussed in Section 5.3. Combined with a Reynolds analogy, it also provides heat-transfer estimates for flat-plates. However, these estimates have documented limitations under strong wall cooling. Hopkins and Inouye [28] reported agreement generally within about 10% for skin-friction and, with specified recovery and Reynolds analogy factors, heat-transfer for the investigated data at $T_w/T_r > 0.3$, but found that the evaluated theories failed to reproduce the data at lower wall-temperature ratios. More recent DNS comparisons likewise reveal deficiencies in van Driest II skin-friction predictions for highly cooled walls [29]. Agreement with these correlations alone therefore does not establish the accuracy of predicted wall shear stress or heat-flux and should be taken cautiously as a sole reference value.

[2] https://gitlab.com/openfoam/core/openfoam/-/work_items/3610

### 2.2.4 Conclusions and implications for this thesis

The literature and the wall-resolved results in Section 7.2 show that SA and SST can struggle to predict wall shear and heat-transfer even in attached hypersonic cold-wall boundary layers. The reviewed studies do not establish a single explanation for these deficiencies or a generally accepted correction. Reliable aerothermal prediction therefore requires further assessment and development of the turbulence and thermal closures, even for these geometrically simple flows.

For the present wall-function assessment, this means that errors in the underlying RANS field must be considered alongside those introduced by the wall closure and its numerical coupling. The thesis first establishes wall-resolved baselines, then evaluates the wall-function closures on prescribed profiles (Section 7.3) and examines their coupled CFD behaviour on different meshes (Section 7.4). These comparisons distinguish the influences of turbulence modelling, wall-function closure and mesh resolution, while allowing for interactions between them. Together, the results identify a need to improve turbulence models, wall-function closures and the associated mesh design.

## 2.3 RANS wall-function fundamentals

This section is a brief intro to the idea of wall-functions and the difference from the universality of the boundary layers between incompressible and compressible turbulent boundary layers. For more details the reader is encourage to read Chapter 8 - "Wall Functions" of Hanjalic and Launder's "Modelling Turbulence in Engineering and the Environment" [30] - this section is far from a complete introduction.

### 2.3.1 Spalding's law of the wall

Wall functions exploit *some* approximate universality of the mean velocity profile in the inner region of smooth-wall turbulent flows. The dimensionless velocity and wall-normal distance are defined as $u^+ = u/u_\tau$ and $y^+ = yu_\tau/\nu$, respectively, where $u$ is the mean tangential velocity, $y$ is the distance from the wall, $\nu$ is the kinematic viscosity, and $u_\tau = \sqrt{\tau_w/\rho}$ is the friction velocity based on the wall shear stress $\tau_w$ and fluid density $\rho$. For incompressible, constant-property flows under approximately equilibrium conditions, profiles expressed in these wall units follow the linear relation $u^+ = y^+$ in the viscous sublayer and a logarithmic relation in the logarithmic region (see Figure 2.3). Spalding [31] combined these limiting behaviours into a single analytical interpolation spanning the viscous sublayer, blending in the buffer layer, and logarithmic region - often referred to as the universal or Spalding's law-of-the-wall. This lack of a universal law in the buffer layer is linked to general caution of using wall-functions in this specific region of $5 \leq y^+ \leq 30$. Figure 2.3 illustrates this blending and compares it with DNS and the two laws relations in the sublayer and log-layer respectively. Beyond the logarithmic region of the inner region of the boundary layer is the wake region, for which again special laws, such as Cole's law-of-the-wake [32].
Such relations provide a basis for wall functions, which infer the wall shear stress from the resolved near-wall velocity and its distance from the wall, reducing the need to resolve the steep near-wall gradients explicitly. Figure 2.4 contrasts a wall-resolved mesh with a conventional wall-function mesh whose first cell centre lies in the logarithmic region. This reduction in near-wall mesh requirement reduces the computational time by reducing the overall cell number and the stiffness of the equation by avoiding the typically thin slivered cells directly at the wall [33]. A formulation spanning all three regions is referred to as adaptive wall-function, and permits the first cell point to lie in different parts of the inner layer.

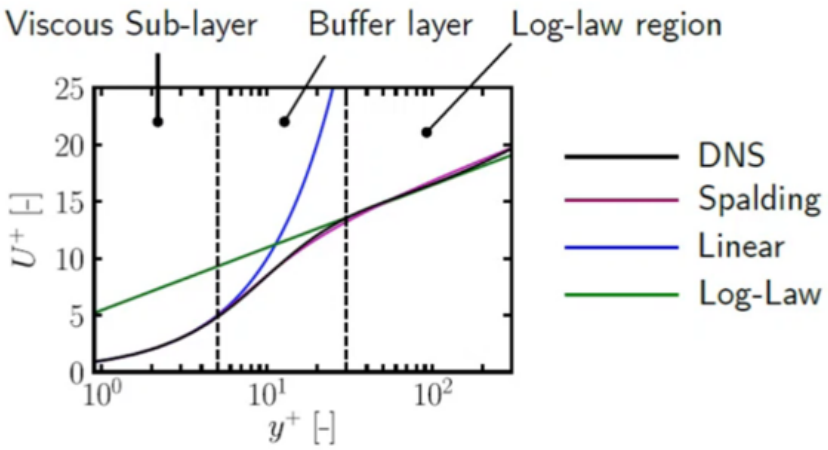



**Figure 2.3** Dimensionless mean velocity profile in wall units: Spalding's law bridges the viscous sublayer, buffer layer and logarithmic region, approaching the linear and logarithmic limits. The DNS profile is shown for comparison. From `https://cfd.blogs.upv.es/turbulence/wall-functions/`[accessed December 2025]

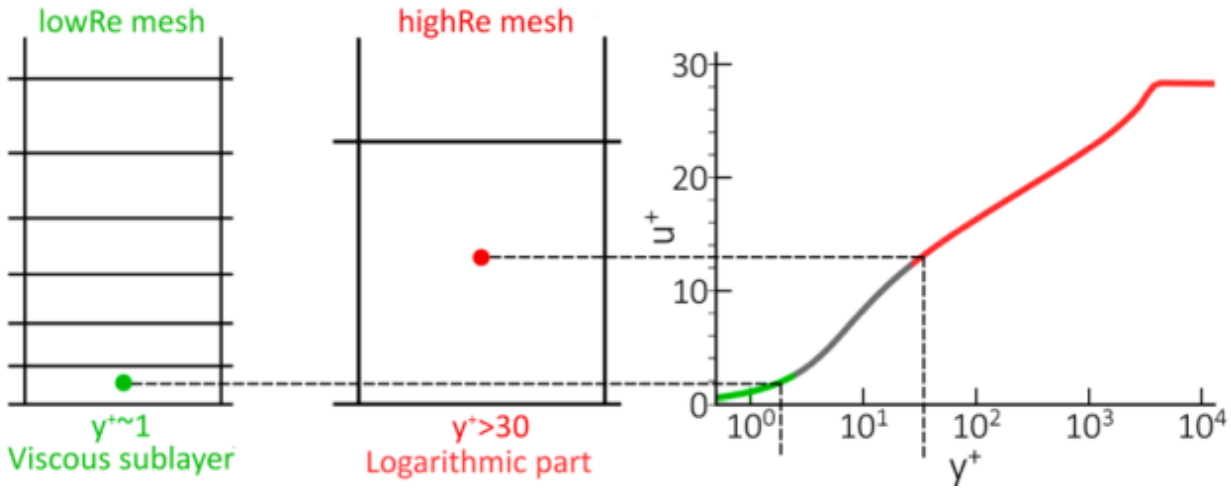



**Figure 2.4** Schematic comparison of a wall-resolved mesh (left), with its first cell centre in the viscous sublayer, and a conventional wall-function mesh (centre), with its first cell centre in the logarithmic region. The profile on the right indicates the corresponding locations in wall units. Reproduced from `https://ansyshelp.ansys.com/public/account/secured?returnurl=/Views/Secured/corp/v251/en/flu_th/section_kbf_dl4_vvb.html`[accessed September 2026]

### 2.3.2 Velocity transformations

The freestream-normalized profiles in Figure 2.1 illustrate differences between the three cases, but do not directly test inner-layer similarity, which concerns $u^+ = u/u_\tau$. Compressible velocity transformations account for mean-property variations to relate the profiles to an approximately common law of the wall. Examples include the van Driest [34] and Trettel–Larsson [35] transformations, and the more recent formulations of Griffin et al. [36] and Hasan et al. [37]. Their accuracy and applicability depend on the flow conditions. The classical van Driest transformation retains $y^+$, whereas the Trettel–Larsson and Griffin transformations use the semi-local coordinate $y^*$. Figure 2.5 illustrates the approximate collapse obtained with the Trettel–Larsson transformation. These transformations are not assessed individually here, the Griffin transformation is assessed indirectly through its role in the wall-function formulation [4]. Further discussion is given by Pecnik and Hasan [9].

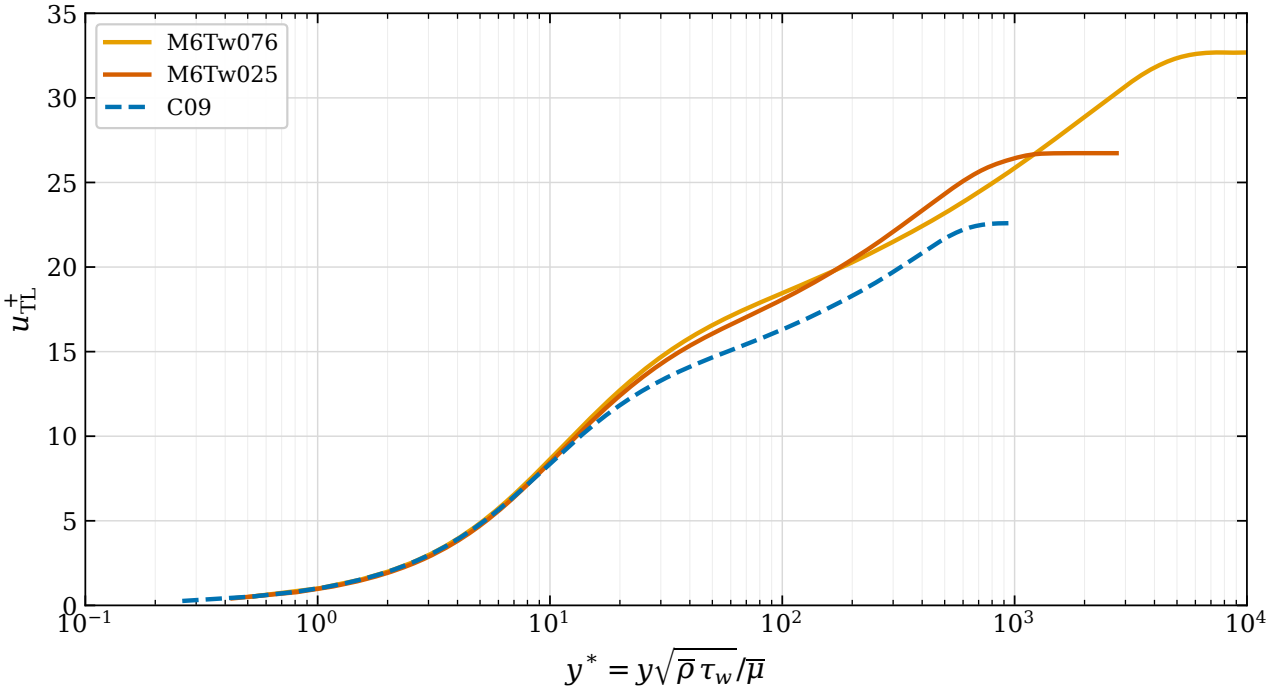



**Figure 2.5** Trettel–Larsson transformed velocity $u^+_{\mathrm{TL}}$ for M6Tw076 and M6Tw025 [5] and the fully turbulent C09 case [6], plotted against the semi-local coordinate $y^* = y\sqrt{\overline{\rho}\tau_w}/\overline{\mu}$. In contrast to the freestream-normalized profiles in Figure 2.1, the transformed profiles approximately collapse onto each other when both velocity and wall distance are transformed.

### 2.3.3 Mean temperature-velocity relations - T(U)

The above mentioned velocity relation, serving as the momentum closure in wall-functions, needs a closure for the energy closure of the transport equations in the boundary layer is necessary. A similarity relation was found in the temperature-velocity relations - $T(U)$. Many of these relations exist, most notably by Crocco–Busemann, the Walz Relations [38], by Duan-Martin [39], or Zhang's Generalized Reynolds Analogy8 [40]. For a calorically perfect gas, the Crocco–Busemann and Walz relations, the Duan-Martin relation, and the mean temperature-velocity relation obtained from the generalized Reynolds analogy share a quadratic form (see Figure 2.6). These analytical relations can be used in wall-functions to close the energy transport in the boundary layer. For the wall-function formulations investigated here, the mean $T(U)$ relation supplies the thermal closure. Its accuracy is therefore an important additional criterion when assessing the RANS turbulence models coupled to these wall functions. It must be considered alongside $U(y)$, $T(y)$, their wall gradients and the resulting wall loads: agreement in $T(U)$ alone does not establish the spatial temperature gradient, since $\mathrm{d}T/\mathrm{d}y = (\mathrm{d}T/\mathrm{d}U)(\mathrm{d}U/\mathrm{d}y)$. Section 7.2.2 applies these complementary assessments.

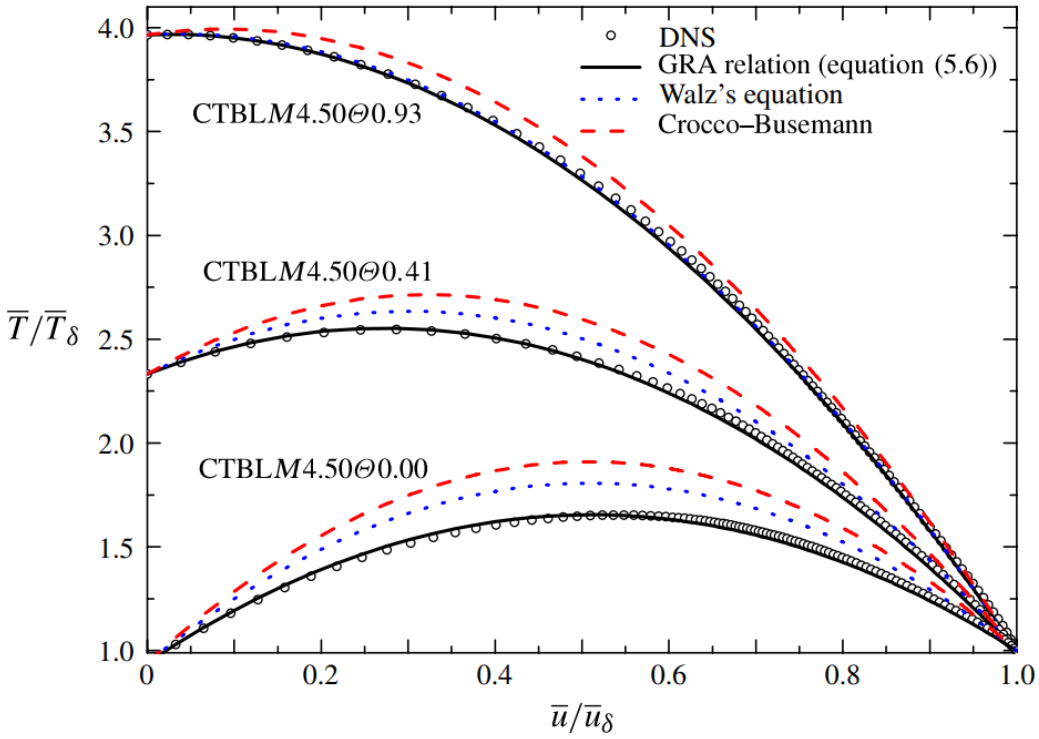



**Figure 2.6** Mean $T(U)$ relations for three Mach-4.5 turbulent flat-plate boundary layers with different wall thermal conditions, comparing the Crocco–Busemann, Walz and generalized Reynolds analogy relations with DNS. Reproduced from Zhang et al. [40].

### 2.3.4 Conclusion and implications to thesis work

The possibility to describe the velocity and temperature boundary layer in universally similar fashion, allows for analytical descriptions of the boundary layer. With such analytical description the near-wall region need not be resolved anymore, but wall-functions based on these laws can be applied on coarse meshes. Conventional coarse wall-function meshes use a large wall-adjacent cell to span the modelled near-wall region (see Figure 2.4). The cold-wall profile shapes introduced in Section 2.1 can make the associated internal-face spacing inadequate, as examined in Section 7.4.2. Special formulations for the velocity and and temperature relations in hypersonic boundary layers are required, two wall-functions designed for these, by Li et al. [3] and Griffin et al. [4], are being investigated in this thesis work (Section 5).

# 3 Numerical framework

This chapter introduces the OpenFOAM/HiSA framework used in this work and summarizes the physical models, numerical methods and implementation changes relevant to aerothermal predictions. Particular attention is given to how wall heat-flux calculation and internal-face transport are handled, which underpin the later wall-resolved mesh assessment and the wall-function mesh-sensitivity analysis. Complete case dictionaries, property coefficients and source modifications are documented in the repository identified in the Data Availability statement.

## 3.1 OpenFOAM and HiSA

### 3.1.1 Open Field Operation And Manipulation - OpenFOAM

**Open F**ield **O**peration **A**nd **M**anipulation (OpenFOAM) is an open-source[1] CFD framework based on the finite volume method (FVM) and developed primarily by OpenCFD Ltd since 2004. This work uses version v2412. A relevant subsequent development is the introduction of Menter's generalized $k$–$\omega$ (GEKO) turbulence model [41] in OpenFOAM v2606. This model should also be considered in future turbulence-model assessments.

OpenFOAM can be adapted to specific user needs through custom libraries that interact with the framework via its application programming interface (API). This allows extensions without modifying the core source code and simplifies upgrades to later versions.

The FVM is the most widely used approach in CFD, adopted by commercial codes such as Ansys Fluent[2] and research codes such as NASA's FUN3D[3] and DLR's TAU[4]. However, access to several solvers designed for compressible flow is restricted, including DLR's TAU, NASA's VULCAN-CFD and NASA's LAURA, the latter developed specifically for aerothermal heating.

OpenFOAM has nevertheless been widely used in research, with applications and validation studies covering a broad range of flow conditions. Examples are documented in the six volumes of the OpenFOAM Journal[5] published to date and in contributions to the annual OpenFOAM Workshops[6].

### 3.1.2 High Speed Aerodynamic solver - HiSA

The **Hi**gh **S**peed **A**erodynamic (HiSA) solver [42] is an open-source[7] solver developed for OpenFOAM to provide efficient and robust simulations of high-speed, compressible viscous flows. It uses a matrix-free, implicit, coupled density-based formulation. The resulting sparse, non-symmetric linear systems are solved using the generalized minimal residual (GMRES) algorithm [43] with lower–upper symmetric Gauss–Seidel (LU-SGS) preconditioning [44]. This work uses HiSA version 1.12.5.

HiSA has been applied to numerous cases.[8] Simulations of supersonic ejector flow [45] and predictions of pressure coefficients on a missile body in supersonic flow [46] showed acceptable agreement with

[1] https://gitlab.com/openfoam/core/openfoam
[2] https://ansys.synopsys.com/products/fluids/ansys-fluent
[3] https://fun3d.larc.nasa.gov/
[4] https://www.dlr.de/en/as/research-and-transfer/software-solutions/aerodynamics/software-tau
[5] https://journal.openfoam.com/
[6] https://openfoamworkshop.org/
[7] https://hisa.gitlab.io/index.html
[8] https://hisa.gitlab.io/doc.html

experiments. The solver has also been used to develop a compressibility-corrected $k$–$\omega$ turbulence model [47]. Its application to separation and heat-flux prediction around a rectangular protrusion in hypersonic flow yielded mixed results [48].

HiSA natively provides the HLLC and AUSM$^+$-up numerical flux schemes. This limited selection is a drawback, given the developments in the AUSM family since the introduction of AUSM$^+$-up in 2006 [49]. This work therefore uses an in-house extension of HiSA that provides an AUSMDV2$^+$-based scheme, as described in Section 3.5.1.

#### Riemann boundary condition

HiSA also provides characteristic, or Riemann-type, boundary conditions for compressible flow. These use a linearised wave analysis to distinguish information entering and leaving the domain. At open boundaries, incoming information follows the prescribed external data, while outgoing information comes from the interior solution, according to the local flow direction and boundary-normal Mach number. The farfield formulation aims to limit artificial reflections and are thus preferably used for supersonic flow [50, Section 2.1]. Ordinary `zeroGradient` conditions simply extrapolate interior values.

At solid walls, `characteristicWallPressure` adjusts pressure using the wall-normal velocity difference between the adjacent cell and the wall. In personal correspondence, Oxtoby [51] explained that this improves robustness when the adjacent flow has a supersonic wall-normal component, for example near corners, where zero-gradient pressure can become unstable. For this HiSA separately overwrites the density stored at the wall face with the value of the first off-wall cell. It was found that especially on non-wall-resolving, coarse meshes, associated with wall-function this simplification results in unphysical wall density readings. This can have implications in post-processing or with wall-function closures that rely on the density at the wall face.

## 3.2 Physical model and thermophysical properties

HiSA solves the compressible conservation equations for mass, momentum and total energy. The turbulent calculations use RANS with the eddy-viscosity and thermal closures introduced in Section 2.2. The fluid is treated as a single-component or frozen-composition ideal gas, without chemical reactions. The boundary conditions and property values are chosen to represent each respective reference case and are stated in their respective section throughout this work.

### 3.2.1 Thermophysical properties in OpenFOAM

OpenFOAM assembles the thermophysical description from separate models for the equation of state, caloric properties, transport properties and energy variable. Their combination is selected in `thermophysicalProperties` and must be registered in the installation. The `perfectGas` equation of state, $p = \rho R T$, can be combined with constant specific heat through `hConst`, or temperature-dependent specific heat through `janaf`. The latter uses piecewise NASA seven-coefficient polynomials to describe the temperature dependent behaviour.

Molecular viscosity $\mu$ and thermal conductivity $\kappa$ must reproduce the reference laws together. Their relation is expressed by the molecular Prandtl number, $Pr = c_p \mu / \kappa$. In particular, the standard `sutherland` transport model couples its viscosity law internally in OpenFOAM to a modified Eucken conductivity relation and it is not possible to independently prescribe a constant $Pr$. This is a limitation of that model combination in OpenFOAM. The additional combinations described in Section 3.5.2 permit separate polynomial descriptions of $\mu(T)$ and $\kappa(T)$. This allows, that for the calorically perfect benchmarks with a constant $Pr$, the conductivity polynomial-coefficients can then be set proportional to the viscosity coefficients so that $Pr$ can remain constant.

### 3.2.2 Turbulent transport

The turbulent Prandtl number $Pr_t$ controls the modelled turbulent heat transport and is distinct from molecular $Pr$. With the gradient-diffusion closure, the effective coefficients are $\mu_{\text{eff}} = \mu + \mu_t$ and $\kappa_{\text{eff}} = \kappa + c_p\mu_t/Pr_t$. Unless stated otherwise, the turbulent calculations use $Pr_t = 0.9$, as specified in Paragraph 6.1.2. These effective coefficients enter both the numerical transport operations below and the later thermal-profile assessment in Section 7.2.3.

## 3.3 Numerical methods

The equations are discretized by a cell-centred finite-volume method. Fluxes across cell faces determine the discrete conservation balances. Table 3.1 summarizes the numerical setup, which unless stated otherwise, is the same numerical method used throughout this work, with the turbulence-equation entries adapted to the selected model. Pseudo-time Courant limits and other convergence controls are adjusted by case. The complete settings are supplied with the simulation setups in the Data Availability statement.

**Table 3.1** Numerical methods in the representative M6Tw025 configuration.

| Operation | Selected method |
|---|---|
| Convective flux | AUSMDV2$^+$-based scheme; dictionary selection `AUSMDV` (Section 3.5.1) |
| Face-state reconstruction | Weighted van Leer limiter (`wVanLeer`) for density, velocity and temperature |
| Cell gradients | Limited Gauss gradients: `cellLimited Gauss linear 1` |
| Momentum and energy diffusion | `Gauss linear compact`: linear coefficient interpolation and HiSA's compact face-normal gradient |
| Other diffusion operators | `Gauss linear corrected`; general face-normal gradient `corrected` |
| Turbulence convection | Bounded first-order upwind for turbulence properties |

### 3.3.1 Convective and diffusive face operations

Convective fluxes use reconstructed states on either side of each face. The reconstruction and its limiting must be distinguished from the face-normal gradients used for viscous stress and thermal diffusion.

For an orthogonal wall-normal stencil, the compact gradient reduces to the slope between neighbouring cell-centre values. The coefficient multiplying that gradient is interpolated to the face with distance weighting

$$D_f\phi = \frac{\phi_R - \phi_L}{h}, \qquad \Gamma_f \simeq \frac{b\Gamma_L + a\Gamma_R}{h}, \qquad a = y_f - y_L, \quad b = y_R - y_f, \quad h = a + b. \tag{3.1}$$

Here $y_L$ and $y_R$ are the neighbouring cell-centre positions, $y_f$ is the face position, and $h = y_R - y_L$ is the full centre separation. The field $\phi$ denotes a transported quantity, such as internal energy, and $\Gamma$ its effective diffusion coefficient. For a perfect gas with constant specific heat, the energy-diffusion term can equivalently be expressed using temperature and effective thermal conductivity. On non-orthogonal meshes, the compact scheme adds a correction using interpolated cell gradients, but in the scope of this work all meshes are 2D, structured, orthogonal grids.

For a face midway between two centres, the secant gradient has a second-order truncation error. Unequal centre-to-face distances introduce an additional asymmetry term. Thus, neither the nominal order nor the wall-function sampling height alone establishes the accuracy of an internal-face flux. The cold-wall profile analysis in Section 7.4.2 examines these gradient and coefficient-interpolation operations and their implications.

## 3.4 Wall heat-flux and shear stress

Wall heat-flux and shear stress depend on local derivatives, so their numerical evaluation is part of the near-wall resolution requirement. This section describes the standard OpenFOAM evaluation of heat-flux and shear stress.

### 3.4.1 Heat-flux and effective thermal transport

The `wallHeatFlux` function object in OpenFOAM evaluates the effective energy diffusivity multiplied by the outward-normal gradient of sensible internal energy in the present HiSA setup. With radiation omitted,

$$q_w^{\mathrm{OF}} = \alpha_{\mathrm{eff},w} \left.\frac{\partial e}{\partial n_{\mathrm{out}}}\right|_w, \qquad \alpha_{\mathrm{eff}} = \frac{c_p}{c_v}(\alpha + \alpha_t), \qquad \alpha = \frac{\kappa}{c_p}. \tag{3.2}$$

Here $e$ denotes sensible internal energy and $c_v$ the specific heat at constant volume. The field `alphat` stores $\alpha_t$, the turbulent contribution to heat transport. It is related to the kinematic eddy viscosity $\nu_t$, stored as `nut`, through the turbulent Prandtl number $Pr_t$:

$$Pr_t = \rho \frac{\nu_t}{\alpha_t}. \tag{3.3}$$

Here $\rho$ is the fluid density.

For constant specific heats, a fixed wall temperature and an orthogonal wall-adjacent cell, the discrete expression for the wall-heat-flux thus becomes

$$q_w^{\mathrm{OF}} = \kappa_{\mathrm{eff},w} \frac{T_w - T_1}{y_1}, \qquad q_w = -q_w^{\mathrm{OF}}. \tag{3.4}$$

The distance $y_1$ is measured from the wall face to the first cell centre.

### 3.4.2 Shear stress and effective viscosity

The `wallShearStress` function object in OpenFOAM calculates wall shear stress from the velocity gradient and effective viscosity. For a flat, stationary no-slip wall, the tangential stress acting on the fluid is approximated by

$$\boldsymbol{\tau}_{w,t}^{\mathrm{OF}} \simeq -\mu_{\mathrm{eff},w} \frac{\boldsymbol{U}_{1,t}}{y_1}, \qquad \mu_{\mathrm{eff}} = \mu + \rho\nu_t. \tag{3.5}$$

It opposes the tangential fluid velocity. From the shear stress $\tau_w$, the skin-friction coefficient $C_f$ is obtained by normalizing this stress with the specified reference dynamic pressure. In wall-resolved simulations, the turbulent contributions vanish at the wall. Wall shear stress and heat-flux therefore depend on the molecular transport coefficients and the resolved temperature and velocity gradients only.

### 3.4.3 Implications for mesh resolution

Equations (3.4) and (3.5) use a wall-to-cell secant to approximate the wall derivative. This one-sided approximation is generally first-order accurate at the wall, even when the internal-face discretization is second-order accurate. A small nominal $y_1^+$ alone does not establish that the secant represents the wall slope. Section 6.2 assesses the required first-cell spacing through mesh-convergence studies.

A wall function instead supplies modelled wall heat-flux and shear stress from a sampled boundary layer state. Their direct substitution into HiSA, described in Section 5.2.3, avoids the wall-gradient approximation above. Transport across internal faces still depends on the numerical methods in Section 3.3, the resulting mesh sensitivity is examined in Section 7.4.2.

## 3.5 Extensions and corrections to the framework

The framework was extended to support the chosen convective flux, reference-consistent molecular transport and additional convergence diagnostics. A separate correction addresses the variable-density SA formulation. The following descriptions identify the purpose and scope of each change. Their source is supplied through the Data Availability statement.

### 3.5.1 HiSA extension with AUSMDV2$^+$

The in-house HiSA extension provides an AUSMDV2$^+$-based flux scheme, selected as `AUSMDV` in the case dictionaries. It blends AUSMD$^+$ and AUSMV$^+$ contributions following Phongthanapanich and Matthujak [52]. The scheme was selected for shock resolution and robustness. The stand-off-distance study in Section 4.2 assesses this implemented variant and documents its remaining carbuncle sensitivity. The later boundary layer assessment separately examines the viscous transport operations introduced in Section 3.3.

### 3.5.2 Additional thermophysical-model combinations

Matching the DNS references requires independent control of the molecular viscosity and conductivity laws. The user library `myThermos` registers OpenFOAM's existing polynomial transport model with the ideal-gas equation of state, sensible internal energy, and either constant or temperature-dependent specific heat. It enables these combinations without changing the underlying property models.

For the calorically perfect benchmarks, seventh-order polynomials represent the prescribed viscosity laws, and proportional conductivity coefficients enforce the selected constant molecular Prandtl number. The fits are checked over the temperatures encountered in each simulation. Section 6.1 documents the reference laws and fit assessment.

### 3.5.3 Density-gradient correction to Spalart–Allmaras

The OpenFOAM v2412 SA implementation places density inside the diffusive flux of its transported working variable. For variable density, this introduces an additional density-gradient term relative to the material-form SA equation [22, 23]. The custom `SpalartAllmarasDensityCorrected` model cancels that term while retaining the other components of the local implementation. Appendix F gives the derivation and source audit. This correction restores the intended continuum diffusion structure and is subsequently used as the only SA model being assessed in Section 7.2.

## 3.6 Convergence assessment and reproducibility

Iterative convergence concerns changes in the solution on a fixed mesh, while mesh convergence concerns changes under spatial refinement. Both must be assessed before differences from a physical reference can be attributed to the flow model.

### 3.6.1 Iterative convergence

The assessment combines residual histories with the evolution of physical outputs. HiSA reports normalized sums of absolute residuals for continuity, momentum and total energy. These are inspected together with the turbulence-equation residuals. For flat-plates, the monitored physical outputs include the wall-averaged heat-flux and the magnitude of the wall-averaged shear vector. It is important to note that besides extrema and averages, it was seen to also be crucial to track the coefficient of variance of the tracked field to assess convergence, as extrema or averages can hide when adjacent cells are still balancing out.

### 3.6.2 Mesh convergence and numerical uncertainty

The mesh studies use the Eça–Hoekstra procedure [53], which fits systematically refined solutions by least squares and accommodates scattered or non-monotonic refinement data. It shares the extrapolation basis of the Grid Convergence Index [54] and the procedure of Celik et al. [55]. The fit estimates the zero-spacing result $\phi_0$ and an uncertainty $U_\mathrm{f}$ for the finest-grid result $\phi_\mathrm{f}$, with a safety factor dependent on the observed convergence behaviour.

Where an independent reference $\phi_\mathrm{ref}$ is available, the signed errors and normalized finest-grid uncertainty are reported as

$$e_j = 100\frac{\phi_j - \phi_\mathrm{ref}}{\phi_\mathrm{ref}}, \quad j \in \{\mathrm{f}, 0\}, \qquad u_\mathrm{f} = 100\frac{U_\mathrm{f}}{|\phi_\mathrm{f}|}. \tag{3.6}$$

The uncertainty estimates spatial-discretization effects and presupposes that iterative and round-off errors are sufficiently small. It does not measure agreement with DNS or account for turbulence-model error. The wall-resolved mesh choices are justified in Section 6.2, with the full convergence results in Appendix B.

### 3.6.3 Reproducibility

The Data Availability statement identifies the repository containing the simulation configurations and supporting data. The case dictionaries specify the numerical schemes, solution controls, physical properties and boundary conditions. The accompanying source modifications and processing scripts document how the reported wall loads and comparisons are obtained. Case-specific departures from the common setup are stated with the corresponding assessment.

# 4 Verification and validation of the numerical framework

Verification assesses whether a mathematical model is implemented correctly and its equations are solved with sufficient numerical accuracy, whereas validation assesses how accurately the model represents physical reality for its intended use.[1] For the OpenFOAM/HiSA framework introduced in Chapter 3, analytical and numerical reference comparisons and convergence studies provide verification checks, while comparisons with measurements provide validation evidence. The assessment covers attached-shock angles and post-shock states (Section 4.1), detached-shock stand-off distances and carbuncle behaviour (Section 4.2), laminar velocity and temperature profiles (Section 4.3), laminar heat-transfer along a flat-plate and through a compression-corner interaction (Section 4.4), and adiabatic turbulent velocity profiles and skin-friction (Section 4.5). Experimental comparisons concern detached-shock stand-off distances and the adiabatic turbulent flat-plate. These checks assess the framework's capabilities and limitations before the cold-wall turbulence-model and wall-function evaluation in Chapter 7.

## 4.1 Shock angle and post-shock conditions

This case of an wedge in hypersonic flow compares the predicted shock angle and downstream state with the weak attached-shock solution of the $\theta$–$\beta$–$M$ and Rankine–Hugoniot relations for a calorically perfect gas with $\gamma = 1.4$. A mesh-refinement study at $M_\infty = 10$, $\theta = 15^\circ$ is complemented by a sweep over $M_\infty = 2, 4, 6, 8, 10$ at wedge angles of $5^\circ$ and $15^\circ$.

The aim is to verify the shock-capturing accuracy of the OpenFOAM/HiSA setup employing the AUSM flux scheme (see Section 3.5.1). The inviscid Euler cases first isolate shock capturing from molecular transport and turbulence modelling. The same meshes are then used with Sutherland viscosity and either Spalart–Allmaras (SA) or $k$–$\omega$ SST. The SA implementation includes the correction described in Section 3.5.3. The inclusion of the turbulence model analysis was chosen because Tian et al [56] showed that especially eddy-viscosity models suffer from serious unphysical increase in turbulence quantities across shock. So this investigation serves to study if this also seriously effects the shock-angle or post-shock conditions. Retaining the slip wall permits a comparison of shock angle and downstream state without the effects of a developing boundary layer. The shared setup and extraction procedure are described below, followed by the Euler and viscous turbulence model results separately.

#### General case setup.

The domain shown in Figure 4.1 extends from $x = 0$ to $0.15\,\mathrm{m}$, with an upper boundary at $y = 0.08\,\mathrm{m}$. The lower wall is horizontal up to the wedge leading edge at $(x, y) = (0.05, 0)\,\mathrm{m}$, followed by the inclined wedge surface. A single cell in the spanwise direction and `empty` front and back boundaries give a two-dimensional calculation. The cases share the freestream and wall settings in Table 4.1. The structured mesh family contains 3 000, 12 000, 48 000 and 192 000 cells, where both x and y directions are refined by a factor of two between successive meshes. A uniform freestream is supplied through characteristic farfield inlet conditions (Section 3.1.2). The rear and upper boundaries use `zeroGradient` extrapolation for velocity, pressure and temperature.

[1] Terminology follows AIAA G-077-1998 as summarized in the NASA NPARC Alliance glossary: `https://www.grc.nasa.gov/www/wind/valid/tutorial/glossary.html`.

**Table 4.1** Freestream and wall conditions shared by the Euler and viscous wedge cases.

| Quantity or condition | Value or specification |
|---|---|
| Mach number $M_\infty$ | $2, 4, 6, 8, 10$; $10$ for mesh refinement |
| Static pressure $p_\infty$ | $101\,325\,\mathrm{Pa}$ |
| Static temperature $T_\infty$ | $300\,\mathrm{K}$ |
| Velocity $\boldsymbol{U}_\infty$ | $(M_\infty a_\infty, 0, 0)$, $a_\infty \simeq 347.250\,\mathrm{m\,s^{-1}}$ |
| Wall velocity and pressure | Impermeable slip; `characteristicWallPressure` |
| Wall temperature | Not prescribed; $\partial T/\partial n = 0$ (`zeroGradient`) |

All cases use the same calorically perfect-gas model (Table 4.2). In the Euler cases, the `const` transport model with $\mu = 0$ gives $\kappa = c_p\mu/Pr = 0$. Together with disabled turbulence modelling, this reduces the flow model to the compressible Euler equations. The viscous cases instead use Sutherland's law, $\mu = A_s T^{3/2}/(T + T_s)$, with $A_s = 1.458 \times 10^{-6}\,\mathrm{Pa\,s\,K^{-1/2}}$ and $T_s = 110.4\,\mathrm{K}$, and the associated OpenFOAM thermal-conductivity model (see Section 3.2.1). The turbulence inflow settings in Table 4.2 follow the recommendations on the Turbulence Modeling Resource website [2][3]. All viscous cases are advanced to 10 000 iterations with post-shock probe histories recorded to assess iterative stability.

**Table 4.2** Gas model, transport models and turbulence inflow settings for the wedge cases.

| Property or model | Specification |
|---|---|
| Gas model | Calorically perfect air (`perfectGas`, `hConst`) |
| Specific heat $c_p$ | $1004.86\,\mathrm{J\,kg^{-1}\,K^{-1}}$ |
| Specific gas constant $R$ | $287.102\,\mathrm{J\,kg^{-1}\,K^{-1}}$, from $W = 28.96\,\mathrm{kg\,kmol^{-1}}$ |
| Heat-capacity ratio $\gamma$ | $c_p/(c_p - R) \simeq 1.4$ |
| Euler transport and turbulence | $\mu = 0$, $\kappa = 0$; turbulence disabled |
| Viscous molecular transport | Sutherland; $\mu_\infty = 1.846 \times 10^{-5}\,\mathrm{Pa\,s}$ |
| Viscous turbulence model | SA or $k$–$\omega$ SST; $Pr_t = 0.9$ |
| SA inflow | $\widetilde{\nu}_\infty/\nu_\infty = 4$ |
| SST inflow | $I = 5.594 \times 10^{-5}$, $\nu_{t,\infty}/\nu_\infty = 10^{-3}$ |

[2] `https://tmbwg.github.io/turbmodels/spalart.html`
[3] `https://tmbwg.github.io/turbmodels/sst.html`

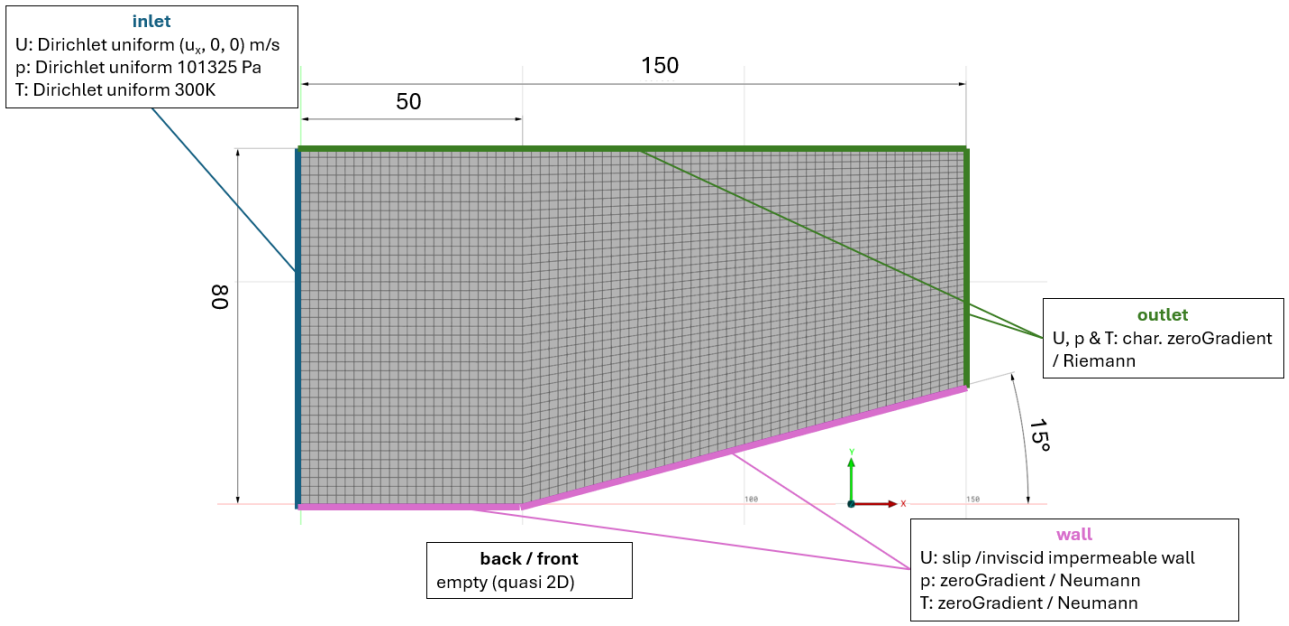
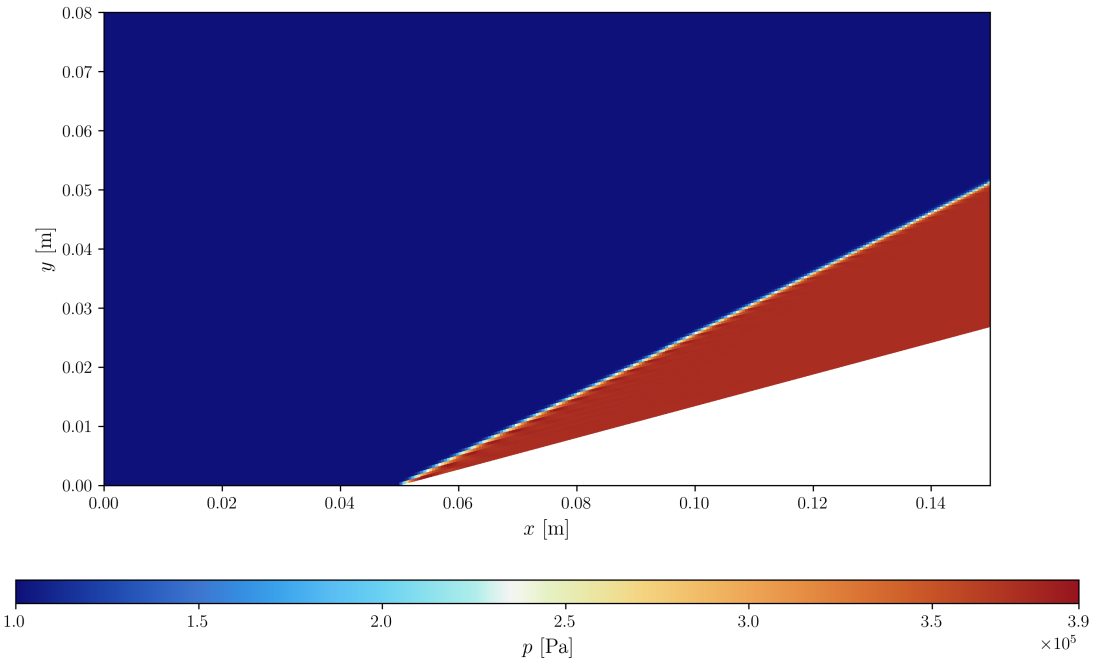


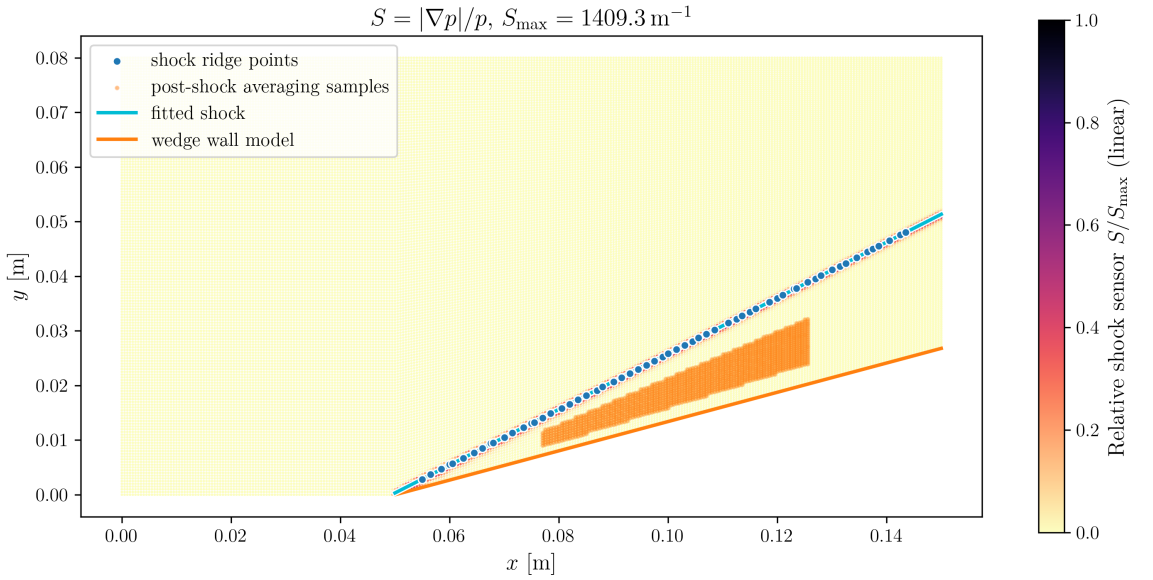

**Figure 4.1** Computational domain and coarse 3 000-cell mesh for the $15^\circ$ wedge, reused for the Euler and viscous cases.

### Extraction of verification quantities.

The extraction procedure described in the following is applied to the exported CFD fields of all three flow models - Euler and both viscous turbulence model cases. Figure 4.2 illustrates it for an Euler case. Local maxima of the normalized pressure-gradient magnitude $S = |\nabla p|/p$ are selected in successive streamwise strips to identify the shock ridge. A robust straight-line fit through these points gives the shock angle. Pressure, temperature and Mach number are averaged over an interior region (orange region in Figure 4.2) between the wedge wall and the fitted shock, with density evaluated pointwise from $\rho = p/(RT)$ before averaging. The subsequently reported quantity-ratios of pre- and post-shock conditions use the inflow quantities as pre-shock reference. On coarse meshes, finite grid spacing and the numerically broadened shock limit the precision of the ridge localization. This is an extraction sensitivity, rather than an independently established uncertainty bound.




(a) Static pressure field.




(b) Shock-ridge fit and post-shock sampling.

**Figure 4.2** Euler CFD result and extraction procedure for $M_\infty = 4$, wedge angle $\theta = 15°$ on the 48 000-cell mesh. Top: static pressure field, showing the pressure rise across the oblique shock. Bottom: the blue points with white outlines denote the detected shock ridge, the cyan line shows its straight-line fit. The orange interior samples define the post-shock region where the post-shock quantities are averaged across. The orange line represents the wedge wall. The background shows the exported point values of $S/S_{\max}$ on a linear scale, where $S = |\nabla p|/p$ and $S_{\max}$ is its maximum over the field. Dark colours identify the resolved shock transition, weak gradients appear pale.

For the following assessment the signed relative errors of a quantity $\phi$from its analytical reference $\phi_{\mathrm{ref}}$ is defined as

$$e(\phi) = 100\,\frac{\phi - \phi_{\mathrm{ref}}}{\phi_{\mathrm{ref}}} \quad [\%]. \tag{4.1}$$

### 4.1.1 Inviscid Euler flow

In the following the investigated $\theta = 5°$ and $15°$ wedges are simulated using inviscid Euler formulation. First the mesh convergence is discussed, then the results across the various Mach numbers quantitativly and qualitativly assessed.

#### Mesh sensitivity and numerical uncertainty.

The four structured meshes give refinement ratios of $h/h_{\mathrm{f}} = 8, 4, 2, 1$, where the subscript $\mathrm{f}$ denotes the finest mesh. Figure 4.3 shows the mesh convergence with mesh refinement. A general reduction in error towards the finest mesh can be observed, with relative errors of the finest mesh for all quantities being well below 1% (Table 4.3 $e_f$).

**Table 4.3** Finest-grid accuracy and Eça–Hoekstra extrapolated mesh uncertainty for $M_\infty = 10$, $\theta = 15°$. The CFD values are from the 192 000-cell mesh. The errors $e_{\mathrm{f}}$ and $e_0$ refer to the finest-grid and extrapolated values, respectively. $u_f$ is the Eça–Hoekstra estimated relative uncertainty of the finest-grid result due to mesh resolution.

| Quantity | Theory | CFD | $e_{\mathrm{f}}$ [%] | $u_{\mathrm{f}}$ [%] | $e_0$ [%] |
|---|---|---|---|---|---|
| $\beta$ [°] | 19.94158 | 19.92027 | –0.107 | 0.251 | –0.018 |
| $p_2/p_1$ | 13.40437 | 13.41125 | +0.051 | 0.576 | +0.298 |
| $T_2/T_1$ | 3.19434 | 3.19485 | +0.016 | 2.413 | –0.787 |
| $\rho_2/\rho_1$ | 4.19628 | 4.19781 | +0.036 | 0.785 | +0.539 |
| $M_2$ | 5.27926 | 5.27879 | –0.009 | 1.411 | +0.461 |

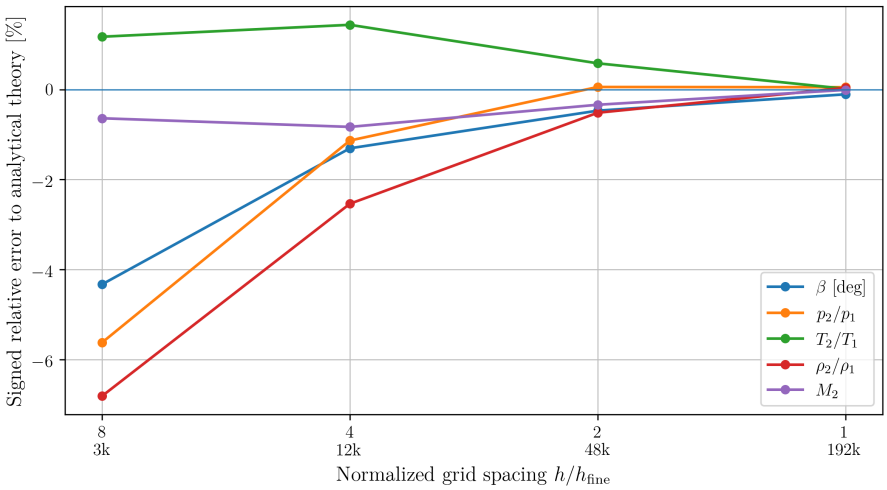



**Figure 4.3** Euler mesh dependence of the shock angle $\beta$ and post-shock conditions for $M_\infty = 10$ and wedge angle $\theta = 15°$. Signed relative error between CFD and theory $e_f$ according to Equation (4.1).

The Eça–Hoekstra assessment (Section 3.6) supports treating the finest mesh as sufficiently converged for this shock-verification study (Table 4.3). Both finest-grid $e_{\mathrm{f}}$ and extrapolated $e_0$ results agree closely with theory, and all analytical values lie within the estimated uncertainty intervals $u_{\mathrm{f}}$. The larger uncertainties for the temperature ratio and downstream Mach number nevertheless show that small finest-grid errors do not imply negligible mesh sensitivity. Sensitivity to the actual extraction, such as the shock angle resolution on the spatial discretization, or from the averaging, limits the interpretation of these intervals as bounds on discretization error alone.

**Accuracy across Mach number and wedge angle.**

The Mach-number sweep uses the 48 000-cell mesh for each of the two wedge angles, as it was shown to be sufficiently mesh converged and allows quick computation. Table 4.4 reports the largest absolute error across the five Mach numbers for each quantity, while Figures 4.4 and 4.5 retain the variation across individual cases. The maximum shock-angle differences are $0.162°$ and $0.094°$ for the $5°$ and $15°$ wedges, respectively. All four post-shock state quantities remain well within $1\%$ of their analytical references throughout the sweep. This supports the accuracy of the simulation setup to resolve oblique shocks.

**Table 4.4** Maximum Euler error magnitudes on the 48 000-cell meshes over $M_\infty = 2, 4, 6, 8, 10$. For the shock angle, the reported value is $\max|\beta_{\mathrm{CFD}} - \beta_{\mathrm{ref}}|$ in degrees, for the post-shock quantities, it is $\max|e(\phi)|$ in percent according to Eq. (4.1). The maxima occur at $M_\infty = 10$, except for the pressure ratio at $\theta = 15°$, whose maximum occurs at $M_\infty = 8$.

| Quantity | Unit | $\theta = 5°$ | $\theta = 15°$ |
|---|---|---|---|
| Shock angle | ° | 0.162 | 0.094 |
| Pressure ratio | % | 0.326 | 0.066 |
| Temperature ratio | % | 0.253 | 0.582 |
| Density ratio | % | 0.579 | 0.516 |
| Downstream Mach number | % | 0.132 | 0.339 |

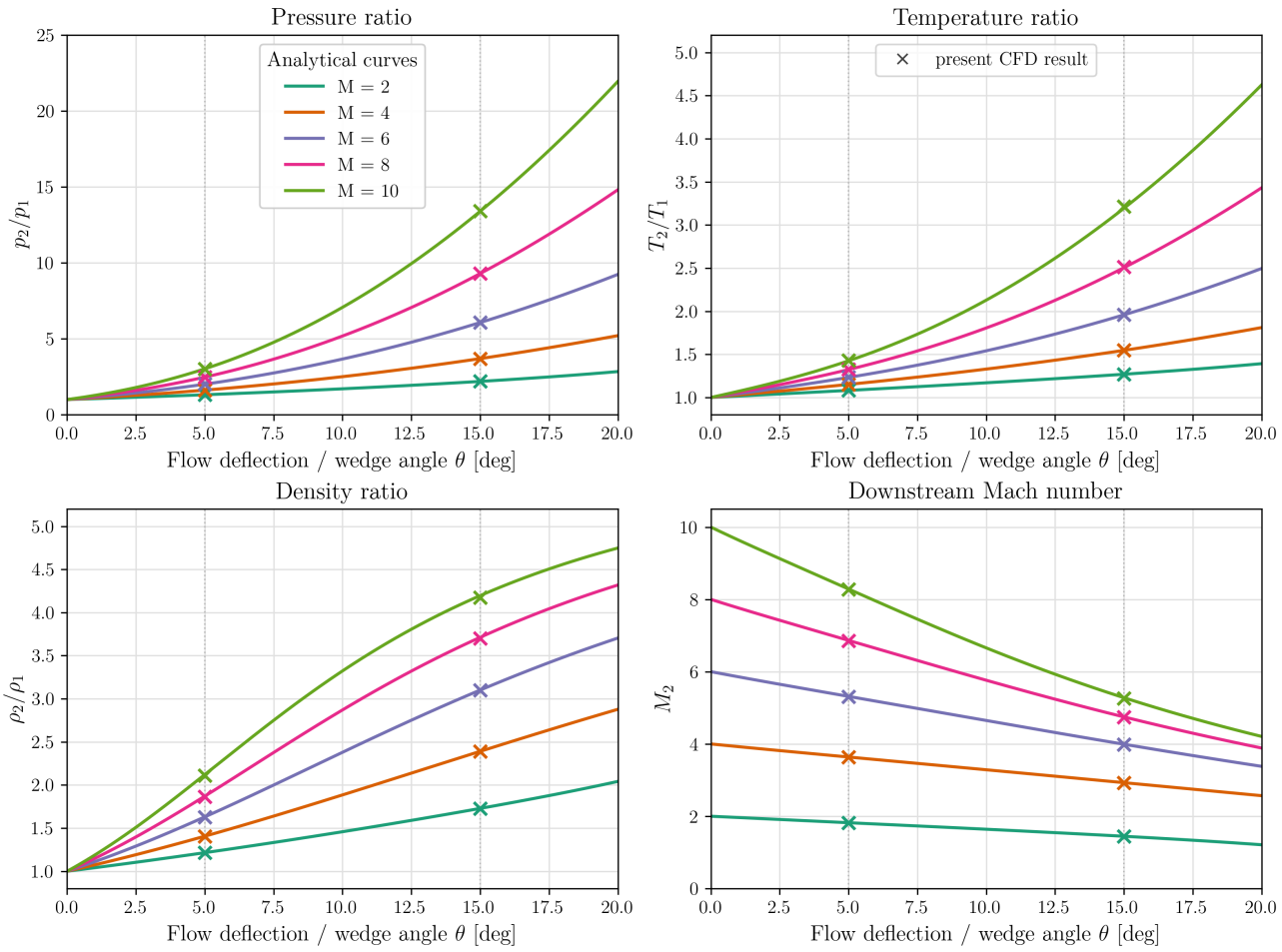



**Figure 4.4** Euler CFD post-shock conditions on the 48 000-cell meshes for $5°$ and $15°$ wedges at freestream Mach numbers from 2 to 10, compared with the weak oblique-shock solution of the Rankine–Hugoniot relations.

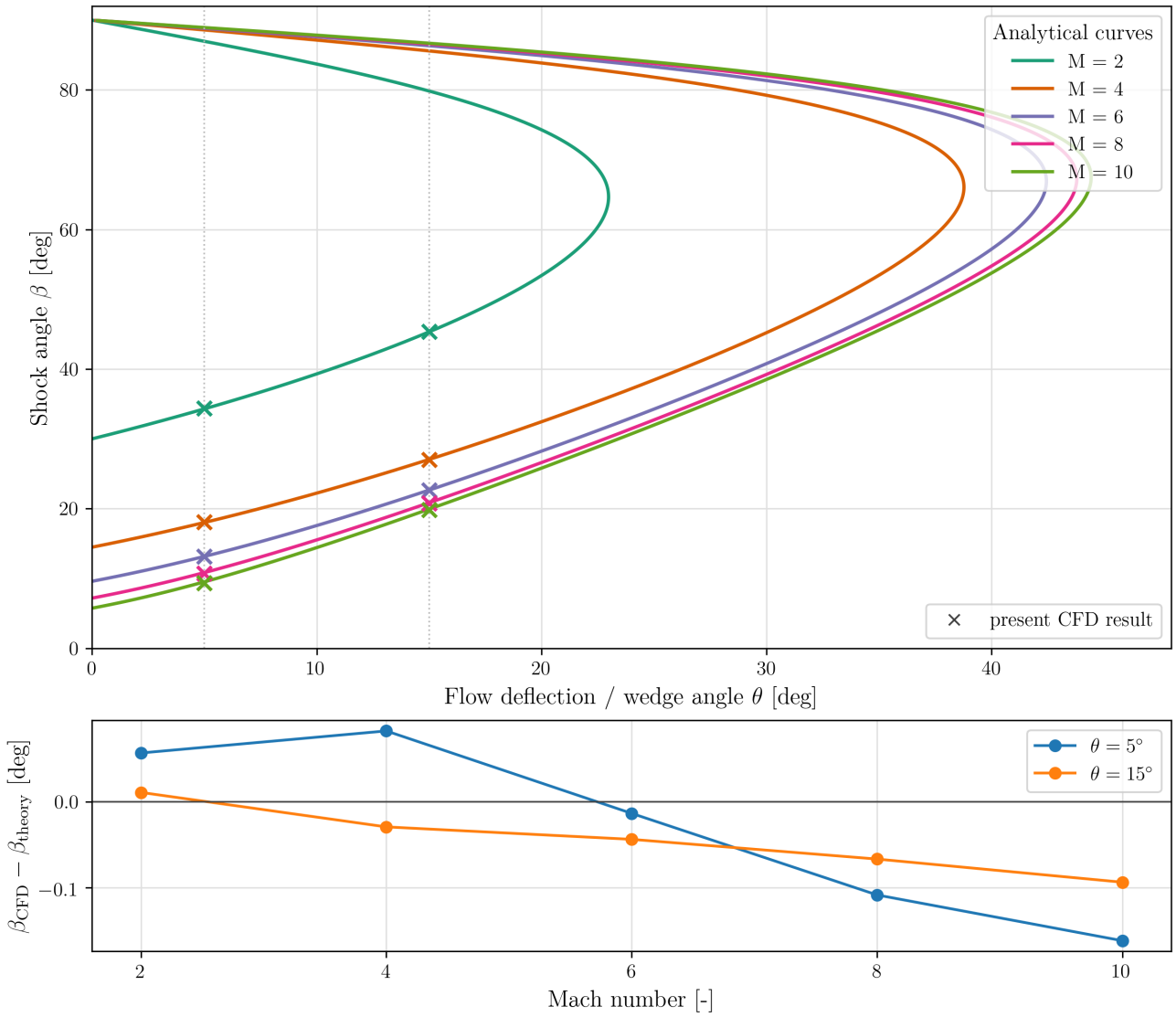



**Figure 4.5** Euler CFD shock angles on the 48 000-cell meshes for $5°$ and $15°$ wedges at freestream Mach numbers from 2 to 10. Top: comparison with the analytical $\theta$–$\beta$–$M$ curves. Bottom: signed angular difference $\beta_{\mathrm{CFD}} - \beta_{\mathrm{ref}}$ in degrees.

### 4.1.2 Viscous flow with SA and SST

In the following the investigated $\theta = 5°$ and $15°$ wedges are simulated including turbulence modelling formulation. First the mesh convergence is discussed, then the results across the various Mach numbers quantitativly and qualitativly assessed.

#### Mesh sensitivity.

Figure 4.6 shows the four-grid results for both viscous setups, Spalart–Allmaras and $k$–$\omega$ SST. Their refinement trends closely follow those of the previous discussed Euler simulation where non-monotonic mesh convergence behaviour therefore persists with molecular and turbulent transport. On the finest mesh, the four post-shock state discrepancies are all below $0.032\%$ in magnitude (Table 4.5). Both models give $\beta = 19.91692°$, a difference of $-0.02465°$ from theory, compared with $-0.02131°$ for Euler. Thus the same finest, 192 000-cell mesh shows equally good mesh convergence behaviour with viscousity and turbulence modelled into the simulation. At both post-shock probes, the eight refinement runs show no recorded variation over the last 1 000 iterations at the output precision of approximately six significant digits. This supports iterative stability at those points.

**Table 4.5** Measured values and signed relative discrepancies on the 192 000-cell mesh at $M_\infty = 10$, $\theta = 15°$. Both models use Sutherland viscosity and the analytical references in Table 4.3. The errors follow Eq. (4.1), including for $\beta$.

| | SA | | SST | |
|---|---|---|---|---|
| Quantity | CFD | $e_\mathrm{f}$ [%] | CFD | $e_\mathrm{f}$ [%] |
| $\beta$ [°] | 19.91692 | –0.124 | 19.91692 | –0.124 |
| $p_2/p_1$ | 13.40856 | +0.031 | 13.40865 | +0.032 |
| $T_2/T_1$ | 3.19463 | +0.009 | 3.19458 | +0.007 |
| $\rho_2/\rho_1$ | 4.19726 | +0.023 | 4.19734 | +0.025 |
| $M_2$ | 5.27901 | –0.005 | 5.27905 | –0.004 |

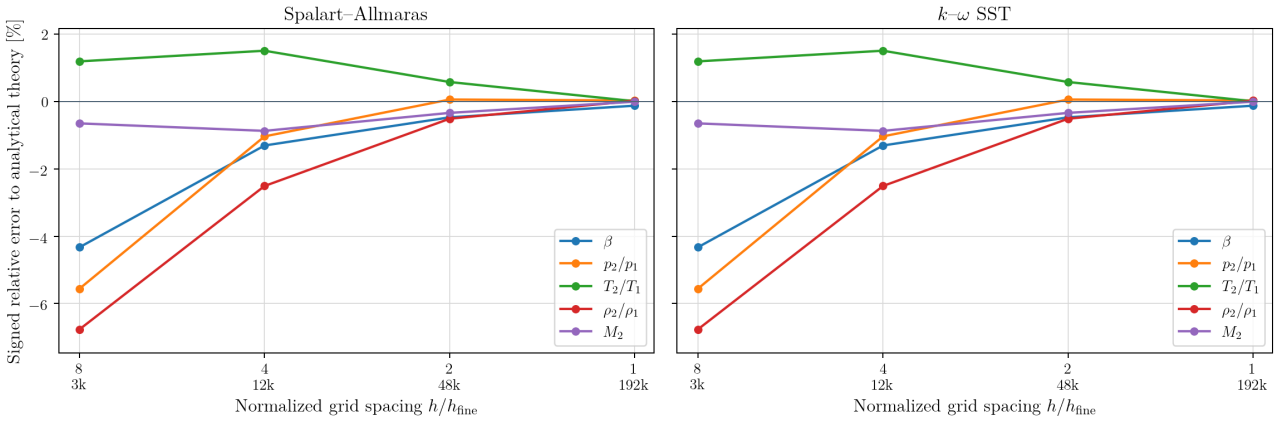



**Figure 4.6** Mesh dependence with Sutherland viscosity and Spalart–Allmaras (left) or SST (right) at $M_\infty = 10$, $\theta = 15°$. Both panels use the same scale and show signed relative discrepancies according to Eq. (4.1).

#### Accuracy across Mach number and wedge angle.

The twenty viscous sweep cases repeat the same conditions as used for the Euler runs on the same 48 000-cell meshes. Figures 4.7 and 4.9 compare the downstream states with the analytical weak oblique-shock solution for SA and SST, respectively. Figures 4.8 and 4.10 show the corresponding shock angles against the $\theta$–$\beta$–$M$ theory, together with their signed angular discrepancies. Table 4.6 summarizes their maximum discrepancies. For both turbulence models, the maximum shock-angle differences round to $0.162°$ at $5°$ and

$0.094°$ at $15°$. Every post-shock state discrepancy is well below $1\%$ in magnitude for either models. These maxima are close to the Euler values, with only small changes across the sweep. Across all twenty runs, the largest recorded peak-to-peak variation over the last 1 000 iterations is $0.0135\%$, in pressure at one probe for the SST case at $M_\infty = 2$, $\theta = 15°$, hence full iterative convergence is ensured for these results. All in all, this shows, that both SA and SST models are equally applicable in capturing and resolving oblique shocks and recovering their post-shock states in the used numerical setup.

**Table 4.6** Maximum error magnitudes over $M_\infty = 2, 4, 6, 8, 10$ on the 48 000-cell meshes. Shock-angle errors are absolute differences in degrees, state errors are $\max |e(\phi)|$ in percent according to Eq. (4.1). Both turbulence models use Sutherland viscosity.

| | | SA | | SST | |
|---|---|---|---|---|---|
| Quantity | Unit | $5°$ | $15°$ | $5°$ | $15°$ |
| Shock angle | ° | 0.162 | 0.094 | 0.162 | 0.094 |
| Pressure ratio | % | 0.324 | 0.064 | 0.324 | 0.064 |
| Temperature ratio | % | 0.247 | 0.580 | 0.246 | 0.580 |
| Density ratio | % | 0.570 | 0.511 | 0.570 | 0.511 |
| Downstream Mach number | % | 0.129 | 0.337 | 0.129 | 0.337 |

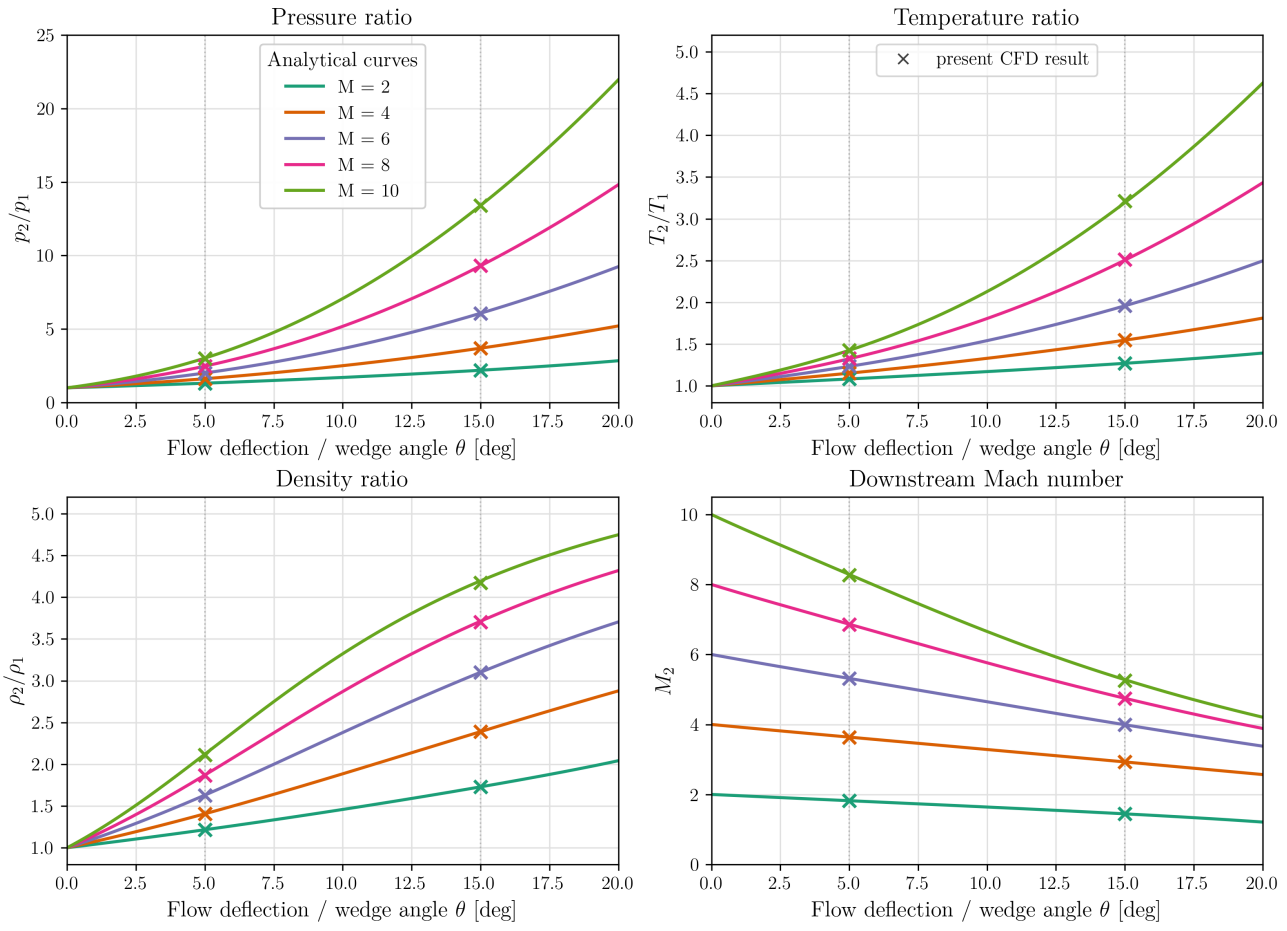



**Figure 4.7** Post-shock conditions with Spalart–Allmaras and Sutherland viscosity on the 48 000-cell meshes for $5°$ and $15°$ wedges at $M_\infty = 2, 4, 6, 8, 10$. Crosses denote the CFD averages, curves show the weak oblique-shock solution of the Rankine–Hugoniot relations. The four panels show pressure ratio, temperature ratio, density ratio and downstream Mach number.

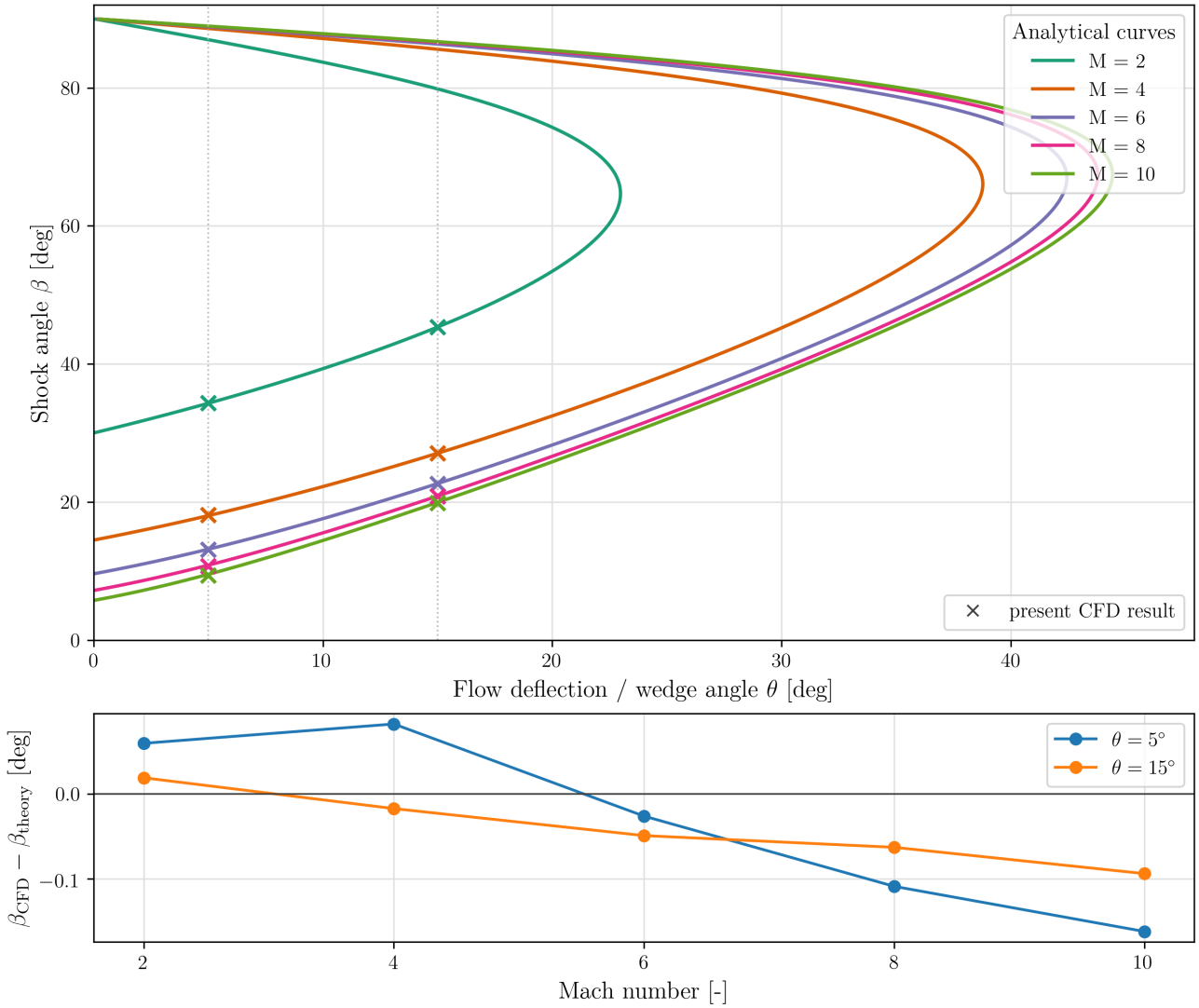



**Figure 4.8** Shock angles with Spalart–Allmaras and Sutherland viscosity on the 48 000-cell meshes for $5°$ and $15°$ wedges at $M_\infty = 2, 4, 6, 8, 10$. Top: CFD fitted angles (crosses) and the analytical $\theta$–$\beta$–$M$ curves. Bottom: signed angular difference $\beta_{\mathrm{CFD}} - \beta_{\mathrm{ref}}$ in degrees, using the weak-shock reference.

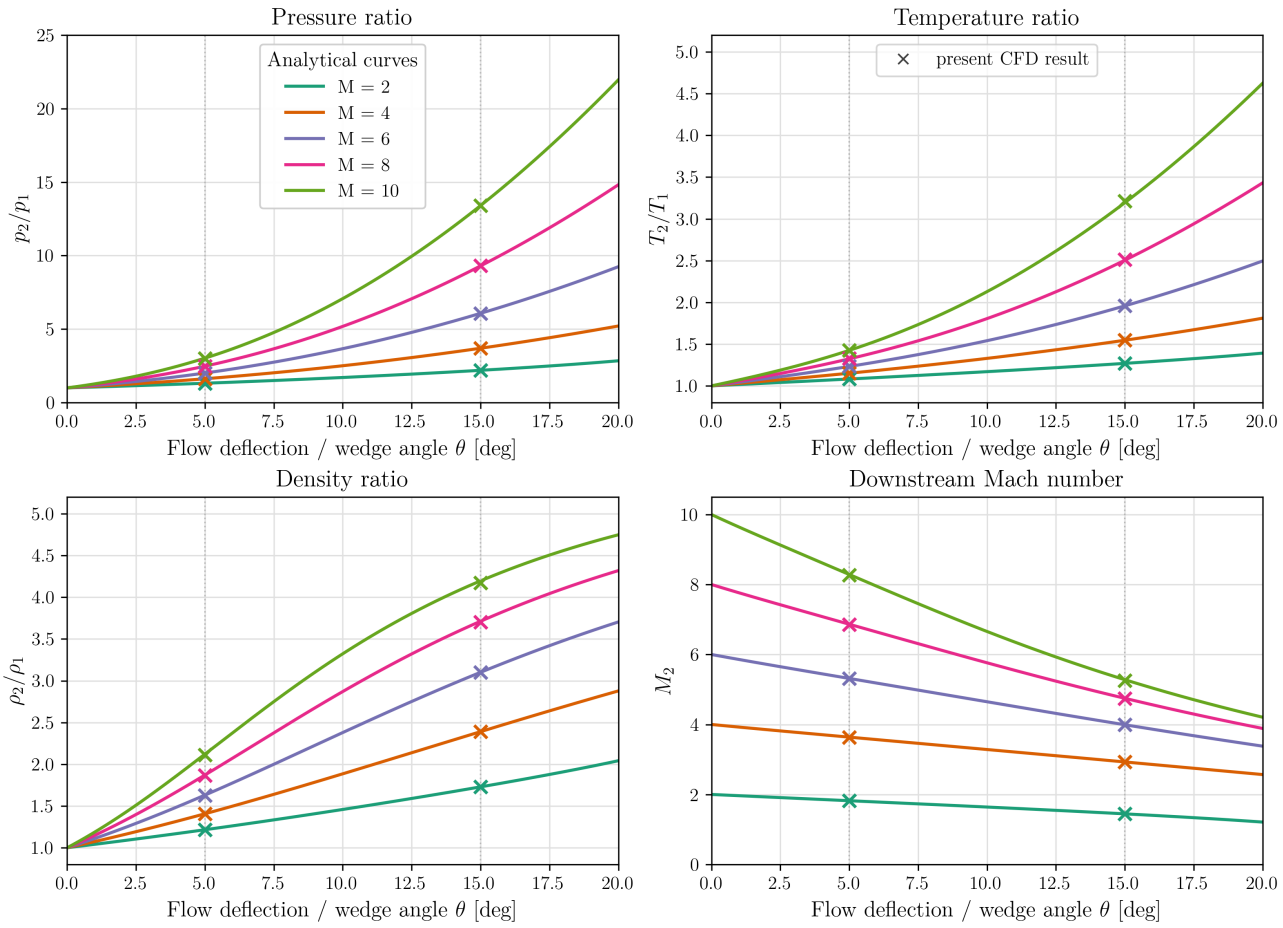



**Figure 4.9** Post-shock conditions with $k$–$\omega$ SST and Sutherland viscosity on the 48 000-cell meshes for $5°$ and $15°$ wedges at $M_\infty = 2, 4, 6, 8, 10$. Crosses denote the CFD averages, curves show the weak oblique-shock solution of the Rankine–Hugoniot relations. The four panels show pressure ratio, temperature ratio, density ratio and downstream Mach number.

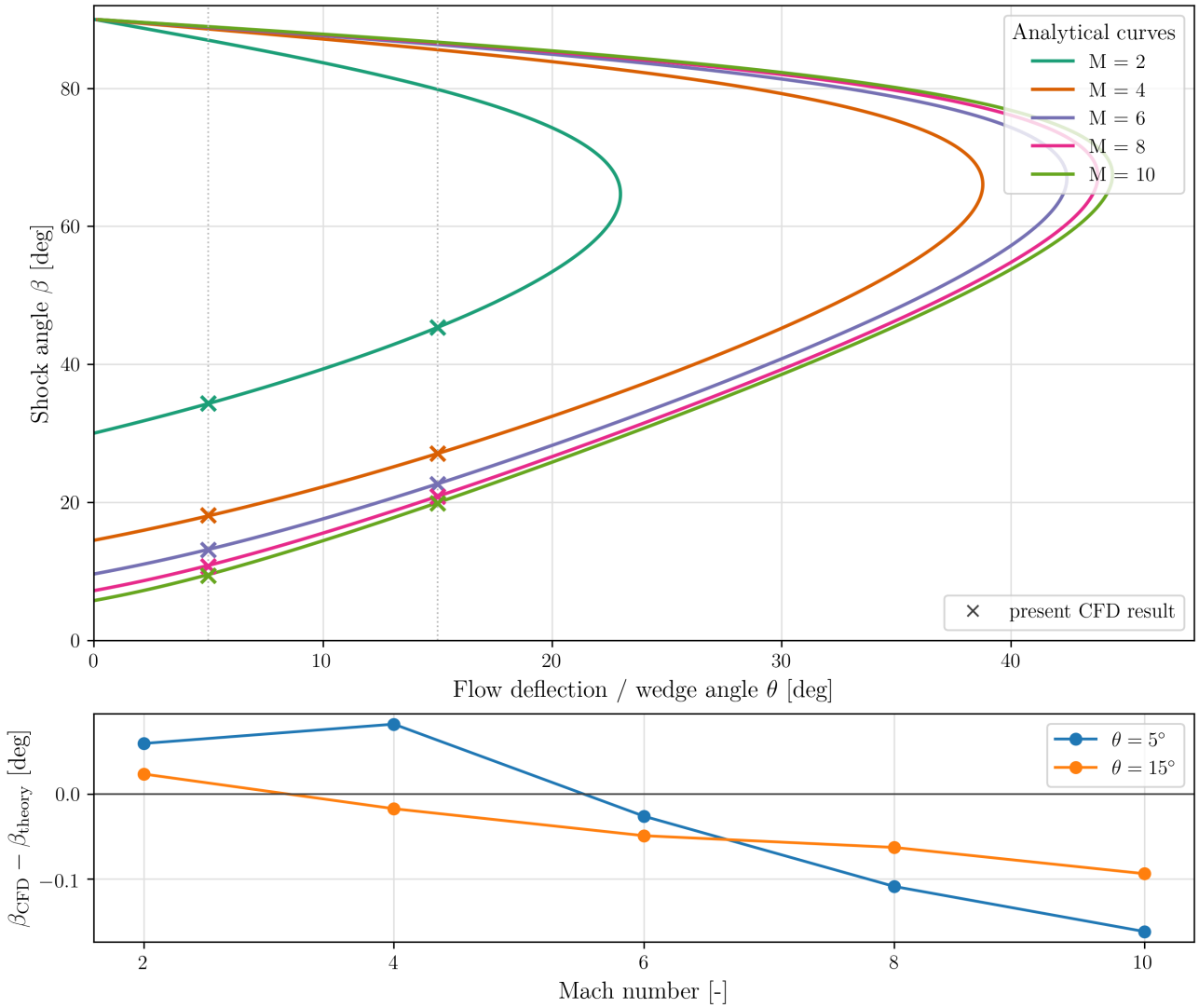



**Figure 4.10** Shock angles with $k$–$\omega$ SST and Sutherland viscosity on the 48 000-cell meshes for $5°$ and $15°$ wedges at $M_\infty = 2, 4, 6, 8, 10$. Top: CFD fitted angles (crosses) and the analytical $\theta$–$\beta$–$M$ curves. Bottom: signed angular difference $\beta_{\mathrm{CFD}} - \beta_{\mathrm{ref}}$ in degrees, using the weak-shock reference.

### 4.1.3 Conclusion and findings

While there may be excess turbulent kinetic energy production associated with the turbulence models as found in [56], this evaluation shows no major negative impact on the resolved oblique shock angle or recovered post-shock conditions. It can be confidentally concluded that the in Chapter 3 outlined numerics, especially regarding the flux scheme, and the Spalart–Allmaras and $k$–$\omega$ SST turbulence model, is capable of solving oblique shock cases correctly.

## 4.2 Shock stand-off distance and carbuncle behaviour

The detached-shock cases in this section compare the predicted shock stand-off distance with the measurements of Kim [57] on a cylinder and the wedge experiments of Hornung and Smith [58]. The aim is to assess the accuracy of the OpenFOAM/HiSA setup with the implemented AUSMDV2$^+$ (see Section 3.5.1) for whether or not it can resolve strong, detached shocks physically correctly. In this study also the so-called carbuncle behaviour was encountered and it's cause, effect and inevitably is discussed. It was chosen to run this validation laminar, to isolate the underlying numerics from potentially influence of unphysical behaviour of turbulence models at stagnation points [59] or across shocks [56]. Both Kim and Hornung and Smith, have tested and found their experiments to be suitably approximated as 2D phenomenons.

**General case setup.**

The Kim configuration consists of a circular cylinder with diameter $D = 10\,\mathrm{mm}$ in air at $M_\infty = 6$. The Hornung–Smith configuration comprises a symmetrical sharp wedge in argon at $M_\infty = 16$, with flank length $w = 51\,\mathrm{mm}$ and face inclination $\delta_w = 46.2°$ to the freestream. Figure 4.11 and Figure 4.12 show the respective geometries and the coarsest meshes investigated respectively alongside the applied boundary conditions. For the Hornung–Smith sharp wedge mesh it was elected to retain the refinement of the structured mesh specifically around the zone along the stagnation line where the shock is expected to sit, rather than systematically refineing the whole domain. It was elected to not fit either meshes in a shock-curved manner, as to keep subsequent mesh refinement steps more easily controllable by simple halving the x and y related spacing. For the radially structured mesh for the Kim et al study, it was ensured, that the stagnation line runs through cell centres and not between cells, so to avoid any unnecessary interpolation to resolve the pressure along the stagnation line, as was necessary for the H-grid of the Hornung–Smith mesh.

Both configurations are represented by two-dimensional, laminar, viscous calculations. The air in the experiment by Kim et al is modelled as thermally perfect, calorically imperfect gases using ideal-gas equation of state with temperature-dependent JANAF caloric properties. The Argon in the experiment by Hornung–Smith et al, is specifically chosen to avoid dissociation and explicitly stated to behave as a perfect gas, thus modelled as thermally and calorically perfect, with constant $\gamma = 5/3$ [58]. Both cases employ Sutherland viscosity, $\mu = A_s T^{3/2}/(T + T_s)$, with constants specific to the respective gas. The prescribed simulation conditions are summarized in detail in Table 4.7. The shock stand-off distance is measured along the stagnation line from the upstream cylinder surface or wedge tip and normalized by $D$ or $w$, respectively, following the definitions used in the reference studies.

**Table 4.7** Geometry, prescribed freestream and wall conditions, and gas-model parameters of the shock stand-off simulations. The dimensional inputs describe the present calculations, with their relation to the experimental references is discussed in the text below.

| Quantity | Kim cylinder | Hornung–Smith wedge |
|---|---|---|
| Characteristic dimension | $D = 10\,\mathrm{mm}$ | $w = 51\,\mathrm{mm}$ |
| Wedge face inclination $\delta_w$ [°] | — | 46.2 |
| Mach number $M_\infty$ | 6 | 16 |
| Static pressure $p_\infty$ [Pa] | 327 | 261 |
| Static temperature $T_\infty$ [K] | 34 | 57 |
| Heat-capacity ratio $\gamma$ | $c_p/(c_p - R)$ | $5/3$ |
| Wall temperature $T_w$ [K] | 295 | 295 |
| Gas | Air $c_p(T)$ | Argon $c_p$ |
| Specific heat $c_p$ | JANAF $c_p(T)$ | Constant $c_p = 520.3$ |
| Specific gas constant $R$ [$\mathrm{J\,kg^{-1}\,K^{-1}}$] | 287.102 | 208.132 |
| Sutherland $A_s$ [$\mathrm{Pa\,s\,K^{-1/2}}$] | $1.458 \times 10^{-6}$ | $1.96 \times 10^{-6}$ |
| Sutherland $T_s$ [K] | 110.4 | 144 |

The prescribed wall temperature is a modelling assumption for both cases, as neither explicitly states it. Kim et al do not specify the particular dimensional freestream state $p_\infty = 327\,\mathrm{Pa}$, $T_\infty = 34\,\mathrm{K}$ in their own publication, but the values must rather be taken from the associated shock tunnel evaluation study in [60, Run 434, Channel 2 (sonic throat at inlet, AR = 28)]. Hornung and Smith report the nominal conditions in the test section for the Argon case as $M_\infty = 16$, $T_\infty = 57\,\mathrm{K}$ and $\rho_\infty = 2.2e^{-5}\,\mathrm{g\,cm^{-3}}$ in their Table 1. The specific gas constant $R$ and heat-capacity ratio $\gamma$ are taken from the NIST database[4]. The Sutherland viscosity law constants for Argon and air are taken from [61, Table 1-1], with a given accuracy of +-2% within the temperature of 120 - 1500K and 170 - 1900K respectively. No source of a farther validated Sutherland could be found. With the extremely low inflow temperatures this results in a certain extrapolation in the viscosity and thus modelling uncertainty for reproducing the experiments. Additionally these extremely low temperatures and pressures for the cold-air experiment by Kim et al are well below the air saturation line (≈45 K under these reservoir conditions for nitrogen [62]), but Tamaki argues supersaturation over the 1 ms test time prevented any observable condensation under same test conditions in the same facility, but a different experiment[60].

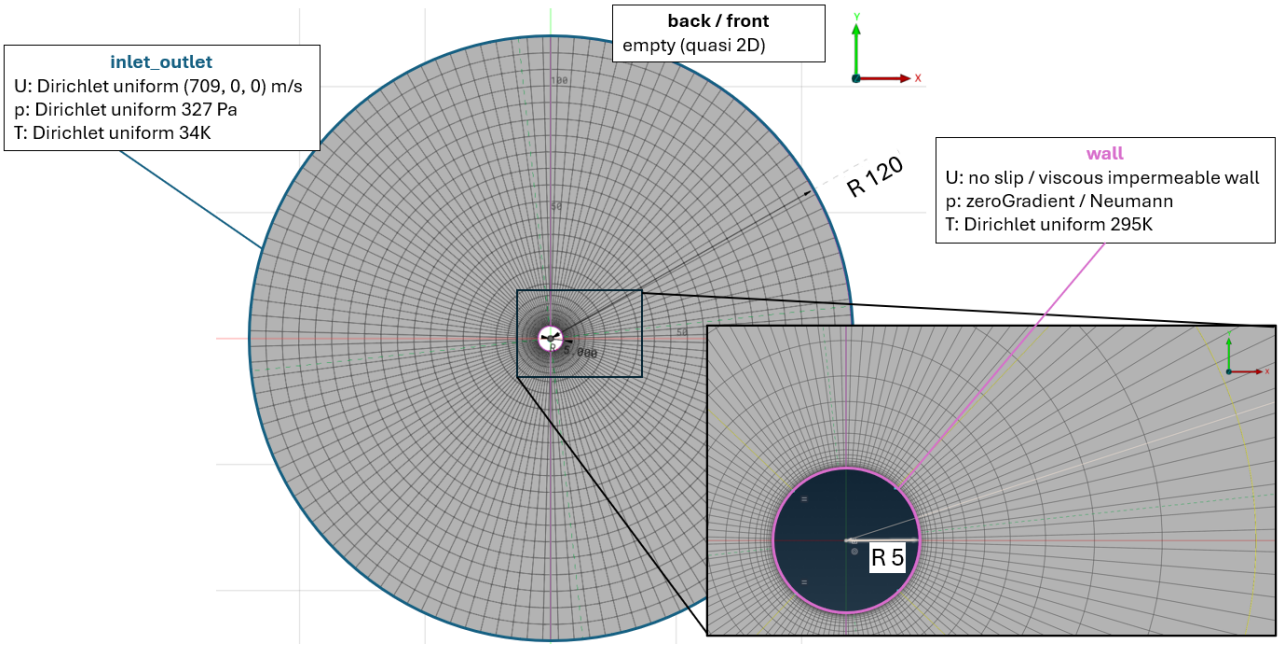



**Figure 4.11** Computational domain and coarse 6 000-cell radially structured mesh for the $D = 10\,\mathrm{mm}$ cylinder of Kim et al.

[4] https://webbook.nist.gov/cgi/inchi/InChI%3D1S/Ar [accessed May 20, 2026]

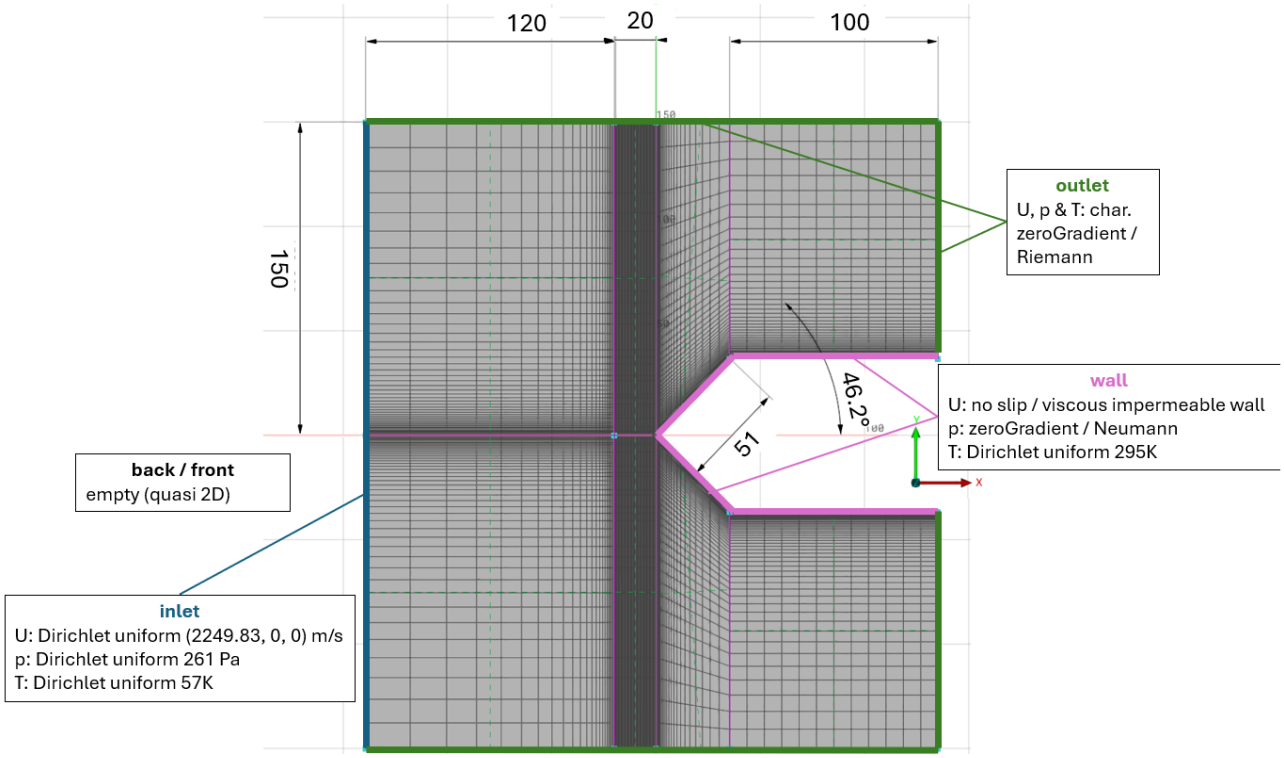
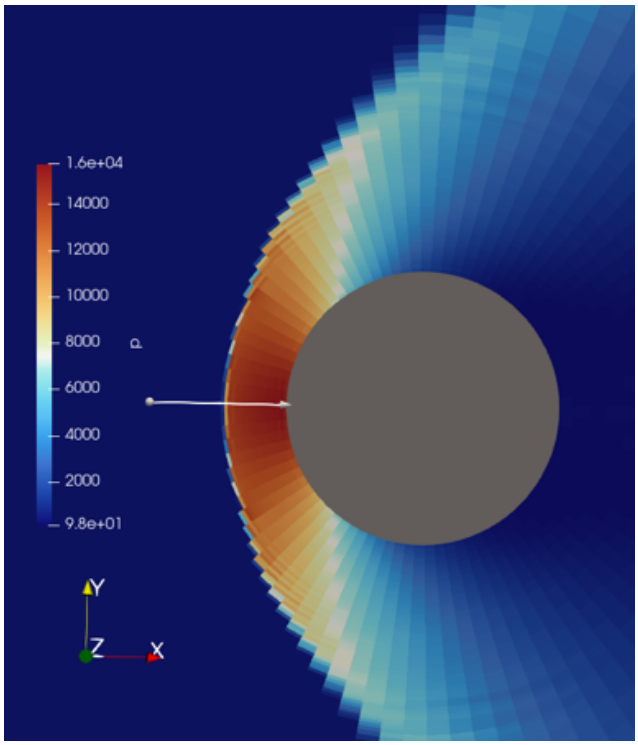


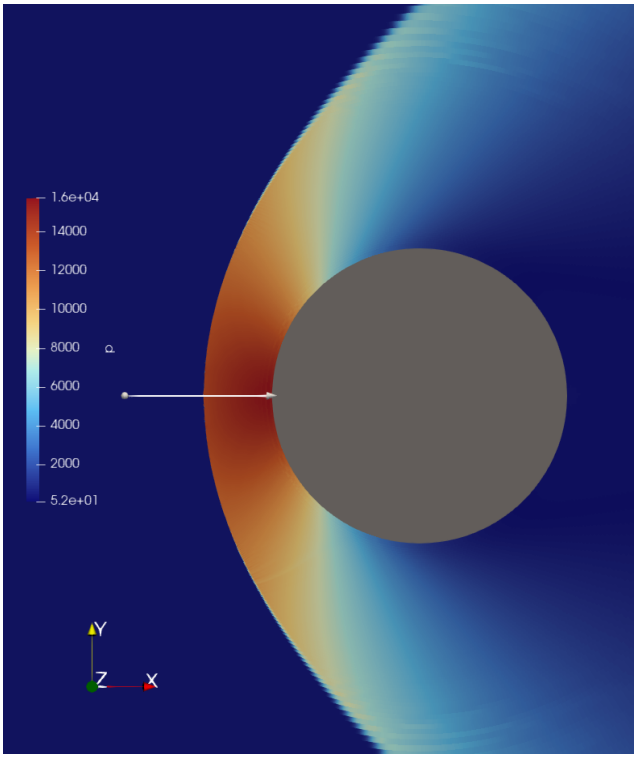

**Figure 4.12** Computational domain and coarse 16 000-cell structured mesh for the $\delta_w = 46.2°$ wedge of Hornung and Smith, with local refinement along the stagnation line where the shock is expected to sit.

### Extraction of verification quantities.

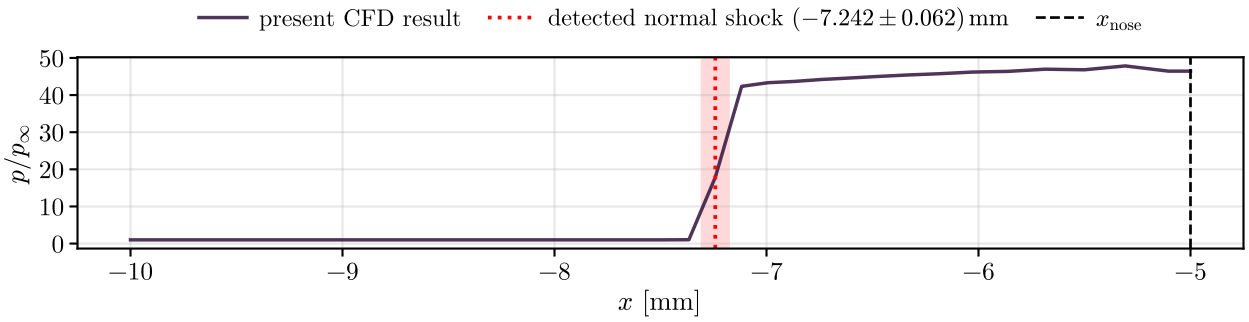
The shock is located using a pressure-gradient maximum, following the same principle as in Section 4.1. Here, fonly the one-dimensional pressure along the stagnation line is exported and analysed, rather than the whole pressure field. The shock position is associated to the cell centre sample location with the largest streamwise pressure gradient $\mathrm{d}p/\mathrm{d}x$. Its distance to the cylinder nose or wedge tip gives the stand-off distance, sometimes given normalized by $D$ or $w$, respectively. Half the local cell width is reported as uncertainty due to spatial discretization. Figure 4.13 illustrates the procedure for the Kim cylinder case on the coarsest mesh ($6\,000$ cells, panels a and c) and the finest mesh ($2.3$ million cells, panels b and d) - this also illustrates the strength of the low-dissipative AUSMDV2$^+$ which is able to capture the shock with a singular intermediate cell.



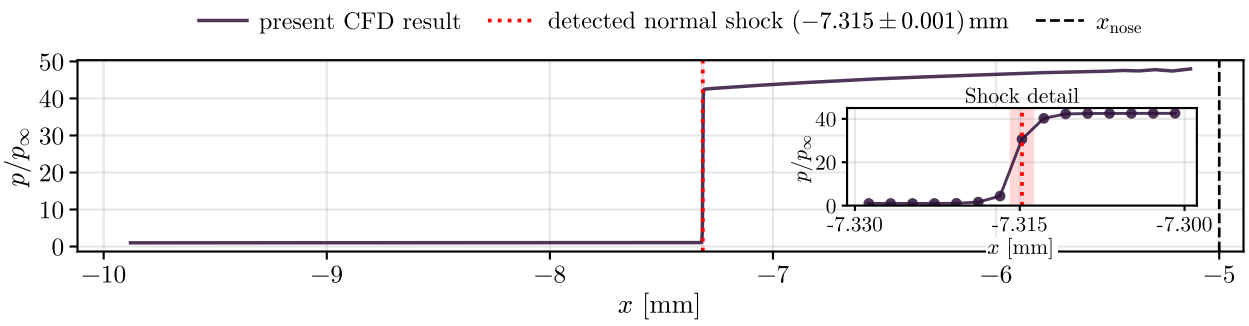

(a) Coarsest mesh: static pressure field [Pa].




(b) Finest mesh: static pressure field [Pa].




(c) Coarsest mesh: stagnation-line pressure profile.




(d) Finest mesh: stagnation-line pressure profile.

**Figure 4.13** Shock stand-off extraction for the Kim cylinder at $M_\infty = 6$. Top (a, b): static pressure fields in $\mathrm{Pa}$ and sampling lines for the coarsest (approximately $6\,000$ cells) and finest (approximately $2.3$ million cells) meshes. Below (c, d): corresponding normalized pressure profiles against physical $x$-position in millimetres. The red dotted line marks the detected shock; the black dashed line marks the cylinder nose. The shaded band spans half the local cell width on either side of the detected shock, indicating spatial extraction uncertainty. The inset in (d) enlarges the shock rise; dots mark exported cell-centre samples. The same extraction procedure was applied to the Hornung–Smith cases.

**Mesh convergence on shock stand-off distance.**

The Kim mesh family comprises seven refinement levels, ranging from approximately $6\,000$ to $2.3$ million cells. Refinement retains the radial mesh structure and the outward growth of cell widths. The Hornung–Smith family comprises six levels, labelled $16\,000$, $65\,000$, $260\,000$, $1$, $4$ and $8$ million equivalent cells. The finer Hornung–Smith calculations use a reduced domain with a symmetry boundary to limit the computational cost. Their labels indicate the equivalent refinement level of the original mesh family, rather than the actual number of cells, for example, the $1$-million-equivalent case contains $\approx 500\,000$ cells.

Figures 4.14 and 4.15 compare the extracted stand-off distances with the experimental references and show the signed relative error to the experimental measurement. The reference values are $d_{\mathrm{ref}} = 2.280\,\mathrm{mm}$ for the Kim cylinder and $\Delta_{\mathrm{ref}} = 12.189\,\mathrm{mm}$ for the Hornung–Smith wedge, corresponding to $d/D = 0.228$ and $\Delta/w = 0.239$, respectively. The error bars represent the extraction resolution associated with half the local cell width as described above in the extraction methodology.

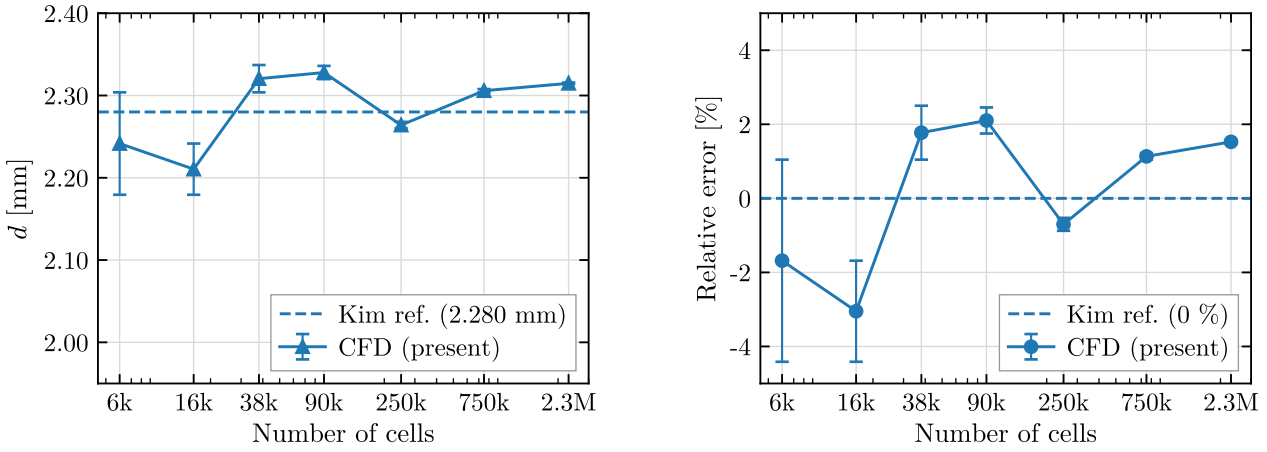



**Figure 4.14** Shock stand-off distance $d$ ahead of the Kim cylinder (left) and signed relative discrepancy from the experimental reference (right). Error bars indicate half the local sampling interval, expressed in millimetres or as a percentage of the reference distance.

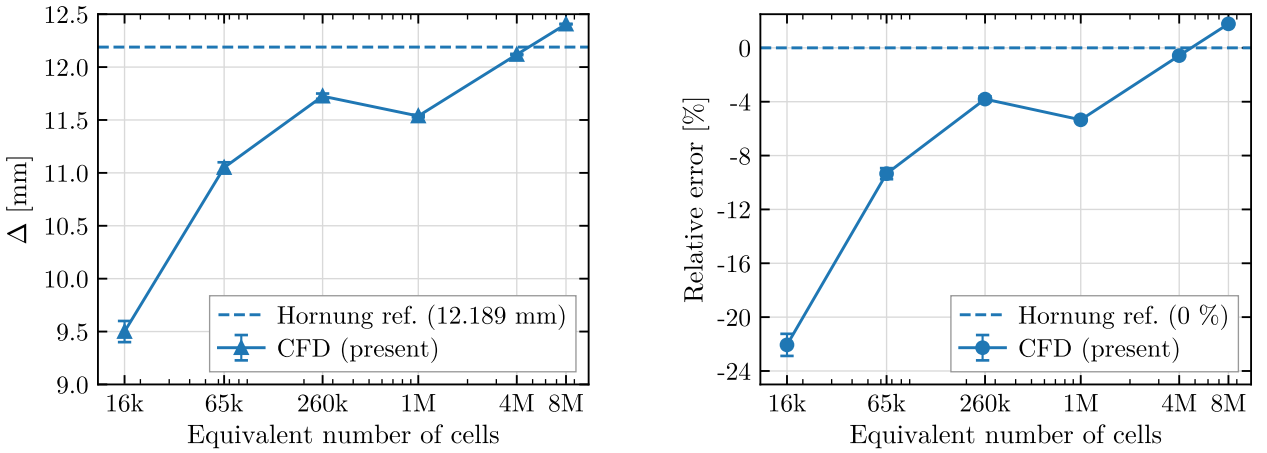



**Figure 4.15** Shock stand-off distance $\Delta$ ahead of the Hornung–Smith wedge tip (left) and signed relative discrepancy from the experimental reference (right). The abscissa gives equivalent cell counts for the refinement family. Error bars indicate half the local sampling interval, expressed in millimetres or as a percentage of the reference distance.

For the Kim cylinder, the discrepancies range from approximately $-3.05\%$ to $+2.10\%$, with $+1.53\%$ on the finest mesh. Refinement changes the sign of the error several times, without a consistently

decreasing amplitude. The Hornung–Smith results show a larger initial mesh dependence: the discrepancy decreases in magnitude from $-22.06\%$ on the coarsest mesh to $-3.81\%$ at $260\,000$ cells, but increases again to $-5.35\%$ on the $1$-million-equivalent mesh. The two finest levels give $-0.58\%$ and $+1.78\%$, respectively. Thus, refinement generally improves the Hornung–Smith result, but neither family establishes a systematic monotonic or damped oscillatory mesh convergence behaviour. A reliable order of convergence or extrapolated mesh-independent value cannot be inferred from these sequences.

The pressure fields provide an explanation for part of this irregular mesh dependence. Figures 4.16 and 4.18 show local distortions of the captured shock wave, instead of a smoothly curved line. It's severity vary between meshes. These distortions are consistent with the carbuncle phenomenon discussed in the following paragraph. For the Kim cylinder, the close-up in Figure 4.17 shows distinct dents away from the stagnation line. For the Hornung–Smith wedge, a local indentation is visible on the lower part of the shock at $65\,000$ cells, while the $260\,000$-cell and $1$-million-equivalent cases exhibit distortions near the stagnation line. The latter is enlarged in Figure 4.19. A deformation in this region directly affects the measured stand-off distance. Consequently, the refinement study combines ordinary resolution effects with changes in shock shape that need not diminish smoothly with cell size.

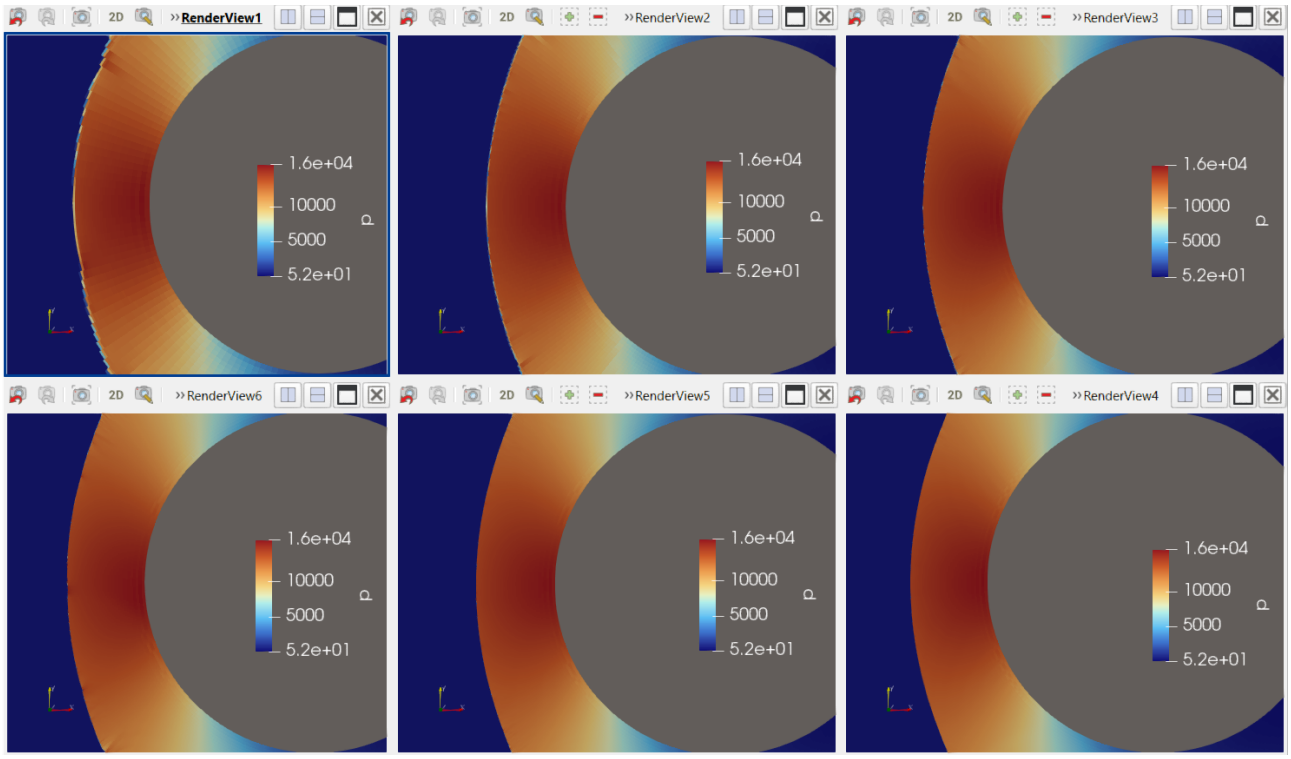



**Figure 4.16** Static pressure fields in $\mathrm{Pa}$ for the Kim cylinder at the final evaluated iterations. From left to right, the top row shows the $16\,000$, $38\,000$ and $90\,000$-cell meshes, and the bottom row the $250\,000$, $750\,000$ and $2.3$-million-cell meshes. Local shock distortions vary between refinement levels.

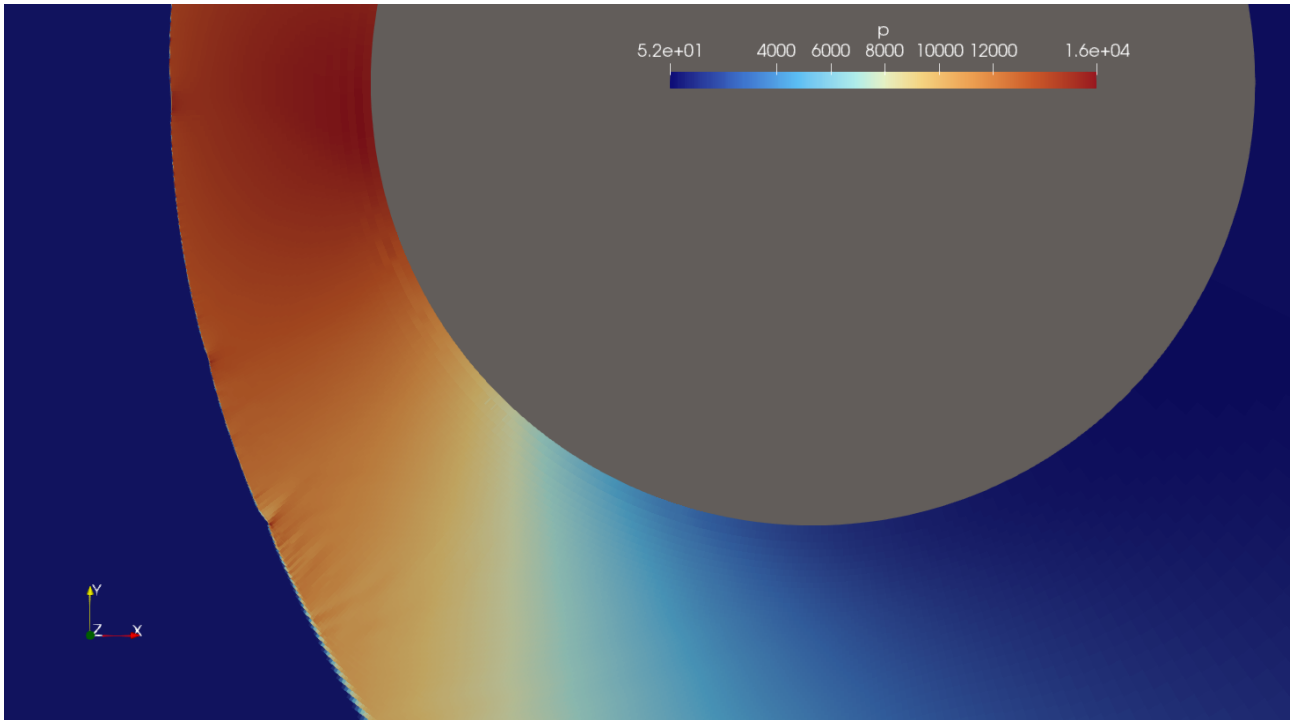



**Figure 4.17** Close-up of the Kim cylinder pressure field in $\mathrm{Pa}$ on the $250\,000$-cell mesh, showing local indentations of the shock away from the stagnation line.

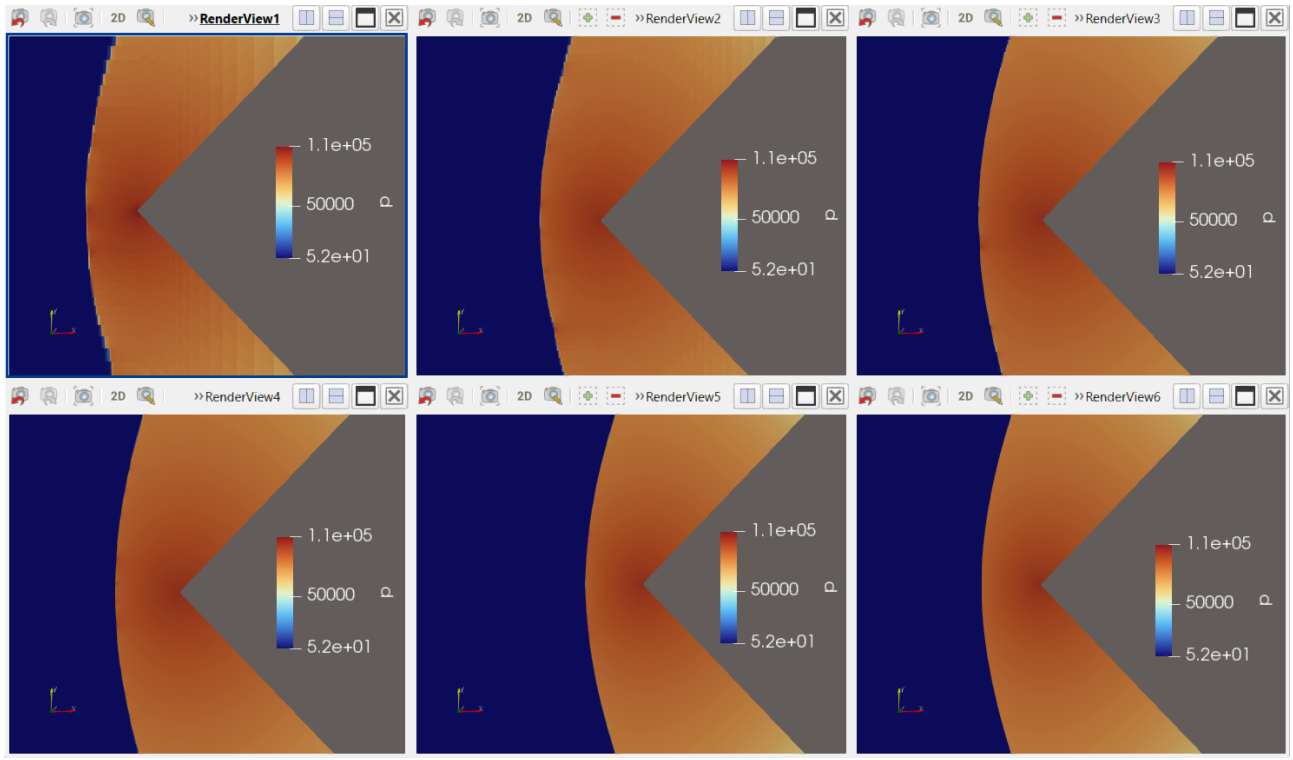



**Figure 4.18** Static pressure fields in $\mathrm{Pa}$ for the Hornung–Smith wedge at the final evaluated iterations. From left to right, the top row shows the $16\,000$, $65\,000$ and $260\,000$-cell meshes, and the bottom row the $1$, $4$ and $8$-million-equivalent meshes. The shock shape does not change smoothly with refinement.




**Figure 4.19** Close-up of the Hornung–Smith pressure field in $\mathrm{Pa}$ on the $1$-million-equivalent mesh, showing an indentation of the shock in the stagnation region.

Iterative convergence must therefore be distinguished from mesh convergence. In the following the iterative convergence of the specific cases with severe carbuncling and subsequently being outliers in the mesh converegence, is discussed in detail, so show that these cases infact reach a stationary, converged state. The residual histories were inspected together with the evolution of the shock position and comparisons of the pressure field between iterations. Representative stand-off histories are shown in Figures 4.20 and 4.21. For the Kim $250\,000$-cell case, the detected shock position remains unchanged from $12\,000$ to $25\,000$ iterations. On the $2.3$-million-cell mesh, the final samples at $43\,000$–$45\,000$ iterations identify the same position. The Hornung–Smith histories likewise approach plateaus, including the visibly distorted $1$-million-equivalent case. For that case, Figure 4.22 additionally shows pressure changes below $0.1\%$ in magnitude in the displayed region between $200\,000$ and $250\,000$ pseudo-iterations. These observations support an effectively stationary shock configuration at the available extraction resolution, even when its shape remains distorted. Continuing the iterations alone therefore does not resolve the observed mesh dependence.

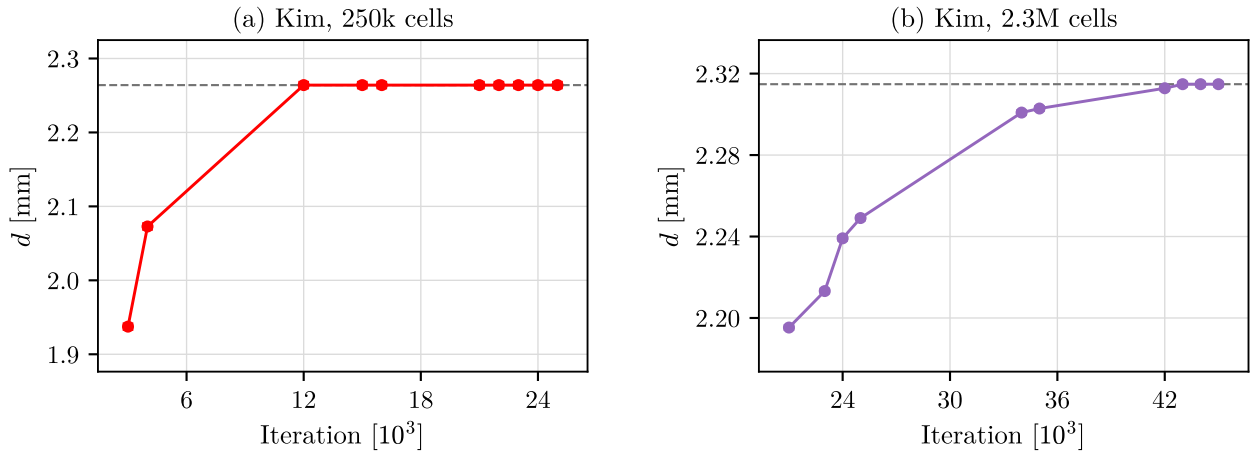



**Figure 4.20** Evolution of the Kim shock stand-off distance for the 250 000-cell (left) and 2.3-million-cell (right) meshes. Dashed lines mark the final extracted distances, error bars indicate half the local sampling interval.

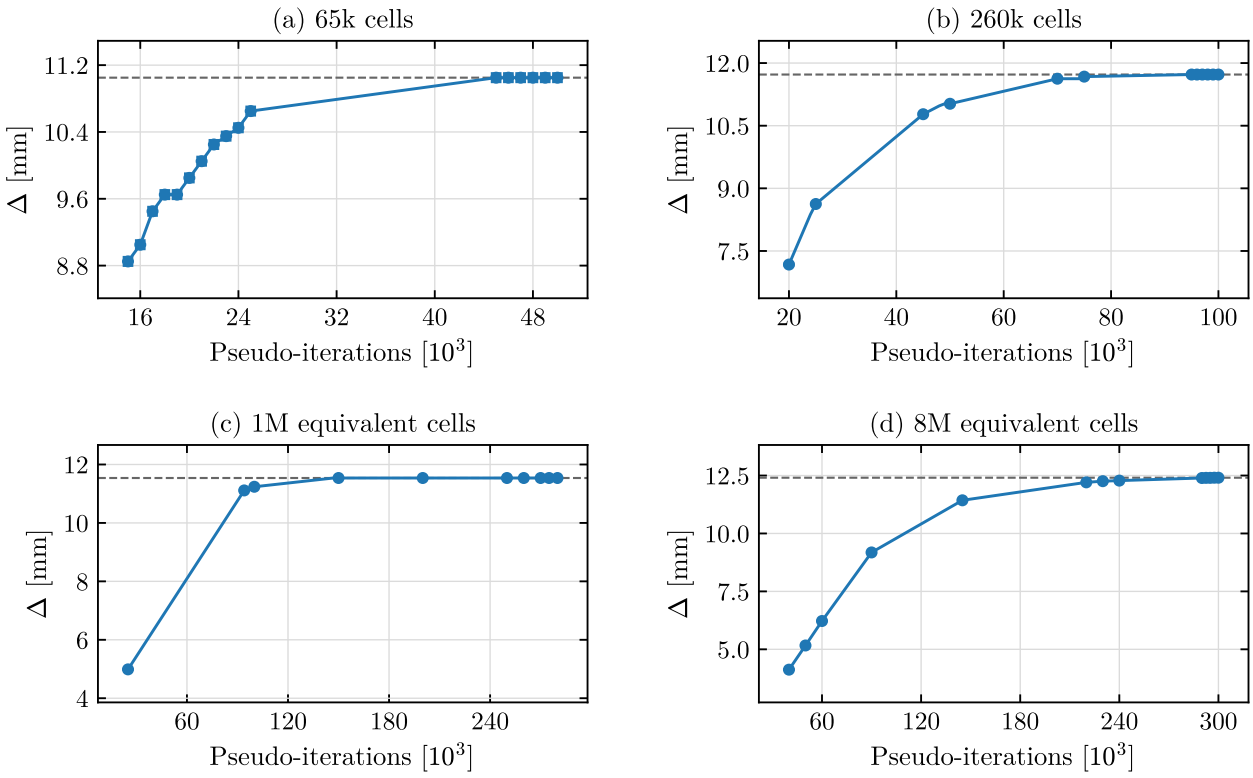



**Figure 4.21** Evolution of the Hornung–Smith shock stand-off distance for the 65 000-cell, 260 000-cell, 1-million-equivalent and 8-million-equivalent meshes. Dashed lines mark the final extracted distances, error bars indicate half the local sampling interval.

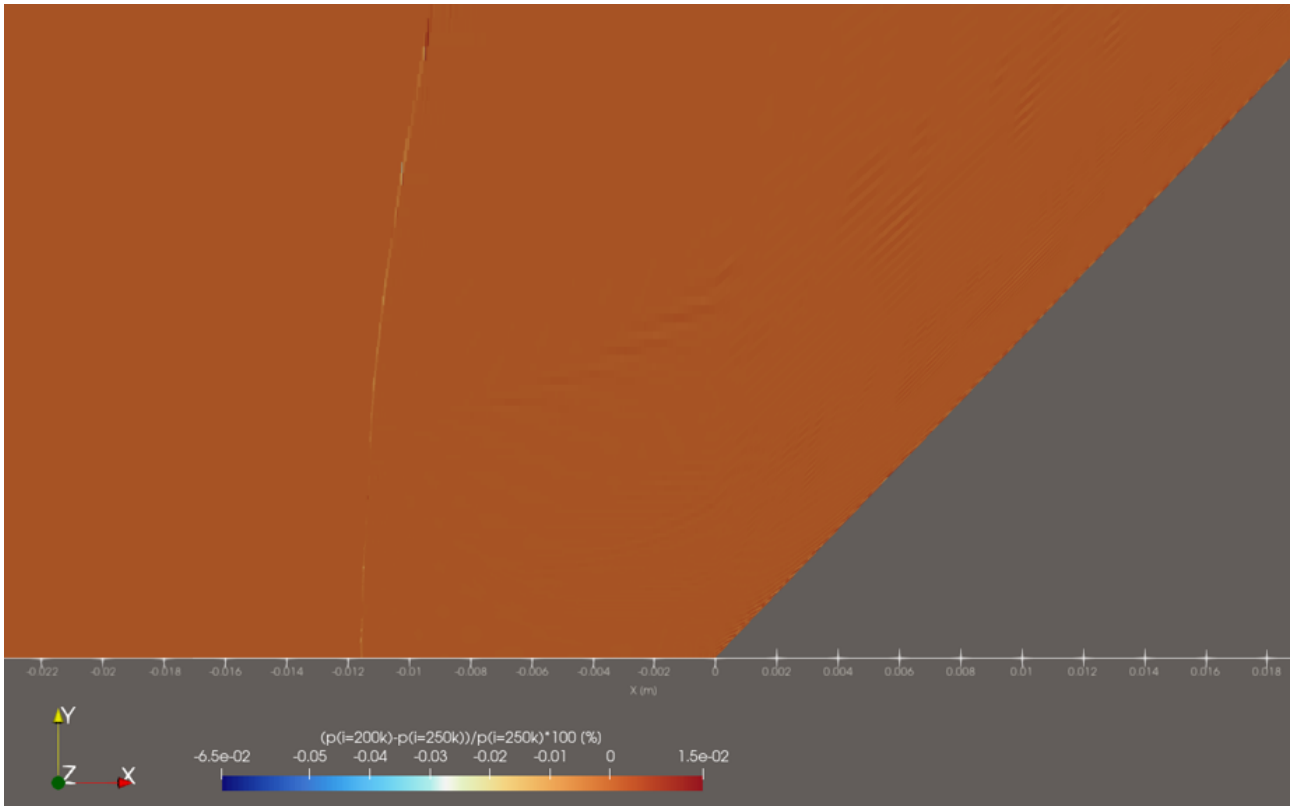



**Figure 4.22** Relative pressure change $100(p_{200\,000} - p_{250\,000})/p_{250\,000}$ in percent for the Hornung–Smith 1-million-equivalent case, where subscripts denote pseudo-iterations. The displayed differences remain below $0.1\%$ in magnitude despite the persistent shock deformation.

The effect of these distortions on stagnation-region heat-transfer is outside the scope of the present stand-off assessment. This region is commonly laminar, while turbulence closures may additionally suffer from a stagnation-point anomaly [59]. The present laminar calculations isolate the shock-capturing behaviour from that additional modelling issue.

**On carbuncles.**

The observed shock distortions are attributed to the so-called carbuncle phenomenon: a shock instability arising from the interaction between the numerical discretization, particularly the flux scheme, and the mesh [63, 64]. A particular difficulty is that the instability can evolve into a stationary, iteratively converged solution while leaving local, potentially asymmetric irregularities in the shock [63, p. 238]. Thus, convergence of the iterative solution does not by itself exclude a carbuncle.

Robinet et al. [63, p. 261] relate this behaviour to the possibility of multiple solutions of the Euler equations and note that even the Navier–Stokes equations need not possess a unique solution at high Reynolds numbers. The laminar, viscous formulation used here therefore offers no automatic protection against carbuncles. The physical interpretation and implication of carbuncles is also not entirely settled: Dumbser et al. [64, p. 648] discuss whether the phenomenon should be regarded as wholly numerical or whether it may have a physical counterpart. For the present validation, the relevant issue is the dependence of the computed shock shape on the mesh and numerical treatment, but no high resolution images of the measured shocks in the experiments are available for comparison.

Carbuncles are particularly associated with strong shocks aligned with grid lines, although their occurrence is not restricted to this configuration [65, 52], thus fitting the mesh to the shock can in fact worsen the carbunlce phenomenon. Their onset, location and severity cannot be inferred reliably from mesh size alone either. Elling [65] argues that a universal cure is not available and that numerical methods can instead be made more resistant to the instability, for example through additional dissipation. Such stabilization must be balanced against the accuracy sought from a scheme with low numerical dissipation. Refinement alone consequently provides no guarantee of suppression, and the distortion need not grow or diminish monotonically as the mesh is refined [65].

The study introducing the AUSMDV2$^{+}$ scheme used here also examines carbuncle distortions in blunt-body shock calculations, including the stagnation region in which the present stand-off distance is measured [52]. Its comparisons illustrate the sensitivity to the flux formulation, while reporting improved shock stability for the proposed scheme in the cases tested. Related work specifically addresses carbuncle suppression for AUSMDV on triangular grids [66]. These results motivate improved robustness, but do not establish freedom from carbuncles for every mesh and flow configuration.

For the present mesh families, changes in carbuncle severity between different meshes prevent a clean interpretation of the stand-off results as a conventional mesh-convergence sequence. The practical validation finding is more limited: the Kim results lie within approximately $\pm 3\%$ of the reference, and the two finest Hornung–Smith results, whose shocks show only weak visible distortions, also lie within this range. This is an observed level of agreement for the tested cases, rather than a general numerical-uncertainty bound. For subsequent simulations containing strong shocks, the shock shape must therefore be inspected in detail and the presence and severity of carbuncles assessed individually.

## 4.3 Laminar boundary layer profiles

The zero-pressure-gradient flat-plate cases investigated in this section compare the predicted laminar velocity and temperature profiles with the reference solutions of Malik and Anderson [2] and Marxen et al. [67]. The Malik–Anderson cases comprise an adiabatic Mach-10 boundary layer with a calorically perfect air and an isothermal Mach-15 boundary layer with either calorically perfect or calorically imperfect air. The "case I" in Marxen et al is a isothermal wall with Mach-10, calorically imperfect gas. The aim is to assess the coupled treatment of viscous momentum transport, heat conduction and temperature-dependent gas properties in OpenFOAM/HiSA. The comparison uses profiles at several streamwise stations to examine both their agreement with the published solutions and their correct downstream development of the boundary layer.

**General case setup.**

All configurations represent two-dimensional flow over a flat, impermeable, no-slip wall, with the freestream parallel to the plate and no externally imposed streamwise pressure gradient. A single cell in the spanwise direction and `empty` side boundaries reduce the computational problem to two dimensions. The mesh is refined from the wall outwards, a downstream spacing sensitivity study exposed no sensitivity. Only the leading edge region was more finely downstream refined. The edge conditions and wall thermal conditions are summarized in Table 4.8, where the subscript $e$ denotes the freestream state used to normalize the profiles. The Mach-10 Malik–Anderson wall is adiabatic, the remaining cases prescribe a uniform wall temperature as per the respective publication.

**Table 4.8** Reference flow conditions and prescribed wall thermal conditions for the three Malik–Anderson cases and Marxen case I.

| Quantity | Malik–Anderson [2] | | | Marxen [67] |
|---|---|---|---|---|
| | Mach 10 perfect gas | Mach 15 perfect gas | Mach 15 imperfect gas | Case I |
| Mach number $M_e$ | 10 | 15 | 15 | 10 |
| Edge temperature $T_e$ [K] | 350 | 700 | 700 | 278 |
| Unit Reynolds number $Re_x/x$ [$\mathrm{m}^{-1}$] | $6.6 \times 10^6$ | $3.4 \times 10^6$ | $3.4 \times 10^6$ | $9.956 \times 10^6$ |
| Thermal wall condition | Adiabatic | Isothermal | Isothermal | Isothermal |
| Wall temperature $T_w$ [K] | Computed | 1225 | 1225 | 1198.18 |
| Prescribed ratio $T_w/T_e$ | — | 1.75 | 1.75 | 4.31 |
| Caloric model | Perfect | Perfect | Imperfect | Imperfect |

The Malik–Anderson calculations considered here prescribe a uniform freestream at the inlet and allow the boundary layer to develop from the plate leading edge. This retains the leading-edge viscous interaction, whose influence on the downstream profiles can be assessed against the self-similar reference solutions. For the Mach-15 calorically imperfect case, the computational domain extends from $x = 0$ to $0.8\,\mathrm{m}$, with a wall-normal height of $0.05\,\mathrm{m}$ and $1000 \times 200$ in-plane cells. Its inlet and upper boundary use characteristic farfield conditions (Section 3.1.2).

In the Marxen case, a pre-developed laminar boundary layer profile is prescribed at the inlet, following the arrangement described by Marxen et al. [67, Section II.B]. The computational domain begins at $x_{\mathrm{in}} = 0.116576\,\mathrm{m}$, extends to $x = 0.690288\,\mathrm{m}$, and has a wall-normal height of $0.07\,\mathrm{m}$. The mesh contains $1000 \times 200$ in-plane cells. At the inlet the velocity and temperature is asigned as a pre-developed laminar boundary layer, alongside with uniform pressure $p_e = 4135.6\,\mathrm{Pa}$. Streamwise positions are measured from the nominal virtual leading edge at $x = 0$, so the inlet represents approximately $0.1166\,\mathrm{m}$ of prior boundary layer development set as the inlet conditions as a non-uniform list.

All cases use an ideal-gas equation of state with fixed composition. The calorically perfect cases retain constant heat capacities and $\gamma \simeq 1.4$, with temperature-dependent Sutherland viscosity. For calorically imperfect cases, $c_p$ varies with temperature, while the specific gas constant remains fixed. Malik–Anderson denote *perfect gas* for thermally and calorically perfect gas models and *imperfect gas* for thermally perfect, i.e. non-reacting, but calorically imperfect gas. Marxen designates Case I as *thermally perfect*: it is the non-reacting, calorically imperfect model described in Sections III.B.1 and IV.A of the paper [67]. The thermodynamic and molecular-transport models specified in the reference papers are summarized separately for each case in Table 4.9.

**Table 4.9** Thermodynamic and molecular-transport models in the reference studies for the four laminar boundary layer cases. Equation numbers refer to the paper cited in the corresponding column heading.

| Property or model | Malik–Anderson [2] | | | Marxen [67] |
|---|---|---|---|---|
| | Mach 10 perfect gas | Mach 15 perfect gas | Mach 15 imperfect gas | Case I |
| Specific heat $c_p(T)$ | Constant | Constant | Eggers, Eq. (59) | Eggers-type, Eq. (32) |
| Dynamic viscosity $\mu(T)$ | Sutherland, Eq. (56) | Sutherland, Eq. (56) | Sutherland, Eq. (56)* | Sutherland, Eq. (22) |
| Thermal conductivity $\kappa(T)$ | Constant-$Pr$ closure, $\kappa = c_p\mu/0.7$ | Constant-$Pr$ closure, $\kappa = c_p\mu/0.7$ | Keyes, Eq. (58) | Keyes, Eq. (33) |
| Molecular Prandtl number $Pr$ | $0.7$, constant | $0.7$, constant | Variable, $Pr = c_p\mu/\kappa$ | Variable, $Pr = c_p\mu/\kappa$ |

*The Sutherland viscosity constant $d_1$ for the Malik–Anderson Mach-15 imperfect-gas case may contain a typographical error (see Appendix E).

In the calorically imperfect simulations the heat-capacity $c_p(T)$, viscosity $\mu(T)$ and thermal conductivity $\kappa(T)$ were described using polynomials fitted to the analytical property laws listed in Table 4.9, utilizing the extension to OpenFOAM's thermophysical model structure described in Section 3.2.1.

**Extraction and normalization.**

For the profile comparisons, the dimensional wall distance $y$ is scaled with the local boundary layer length $L_r = \sqrt{1/Re_x}$ into the nondimensional

$$Y = \frac{y}{L_r(x)}, \qquad L_r(x) = \sqrt{\frac{\nu_e x}{U_e}}, \qquad Re_x = \frac{U_e x}{\nu_e}, \tag{4.2}$$

where $\nu_e = \mu_e/\rho_e$. Velocity and temperature are normalized by $U_e$ and $T_e$, respectively. This normalization follows Malik–Anderson definition and allows the self-similar boundary layers to collapse onto the same velocity and temperature curves independant of the streamwise sampling stations. Departures from this collapse reveal changes in profile shape beyond the expected $\sqrt{x}$ growth of the layer.

### 4.3.1 Malik–Anderson cases

In the follwing, first, the implication and quantification are discussed of the viscous interaction of the leading edge shock that the CFD is resolving, because no inlet profile is being prescribed. Then the results of the three Malik–Anderson cases are discussed in the following order: calorically perfect Mach-10, calorically perfect Mach-15, calorically imperfect Mach-15.

#### Leading-edge shock and viscous interaction.

The uniform-inlet calculations of the Malik–Anderson case, in contrast to the Marxen cases which are setup with an inlet profile, include the development of the boundary layer from the plate leading edge. The displacement of the boundary layer this leading edge turns and compresses the external flow, producing a leading-edge shock and a streamwise variation in pressure. This pressure field in turn affects the boundary layer development, even though the plate itself imposes no turning angle. Such viscous–inviscid interaction is a physical difference between the present fulldomain CFD and a zero-pressure-gradient similarity solution [68].

For more information on this effect of vicsous interaction and shocks on leading edges of flat-plates the reader is referred to Chapter 7 in Anderson's *Hypersonic and high temperature gas dynamics* [69]. In there Anderson also introduces the parameter $\bar{\chi}$ that governs laminar viscous interaction and is defined as

$$\bar{\chi}(x) = \frac{M_e^3}{\sqrt{Re_x}}\sqrt{C}, \qquad C = \frac{\rho_w \mu_w}{\rho_e \mu_e}, \tag{4.3}$$

where for an adiabatic-wall, zero-pressure-gradient flat-plate $C = 0$. For a non-adiabatic-wall zero-pressure-gradient flat-plate this is not the case, but for all zero-pressure-gradient flat-plates edge quantities can be estimated with the nominal freestream quantities as reported in Table 4.8. For the Mach-10 stations, $\bar{\chi}$ falls from $1.74$ at $x = 0.050\,\mathrm{m}$ to $0.50$ at $x = 0.606\,\mathrm{m}$. For both Mach-15 cases it falls from $5.79$ at $x = 0.100\,\mathrm{m}$ to $2.05$ at $x = 0.800\,\mathrm{m}$. The slow $x^{-1/2}$ decay and strong Mach-number dependence explain why the leading-edge influence can remain relevant futher downstream, particularly for the Mach-15 cases, and thus why the boundary layer profiles close to the leading edge may not perfectly collpse to their self-similiarity profile. Hence, for each plotted boundary layer profile extracted from the CFD results, the associated $\bar{\chi}(x)$ is provided with it, alongside the physical station.

#### Calorically perfect gas.

Figure 4.23 shows the adiabatic Mach-10 case. The velocity and temperature profiles approach the reference with further downstream sampling, to a good degree. The adiabatic wall temperature is ever so slightly underpredicted, with $T_w/T_e$ approximately $17$ compared to the references $\approx 17.4$. Thus the principal profile shapes are reproduced by the CFD simulation.

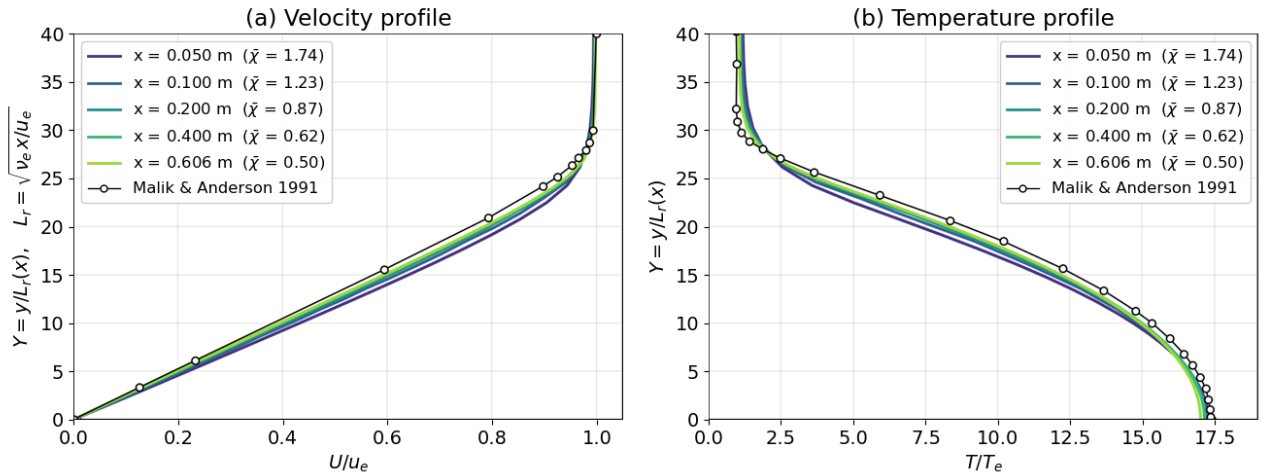



**Figure 4.23** Velocity and temperature profiles for the adiabatic Mach-10 perfect-gas case. Coloured curves show successive OpenFOAM/HiSA extracted stations; symbols denote the digitized reference from Malik and Anderson [2, Fig. 1]. Each station uses its local similarity length $L_r(x)$ (see Eq. (4.2)) for normalization. Each profile is named by it's physical downstream extraction station $x$ alongside it's viscous interaction parameter $\bar{\chi}$ (see Eq. (4.3)). The downstream stations approach the reference as the estimated interaction parameter decreases.

The isothermal Mach-15 perfect-gas case in Figure 4.24 exhibits a qualititively good agreement to the reference. The imposed cold-wall produces an interior temperature maximum, which is captured at approximately $T/T_e = 10$ and $Y = 7$. The profile similarity also improves with the weakening leading-edge interaction $\bar{\chi}$.

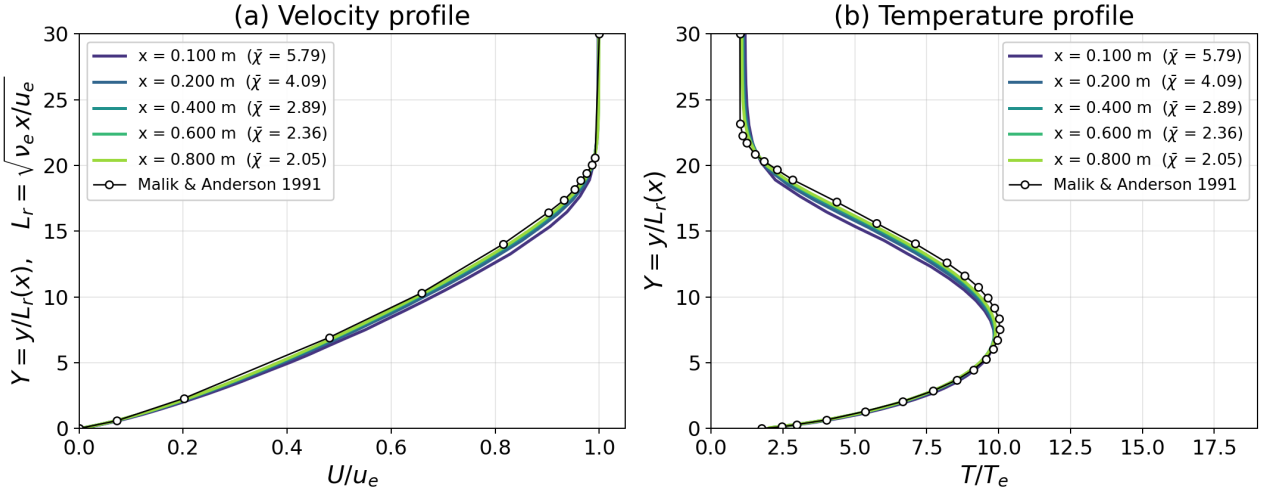



**Figure 4.24** Velocity and temperature profiles for the isothermal Mach-15 perfect-gas case, compared with the digitized profiles from Malik and Anderson [2, Fig. 3]. Coloured curves show successive OpenFOAM/HiSA extracted stations; symbols denote the digitized reference from Malik and Anderson [2, Fig. 1]. Each station uses its local similarity length $L_r(x)$ (see Eq. (4.2)) for normalization. Each profile is named by it's physical downstream extraction station $x$ alongside it's viscous interaction parameter $\bar{\chi}$ (see Eq. (4.3)). The downstream stations approach the reference as the estimated interaction parameter decreases.

**Calorically imperfect gas and reference uncertainty.**

For the Mach-15 imperfect-gas case (Figure 4.25), the reduced temperature maximum relative to the perfect-gas case is recovered qualitatively. However, the predicted peak is approximately $T/T_e = 8.4$, compared with about $9.0$ in the reference, and the outer velocity and temperature profiles are broader in $Y$. This discrepancy persists downstream, although the viscous interaction even at the most downstream station $x = 0.8\mathrm{m}$ is still substantial at $\bar{\chi} = 2.05$, where Anderson describes $\bar{\chi} \geq 3$ as strong interaction [69, p. 388]. But this was a non-issue for the Mach-15 perfect-gas cases. A reduction in $\bar{\chi}$ alone therefore does not explain the observed differences. Rather the remaining temperature-peak and profile-shape discrepancies must be interpreted alongside the transport-coefficient uncertainty and potential under-convergence of the reference examined in Appendix E. Hence in the following another hypersonic, calorically imperfect laminar boundary layer case is investigated, with the Marxen Case I.

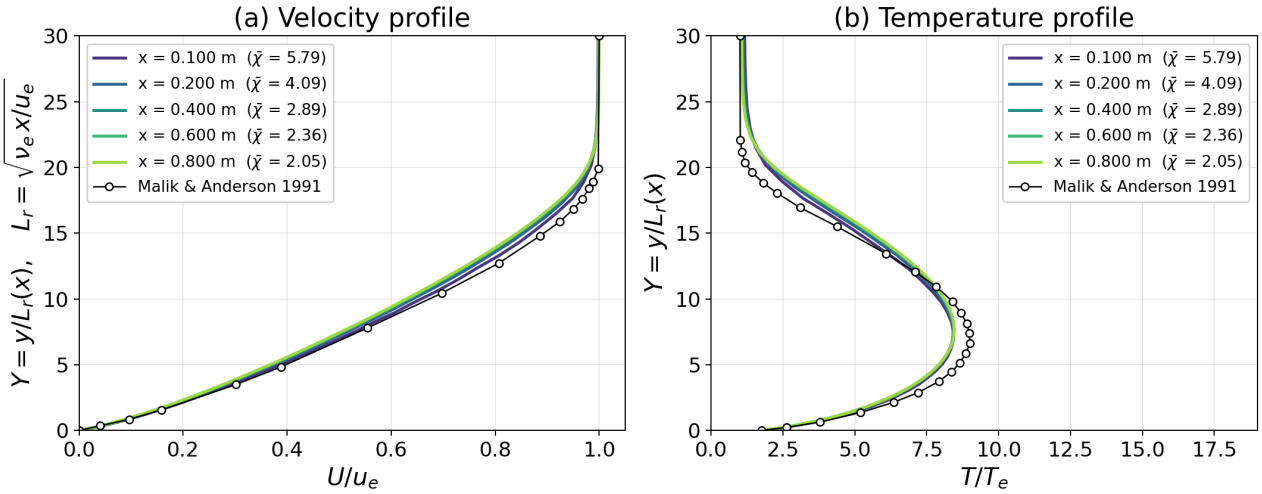



**Figure 4.25** Velocity and temperature profiles for the isothermal Mach-15 calorically imperfect case. The reference is digitized from Malik and Anderson [2, Fig. 3]. The remaining temperature-peak and profile-shape discrepancies must be interpreted alongside the transport-coefficient uncertainty and potential under-convergence of the reference examined in Appendix E.

### 4.3.2 Marxen case

The uncertainty associated with the former investigated Malik–Anderson Mach-15 imperfect-gas reference makes an additional benchmark necessary to assess the temperature-dependent thermophysical implementation. Marxen Case I was selected for this purpose: it also combines temperature-dependent heat capacity, Sutherland viscosity and Keyes conductivity, with the viscosity coefficients explicitly specified in Eq. (22) of the paper [67]. It therefore provides a separate reference unaffected by the disputed Malik–Anderson coefficient and convergence discussed in Appendix E. Prescribing a developed inlet profile also allows the downstream laminar development to be assessed without including the plate's leading edge shock in the computational domain.

#### Inlet-profile comparison.

Figure 4.26 compares the Fortran similarity solution used to construct the inlet profile for the CFD simulation with the digitized profiles in Fig. 11 of Marxen et al. [67]. The velocity closely follows the reference symbols. The temperature distribution reproduces the wall value, the interior maximum near $Y = 5$, and the return to the freestream temperature, with a small overprediction of the peak. This comparison checks the profile supplied to the inlet of $x = 0.1166\,\mathrm{m}$ is accurate, leaving the subsequent downstream development to the CFD simulation.

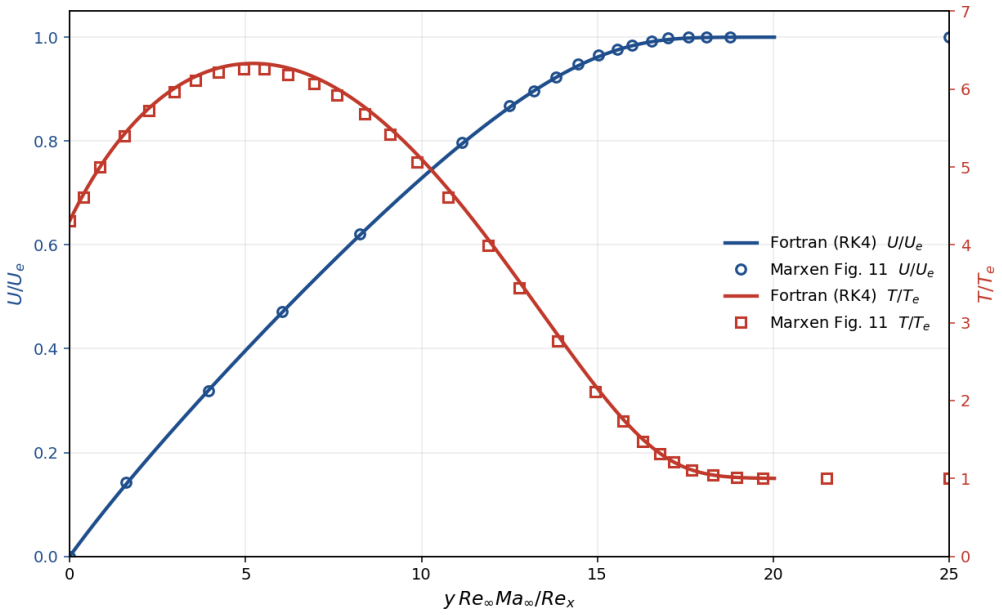



**Figure 4.26** Similarity solution used to construct the Marxen Case-I inlet. Solid curves denote the Fortran solution obtained with fourth-order Runge–Kutta integration; symbols denote digitized data from Marxen et al. [67, Fig. 11]. Velocity and temperature use the left and right ordinates, respectively. The abscissa retains the paper's nondimensional notation and corresponds to the scaled wall distance $Y$ in Eq. (4.2).

#### Downstream profiles.

Figure 4.27 shows the boundary layer profiles at the four extracted downstream stations of $x = 0.257$, $0.325$, $0.402$ and $0.486\,\mathrm{m}$. They correspond, to the plotted precision, to $\sqrt{Re_x} = 1600$, $1800$, $2000$ and $2200$. The stations thus have equal increments in $\sqrt{Re_x}$, following the streamwise scaling used by Marxen et al. [67, Section II.B]; this explains their non-round dimensional positions and increasing physical spacing. Here $Re_x$ has the conventional definition in Eq. (4.2). The positions are measured from the nominal virtual origin and include the inlet's approximately $0.1166\,\mathrm{m}$ of prior development, rather than representing distances travelled since entering the mesh.

The four velocity and temperature profiles collapse closely to the reference. The temperature maximum is approximately $T/T_e = 6.3$ and remains ever so slightly above the reference, as already seen in the inlet comparison. There is little additional streamwise spread over the sampled interval. These results support the consistent downstream development of the prescribed laminar profile with temperature-dependent properties. Since the reference comparisons use digitized curves and the inlet already approximates that reference, the evidence concerns profile preservation and development under the stated setup.

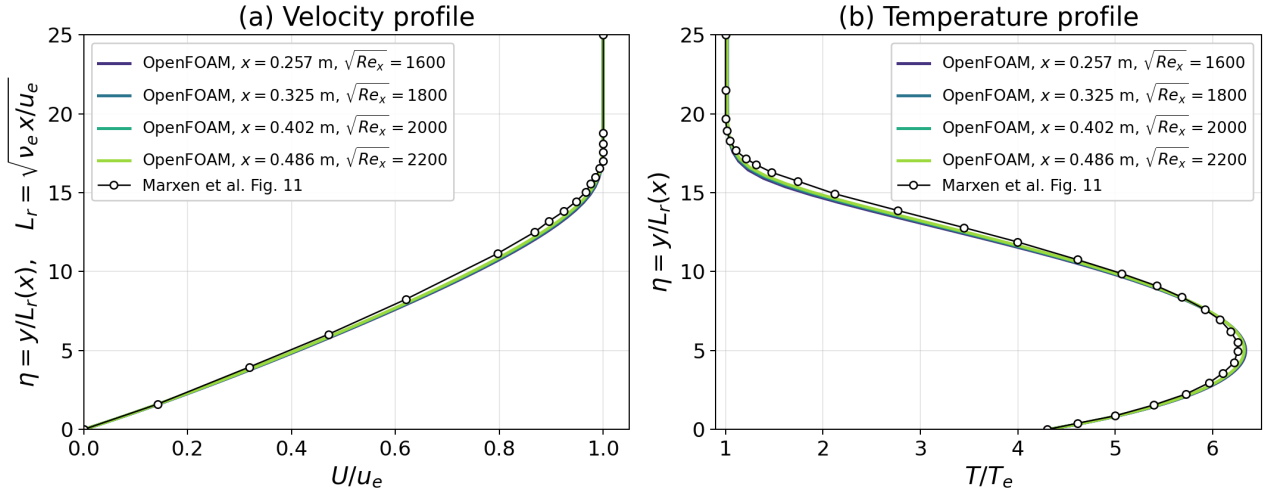



**Figure 4.27** OpenFOAM/HiSA velocity and temperature profiles downstream of the prescribed inlet, compared with Marxen et al. [67, Fig. 11]. The four stations are ordered by increasing $x$, with $\sqrt{Re_x}$ included in the legends. The ordinate labelled $\eta = y/L_r(x)$ in this plot is the coordinate denoted by $Y$ in Eq. (4.2).

## 4.4 Laminar heat-transfer along a flat-plate and compression corner

The compression-ramp cases of Cao et al. extend the previous laminar validation from attached boundary layer profiles to wall heat-transfer on an attached zero-pressure-gradient flat-plate as well as through a strong shock-wave/boundary layer interaction. The geometry combines a leading flat-plate with a subsequent compression corner. Upstream of the interaction, the attached layer provides a further check of nominally zero-pressure-gradient flat-plate flow. The adverse pressure gradient imposed by the ramp then causes separation ahead of the corner and reattachment on the ramp, with a pronounced redistribution of surface heating. This tests the coupled prediction of viscous flow and heat-flux calcualtion in attached and separated regions within the same configuration.

The heat-transfer references are the two-dimensional laminar base flows in Cao et al. [70, Fig. 4(b)] and Cao et al. [71, Fig. 8(a), sharp-edge case R0]. These correspond to same geometry, just under different freestram pressures of $1550\,\mathrm{Pa}$ and $760\,\mathrm{Pa}$ respecitively, thus also higher and lower Reynolds numbers respectively.

#### General case setup.

The sharp-edged flat-plate has length $L = 100\,\mathrm{mm}$ and is followed by a $15°$ compression ramp. The leading edge is at $x = 0$, the corner at $x/L = 1$, and the computational outlet at $x = 400\,\mathrm{mm}$. The upper boundary lies at $y = 150\,\mathrm{mm}$. Here $x$ is the horizontal streamwise coordinate, including on the inclined ramp. One spanwise cell with `empty` side boundaries enforces two-dimensional flow, and the calculations use laminar molecular transport without a turbulence model.

The uniform freestream defines the characteristic farfield conditions at the inlet and upper boundary. The downstream boundary uses characteristic pressure inlet/outlet conditions (Section 3.1.2). The plate and

ramp form an impermeable, no-slip wall with prescribed uniform temperature and zero pressure gradient. Table 4.10 summarizes the two flow conditions and the common thermal model.

**Table 4.10** Freestream conditions and thermal/transport models for the two Cao compression-ramp cases. Unit Reynolds numbers are evaluated from the OpenFOAM/HiSA freestream properties; the reference studies report $8.6 \times 10^6$ and $4.2 \times 10^6\,\mathrm{m}^{-1}$.

| Quantity | Higher Reynolds number | Lower Reynolds number |
|---|---|---|
| Heat-transfer reference | Cao et al. [70, Fig. 4(b)] | Cao et al. [71, Fig. 8(a), R0] |
| Mach number $M_\infty$ | 7.7 | 7.7 |
| Static pressure $p_\infty$ [Pa] | 1550 | 760 |
| Static temperature $T_\infty$ [K] | 125 | 125 |
| Unit Reynolds number $Re_x/x$ [$\mathrm{m}^{-1}$] | $8.620 \times 10^6$ | $4.227 \times 10^6$ |
| Wall temperature $T_w$ [K] | 293 | 293 |
| Equation of state | Calorically perfect ideal gas | |
| Gas constant $R$ [$\mathrm{J\,kg^{-1}\,K^{-1}}$] | 287.05 | |
| Heat capacity $c_p$ [$\mathrm{J\,kg^{-1}\,K^{-1}}$] | 1004.703, constant | |
| Heat-capacity ratio $\gamma$ | $\simeq$1.4 | |
| Dynamic viscosity $\mu(T)$ | Polynomial fit to Sutherland's law | |
| Thermal conductivity $\kappa(T)$ | $c_p\mu(T)/Pr$ | |
| Molecular Prandtl number $Pr$ | 0.71, constant | |

The reference calculations use Sutherland viscosity and constant $Pr = 0.71$. In OpenFOAM/HiSA, the temperature-dependent viscosity is represented by a seventh-order polynomial fit to Sutherland's law over $40$–$1000\,\mathrm{K}$. The conductivity polynomial is obtained by multiplication with $c_p/Pr$. Thus the fitted transport laws retain the prescribed constant molecular Prandtl number that the reference case requires.

**Mesh families.**

Figure 4.28 shows the geometry and the structured mesh. Refinement is concentrated near the wall, the leading edge, and in the region of the expected separated interaction. The higher-Reynolds-number case is used for the mesh study. Two distinct mesh convergence investigations need to be performed: once on the first-cell height, and second on the downstream spacing, especially in the separated interaction region. The wall-normal comparison holds the nominal downstream spacing at $\Delta x = 0.15\,\mathrm{mm}$ and varies the first-cell height through $\Delta y_1 = 0.1, 0.01, 0.002$ and $0.0005\,\mathrm{mm}$. The streamwise comparison holds $\Delta y_1 = 0.002\,\mathrm{mm}$ and uses $\Delta x = 0.6$, $0.3$ and $0.15\,\mathrm{mm}$. The $\Delta x$ values in the figures describe the local refinement in the separation region, rather than a uniform spacing over the full plate and ramp.

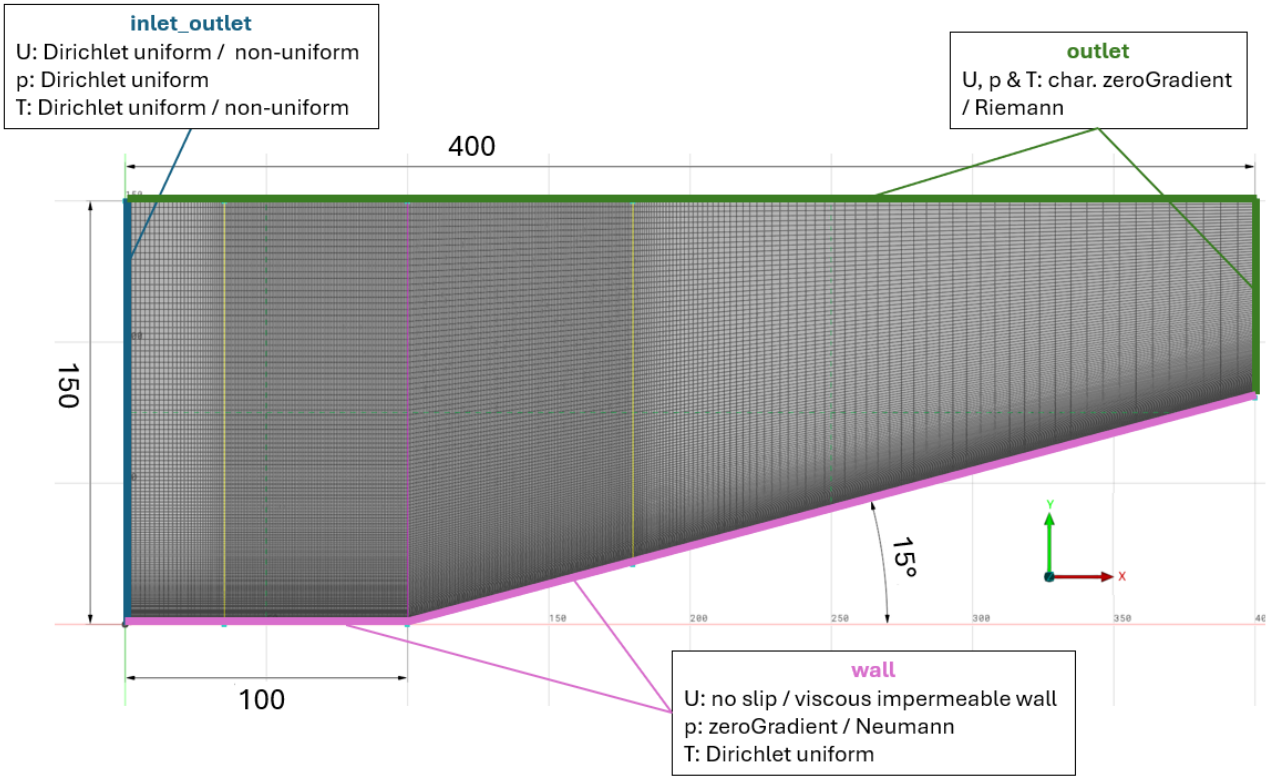



**Figure 4.28** Compression-ramp geometry and mesh layout. The flat-plate extends to $x = 100\,\mathrm{mm}$, followed by a $15°$ ramp. Local streamwise refinement around the corner interaction region (between the yellow lines).

### Extraction and normalization.

The two surface quantity of interest and published is the Stanton Number$St$, which are expressed as

$$St = \frac{q_w}{\rho_\infty U_\infty c_p (T_r - T_w)}, \tag{4.4}$$

with $q_w > 0$ denoting heat transferred into the wall. The recovery temperature is $T_r = T_\infty[1+r(\gamma-1)M_\infty^2/2]$, with the recovery factor $r = \sqrt{Pr} \simeq 0.8426$, giving $T_r \simeq 1374\,\mathrm{K}$. The attached, laminar velocity and temperature profiles are extracted at $x = 40\,\mathrm{mm}$, or $x/L = 0.4$. This station lies in the attached flat-plate region, far enough upstream of the corner-induced separation. The boundary layer profile figures use $U_x/U_\infty$, $T/T_\infty$ and $\eta = (y/x)\sqrt{Re_x}$, equivalent to the wall coordinate introduced in the previous Section 4.3 and it's Eq. (4.2).

Because such separation bubbles represent genuine physical unsteady phenomena, which the stationary RANS cannot capture [72]. The wall pressure and heat-flux are thus averaged over the final 15 000 steady-solver pseudo-iterations. These are numerical iteration-window averages, not physical-time means.

### Results: attached flat-plate region.

Figure 4.29 compares the upstream, attached laminar boundary layer profiles at both Reynolds numbers with the digitized profiles reported by Cao et al. Figure [70, Fig. 7(a)] and Cao et al. [73, Fig. 2(b)] respectively. The CFD simulation with fine enough leading edge refinement, is capable of reproducing both case's laminar boundary layer profiles well. These comparisons are supportive to the previous, arguably more thorough laminar-profile assessment in Section 4.3.

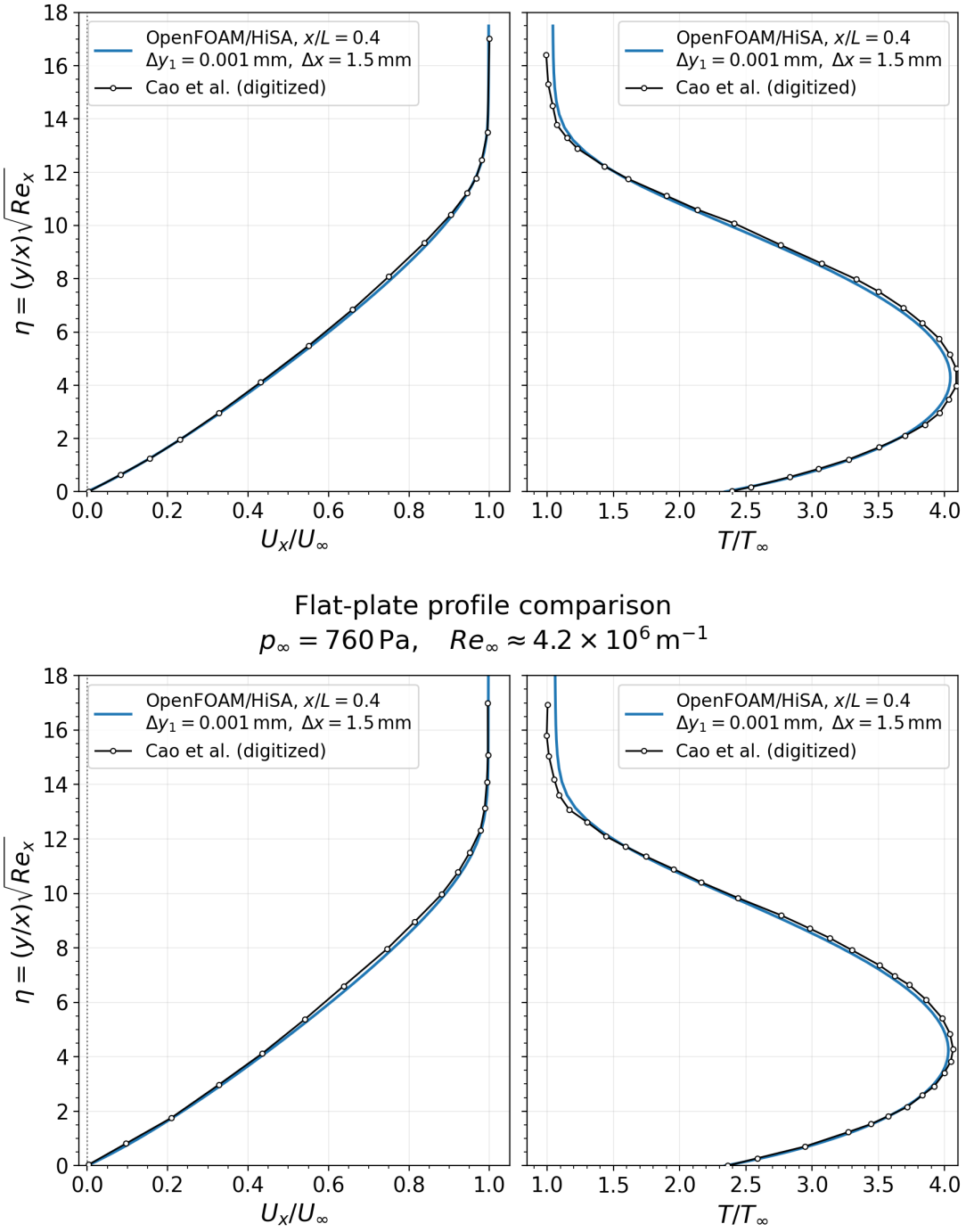



**Figure 4.29** Attached boundary layer profiles at $x/L = 0.4$: $p_\infty = 1550\,\mathrm{Pa}$ above and $760\,\mathrm{Pa}$ below. Symbols are digitized from Cao et al. [70, Fig. 7(a)] and Cao et al. [73, Fig. 2(b)], respectively. This sample station lies sufficiently upstream of the separation and transition as to represent a zero-pressure-gradient laminar boundary layer, also shown in the respective Cao et al. figures and stated in their captions, and far enough downstream of the leading edge to have weak viscous interaction.

**Mesh sensitivity across whole domain.**

The Stanton-number windows in Figures 4.30 and 4.31 are separate the attached upstream region, the separation in the corner region, and reattachment with the downstream ramp.

The Stanton-number in the attached upstream region shows a significant sensitivity to the wall-normal spacing, with a fine enough mesh achieving good agreement to the reference (see Figure 4.30 (a)). As

expected, virtually no downstream mesh requirement exists for this effectlvely zero-pressure-gradient flat-plate (see Figure 4.31 (a)).

The separated interaction is more demanding. The wall-normal coarser meshes underpredict the separation bubble with too late detachment and too early reattachment (see Figure 4.30 (b)). But pairing a fine enough wall-normal mesh, with too coarse downstream spacing results in unphysical oscillations (see Figure 4.31 (b)) - it should be restated here, that the plotted results already represent averages over the last 15 000, oscillating converged, pseudo iterations. So the oscialltory over and undershoot of visbile in Figure 4.31 (b) and somewhat also visible in Figure 4.31 (c) of the coarsest $\Delta x = 0.6\mathrm{mm}$ mesh does not represent unsteady oscillatory behaviour across the pseudo-iterations.

Throughout all wall-normal and streamwise mesh refinement an overshoot in the heat-flux at the reattachment point remains, despite position of the reattachment being solved quite accuratly (see Figure 4.30 (c) and Figure 4.31 (c)).

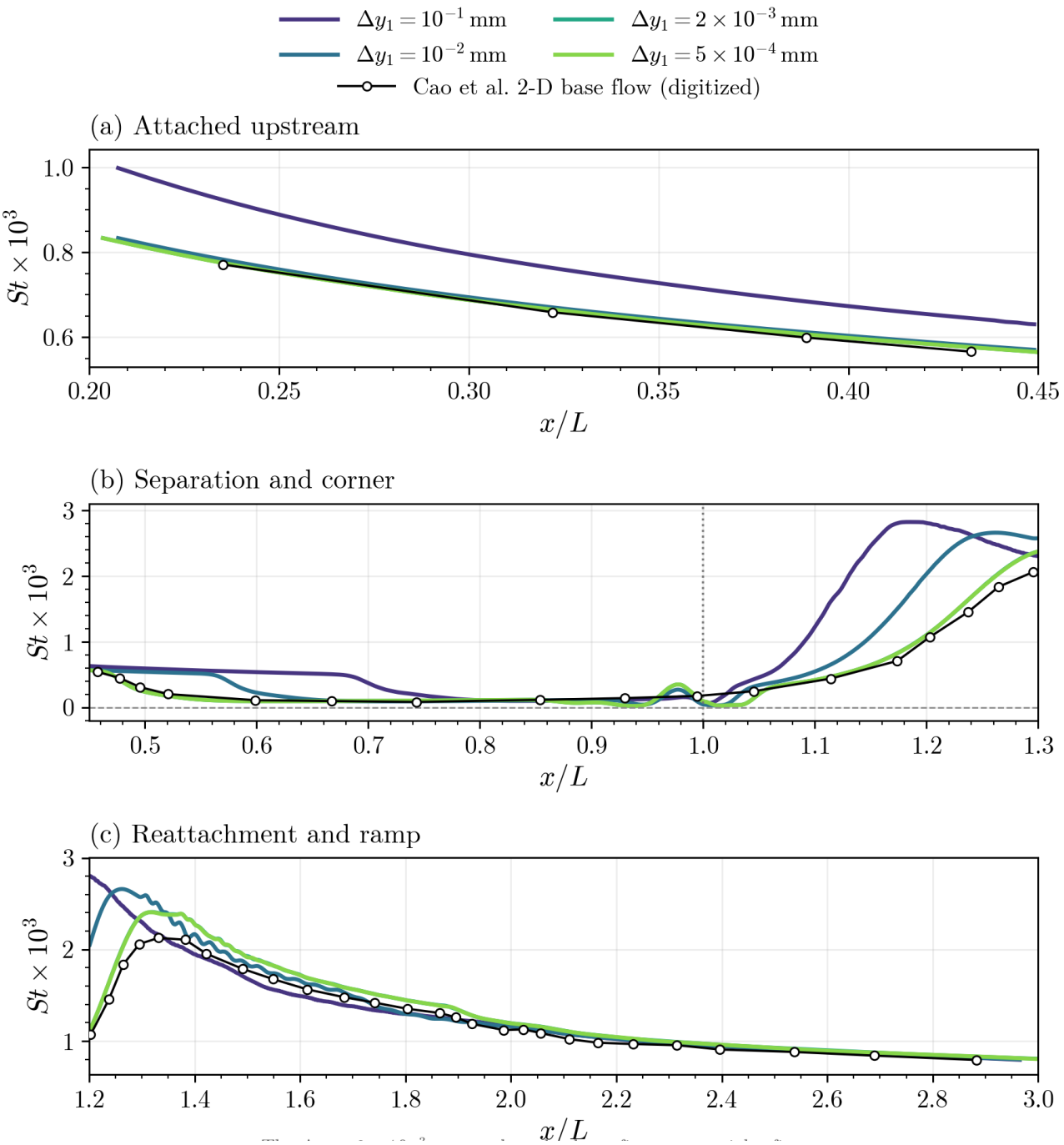



**Figure 4.30** Wall-normal mesh sensitivity at $p_\infty = 1550\,\mathrm{Pa}$ and nominal separation-region spacing $\Delta x = 0.15\,\mathrm{mm}$. The reference is the two-dimensional base flow of Cao et al. [70, Fig. 4(b)]. The two finest wall-normal meshes are visibly not distinguishable in this plot.

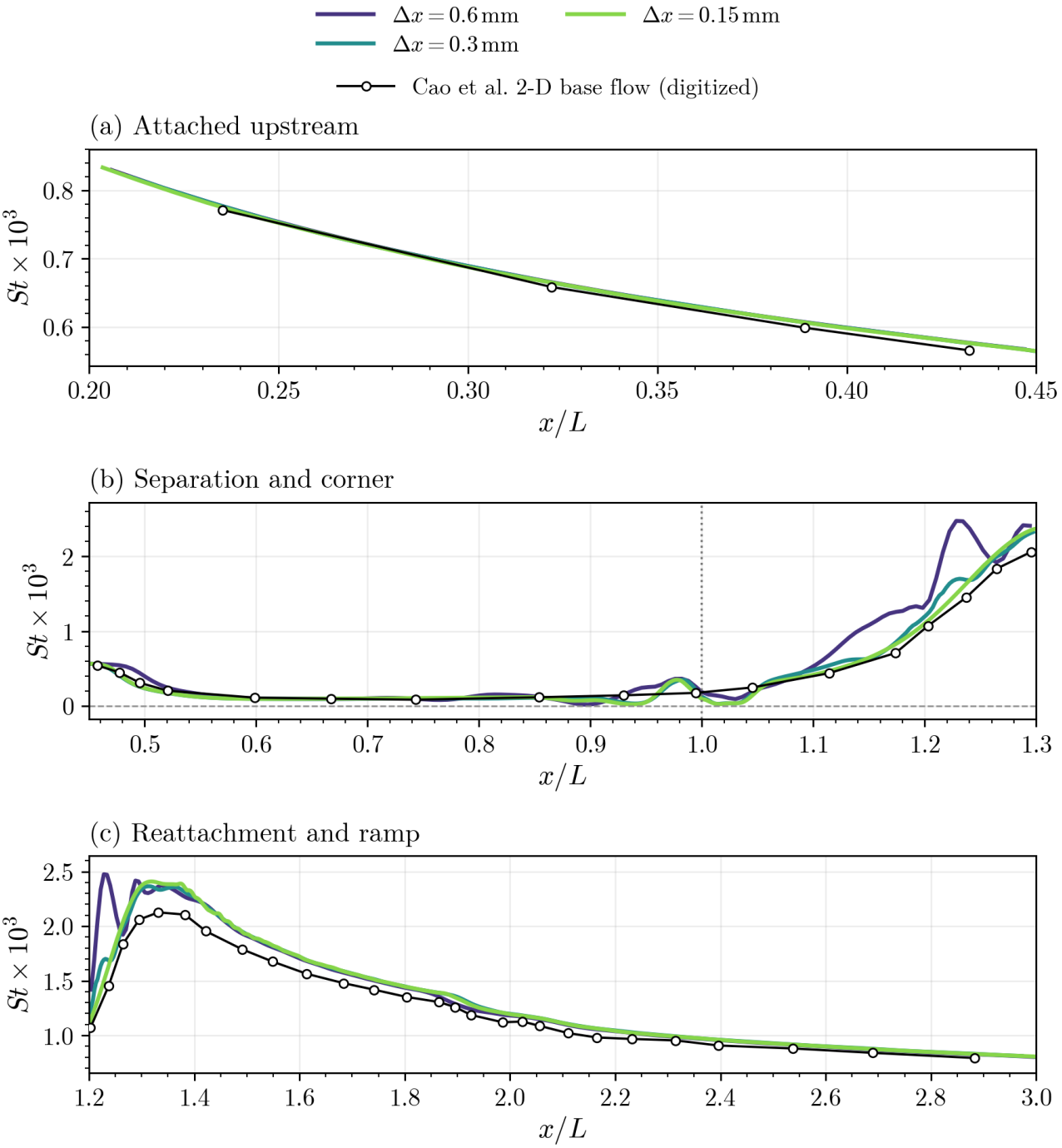



**Figure 4.31** Streamwise mesh sensitivity at $p_\infty = 1550\,\mathrm{Pa}$ and $\Delta y_1 = 0.002\,\mathrm{mm}$. The stated $\Delta x$ is the local spacing in the separation region. Symbols denote the two-dimensional base-flow Stanton number from Cao et al. [70, Fig. 4(b)]; computed curves are final pseudo-iteration-window means.

To put the wall-normal spacing into context, the later in Section 6.2 more closely discussed first-cell acoustic Reynolds number,

$$Re_{\mathrm{cell1}} = \frac{\rho_w a_w \, \delta_n}{\mu_w}, \tag{4.5}$$

is investigated along the compression corner. As will be discussed in Section 6.2 and in that section shown through extensive mesh convergence study on fully turbulent zero-pressure-gradient flat-plates, $Re_{\mathrm{cell1}}$ has been shown and promises a good assessment of the required wall-normal mesh spacing for aerothermal heating simulations. Given that only after the $\Delta y_1 = 2 \times 10^{-3}$ mesh convergence of the Stanton-number along the whole domain was reached (see Figure 4.30 (a-c)), Figure 4.32 indicates a similar finding, that

the $Re_{\text{cell1}}$ shall be kept around unity, with, in this case, a peak of $Re_{\text{cell1}} \approx 3$ being acceptable. This finding can also be broken down into the different regions of this compression corner case: the attached flat-plate upstream region showed mesh convergence with the $\Delta y_1 = 10^{-2}$ wall-normal mesh already, which, in this region also sees no higher than a $Re_{\text{cell1}} \approx 2$ (see Figure 4.32 $x/L = 0\text{to}0.5$). The study therefore supports that both first-cell height as well as streamwise spacing underly a certain minumim requirement to achive full mesh convergence, with the study indicating that the first-cell height requirement being well suited to be described using the first-cell acoustic Reynolds number.

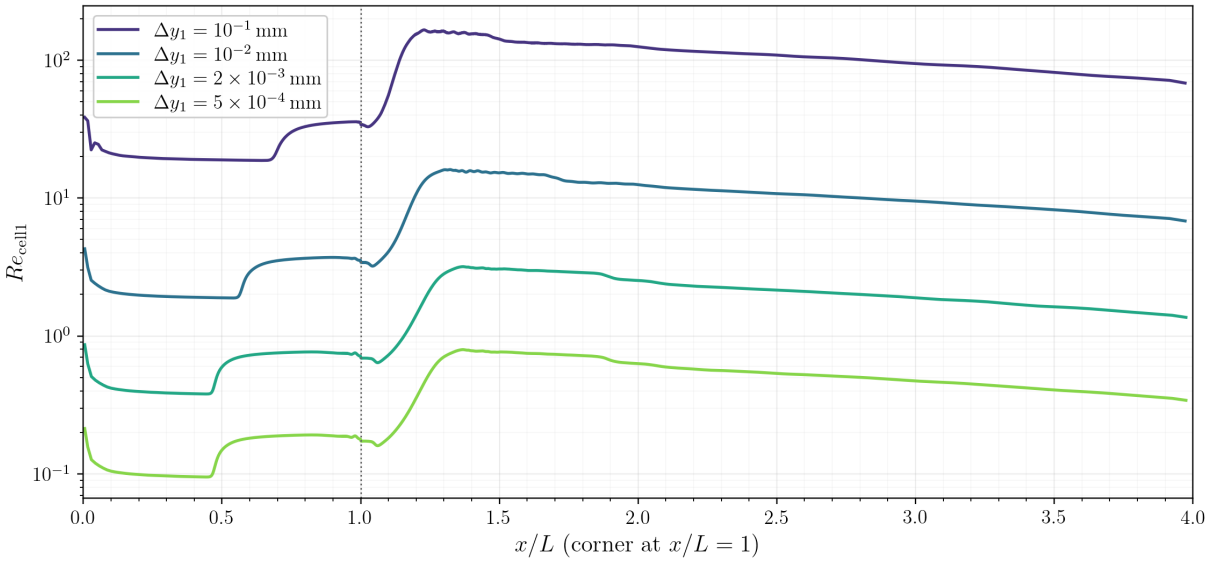



**Figure 4.32** First-cell acoustic Reynolds number for the wall-normal mesh family at nominal $\Delta x = 0.15\,\text{mm}$, averaged over the same final pseudo-iteration windows as the surface comparison. The distance used is the wall-to-cell-centre distance defined in Eq. (4.5).

### Lower-Reynolds-number case and scope of validation.

The $760\,\text{Pa}$ case in Figure 4.33 provides a second comparison. Applying the findings of wall-normal and streamwise mesh spacing requirements, the Stanton number and detachment and reattachment points are predictable reasonably well. But the overshoot in the heat-flux at the reattachment point preveils for this case as well.

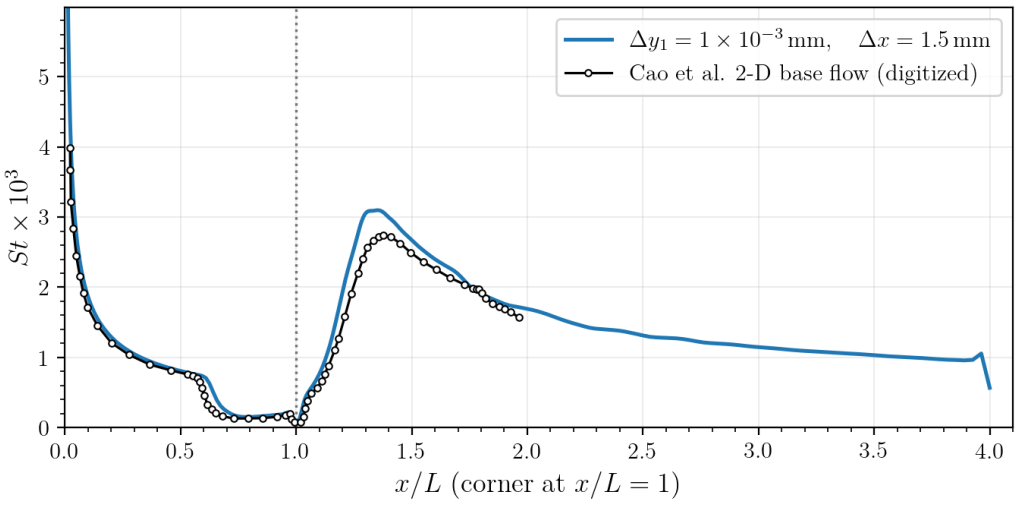



**Figure 4.33** Stanton number of the Low-Reynolds, $p_\infty = 760\,\text{Pa}$, case. The heat-transfer reference is the sharp-edge R0 two-dimensional base flow in Cao et al. [71, Fig. 8(a)].

The two cases support the qualitative prediction of attached laminar heating and its redistribution by separation and reattachment. The mesh study also identifies the corner interaction and peak heating as the remaining sensitive parts of the solution. Agreement in the upstream profiles and attached heat-transfer curve therefore complements, but does not replace, the separated-flow assessment. These results establish the scope of the laminar comparison without assigning a quantitative accuracy bound.

## 4.5 Adiabatic turbulent boundary layer profiles

The Mach-4.5 flat-plate case investigated in this section assesses the predictions of the $k$–$\omega$ shear-stress-transport (SST) and Spalart–Allmaras (SA) models in an attached, compressible turbulent boundary layer. The reference solutions are the Wind-US calculations distributed through the NASA Glenn NPARC Alliance Validation Archive [74]. The experimental basis is the flat-plate investigation of Coles [75], with the experimental data used here are taken from the NASA archive[5]. The principal objective is to compare OpenFOAM/HiSA with NASA's implementation of the corresponding turbulence models. This comparison checks whether the predicted velocity profiles and wall shear reproduce the reference behaviour qualitatively. The measurements provide an additional assessment against physical observations.

The selected NASA solutions use Wind-US 3.162 on a structured mesh with its unstructured solver, denoted *Str-Uns* in the archive and figure legends. Both reference calculations assume a fully turbulent boundary layer. The comparison therefore concerns the development of an attached turbulent layer, while the different transition and tripping histories in the experiment remain relevant when interpreting agreement with Coles' experimental results. The present section considers the OpenFOAM native implementation of $k$–$\omega$ SST and the density-corrected SA implementation (see Section 3.5.3).

**General case setup.**

The flow is parallel to a flat-plate at $M_\infty = 4.512$, with no imposed geometrical turning or external streamwise pressure gradient. The lower boundary upstream of $x = 0$ is a slip wall, where the subsequent no-slip, adiabatic plate begins at $x = 0$. No transition model or explicit trip is included, the no-slip wall is solved as fully turbulent. Both OpenFOAM/HiSA cases use one spanwise cell with `empty` side boundaries. The inlet state is fixed to the prescribed freestream, corresponding to the frozen inlet of the NASA case. The upper and rear boundaries use characteristic farfield conditions (Section 3.1.2). At the viscous wall, velocity is zero and pressure and temperature have zero normal gradient, so the adiabatic wall temperature is obtained as part of the solution. The upstream slip surface is excluded from the viscous-wall distance calculation.

Air is represented by a calorically perfect ideal gas with $c_p = 1004.5002\,\mathrm{J\,kg^{-1}\,K^{-1}}$, $R = 287.00005\,\mathrm{J\,kg^{-1}\,K^{-1}}$, and $\gamma = 1.4$. The molecular viscosity matches the Wind-US Sutherland law after conversion to SI units,

$$\mu(T) = A_s \frac{T^{3/2}}{T + T_s}, \qquad A_s = 1.49610554 \times 10^{-6}\,\mathrm{Pa\,s\,K^{-1/2}}, \qquad T_s = 120\,\mathrm{K}. \tag{4.6}$$

A constant molecular Prandtl number is imposed through $\lambda = c_p \mu / Pr$, with $Pr = 0.72$. Both turbulence models use a constant $Pr_t = 0.9$, as documented in the NASA Wind-US user guide, thus a reasonable assumption to also have been used for the NASA reported results. Table 4.11 summarizes the matched physical conditions. The freestream viscosity is $\mu_\infty = 3.89802 \times 10^{-6}\,\mathrm{Pa\,s}$, giving $Re_x/x = 6.98354 \times 10^6\,\mathrm{m^{-1}}$.

[5] `https://www.grc.nasa.gov/WWW/wind/valid/fpmach45/fpmach45.html` [accessed June 11th 2026]

**Table 4.11** Freestream and thermal-model quantities prescribed for the OpenFOAM/HiSA Coles flat-plate cases.

| Quantity | OpenFOAM/HiSA |
|---|---|
| Mach number $M_\infty$ | 4.512 |
| Static pressure $p_\infty$ [Pa] | 671.719 |
| Static temperature $T_\infty$ [K] | 60.4672 |
| Freestream velocity $U_\infty$ [$\mathrm{m\,s^{-1}}$] | 703.289 |
| Heat-capacity ratio $\gamma$ | 1.4 |
| Unit Reynolds number $Re_x/x$ [$\mathrm{m^{-1}}$] | $6.984 \times 10^6$ |
| Molecular Prandtl number $Pr$ | 0.72 |
| Turbulent Prandtl number $Pr_t$ | 0.9 |
| Wall thermal condition | Adiabatic |

**Mesh and turbulence-model settings.**

Both cases use the structured mesh taken from the NASA Str-Uns flat-plate validation archive [74] and converted to OpenFOAM, with coordinates converted from feet to metres. Figure 4.34 shows the mesh and boundary-condition layout shown in the archive.

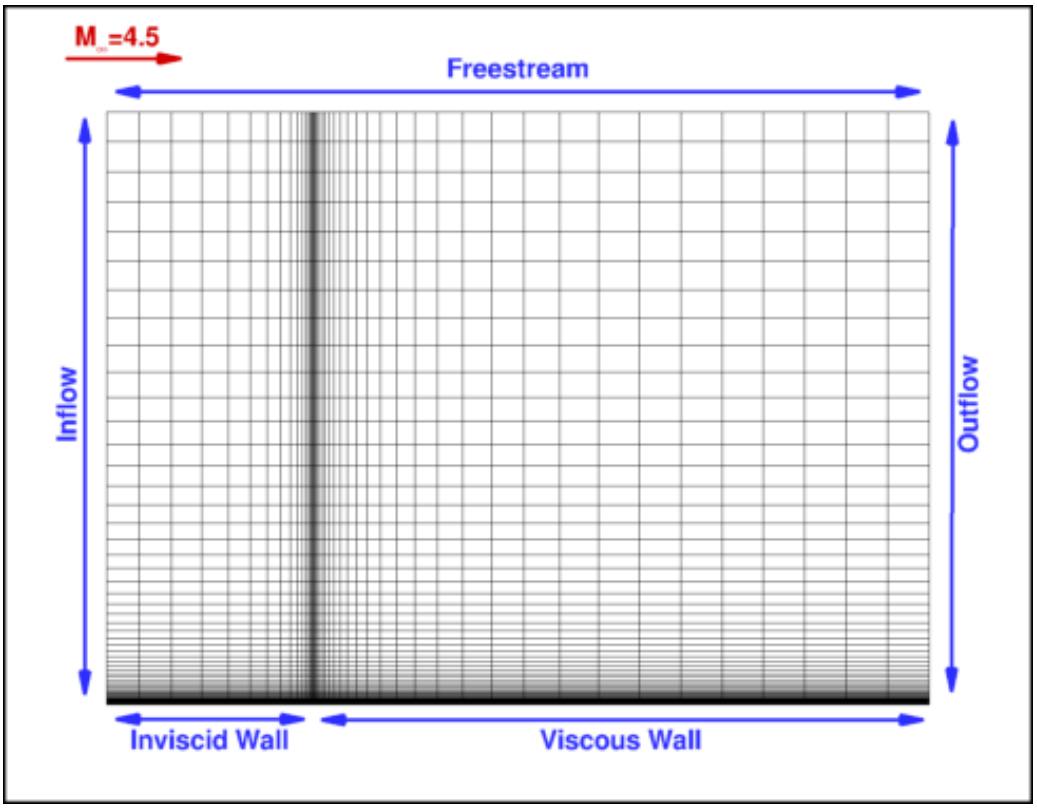



**Figure 4.34** Structured mesh and boundary-condition layout for the Mach-4.5 flat-plate case. Taken from the NASA NPARC Alliance Validation Archive [74].

The two models use the inlet quantities and near-wall conditions listed in Table 4.12. These values match the NASA inlet settings.

**Table 4.12** Turbulence inputs and recorded wall resolution of the two OpenFOAM/HiSA cases. Inlet viscosity ratios refer to the molecular kinematic viscosity used in the inlet setup. Wall $y^+$ statistics are from the saved iteration 20 000.

| Quantity | SST | SA |
|---|---|---|
| $k_\infty$ [$\mathrm{m^2\,s^{-2}}$] | $2.32369 \times 10^{-3}$ | — |
| $\omega_\infty$ [$\mathrm{s^{-1}}$] | $2.30738 \times 10^{4}$ | — |
| $\widetilde{\nu}_\infty$ [$\mathrm{m^2\,s^{-1}}$] | — | $5.03533 \times 10^{-4}$ |
| Inlet viscosity ratio | $\nu_{t,\infty}/\nu_\infty = 0.001$ | $\widetilde{\nu}_\infty/\nu_\infty = 5$ |

Iterative convergence is assessed from the stability of the physical wall shear stress. Across the six saved solutions from iterations 18 000 to 20 000, the maximum peak-to-peak downstream skin-friction variation for $x \geq 0.1\,\mathrm{m}$ is $0.0875\%$ for SA and $0.00352\%$ for SST. At the profile station, the corresponding variations are below $0.0002\%$, while changes in the streamwise velocity profiles are below $0.00005\%$ of $U_\infty$. The remaining flow-residual jitter is treated as numerical; it does not produce material changes in these comparison quantities. The figures therefore use the accepted iteration-20 000 solutions as fully converged.

**Extraction and normalization.**

The streamwise and wall-normal velocities are denoted by $u$ and $v$ respectively. Both are normalized with $U_\infty = 703.289\,\mathrm{m\,s^{-1}}$. The streamwise skin-friction coefficient is evaluated as

$$C_{f,x}(x) = \frac{|\tau_{w,x}(x)|}{\frac{1}{2}\rho_\infty U_\infty^2}, \qquad \rho_\infty = 0.0387067\,\mathrm{kg\,m^{-3}}. \tag{4.7}$$

NASA lengths and velocities are converted from feet and feet per second to SI units.

### 4.5.1 Velocity profiles and skin-friction

Figure 4.35 compares both velocity components at $x = 0.5461\,\mathrm{m}$. The streamwise profiles closely follow the NASA solutions through the boundary layer and approach the same outer velocity. the logarithmically plotted velocity profile in Figure 4.36 gives an even better view of the close matching between the numerical results by the NASA Wind-US Code and the herin used OpenFOAM/HiSA framework.

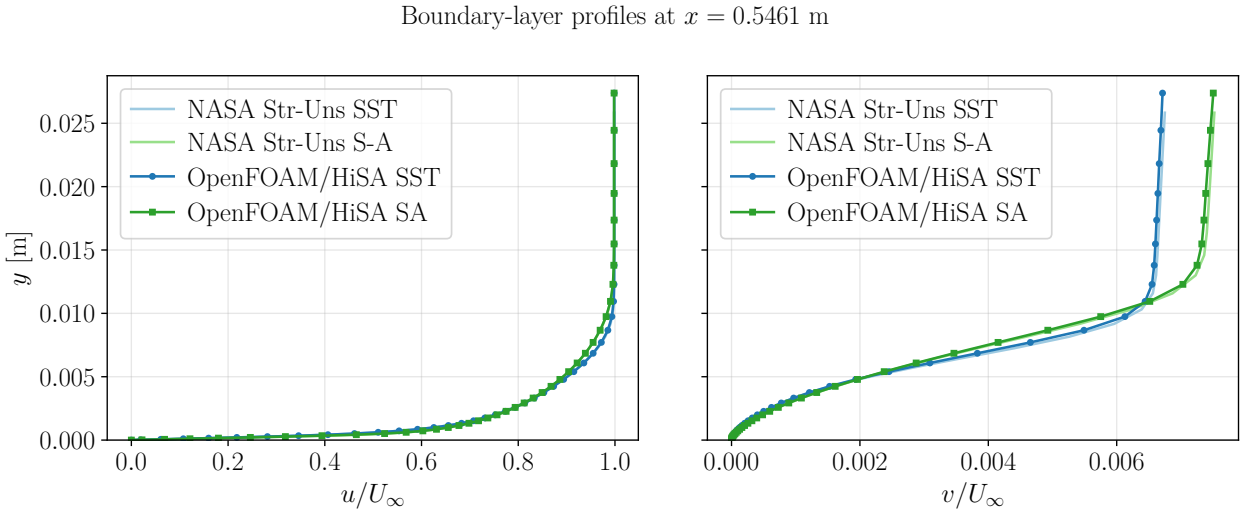



**Figure 4.35** Streamwise and wall-normal velocity profiles at $x = 21.5\,\mathrm{in} = 0.5461\,\mathrm{m}$. Dark curves with symbols show present OpenFOAM/HiSA results; pale curves show the corresponding NASA Str-Uns SST and SA solutions.

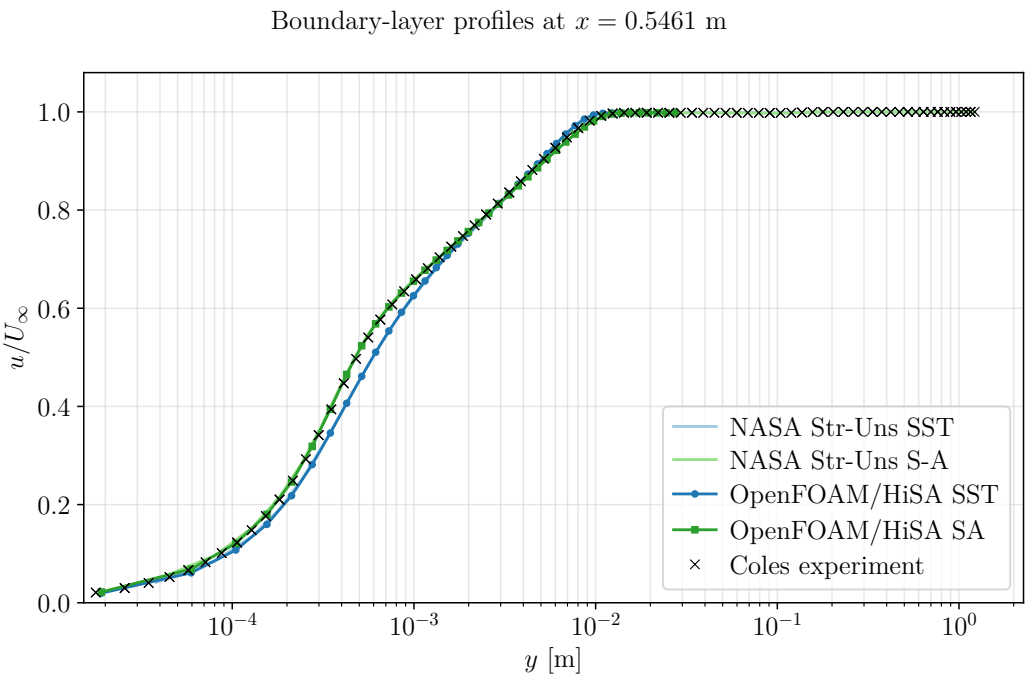



**Figure 4.36** Streamwise velocity at $x = 0.5461\,\mathrm{m}$, plotted against logarithmic wall distance. The experimental reference is the *Experiment* profile supplied with the NASA archive; the numerical curves are the same SST and SA cases as in Figure 4.35.

Figure 4.37 presents the skin-friction coefficient, including separate leading-edge and downstream views. Similarly to the velocity boundary layer profiles, the OpenFOAM/HiSA results match the NASA results, showing that shear-stress calculations are consistent between both codes. The three Coles fence-trip series provide experimental context for the downstream trend.

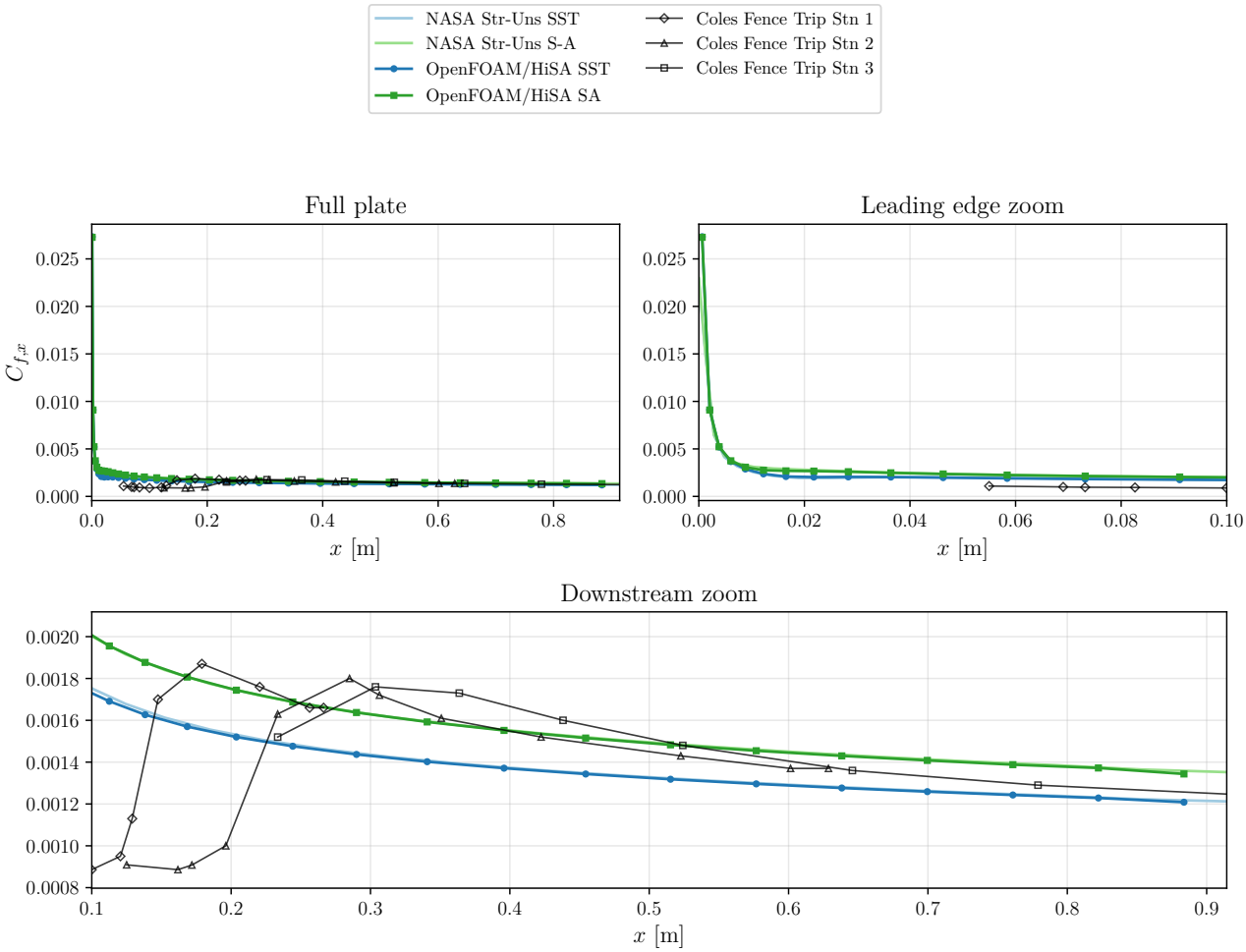



**Figure 4.37** Streamwise skin-friction coefficient: full plate (top left), leading-edge detail (top right) and downstream detail (bottom). Black symbols denote the three archived Coles fence-trip series [75].

### Assessment.

With matched physical mesh dimensions, freestream Reynolds number and molecular and turbulent Prandtl numbers, OpenFOAM/HiSA reproduces the NASA velocity profiles and downstream wall shear closely. The result supports the consistency of the turbulence-model implementations for an attached, adiabatic compressible boundary layer.

# 5 Wall-function formulations and implementation

The investigated wall functions combine momentum and thermal closures to infer wall shear stress and heat-flux from a sampled boundary layer state. Figure 5.1 shows this coupling: the sampled velocity and thermodynamic state, the physical wall temperature and the required reference information determine the wall loads. Their delivery to the CFD equations is distinct from the closure itself. Coupling to a RANS turbulence model additionally requires a compatible near-wall turbulence constraint.

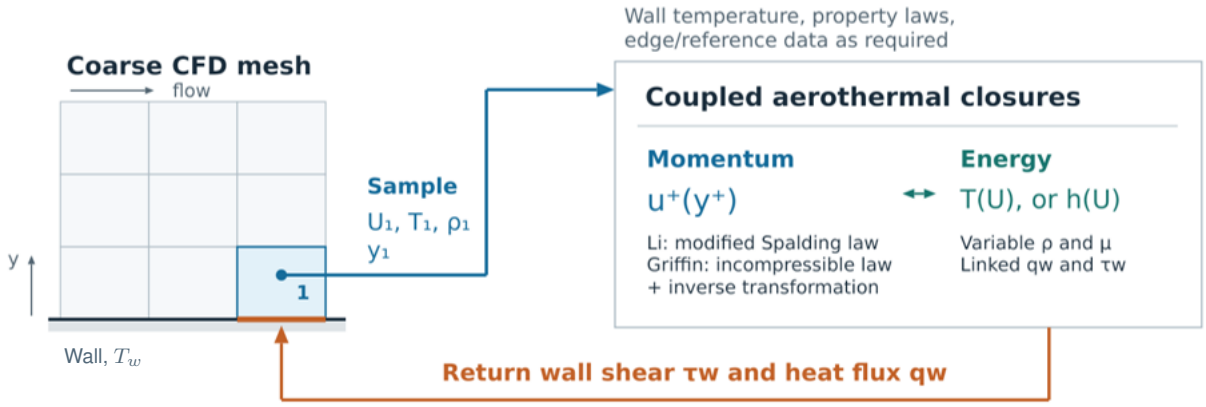



**Figure 5.1** Wall-function coupling using the first cell as the sample. Momentum and thermal closures infer wall shear stress and heat-flux from the sampled state and the wall and reference conditions, and return these loads to the CFD solver.

## 5.1 Wall-function sampling and mesh geometry

The wall-function sampling height is denoted by $y_s$, and the wall-to-first-cell-centre distance by $y_1$. On a conventional wall-function mesh, the first cell supplies the sample, so $y_s = y_1$. Increasing the first-cell height consequently moves the sample outwards and coarsens the neighbouring internal-face stencils (Figure 5.2). The term *conventional coarse wall-function mesh* refers to this coupled change in sampling height and mesh spacing throughout this work.

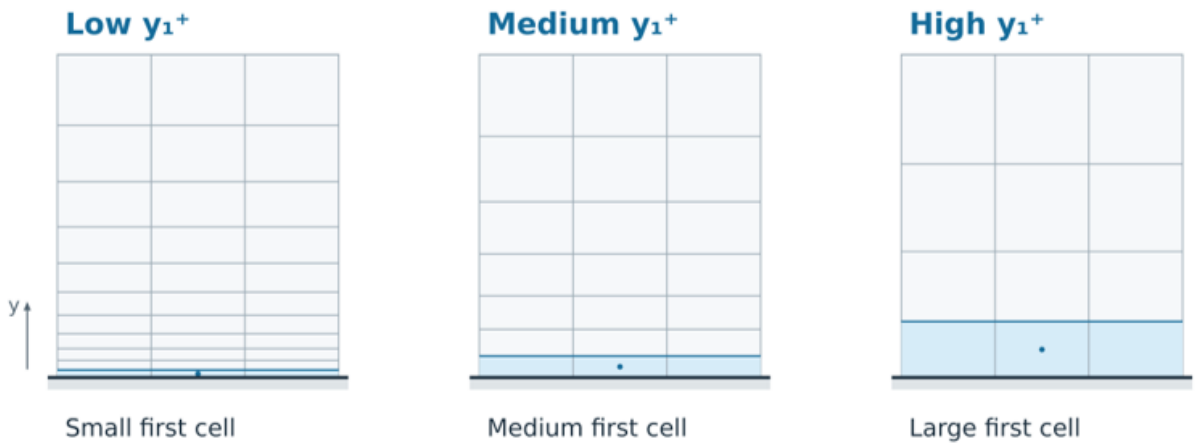



**Figure 5.2** Conventional wall-function mesh strategy. Enlarging the wall-adjacent cell increases both the first-cell sampling height $y_s = y_1$ and the spacing of the nearby internal-face stencils. The indicated wall-unit heights refer to the cell centres, not the full cell heights.

The implemented Li and Griffin wall functions can also sample an interior cell while retaining finer cells below it. The selected cell supplies the closure inputs and its wall-normal distance supplies $y_s$, denoted $y_m$ in Griffin's formulation. The associated SA constraint is applied at that same cell, while the predicted loads are delivered at the physical wall. The cells below the sample remain part of the CFD solution. This option separates closure sampling from the first-cell distance $y_1$ and the centre separation $h$ across an internal face at $y_f$, defined in Equation (3.1). Because the wall loads are inferred from modelled wall slopes, the sample need not meet the wall-resolved secant requirement of Section 3.4. Its location must nevertheless lie within the closure's applicable range. Moreover, prescribing the wall loads does not replace internal-face gradient reconstruction or transport-coefficient interpolation. Section 7.4.3 uses the independent sampling option to assess the additional mesh requirement within the retained boundary layer. The following section first explains the common flux-delivery mechanism and subsequent sections present the individual closures and their RANS coupling.

## 5.2 Wall-flux delivery in OpenFOAM/HiSA

The standard wall-load evaluation in Section 3.4 combines a wall-to-cell gradient with an effective transport coefficient. Encoding a wall-function target through that coefficient is convenient, but constrains which loads can be delivered. This section motivates the direct wall-flux interface used for the investigated closures.

### 5.2.1 Limitations of transport-coefficient boundary conditions

A conventional implementation adjusts wall `nut` and `alphat` until the discrete gradient produces the target stress or heat-flux. For example, inversion of the planar-wall shear relation gives

$$\nu_{t,w} = \frac{\tau_{w,\mathrm{target}} y_1}{\rho_w U_{1,t}} - \nu_w, \tag{5.1}$$

where $U_{1,t} > 0$ denotes the tangential velocity magnitude. An analogous inversion determines the thermal coefficient from the target heat-flux and the wall-to-cell energy difference.
Requiring non-negative turbulent transport leaves the molecular contribution as the smallest representable flux magnitude for a given nonzero secant. It also fixes the flux direction to that of the secant. Neither condition is guaranteed to agree with a wall-function target. The thermal restriction is particularly relevant to non-monotonic temperature profiles: if a sample beyond the temperature maximum is colder than the wall while the physical wall remains cooled, a non-negative conductivity cannot deliver the required heat-flux through that wall-to-cell temperature difference. Even when the signs agree, the target can lie below the

molecular secant contribution. These are limitations of the delivery mechanism, separate from closure accuracy.

### 5.2.2 Limitations of a prescribed-gradient temperature condition

A prescribed-gradient temperature condition can represent either heat-flux sign, but matching the target through the discrete gradient lets the patch temperature depart from the physical wall temperature used by the closure. On a coarse mesh, the resulting extrapolated face state can differ substantially from that physical state. Evaluating density and viscosity at this numerical face temperature then changes the wall transport properties and can make the molecular shear contribution exceed the target in Equation (5.1). This approach therefore introduces an additional coupling between thermal flux delivery and momentum transport. Retaining the physical prescribed wall temperature while replacing the wall flux avoids this inconsistency.

### 5.2.3 Direct wall-flux substitution

The implemented interface replaces the explicit viscous wall-face contributions in HiSA's momentum and total-energy residuals. Each wall function publishes its target traction, target heat-flux and an active-face mask. HiSA first assembles the standard wall contributions, then adds the difference between the targets and those contributions on active faces. In a common convention, with traction acting on the fluid and heat-flux $q_w$ positive into the wall, the corrections per unit area are

$$\Delta\boldsymbol{\tau} = \boldsymbol{\tau}_{w,\mathrm{target}} - \boldsymbol{\tau}_{\mathrm{std}}, \qquad \Delta q = -q_{w,\mathrm{target}} - q_{\mathrm{std}}. \tag{5.2}$$

The standard terms are the actual terms assembled by HiSA, including the full viscous traction and $q_{\mathrm{std}}$ is positive into the fluid, following OpenFOAM's convention. The target traction is tangential, as assumed by the boundary layer wall closures. The energy correction includes the corresponding viscous work, which vanishes at a stationary no-slip wall. Inactive faces retain the standard evaluation.
For the isothermal cases, this interface keeps the physical fixed wall temperature and sets wall `nut` and `alphat` to zero; these fields no longer encode the target loads. The near-wall turbulence constraint remains a separate part of the RANS coupling. Direct substitution changes neither the wall-function closure nor the internal-face transport operators, and does not impose an exact off-wall profile. It removes the coefficient-inversion restriction while leaving closure error, turbulence-model error and spatial-discretization error to be assessed in Chapter 7.
The imposed loads are recorded in the wall-function diagnostics. Li's native heat-flux is positive into the wall, whereas Griffin's and Wang's are positive into the fluid; the interface and comparison scripts convert these signs consistently. Standard `wallHeatFlux` and `wallShearStress` function objects do not include the residual correction and therefore do not report the imposed targets. The coupled wall-function comparisons use the diagnostic targets on active faces; the wall-resolved comparisons use the standard function-object output. The source code and case configurations are identified in the data-availability statement.

## 5.3 Li wall function

Li et al. developed an improved wall function for hypersonic reacting turbulent boundary layers, addressing strong wall cooling and variable gas properties [3]. Their comparison with the precursor formulation of Gao et al. shows substantially smaller skin-friction and heat-flux errors for the investigated cold-wall flat-plates [3, Section IV.B, Figures 17–20]. The improvements concern both the thermal relation and the treatment of variable properties in the velocity law. This motivates investigating the Li formulation here, while the earlier Gao formulation is not additionally being evaluated.

### 5.3.1 Momentum and thermal closures

The thermal closure is a quadratic relation between static enthalpy and velocity [3, Eqs. (12)–(15)]. It extends the generalized temperature–velocity relation of Zhang et al. by using enthalpy to accommodate reacting-gas thermodynamics. Its generalized recovery enthalpy determines the wall slope and thereby relates heat-flux to wall shear, extending the fixed-recovery-factor prescription of the earlier Walz-based construction.

The momentum closure combines a viscosity-corrected viscous-sublayer relation with a compressible logarithmic law in a unified expression [3, Eqs. (27), (30)–(32)]. It assumes approximately constant total shear stress in the inner layer and blends through the buffer layer. Sampling below the logarithmic region is therefore permitted by construction, although accuracy can still vary with sampling height.

The formulation retains $\kappa = 0.41$ and $B = 6.0$ and approximates the viscosity variation by an enthalpy power law. Li et al. fit its exponent, denoted $\omega_H$ here, to approximately $0.72$ in their reacting viscous-sublayer profiles [3, Eq. (25) and Figure 12]. This transport approximation is calibrated for those conditions; its transfer to other gas models and wall temperatures is not automatically given and open to scrutiny.

### 5.3.2 Non-reacting implementation and reference-state choice

The present implementation uses fixed composition and omits chemical reactions and species-diffusion contributions. The enthalpy-based closure is retained. With the additional calorically perfect-gas assumption, $h = c_p T$, its thermal relation reduces to the Zhang temperature–velocity relation reproduced by Li et al. [3, Eqs. (8)–(13)]. This equivalence concerns the thermal closure; the momentum closure remains Li's unified velocity law. Removing chemistry alone does not require constant specific heat.

A useful feature is that the reference velocity and enthalpy need not be taken at the boundary layer edge. Li et al. reformulate both closures using an interior reference point and select the second off-wall grid point in their implementation [3, Section IV.C, Eqs. (40)–(42)]. Here, the matching cell and a second cell farther from the wall provide the two velocity–enthalpy states. Together with the physical wall state, they determine the thermal relation. The unified velocity law is then inverted for the wall shear at the matching height, and the inferred thermal wall slope supplies the heat-flux. A prescribed-edge option is retained for comparison.

This local reference construction avoids explicit edge detection, whose mesh-dependent reconstruction introduces additional sensitivity into Griffin's closure inputs (Section 5.4.2). The second reference point remains distinct from the matching point; neither location should be confused with the full height of the wall-adjacent cell.

The recovered shear stress and heat-flux are delivered directly to HiSA (Section 5.2.3). Following Li et al. [3, Eqs. (36)–(39)], the velocity-law derivative also supplies a compatible eddy viscosity and hence a constraint on the SA variable at the selected near-wall layer. This turbulence constraint is separate from prescribing the wall fluxes.

### 5.3.3 Scope of the published assessment

The non-reacting flat-plate case in Li et al. checks the underlying solver and SA model against an experimental data and van Driest II theory [3, Section IV.B and Figure 2]. There it does not assess the improved wall function. The latter is tested on reacting flat-plates against resolved RANS solutions; the second-reference-point variant is demonstrated at first-point $y^+ = 4$ and $8$, whereas the coarser logarithmic-layer tests use edge quantities [3, Sections IV.B–C]. These results do not provide an independent DNS or experimental validation of the non-reacting wall function used here and can thus not be used as a way to validate the code implementation.

Direct reproduction also requires care because Li et al. use a finite-difference solver. They define the wall distance to the first off-wall grid point, while their coupling acts on viscous fluxes between that point and the wall [3, Section II.A, Eq. (17) and Figure 16]. Transfer to a finite-volume method requires matching

the physical sampling and flux locations; equal reported $y^+$ values alone do not establish an equivalent discretization. The published results therefore provide context rather than a direct verification case for the complete implementation.

### 5.3.4 Reference sensitivity and assessment in this thesis

Avoiding edge detection does not remove all reference-state sensitivity. Inferring the thermal relation becomes poorly conditioned when the sampled and reference velocities nearly coincide, and both points must remain within the region represented by the closure. The prescribed-profile tests in Section 7.3.2 compare the second-point and prescribed-edge choices on the M6Tw025 DNS profile and the LowH_M3 resolved SA profile. They show no systematic loss of accuracy from using the second point, supporting this practical advantage while retaining appreciable closure errors. These tests establish the standalone closure response before the coupled mesh sensitivity is examined in Section 7.4.

## 5.4 Griffin inverse-transformation wall function

Griffin et al. construct a compressible near-wall profile by applying an inverse velocity transformation to an incompressible wall law and closing the temperature profile through an algebraic temperature–velocity relation [76, 4]. The formulation accounts for density and viscosity variations associated with wall cooling while retaining a unified description of the viscous sublayer, buffer layer and logarithmic region. Its application here as a RANS wall function requires both the local matching state and information about the boundary layer edge.

### 5.4.1 Momentum and thermal closures

The momentum closure builds on the total-stress velocity transformation of Griffin et al. [36, Eqs. (4)–(5)]. It combines viscous scaling near the wall with turbulence-equilibrium scaling in the logarithmic region, blending between them through the buffer layer. The wall model uses its approximately constant-total-shear-stress form. Inverting this transformation converts the mean shear of a damped incompressible eddy-viscosity model into the compressible velocity gradient, including local density and viscosity variations [4, Eq. (2.2)]. The viscous limit recovers the linear wall law; the model is therefore not restricted by construction to logarithmic-layer sampling.

The thermal closure is a quadratic temperature–velocity relation based on the generalized Reynolds analogy [4, Eq. (2.5)]. It satisfies the wall temperature and the sampled temperature at the matching velocity, while its wall slope is prescribed through the Reynolds analogy factor and the recovery temperature. The latter depends on the edge velocity and temperature. This wall slope links heat-flux directly to wall shear, without solving a separate temperature equation with an assumed constant turbulent Prandtl number. In Griffin's convention, heat-flux is positive into the fluid and negative for a cooled wall.

For a prescribed wall temperature, the inputs are the velocity, temperature and density at the physical matching height $y_m$, the edge state, an equation of state and a viscosity law. The inner reconstruction uses the ideal-gas law with approximately constant pressure and constant specific heat. A trial wall shear stress is used to integrate the inverse relation from the wall region to $y_m$, updating temperature and transport properties along the reconstructed profile. The stress is iterated until the integrated velocity reproduces the sampled velocity; the thermal relation then supplies the heat-flux [4, Section 2.3]. Both wall loads thus follow from reconstructed wall slopes rather than a wall-to-cell secant.

The implementation retains the Reynolds-number-dependent calibration of the incompressible coefficients $\kappa$ and $A^+$ from the earlier research brief [76, Section 2]. These coefficients are selected using the edge-based semi-local friction Reynolds number, so the estimated thickness also enters the closure. The journal article examines this calibration but adopts fixed values $\kappa = 0.41$ and $A^+ = 17$ for its subsequent simulations [4, Section 3]. This distinction matters when comparing the implemented variant with published results.

### 5.4.2 Boundary layer edge reconstruction and sensitivity

The edge quantities are obtained with Griffin et al.'s local-reconstruction method [77]. It reconstructs a local inviscid velocity $U_I$ from the stagnation enthalpy and identifies $\delta_{99}$ where the actual tangential velocity first satisfies $U/U_I = 0.99$ [77, Eq. (9) and Appendix D, Eqs. (D1)–(D2)]. This replaces comparison with a prescribed freestream velocity and permits a varying inviscid outer profile. The compressible construction assumes an inviscid, non-heat-conducting outer flow.

In the present finite-volume implementation, cell-centre states are sampled outward from the wall, and their maximum stagnation enthalpy supplies the reconstruction reference. The first crossing of the squared velocity-ratio criterion brackets the edge. Its position, velocity and temperature are obtained by linear interpolation between the bracketing samples, using the squared ratio as interpolation coordinate. This interpolation is an implementation choice; the continuous edge criterion does not prescribe a discrete interpolation scheme.

Figure 5.3 illustrates the sensitivity for M6Tw025. The extracted thickness and edge temperature vary appreciably between conventional wall-function meshes, whereas the relative variation in edge velocity is smaller. These differences reflect both changes in the computed profiles and their discrete reconstruction; they do not isolate interpolation error. The two DNS markers further show that edge quantities must be compared using consistent definitions.

These variations affect model inputs: edge velocity and temperature enter the thermal closure through the recovery temperature, and thickness enters the coefficient calibration described above. The physical matching height remains a separate input. The figure identifies an additional mesh dependence, but does not establish which edge quantity dominates the wall-flux error. This motivates separating the prescribed-profile closure tests in Section 7.3 from the coupled mesh assessment in Section 7.4.

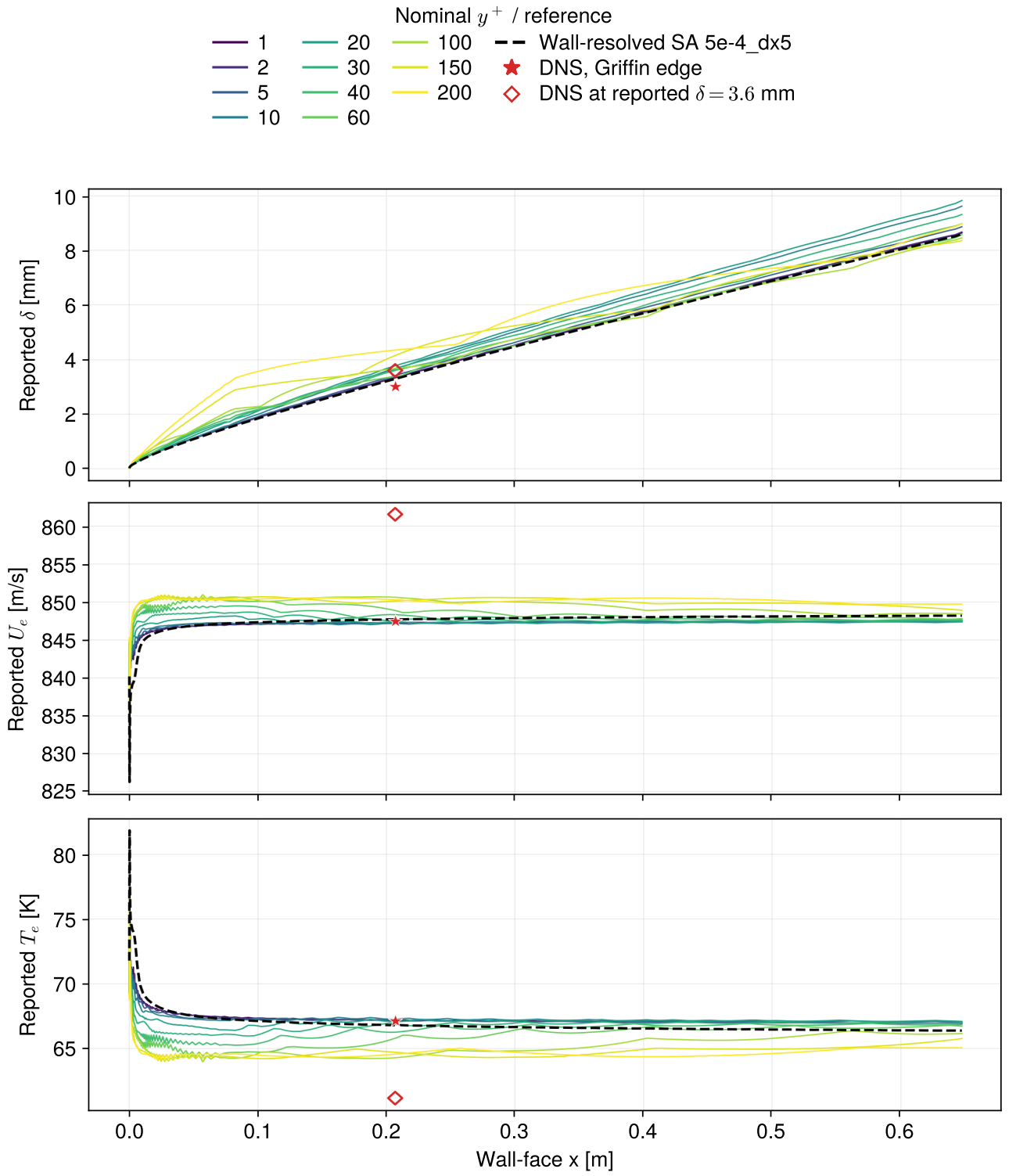



**Figure 5.3** Boundary layer thickness $\delta$, edge velocity $U_e$ and edge temperature $T_e$ for M6Tw025. Colours identify conventional meshes by nominal first-cell $y_1^+$; the dashed curve is the wall-resolved SA reference evaluated with the same criterion. Stars apply the Griffin criterion to the DNS profile, taking the unavailable mean wall-normal velocity as zero. Diamonds show the reported DNS thickness $\delta = 3.6\,\mathrm{mm}$ and the velocity and temperature interpolated at that height [5].

### 5.4.3 Application to RANS and limits of the published validation

Griffin et al. describe the model as suitable for coupling to either LES or RANS. Their assessment combines a-priori tests on prescribed DNS profiles with a-posteriori WMLES of turbulent channels; it contains no coupled RANS validation [4, Sections 3–5]. The principal comparisons use $y_m = 0.3\delta$, but the journal article also varies matching height during mesh refinement, including $0.15\delta$ and $0.075\delta$ [4, Section 4.1]. These are matching locations, not prescribed first-cell-centre heights. As in WMLES it is common practice to retain off-wall resolutions below the point at which the wall model samples the boundary layer and then select the sample station adaptively dependant on the local boundary layer behaviour, i.e. if detachment is detected [78]. In addition to LES commonly using higher-order reconstruction methods, this may explain why Griffin et al. most likely didn't stumble upon same mesh requirement issue (ref. Section 7.4).

Here, the closure is coupled to SA through the near-wall turbulence constraint, with its wall shear and heat-flux delivered directly to HiSA as described in Section 5.2.3. This RANS coupling and the reconstructed edge inputs require their own assessment. The published WMLES performance consequently provides no direct validation of the complete implementation. Nevertheless, the standalone closure can be checked against the published a-priori results, as shown in Table 7.4. That consistency check supports the subsequent error-isolation study without implying equivalent accuracy of the coupled RANS calculations.

## 5.5 Wang–Craft analytical wall function

The third an final wall function investigated in this work is by Wang et al [79]. It is unique compared Li et al and Griffin et al in that it does not rely on analytical momentum and energy closures, but rather on an analytical result of a proposed integral across the boundary layer. The downside of this approach being that it is not possible to run CFD independent analyses like the a-priori solves of the Li and Griffin wall functions (see Section 7.3). The implementation of this wall function raised several issue and ambiguities. Most notably on the streamwise integration necessary. While in the thesis work of Wang [80] it explicitly was implemented and tested on OpenFOAM there is no publicly available resource of the specific implementation. Contacting Wang unfortunately yielded in no response.

The final implementation that was able to be achieved in the scope of this work showed some significant instabilities, most likely stemming from the downstream integration process at the leading edge of the flat-plates. Additionally the heat-flux and skin-friction output were unphysical. The afformentioned lack of a possibility to perform any kind of CFD-independent analysis further complicates debugging and identifying whether or not the issue is with the implementation or fundamentally with the wall function model - also because throughout the works of Wang on this wall function no investigation on a ZPG flat-plate was ever published. Thus it was decided to cut further debugging and development work in favour of spending the time on the analysis of the turbulence models and other two wall functions.

But for future research this wall function should anything but be disregarded, as it was the only investigated wall function that focused on, and proposed, good heat-flux and skin-friction prediction capability on SWBLI and complex geometries including separation.

# 6 Benchmark cases and evaluation methodology

Assessing wall-function performance requires consistent reference data, sufficiently resolved RANS baselines and explicit accuracy targets. This chapter introduces the selected zero-pressure-gradient (ZPG) flat-plate (FP) DNS benchmarks and their limitations, specifies the matching procedure and wall-resolved mesh requirements, and defines the criteria used to judge the results. These choices underpin the successive comparisons of turbulence modelling, wall-function closure and mesh sensitivity in Chapter 7.

## 6.1 Selected zero-pressure-gradient flat-plate DNS reference cases

In this chapter the selected fully turbulent zero-pressure-gradient (ZPG) flat-plate (FP) cases used as references for the assessment of the aerothermal heating are being introduced as well as argued why they were chosen, or certain cases neglected. Furthermore in the respective subsections to each of the four primary sources of the DNS references, specific considerations that must be taken into account are highlighted.

The four primary DNS references are from part 2 [81] and part 4 [39] of Martin, Duan et al's four part series (further noted as "Martin-Duan Part 2" and "Martin-Duan Part 4" respectively) , Zhang et al [5] and Huang et al [82]. The selected benchmarks comprise 12 distinct ZPG flat-plate cases, summarized in Table 6.1.

**Table 6.1** Selected ZPG flat-plate DNS benchmark cases.

| DNS reference | Selected cases |
|---|---|
| Martin-Duan Part 2 | M5T1, M5T2, M5T3, M5T4, M5T5 (adiabatic) |
| Martin-Duan Part 4 | LowH_M3 |
| Zhang | M2p5 (adiabatic), M6Tw025, M6Tw076*, M8Tw048, M14Tw018 |
| Huang | M11Tw020 (stations BL1, BL2, and BL3) |

Case names are retained from the respective publications. The number after $M$ indicates the nominal freestream Mach number, $T$ distinguishes the wall-temperature variants in Martin–Duan Part 2. Their thermal conditions are taken from the reported data rather than inferred from the case names. For Zhang and Huang, $Tw$ denotes the wall-to-recovery temperature ratio $T_w/T_r$, with recovery factor $r = 0.89$ [5, Nomenclature]. Huang's M11Tw020 case is sampled at three downstream stations, BL1–BL3. The asterisk marks the M6Tw076 temperature note below.

These cases were selected as suitable references because they span a wide range of super- to hypersonic cold-wall regime as well as varying degrees of compressibility as indicated in Figure 6.1 range on the heat-transfer coefficient $-B_q$ and friction Mach number $M_\tau$ respectively. $M_\tau$ is discussed by Smits and Dussauge [7] and used here as an indicator for intrinsic compressibility effects, following Hasan et al. [8]. With cold-wall and compressibility effects seemingly being the dominant factors differentiating hypersonic cold-wall boundary layers from incompressible ones, it was chosen to plot Figure 6.1 using $-B_q$ and $M_\tau$. The selected DNS cases don't span the whole $-B_q$-$M_\tau$ domain and mainly lie clustered on a line. This could partly be attributed to the fact that the friction velocity $u_\tau$ is scaling both axes, or it may represent a genuine gap in DNS references which may yield interesting insights. To fully judge and address the real-world extend findings based on these DNS references can be applied, further variables such as a Reynolds number might add important value too, so as not to extrapolate findings of this work into non-applicable domains.

For the cases by Zhang et al it should be positively highlighted that the data has been additionally published digitally at `https://tmbwg.github.io/turbmodels/Other_DNS_Data/supersonic_hypersonic_flatplate.html` [accessed 14 September 2026], although the first reported off-wall data point is already at a $y^+ = 0.5$.

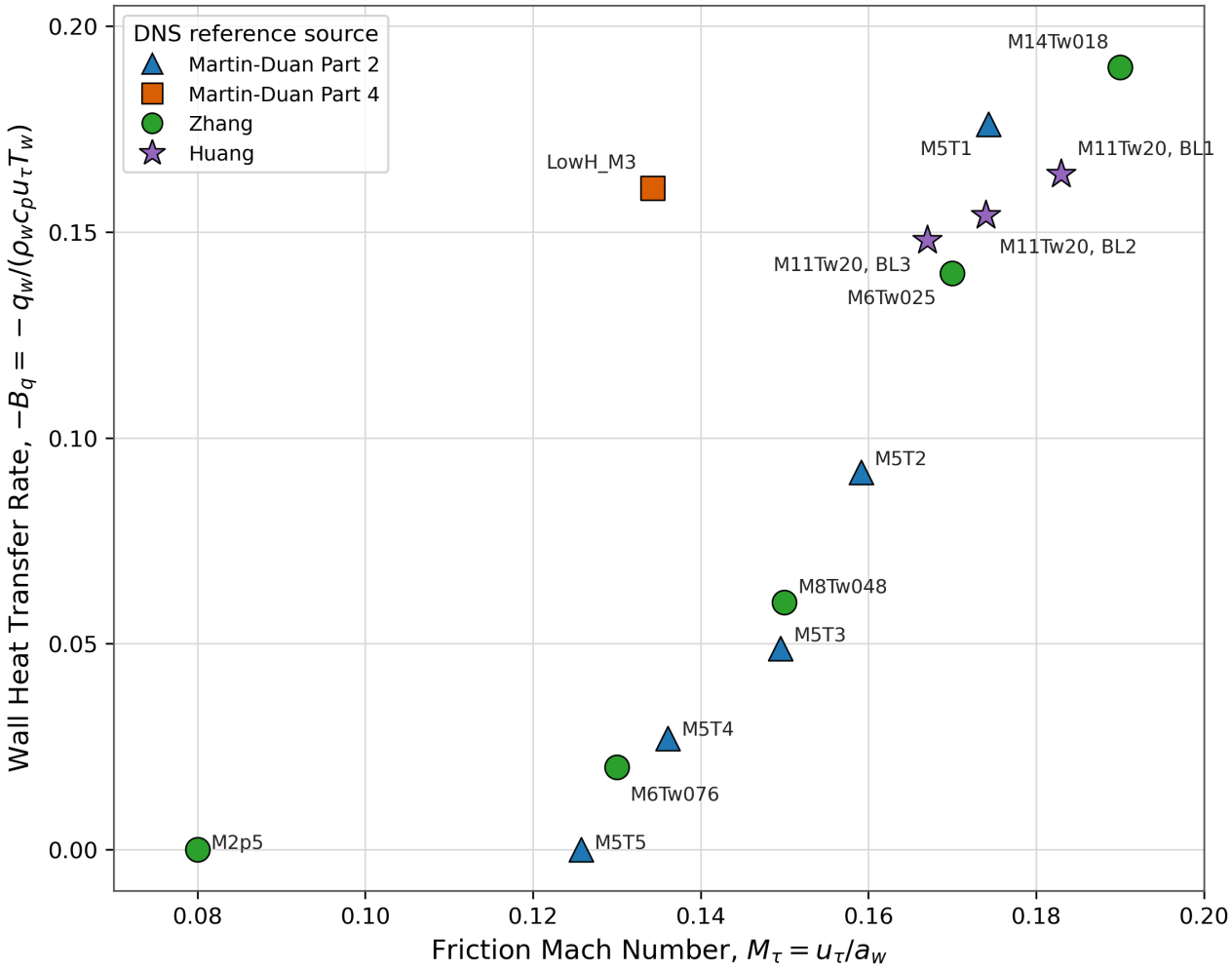



**Figure 6.1** Selected DNS benchmark cases in terms of the dimensionless wall heat-transfer rate $-B_q$ and friction Mach number $M_\tau$.

To put the cold-wall strengths of the twelve investigated ZPG flat-plates into perspective of real world flight conditions, the fraction $\frac{T_w}{T_r}$ of the DLR's flight experiment STORT is looked at, where $T_r$ is the adiabatic recovery temperature. The heat-transfer rate $-B_q$ is arguably a more comparable assessment of the cold-wall effects, but with flight experiments the skin-friction is rarely if ever available. For the calculation of the $\frac{T_w}{T_r}$ during the STORT flight, the published thermo couple readings of the surface temperature on the nose cone at 40, 152.1 and 150.5 mm from the nose tip [83, Figure 10] and the flight Mach numbers [83, Figure 9 (b)] are used. For simplification, and as this shall only serve as a rough estimation and insight to real world flight, perfect gas with $\gamma = 1.4$ and a recovery factor of $r = 0.89$ is assumed. Furthermore the calculations are simplified in assuming the freestream conditions as the boundary layer edge quantities, which is fine for ZPG flat-plates, but with the temperature probes sitting on the nosecone and somewhat close to the nose's bow shock it is arguably a rough engineering approximation, but measurements or high fidelity simulation results of the actual temperature boundary layer are not available. Thus figure should be understood as an indicative plot and not as an exact categorization of the cold-wall strength the STORT flight experienced. Never the less Figure 6.2 shows that $\frac{T_w}{T_r}$ ratios of e.g. 0.25, like herein investigated in the M6Tw025 ZPG flat-plate, are not extraordinarily cold-walls, and well within what a hypersonic vehicle

can experience. But when looking at validation of RANS models, adiabatic or even slightly warm walls (i.e. negative heat-flux, from the wall to the fluid) are to be looked at, but then again it all is very case-by-case dependant on the material, or trajectory.

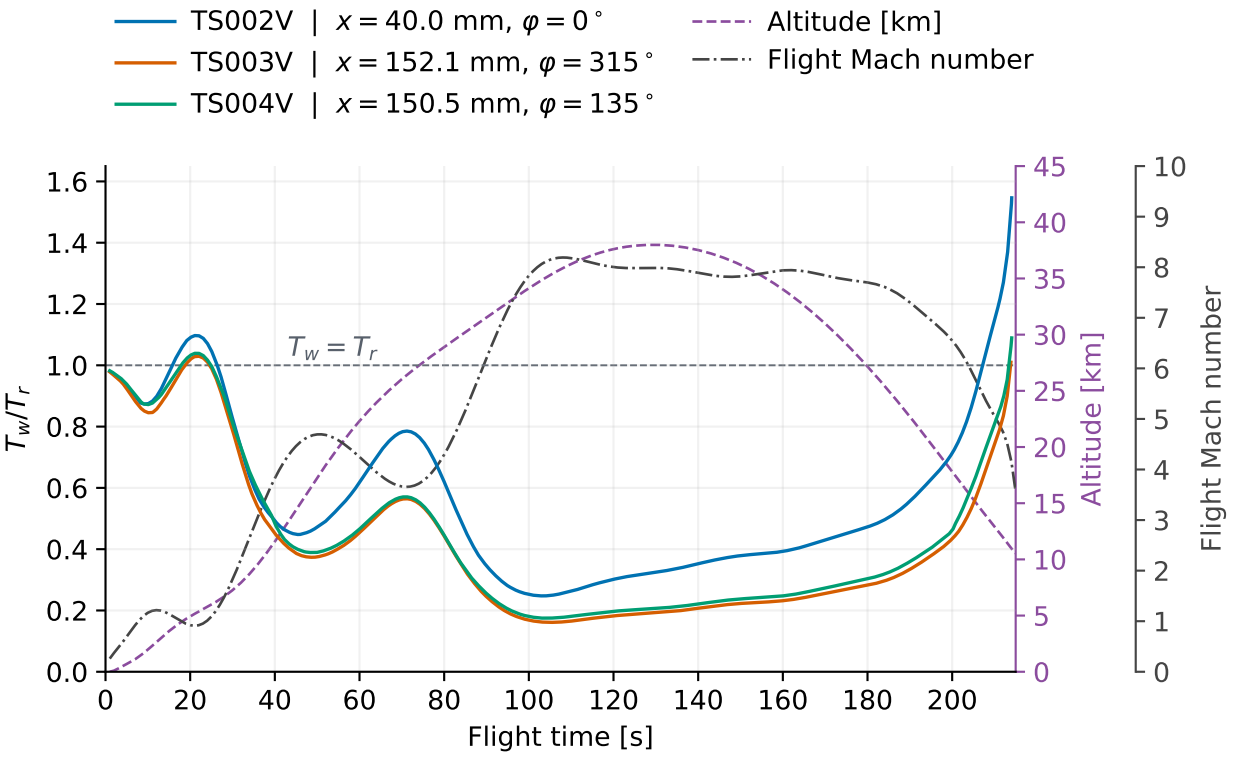



**Figure 6.2** Indicative $T_w/T_r$ ratio experienced by the DLR's STORT hypersonic flight experiment [84]. TS002V, TS003V and TS004V are the surface thermal couples as reported in [83], with their location on the nose cone, as measured in millimetres from the nose tip.

### 6.1.1 Inflow and wall conditions

In this section the inflow and wall conditions of the reference DNS cases are extracted from the respective sources and discussed. For this work it was elected to assign the reported freestream conditions, as reported in the tables below, at the inlet in a uniform manner. In 7.2.4 a publication by Vaughen argued for using back calculated pre-shock conditions instead - a sensitivity analysis was performed and revealed limited affect and no clear claim of which approach, DNS freestream or back calculated pre-shock values are a more correct replication of the DNS cases.

#### Martin-Duan

Martin-Duan Part 2 and Martin-Duan Part 4 are both created using an initial RANS simulation with added transformed DNS fluctuations followed by a recycling scheme to develop the boundary layer along the flat-plate temporally with streamwise periodic boundary conditions [85]. The inflow conditions selected are taken from the reported freestream conditions for the Mach 5 case in Table 1 of [85], which are reiterated or derived from it here in Table 6.2, similarly for the LowH_M3 case the freestream conditions are defined in [39, Table 1]. The only difference in the M5T1 to M5T5 cases are the isothermal wall temperature.

Martin-Duan Part 4 contains additional reacting references cases, which are beyond the scope of this work, and additionally a calorically and thermally perfect case, named LowH_M10. This case is intentionally omitted because of it represents an incredibly far downstream station as a result of multiple recycling which leads to an over 1.3 metre thick boundary layer at the matching station [39, Table 2]. This is also highlighted by the authors themselves, and noted that "the flow conditions themselves might not be realistic" [39].

**Table 6.2** Freestream inflow and wall conditions for the Martin-Duan cases.

| Case | $M_\infty$ | $\rho_\infty$ [kg/m$^3$] | $T_\infty$ [K] | $T_w$ [K] |
|---|---|---|---|---|
| M5T1 | 5.00 | 0.0889 | 216.65 | 228.1 |
| M5T2 | 5.00 | 0.0889 | 216.65 | 433.30 |
| M5T3 | 5.00 | 0.0889 | 216.65 | 649.95 |
| M5T4 | 5.00 | 0.0889 | 216.65 | 866.60 |
| M5T5 | 5.00 | 0.0889 | 216.65 | 1191.6 |
| LowH_M3 | 3.5 | 0.0184 | 226.5 | 125 |

For M5T1, the prescribed wall temperature $T_w = 228.1\,\mathrm{K}$ follows Part 2 Table 1, which gives $T_w/T_\delta = 1.00$ and $T_\delta = 228.1\,\mathrm{K}$ [81]. The separate Reynolds-number inconsistency is examined in Appendix A.

### Zhang

Zhang et al.'s DNS database comprises spatially developing flat-plate turbulent boundary layers with turbulent inflow supplied by a modified recycling/rescaling method. An extended downstream development region allows artificial effects of inflow generation to decay before statistics are evaluated [5]. The cases were selected in a way to represent various experimental hypersonic wind tunnel experimental conditions. The freestream inflow quantities are taken from [5, Tbale 1] and reiterated here in Table 6.3.

**Table 6.3** Freestream inflow and wall conditions for the Zhang cases.

| Case | $M_\infty$ | $\rho_\infty$ [kg/m$^3$] | $T_\infty$ [K] | $T_w$ [K] |
|---|---|---|---|---|
| M2P5 | 2.50 | 0.100 | 270.0 | (adiabatic) |
| M6Tw025 | 5.84 | 0.044 | 55.2 | 97.5 |
| M6Tw076 | 5.86 | 0.043 | 55.0 | 300.0* |
| M8Tw048 | 7.87 | 0.026 | 51.8 | 298.0 |
| M14Tw018 | 13.64 | 0.017 | 47.4 | 300.0 |

The asterisk "*" on the M6Tw076 case's $T_w$ comes from an inconsistency between the paper's Table 1 reported $T_w = 300K$ and the digitally published boundary profile data which at the wall face ($z = 0$), reports a $T/T_\infty = 5.38$, which, with the paper's Table 1 reported $T_\infty = 55.0\mathrm{K}$, results in a $T_w = 295.9\mathrm{K}$. This $4\mathrm{K}$ is a minor inconsistency, but in some Temperature related plots is a visible. Furthermore, with the M6Tw076 being almost adiabatic with a reported $-B_q = 0.02$ [5, Table 2], this wall temperature deviation can carry a higher than expected uncertainty for the wall heat-flux.

### Huang

Huang et al use a precursor RANS simulation similar to Duan and Martin, with turbulent inflow generated by recycling/rescaling and subsequently spatially developing boundary layer through three overlapping DNS boxes. The freestream and isothermal wall-temperature conditions are taken from the reported Table 1 in [82].

**Table 6.4** Freestream inflow and wall conditions for the Huang case.

| Case | $M_\infty$ | $\rho_\infty$ [kg/m$^3$] | $T_\infty$ [K] | $T_w$ [K] |
|---|---|---|---|---|
| M11Tw020 | 10.9 | 0.103 | 66.5 | 300 |

### 6.1.2 Thermal and transport models

All of the investigated DNS references herein represent thermally and calorically perfect gases. With the exception of the M8Tw048 case, which represent pure Nitrogen, all of the DNS references represent air.

**Table 6.5** Thermal models defined in or reasonably inferrable from the DNS references.

| Case | $c_{p,\infty}$ $[\mathrm{J/kg\,K}]$ | $\gamma_\infty$ | $R$ $[\mathrm{J/kg\,K}]$ | $Pr$ |
|---|---|---|---|---|
| Martin-Duan Part 2/4 | 1004.5 | 1.4 | 287 | (not defined) |
| Zhang | 1004.5 | 1.4 | 287 | 0.71 |
| Zhang - M8Tw048 | 1039.68* | 1.4* | 297.05* | 0.71 |
| Huang | 1004.5 | 1.4 | 287 | 0.71 |

The Martin-Duan cases leave the most ambiguity, as throughout their four part series they only ever state "the working fluid is air" [81, p. 422]. For the molecular Prandtl number, Martin and Duan give no explicit value or a law for the conductivity law to infer it. It was chosen to go with a $Pr = 0.71$ as the other references also do so and it is a common value to use for perfect air.

The asterisk "*" on the M8Tw048 case is because it suffers similar ambiguity, where the $\gamma$, $R$ or $c_p$ are not defined, rather the $\gamma$ must be derived through the given recovery temperature ratio. Subsequently, through $\frac{M}{U_\infty} = \sqrt{\gamma R T}$ and $c_p = \frac{\gamma R}{\gamma - 1}$, the thermal gas model can be fully inferred, with the specific gas constant for diatomic Nitrogen given as $R = 296.803\mathrm{J/kg\,K}$ by NIST[1].

**Table 6.6** Viscosity laws defined in or reasonably inferrable from the DNS references.

| Case | Model | Dynamic viscosity $\mu(T)$ |
|---|---|---|
| Martin-Duan Part 2/4 | Power law* | $\mu_\delta \left(\frac{T}{T_\delta}\right)^{0.76}$ |
| Zhang | Sutherland | $1.458 \times 10^{-6} \frac{T^{3/2}}{T + 110.4}$ |
| Zhang - M8Tw048 | Keyes | $1.418 \times 10^{-6} \frac{T^{3/2}}{T + 116.4 \times 10^{-5/T}}$[2] |
| Huang | Sutherland | $1.458 \times 10^{-6} \frac{T^{3/2}}{T + 110.4}$ |

The asterisk "*" on Martin-Duan is because in their initial case description they do not state more than just "is the temperature dependent viscosity and is computed using a power law" [85, p. 348], in Part 4 then explicitly stated as power law with exponent $0.76$ and using edge viscosity and Temperature as reference - with $\mu_\delta$ calculated from Sutherland if $T_\delta \geq 170\mathrm{K}$, otherwise Keyes model [39, Eq. 2.14 and following text] - all Martin-Duan cases are modelled after this concept, as it is reasonable to assume that throughout the four part series this wasn't changed.

#### Implementation of viscosity law and constant molecular Prandtl number

The registered thermophysical combinations described in Section 3.5.2 permit the reference viscosity laws in Table 6.6 to be represented by seventh-order polynomials with independently specified conductivity. For the constant-$c_p$ benchmarks, setting the conductivity coefficients to $c_p/Pr$ times the viscosity coefficients enforces the selected constant molecular Prandtl number. This reproduces the required combination of

[1] [accessed 14 September 2026] `https://webbook.nist.gov/cgi/cbook.cgi?ID=7727-37-9`

[2] As stated in digitally published `M8Tw048_Stat.dat` file.

viscosity and conductivity laws without adopting the conductivity relation of the standard `sutherland` transport model.

Over the minimum and maximum temperature range reported by the solver, the fitted viscosity polynomials were a-priori evaluated against the stated laws, with the assessment present in every case's `/constant/`-folder, like e.g. in figure 6.3 for the M6Tw025 case.

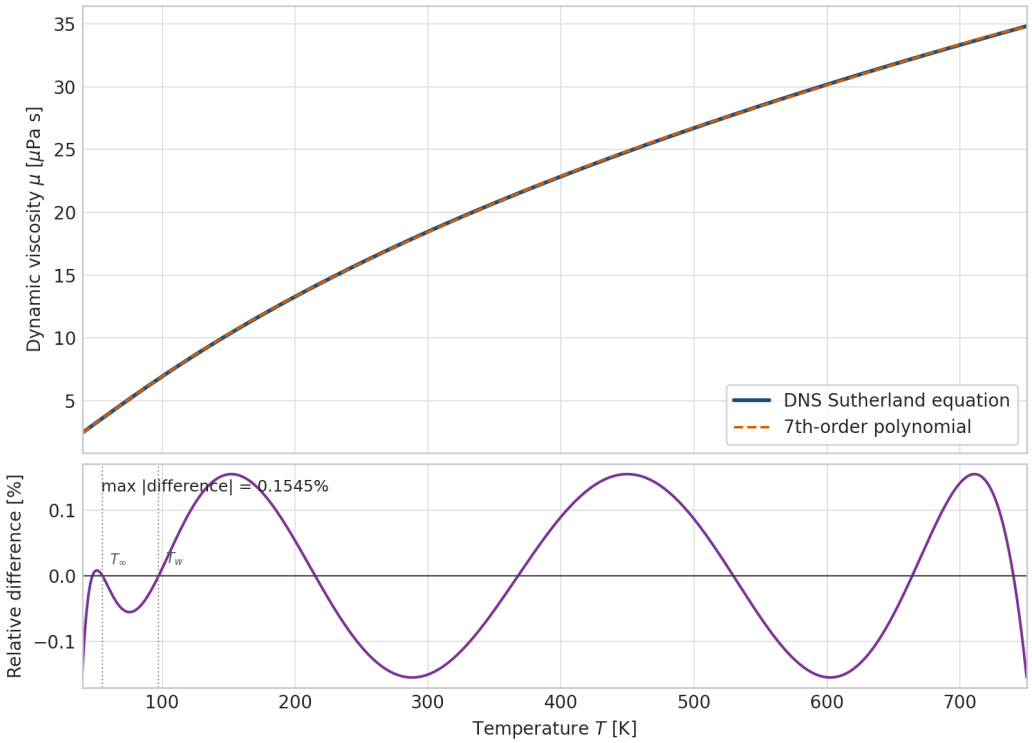



**Figure 6.3** Assessment of the custom 7th-order polynomial fit for the dynamic viscosity on the M6Tw025 case, against the DNS reference's defined viscosity law. With fixed points at the $T_w$ and $T_\infty$.

#### Turbulent Prandtl number

Unless explicitly stated otherwise, the SA and $k$–$\omega$ SST simulations use a constant turbulent Prandtl number $Pr_t = 0.9$. This follows common RANS practice beyond OpenFOAM: the NATO AVT-352 cone–flare comparison reports constant values of $0.9$ for DLR TAU and NASA VULCAN, $0.89$ for STAR-CCM+, and $0.85$ for the Aselsan Ansys Fluent calculations [20, Section IV]. NASA's LAURA manual likewise specifies $Pr_t = 0.9$ as the default [86, Section 5.4.17]. The present choice therefore provides a conventional baseline, retained across the benchmark set to compare the models under a common thermal-closure assumption and it is not fitted separately to each DNS case. DNS turbulent-Prandtl-number distributions and a sensitivity calculation with other constant values are discussed with the wall-resolved profile results in Section 7.2.3.

### 6.1.3 Matching station, quantities and figures of interest

#### On the definition of the matching station between RANS result and DNS reference

The 12 DNS references only provide information about the boundary layer profile and the wall heat-flux and shear stress at a singular downstream station - the Huang case being the exception with providing three downstream stations. Still some studies using the same ZPG FPs, e.g. [87], [88], or [89] plot multiple downstream stations, although it is not traceable from these publications where exactly they got these values from. For the later sections, it is thus abstained form including them, and it is strictly adhered to comparing to the primary source's values. Still plotting the downstream development of the quantities is

crucial to judging their development rather than just one snapshot. This is will become apparent on some later figures like Figure 6.4.

Another crucial point to assess, is at what spatial location to compare the RANS results with the DNS reference. This matching point can be determined by e.g. matching a physical downstream x-station, Reynolds number based on shear velocity $Re_\tau$, Reynolds number based on momentum thickness and wall viscosity $Re_{\delta 2}$, or Reynolds number based on momentum thickness and freestream viscosity $Re_\theta$.

Matching on a physical downstream x-station is not possible because the DNS references do not provide this information with regards to the actual beginning of the boundary layer development. As the ZPG DNS cases investigated in here represent fully turbulent boundary layers and no natural or forced tripping, they are set up with predetermined turbulent boundary layers at the inlet. They subsequently report the boundary layer thickness assigned at the inlet as $\delta_i$, and either the physical downstream sample station in the DNS domain, $x_a$, or the ratio $\frac{x_a}{\delta_i}$.Neither $x_a$ or it's non-dimensionless ratio with $delta_i$ sufficiently defines the physical downstream station in a RANS domain where the boundary layer is being developed from the leading edge onwards. Also the inlet profiles used for the DNS' are not available to exactly reproduce the domain from the same starting point.

Just five weeks before hand-in of this work, Cheng et al in [90] investigated the same Zhang and Huang ZPG flat-plates on RANS energy closure (see previous paragraph 6.1.2). Cheng et al used a novel matching station approach for their work, where they first determined the physical streamwise station at which the RANS solution produces the same boundary layer thickness as reported inflow boundary layer thickness $\delta_i$ in the DNS references. Subsequently the DNS matched sample station is then determined by the DNS reported ratio $\frac{x_a}{\delta_i}$, where $x_a$ the physical downstream location in the DNS domain, from this matched boundary layer thickness $\delta_i$. Tables 6.7 and 6.8 however shows that the $Re_{\delta 2}$ matching chosen for this work, and a physical boundary layer distance shifted matching like done by Cheng et al., agree within 1% for $-B_q$ and less than 2% for $C_h$.

**Table 6.7** Comparison of the dimensionless wall heat-transfer rate $-B_q$ from DNS (D), RANS at the shifted physical matching station as proposed by Cheng et al. (S), and RANS at the $Re_{\delta 2}$-matched station (R). Relative differences are given in percent. The RANS solutions are from the wall-resolved $k$–$\omega$ SST simulations as introduced later in Section 7.2.1.

| Case | DNS (D) $-B_q$ | Shifted (S) $-B_q$ | $Re_{\delta 2}$ (R) $-B_q$ | S vs D [%] | R vs D [%] | S vs R [%] |
|---|---|---|---|---|---|---|
| M6Tw025 | 0.14 | 0.138455697 | 0.137434751 | –1.10 | –1.83 | +0.74 |
| M6Tw076 | 0.02 | 0.017508213 | 0.017582069 | –12.46 | –12.09 | –0.42 |
| M8Tw048 | 0.06 | 0.058947522 | 0.058587750 | –1.75 | –2.35 | +0.61 |
| M14Tw018 | 0.19 | 0.186542796 | 0.185085083 | –1.82 | –2.59 | +0.79 |

**Table 6.8** Comparison of the Stanton number $C_h$ from DNS (D), RANS at the shifted physical matching station as proposed by Cheng et al. (S), and RANS at the $Re_{\delta 2}$-matched station (R). Relative differences are given in percent. The RANS solutions are from the wall-resolved $k$–$\omega$ SST simulations as introduced later in Section 7.2.1.

| Case | DNS (D) $C_h$ | Shifted (S) $C_h$ | $Re_{\delta 2}$ (R) $C_h$ | S vs D [%] | R vs D [%] | S vs R [%] |
|---|---|---|---|---|---|---|
| M6Tw025 | 0.001021456 | 0.001067416 | 0.001050667 | +4.50 | +2.86 | +1.59 |
| M6Tw076 | 0.000633588 | 0.000546307 | 0.000551478 | –13.78 | –12.96 | –0.94 |
| M8Tw048 | 0.000448561 | 0.000462389 | 0.000456311 | +3.08 | +1.73 | +1.33 |
| M14Tw018 | 0.000243366 | 0.000282753 | 0.000277646 | +16.18 | +14.09 | +1.84 |

$$Re_\tau = \frac{\rho_w u_\tau \delta}{\mu_w} \tag{6.1}$$

Matching on the $Re_\tau$ is seen rather often, e.g. [88], or [89]. It has the advantage that by definition of $Re_\tau$ being calculated with the boundary layer thickness $\delta$, this matching ensure equally thick developed boundary layer. The drawback however is that $Re_\tau$, also relies on the shear velocity $u_\tau$, thus any matching with $Re_\tau$ is also an indirect matching on the skin-friction, which undermines subsequent findings on the wall shear stress behaviour. A potential ambiguity in the definition of $Re_\tau$, is the definition of the boundary layer thickness. Whereas most often the boundary layer thickness is defined as the wall-normal distance where $U = 99\%U_\infty$, often noted as $\delta_{99}$, as discussed in Griffin Section 5.4.2, this doesn't necessary always have to be the case, or e.g. in Huang's reference where it is rather defined as "99.5% of the freestream total enthalpy" [82, Nomenclature].

$$Re_\theta = \frac{\rho_\infty u_\infty \theta}{\mu_\infty}, \qquad \text{or} \qquad Re_{\delta 2} = \frac{\rho_\infty u_\infty \theta}{\mu_w}, \tag{6.2}$$

with

$$\theta = \int_0^\delta \frac{\rho u}{\rho_e u_e} \left(1 - \frac{u}{u_e}\right) \mathrm{d}y \tag{6.3}$$

Matching on $Re_{\delta 2}$ or $Re_\theta$ only differs in the wether one uses the wall or freestream viscosity. The big ambiguity herein lies that sometimes the subscript-$\infty$ freestream quantities is defined as the edge quantities, such as in the Martin-Duan cases [81, Section 2.1]. While for an ideal ZPG FP boundary layer edge and freestream should be identical, it might not be the case for more complex flows, or even for CFD where a leading edge shock imposes an ever so slight wall normal pressure gradient. The advantage is, that this matching of the momentum thickness $\theta$, does not explicitly contain shear stress of heat-flux like $Re_\tau$ does. Martin and Duan explicitly control for the $Re_{\delta 2}$ [81]. Fernholz & Finley explicitly favour $Re_{\delta 2}$ over the conventional $Re_\theta$ for comparing compressible boundary layers [91]. Although they describe it soley as a physically motivated, empirical choice and base the momentum thickness Reynolds number on the boundary layer edge quantities.

The limitation of both $Re_\tau$ and $Re_{\delta 2}$ or $Re_\theta$, is that neither boundary layer quantities, nor integrals across the boundary layer can be reliably determined on a coarse wall-function grid. Thus it was decided to use the $C_f$-independent $Re_{\delta 2}$ for matching the wall-resolved RANS turbulence model results to the DNS reference, and subsequently match the physical downstream station between wall-function and wall-resolved RANS, as ideally they both should develop the boundary layer identically. Furthermore the $Re_{\delta 2}$ equation was case-by-case carefully matched to the respective DNS paper's definition.

#### Quantitative DNS reference data

Table 6.9 collects the matching Reynolds number $Re_{\delta 2}$, skin-friction coefficient $C_f$, Stanton number $C_h$, and dimensionless wall heat-transfer rate $-B_q$. The directly reported quantities retain the definitions, signs, and precision used in each source. Zhang's $C_f$ values are reconstructed as described in the table notes. No conversion between $C_h$ and $-B_q$ is applied. Huang's three sampling stations are listed separately for the single M11Tw020 case. The corresponding CFD results and the DNS coefficient reconstructions used for their signed errors are given in Table 7.2; differences between those numerical reference values and this published-data overview are stated there.

**Table 6.9** Published and reconstructed DNS reference values for the 12 selected benchmark cases. Source keys identify the paper and the location of each quantity below the table.

| Source | Case / station | $Re_{\delta 2}$ | $10^3 C_f$ | $10^3 C_h$ | $-B_q$ |
|---|---|---|---|---|---|
| D2 | M5T1 | 1538.4 | 2.46 | 1.39 | |
| D2 | M5T2 | 1521.5 | 2.05 | 1.14 | |
| D2 | M5T3 | 1524.7 | 1.81 | 0.977 | |
| D2 | M5T4 | 1526.4 | 1.50 | 0.839 | |
| D2 | M5T5 | 1536.8 | 1.28 | | |
| D4 | LowH_M3 | 1554.5 | 3.12 | 1.82 | |
| Z | M2p5 | 1657 | 2.31† | | 0 |
| Z | M6Tw025 | 1135 | 1.70† | | 0.14 |
| Z | M6Tw076 | 1746 | 0.998† | | 0.02 |
| Z | M8Tw048 | 1990 | 0.766† | | 0.06 |
| Z | M14Tw018 | 2354 | 0.404† | | 0.19 |
| H | M11Tw020 (BL1) | 2204 | 0.61 | 0.36 | 0.164 |
| H | M11Tw020 (BL2) | 3419 | 0.55 | 0.32 | 0.154 |
| H | M11Tw020 (BL3) | 4390 | 0.51 | 0.29 | 0.148 |

**D2:** Martin-Duan Part 2 [81]: $Re_{\delta 2}$ in Table 1 (p. 421); $C_f$ and $C_h$ in Table 3 (p. 425).
**D4:** Martin-Duan Part 4 [39]: $Re_{\delta 2}$ in Table 2 (p. 31); $C_f$ and $C_h$ in Table 4 (p. 42).
**Z:** Zhang et al. [5]: $Re_{\delta 2}$, $-B_q$, and $u_\tau$ from Table 2; $U_\infty$ from Table 1 (p. 4299). Entries marked † use $C_f = 2(\rho_w/\rho_\infty)(u_\tau/U_\infty)^2$, with $\rho_\infty/\rho_w$ from the outermost digital DNS profile point. Coefficients are rounded to three significant figures.
**H:** Huang et al. [82]: $Re_{\delta 2}$ and $-B_q$ from Table 3; $C_f$ and $C_h$ from Table 4 (both p. 11).

Zhang et al. report $-B_q$ to only two decimal places in both the paper [5, Table 2, p. 4299] and the accompanying digital data headers. Assuming rounding to the nearest $0.01$, this reporting precision implies an absolute rounding bound of $\pm 0.005$. Table 6.10 gives the corresponding intervals and relative bounds for the four nonzero values. The effect is particularly pronounced for M6Tw076: its reported value of $-B_q = 0.02$ has a rounding bound of $\pm 25\%$ relative to that value. Since the reconstructed wall heat-flux and Stanton number are proportional to $B_q$ when the other inputs are held fixed, the same relative bound applies to these quantities. This reporting precision therefore limits quantitative heat-flux validation based on the tabulated $-B_q$ values, particularly for M6Tw076, but also a $\pm 3.57\%$ is significant, when the goal is to achieve a $\pm 5\%$ heat-flux prediction accuracy 6.3. Thus and because of the aforementioned ambiguity and mismatch in the reported wall temperature, the M6Tw076 is being excluded from heat-flux assessments.

**Table 6.10** Rounding bounds for Zhang et al.'s reported $-B_q$ values, assuming rounding to the nearest $0.01$.

| Case | Reported $-B_q$ | Rounding interval | Relative bound |
|---|---|---|---|
| M6Tw025 | 0.14 | 0.135 – 0.145 | ±3.57% |
| M6Tw076 | 0.02 | 0.015 – 0.025 | ±25.0% |
| M8Tw048 | 0.06 | 0.055 – 0.065 | ±8.33% |
| M14Tw018 | 0.19 | 0.185 – 0.195 | ±2.63% |

Relative bounds are calculated as $0.005/|B_{q,\mathrm{reported}}|$ and expressed as percentages of the reported value.

### Qualitative DNS reference figures

The reference figures of interest for this work are particularly the boundary layer profiles. For the Zhang cases these are taken directly from the downloadable digital data, thus providing a high quality reference, with no digitization errors, and very fine resolution.

The M11Tw0120 reference case by Huang et al has been published with the momentum and thermal boundary layer profile in [82, Figure 12(a), Figure 13(a)], but only for the BL2, downstream station.

For Martin–Duan Part 2, the van Driest–transformed velocity profiles and $T(U)$ relations are digitized from Figures 5 and 6, respectively [81]. The wall levels in Figure 6, interpreted alongside Table 1, suggest mixed temperature normalization: $T/T_\delta$ for M5T1, with $T_\delta = 228.1\,\mathrm{K}$, and $T/T_\infty$ for M5T2–M5T5, with nominal $T_\infty = 216.65\,\mathrm{K}$ [85, Table 1]. Velocity remains normalized as $U/U_\delta$ for all five cases and is dimensionalized using each case's tabulated edge state. This mixed interpretation is inferred here from the wall levels and tabulated conditions. Contacting the authors to confirm this accidentally mixed normalization yielded no response, from the clear requirement of the thermal boundary layer profiles to match the prescribed wall temperature at the wall, this inferred correction is reasonable. For Part 4 LowH_M3, Figure 7 supplies the transformed velocity profile, while Figure 8 is dimensionalized using the tabulated edge temperature and velocity [39].

For future reusing of these DNS reference cases it should also be noted, that Zhang and Huang define the dimensionless wall shear stress coefficient $C_h$ and the Stanton number $C_h$ via the freestream quantities [5, 82, Nomenclature], whereas Martin and Duan use the boundary layer edge quantities [81, p.424], for their cases the quantities at 99% freestream velocity [81, p. 421]

The separate inconsistency between the Martin-Duan Part 2 Reynolds-number columns and the resulting limitation of $Re_{\delta 2}$ as a nominal matching target are discussed in Appendix A.1.

## 6.2 Wall-resolved mesh requirements

Wall-resolved RANS meshes are commonly designed to place the first cell centre at $y_1^+ \leq 1$. The NASA Turbulence Modeling Resource describes this as conventional practice for simulations that integrate the turbulence model to the wall.[3] A general basis for treating this rule as sufficient for hypersonic cold-wall heat-flux convergence was not identified in the literature reviewed here. Fletcher [92, p. 401], for example, recommends $y^+ < 5$ to resolve the near-wall velocity profile and notes that compressible temperature profiles may require $y^+ < 2$. These recommendations motivate placing the first point within the viscous sublayer, but do not establish a convergence criterion for heat-transfer under hypersonic cold-wall conditions.

The first-cell centre lies at $y_1 = \delta_n = \Delta y_1/2$ for the wall-normal cells used here, where $\Delta y_1$ is the full first-cell thickness. Its distance in wall units is $y_1^+ = \rho_w u_\tau y_1/\mu_w$, with friction velocity $u_\tau = \sqrt{|\tau_w|/\rho_w}$. The alternative resolution indicator is the first-cell acoustic Reynolds number

$$Re_{\mathrm{cell1}} = \frac{\rho_w a_w\, \delta_n}{\mu_w}. \tag{6.4}$$

Here $\rho_w$, $a_w$ and $\mu_w$ are the density, speed of sound and dynamic viscosity evaluated at the wall.
Dilley [93] found that a cell Reynolds number of approximately unity was required for grid-converged hypersonic heating predictions with the Baldwin–Lomax model. His definition uses first-cell properties and the full cell height, so its numerical threshold cannot be transferred directly to Equation (6.4), but it gives an indication. Related criteria have also been used for laminar stagnation-point heating [94], aeroheating calculations for the X-38 [95] and mesh adaptation in LAURA [86].

Figures 6.4 and 6.5 illustrate the effect of reducing the first-cell height for M6Tw025. They show the streamwise development of the skin-friction coefficient $C_f$ and Stanton number $C_h$, expressed in terms of $Re_\tau$, at fixed downstream spacing. For SA, the curves nearly coincide from $y_1^+ = 1.24$ down to the finest mesh. The $k$–$\omega$ SST model requires refinement to approximately $y_1^+ = 0.129$ before mesh convergence is reached. The weaker first-cell sensitivity of SA is qualitatively consistent with the NASA flat-plate study cited above, which reports greater sensitivity for models using an $\omega$ equation.

[3] https://tmbwg.github.io/turbmodels/flatplate_val_ypluseffect.html

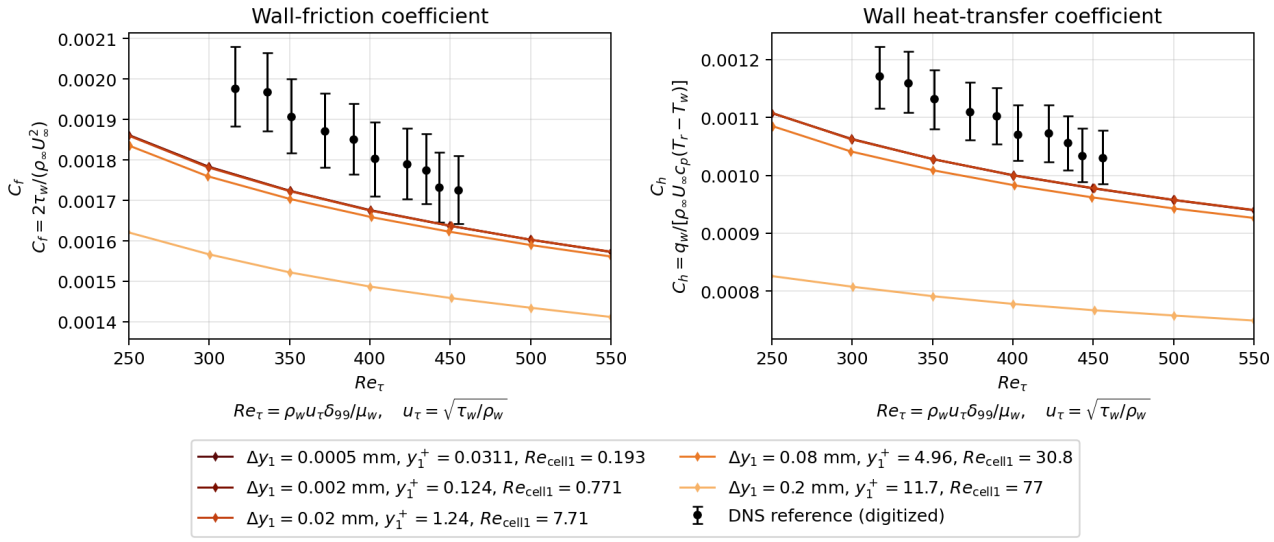



**Figure 6.4** First-cell mesh sensitivity of the SA predictions for M6Tw025: skin-friction coefficient $C_f$ (left) and Stanton number $C_h$ (right) versus $Re_\tau$, with fixed downstream spacing $\Delta x = 5\,\text{mm}$. Coloured curves show the different first-cell thicknesses $\Delta y_1$; black symbols denote the digitized DNS reference [5]. The $y_1^+$ values reported in the legend are evaluated at each mesh's DNS-matched station, defined by $Re_{\delta 2} = 1135$, rather than representing a constant value along the plate.

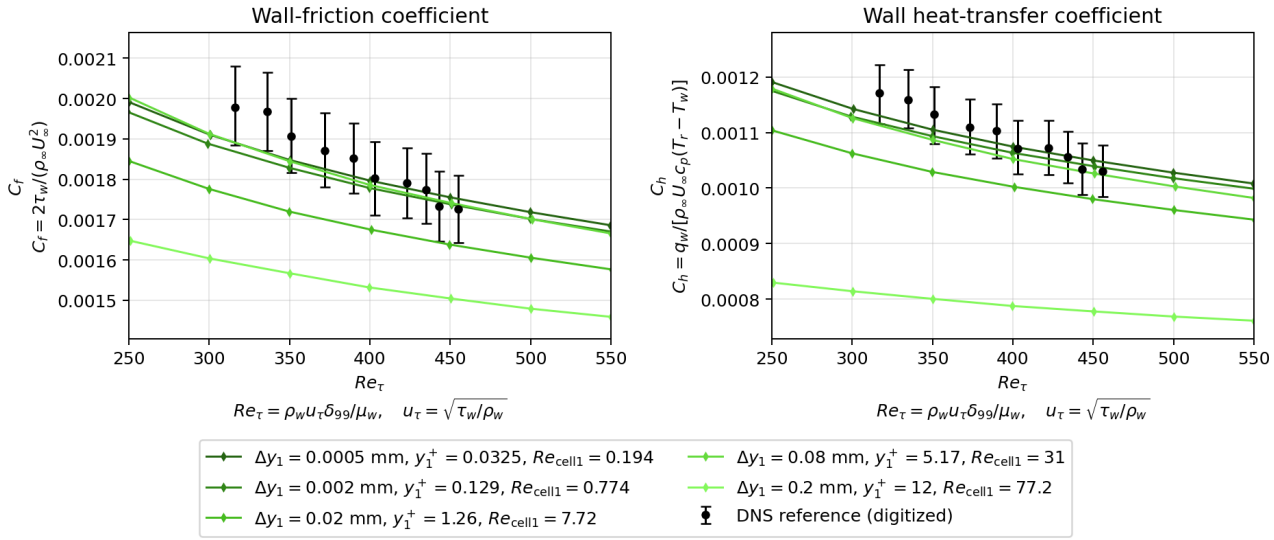



**Figure 6.5** First-cell mesh sensitivity of the $k$–$\omega$ SST predictions for M6Tw025: skin-friction coefficient $C_f$ (left) and Stanton number $C_h$ (right) versus $Re_\tau$, with fixed downstream spacing $\Delta x = 5\,\text{mm}$. Coloured curves show the different first-cell thicknesses $\Delta y_1$; black symbols denote the digitized DNS reference [5]. The $y_1^+$ values reported in the legend are evaluated at each mesh's DNS-matched station, defined by $Re_{\delta 2} = 1135$, rather than representing a constant value along the plate.

Appendix B extends the wall-normal refinement study to all twelve ZPG flat-plate cases. Streamwise-refinement tests for M6Tw025 showed negligible sensitivity of heat-flux and skin-friction to the downstream spacing when the leading edge was refined. The remaining mesh studies therefore focus on the wall-normal direction. They report skin-friction and, for the isothermal cases, heat-transfer against $y_1^+$ and $Re_{\text{cell1}}$ at the DNS-matched downstream stations. For the investigated hypersonic cold-wall cases, these results support $Re_{\text{cell1}} \leq 1$ as a practical resolution guideline, particularly for SST, in preference to relying on $y_1^+ \leq 1$ alone.

The wall-gradient approximation provides a plausible explanation for this stringent resolution requirement. As discussed in Section 3.4, OpenFOAM/HiSA evaluates the wall gradients from the difference between the

wall value and the first off-wall cell value. Accurate wall fluxes therefore require this secant to approximate the local wall derivative. Placing the first cell within the nominal viscous sublayer is insufficient if the velocity or temperature gradient varies appreciably across that distance. The cold-wall profile shapes in Figures 2.1 and 2.2 motivate this distinction between resolving a profile and resolving its wall slope. The mesh studies establish the numerical basis for the wall-resolved model comparisons in Section 7.2, where agreement with the DNS reference is assessed separately there.

## 6.3 Criteria for successful performance

Section 6.1 identifies uncertainties and ambiguities in the DNS references used for this assessment. In comparing RANS predictions with these references, as in related assessments such as that of Xue et al. [88], a quantitative accuracy target is needed. This work adopts a $\pm 5\%$ DNS-relative error band as the target for wall-resolved heat-flux predictions. This band is an engineering acceptance criterion and neither represents the uncertainty of the individual DNS references nor defines a statistical confidence interval. The reference limitations discussed above must therefore be considered when judging agreement within this band.

The target is motivated by the relative simplicity of attached ZPG flat-plate boundary layers and the accuracy already achieved by analytical approaches. Hasan et al. [96, Figure 2], for example, report heat-transfer errors mostly within $\pm 8\%$, with a maximum of $10.3\%$, for a dataset that includes several of the benchmarks investigated here. Wall-resolved RANS should therefore aim to improve on this level of accuracy. For wall-function calculations, $\pm 10\%$ is adopted as the minimum acceptable heat-flux accuracy: for these ZPG cases, larger errors would leave the additional computational cost of CFD without a corresponding gain in heat-flux accuracy over the analytical approach.

Beyond error magnitude, engineering use also requires an assessment of how consistently a method behaves. A systematic bias is easier to account for than an error that varies strongly and non-monotonically with sampling height or mesh spacing. This distinction helps frame the three analyses that follow: wall-resolved RANS discrepancies in Section 7.2, wall-function closure errors in Section 7.3, and spatial-discretization errors in Section 7.4. For example, the wall-resolved heat-flux discrepancy for a given case may act primarily as a bias, while the closure and spatial discretization may introduce more irregular variations as the sampling height and mesh spacing change. The following sections examine these behaviours alongside the quantitative accuracy targets.

# 7 Isolation of errors in wall-function predictions

The motivation for using wall functions is to reduce the near-wall resolution required for aerothermal predictions without sacrificing the accuracy needed for thermal design. As outlined in Chapter 1, assessing that accuracy requires more than comparing a wall-function CFD result with a reference heat-flux. The result depends on the wall-function closure, the turbulence model supplying its off-wall state, and the discretization of the remaining boundary layer. A discrepancy in any one of these components can obscure the performance of the others.

This chapter therefore examines the zero-pressure-gradient (ZPG) flat-plate benchmarks introduced in Chapter 6 through successive diagnostic comparisons. Section 7.2 first establishes the wall-resolved RANS results and distinguishes accuracy in wall shear and heat-transfer from agreement in boundary layer profiles. Section 7.3 then evaluates the wall-function closures on prescribed profiles, removing feedback from the coupled CFD calculation. Section 7.4 examines the additional sensitivity introduced by conventional wall-function meshes and develops a boundary layer-informed spacing criterion. Finally, Section 7.4.5 combines controlled mesh resolution with reference-fitted closures to determine how closely the wall-function calculation can recover its wall-resolved baseline.

Figure 7.1 summarizes this diagnostic sequence and the influences examined at each step. The implemented Li and Griffin wall functions are coupled to Spalart–Allmaras (SA), as described in Sections 5.3 and 5.4. Their coupled results are consequently compared with the corresponding wall-resolved SA calculation. The wall-resolved assessment additionally includes the widely used $k$–$\omega$ SST model to examine whether the observed limitations are specific to SA. The objective is to identify the dominant discrepancies and their interactions, rather than to assume that the total error can be separated into independent additive contributions.

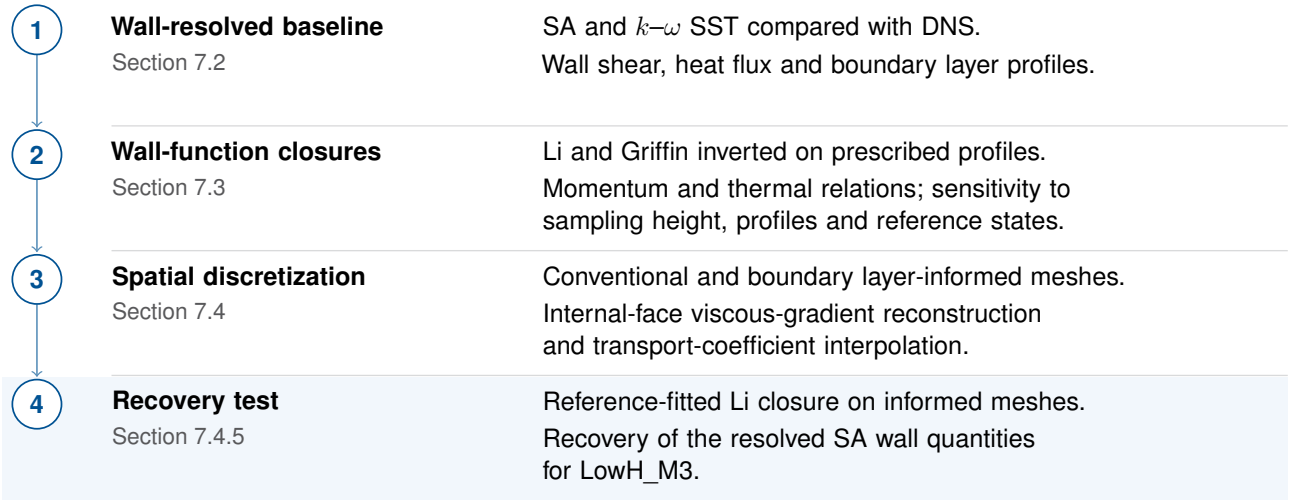



**Figure 7.1** Diagnostic sequence for the attached ZPG flat-plate assessment. The comparisons distinguish interacting influences rather than independent additive error contributions. The final, reference-conditioned test assesses recovery of the resolved SA result for LowH_M3; its discrepancy from DNS remains.

## 7.1 Evaluation logic and reference hierarchy

The wall-resolved assessment uses the twelve ZPG flat-plate cases listed in Table 6.1. Their gas and transport models, wall conditions, and reference-data limitations are specified in Section 6.1. The detailed wall-function comparisons focus on M6Tw025 and LowH_M3. The reason for this selection follows from the profile assessment in Section 7.2.5. These two cases provide complementary evidence, rather than a controlled variation of a single physical parameter outlined in detail in Section 7.2.5.

Three comparisons are distinguished throughout the chapter, as summarized in Table 7.1. The first assesses the wall-resolved RANS calculation against the published DNS reference. The second supplies a fixed boundary layer profile to a standalone wall-function solver and compares the recovered wall quantities with those associated with that profile. The third evaluates the coupled wall-function CFD result against both DNS and the corresponding wall-resolved SA solution.

**Table 7.1** Reference hierarchy used to distinguish RANS discrepancies, conditional wall-function closure errors, and the additional discrepancies of the coupled calculation.

| Calculation | Comparison reference | Purpose |
|---|---|---|
| Wall-resolved RANS | Published DNS wall quantities and profiles | Establish the resolved baseline and assess turbulence and thermal modelling. |
| Standalone wall function | Wall quantities belonging to the prescribed profile; common DNS reference for the paired M6Tw025 test | Assess closure accuracy and sensitivity to the supplied profile without coupled-flow feedback. |
| Coupled wall-function CFD | DNS and wall-resolved SA at the comparison station | Assess overall accuracy and departures from the resolved baseline as mesh and closure treatment change. |

The comparison stations follow the matching procedure in Section 6.1.3. Wall-resolved RANS is matched to the DNS through the reported momentum-thickness Reynolds number, with definitions retained consistently for each dataset. Wall-function CFD is then evaluated at the same physical downstream station as its wall-resolved SA baseline. Matching the coarse wall-function solution independently through a boundary layer integral would introduce an additional dependence on how that integral is reconstructed on the coarse mesh. Holding the downstream station fixed instead exposes changes in boundary layer development caused by the coupled wall treatment.

For a quantity $Q$, the signed relative error is the difference between the calculated value $Q_A$ and the reference value $Q_B$, divided by the reference value:

$$e_{A/B}(Q) = \frac{Q_A - Q_B}{Q_B}, \qquad Q_B \neq 0, \tag{7.1}$$

Both values must use the same definition and sign convention. The plots express this error as a percentage, $100e_{A/B}$. A dimensional wall quantity and its corresponding coefficient have the same relative error only if the calculated and reference values use the same normalization. The published coefficient definitions and reconstructed DNS values are given in Table 6.9. Relative heat-flux errors are undefined for an exactly adiabatic reference, and the M6Tw076 heat-flux comparison is excluded from quantitative assessment because of the reference limitations discussed in Section 6.1.

The wall-resolved mesh studies in Section 6.2 and Appendix B provide the resolution evidence for the wall-resolved baselines. Their interpretation also relies on the numerical methods and convergence criteria in Sections 3.3 and 3.6.

## 7.2 Wall-resolved turbulence-model results

The first comparison establishes the accuracy of the flow calculation before a wall function is introduced. Wall-resolved Spalart–Allmaras (SA) and $k$–$\omega$ SST simulations are evaluated against the selected DNS references using the matching procedure and transport properties in Section 6.1. Both momentum closures use the same gradient-diffusion thermal closure with $Pr_t = 0.9$, unless explicitly stated otherwise. The SA calculations employ the corrected compressible implementation described in Section 3.5.3. SA is also the underlying momentum closure for the coupled Li and Griffin calculations investigated subsequently, whereas SST provides a general comparison with another widely used RANS model.

The mesh studies in Section 6.2, provide the basis for using these solutions as wall-resolved references. Their differences from DNS measure the accuracy of the complete resolved RANS calculation, including its thermal closure and remaining numerical and reference-data uncertainties. In the later wall-function comparisons, the corresponding SA solution serves as the baseline against which the additional effects of the wall closure and its coupling to the discretized flow are assessed.

### 7.2.1 Prediction of wall heat-flux and skin-friction

Table 7.2 reports the, wall coefficients and signed relative errors for both models, across the investigated ZPG FPs at the respectively $Re_{\delta 2}$ matched station. DNS rows give the reference values used to calculate the errors, complementing the published-data overview in Table 6.9. The CFD stations agree with the nominal DNS targets to within $0.1\%$. Overall Table 7.2 shows a relative errors against DNS range from $-14.8\%$ to $+13.7\%$ in heat-flux and from $-18.5\%$ to $+12.5\%$ in wall shear stress for SA, compared with $-12.8\%$ to $+20.6\%$ and $-17.1\%$ to $+15.6\%$, respectively, for $k$–ω SST, across the investigated cases. Neither turbulence model consistently outperforms the other across the investigated cases and wall quantities.

**Table 7.2** Wall-resolved SA and $k$–$\omega$ SST results at $Pr_t = 0.9$, with the DNS values used for comparison (see Table 6.9). The $Re_{\delta 2}$ column gives the nominal target for DNS and the actual sampled value for each CFD solution. Errors $\varepsilon_f$ and $\varepsilon_h$ are $100(C_{\mathrm{RANS}} - C_{\mathrm{DNS}})/C_{\mathrm{DNS}}$ for $C_f$ and $C_h$, respectively. The $-B_q$ column gives Zhang [5, Table 2] and Huang [82, Table 3] DNS values and corresponding RANS values $q_{\mathrm{to\,wall}}/(\rho_w c_p u_\tau T_w)$, using heat-flux positive into the wall and each solution's wall state. Entries for $C_h$ and $\varepsilon_h$ are omitted for the adiabatic references M5T5 and M2p5 and for M6Tw076; these omissions do not imply zero RANS heat-flux.

| Case / station | Model | $Re_{\delta 2}$ | $10^3 C_f$ | $\varepsilon_f\,[\%]$ | $10^3 C_h$ | $\varepsilon_h\,[\%]$ | $-B_q$ |
|---|---|---|---|---|---|---|---|
| *Martin–Duan Part 2* | | | | | | | |
| | **DNS** | **1538.4** | **2.46** | **—** | **1.39** | **—** | |
| M5T1 | SA | 1538.5 | 2.0055 | –18.48 | 1.1839 | –14.83 | |
| | SST | 1538.4 | 2.2824 | –7.22 | 1.3429 | –3.39 | |
| | **DNS** | **1521.5** | **2.05** | **—** | **1.14** | **—** | |
| M5T2 | SA | 1520.8 | 1.7334 | –15.45 | 1.0144 | –11.02 | |
| | SST | 1522.2 | 1.7967 | –12.35 | 1.0531 | –7.62 | |
| | **DNS** | **1524.7** | **1.81** | **—** | **0.977** | **—** | |
| M5T3 | SA | 1525.0 | 1.5229 | –15.86 | 0.8763 | –10.30 | |
| | SST | 1525.5 | 1.5003 | –17.11 | 0.8697 | –10.98 | |
| | **DNS** | **1526.4** | **1.5** | **—** | **0.839** | **—** | |
| M5T4 | SA | 1525.9 | 1.3674 | –8.84 | 0.7545 | –10.07 | |
| | SST | 1526.2 | 1.3018 | –13.22 | 0.7320 | –12.76 | |
| | **DNS** | **1536.8** | **1.28** | **—** | **—** | **—** | |
| M5T5 | SA | 1537.3 | 1.1912 | –6.94 | — | — | |
| | SST | 1536.8 | 1.1032 | –13.81 | — | — | |

Continued on next page

Table 7.2 (continued)

| Case / station | Model | $Re_{\delta 2}$ | $10^3 C_f$ | $\varepsilon_f\,[\%]$ | $10^3 C_h$ | $\varepsilon_h\,[\%]$ | $-B_q$ |
|---|---|---|---|---|---|---|---|
| *Martin–Duan Part 4* | | | | | | | |
| | **DNS** | **1554.5** | **3.12** | **—** | **1.82** | **—** | |
| LowH_M3 | SA | 1554.4 | 2.8950 | –7.21 | 1.7187 | –5.56 | |
| | SST | 1555.0 | 3.5183 | +12.77 | 2.0712 | +13.80 | |
| *Zhang* | | | | | | | |
| | **DNS** | **1657** | **2.3125** | **—** | **—** | **—** | **—** |
| M2p5 | SA | 1658.0 | 2.1858 | –5.48 | — | — | — |
| | SST | 1656.5 | 2.0949 | –9.41 | — | — | — |
| | **DNS** | **1135** | **1.7043** | **—** | **1.0215** | **—** | **0.14** |
| M6Tw025 | SA | 1135.3 | 1.6172 | –5.11 | 0.9659 | –5.44 | 0.1320 |
| | SST | 1135.8 | 1.7563 | +3.05 | 1.0507 | +2.86 | 0.1374 |
| | **DNS** | **1746** | **0.99785** | **—** | **—** | **—** | **0.02** |
| M6Tw076 | SA | 1746.2 | 1.0120 | +1.42 | — | — | — |
| | SST | 1746.2 | 0.9419 | –5.60 | — | — | — |
| | **DNS** | **1990** | **0.76629** | **—** | **0.44856** | **—** | **0.06** |
| M8Tw048 | SA | 1989.9 | 0.8010 | +4.53 | 0.4710 | +5.01 | 0.0592 |
| | SST | 1990.0 | 0.7691 | +0.37 | 0.4563 | +1.73 | 0.0586 |
| | **DNS** | **2354** | **0.40395** | **—** | **0.24337** | **—** | **0.19** |
| M14Tw018 | SA | 2354.6 | 0.4545 | +12.52 | 0.2694 | +10.70 | 0.1825 |
| | SST | 2354.0 | 0.4668 | +15.57 | 0.2776 | +14.09 | 0.1851 |
| *Huang M11Tw020* | | | | | | | |
| | **DNS** | **2204** | **0.61** | **—** | **0.36** | **—** | **0.164** |
| BL1 | SA | 2204.5 | 0.6332 | +3.80 | 0.3760 | +4.44 | 0.1647 |
| | SST | 2203.2 | 0.6623 | +8.57 | 0.3941 | +9.46 | 0.1683 |
| | **DNS** | **3419** | **0.55** | **—** | **0.316** | **—** | **0.154** |
| BL2 | SA | 3418.8 | 0.5820 | +6.40 | 0.3458 | +9.44 | 0.1587 |
| | SST | 3418.8 | 0.6140 | +12.24 | 0.3652 | +15.57 | 0.1627 |
| | **DNS** | **4390** | **0.51** | **—** | **0.29** | **—** | **0.148** |
| BL3 | SA | 4389.9 | 0.5545 | +8.73 | 0.3296 | +13.66 | 0.1553 |
| | SST | 4389.2 | 0.5884 | +15.37 | 0.3499 | +20.64 | 0.1595 |

Figures 7.2 and 7.3 summarize the signed errors in the skin-friction coefficient $C_f$ and Stanton number $C_h$. The two abscissae characterize different aspects of the flow: $-B_q$ represents wall heat-transfer, while $M_\tau$ indicates the strength of compressibility associated with the friction velocity. The Figures 7.2 and 7.3 do not reveal any clear correlation in reduction in error with either cold-wall effect or compressibility. Thus from this investigation it is unclear what causes the turbulence models to have such bad performance in predicting the heat-flux and skin-friction on hypersonic flat-plates. These comparisons therefore establish a substantial baseline discrepancy on some of the simplest attached turbulent flows, but do not by themselves identify its physical cause.

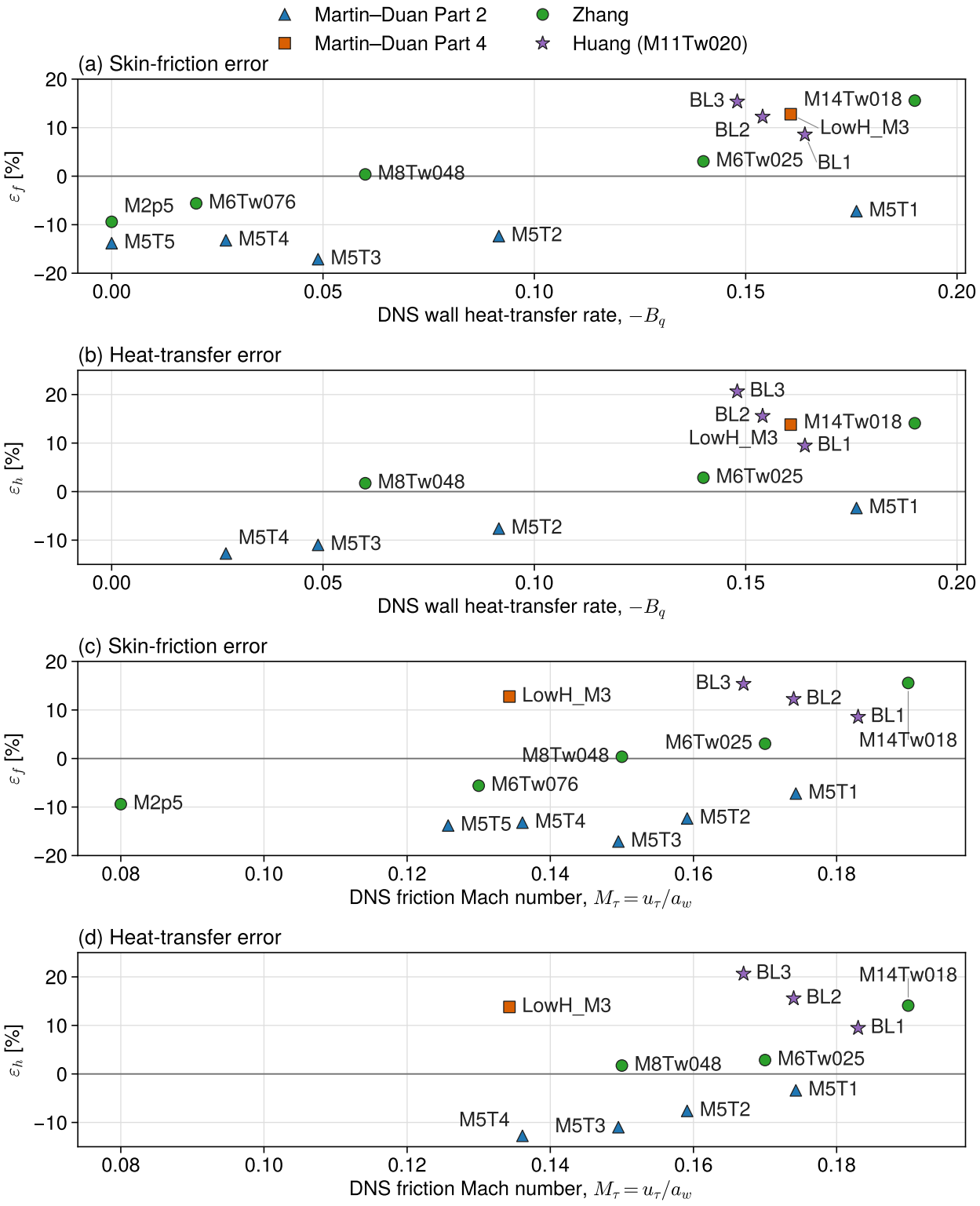



**Figure 7.2** Signed relative errors, $100(C_{\mathrm{RANS}} - C_{\mathrm{DNS}})/C_{\mathrm{DNS}}$, for wall-resolved $k$–$\omega$ SST at $Pr_t = 0.9$. Panels (a,b): errors in the skin-friction coefficient $C_f$ and Stanton number $C_h$, respectively, versus the DNS dimensionless wall heat-transfer rate $-B_q$. Panels (c,d): the corresponding errors versus the DNS friction Mach number $M_\tau = u_\tau/a_w$. Symbols identify the DNS sources and annotations the cases; BL1–BL3 are three stations of the single Huang case. Heat-transfer errors are omitted for the adiabatic references and M6Tw076. Definitions, matching stations and reference-data limitations follow Section 6.1.

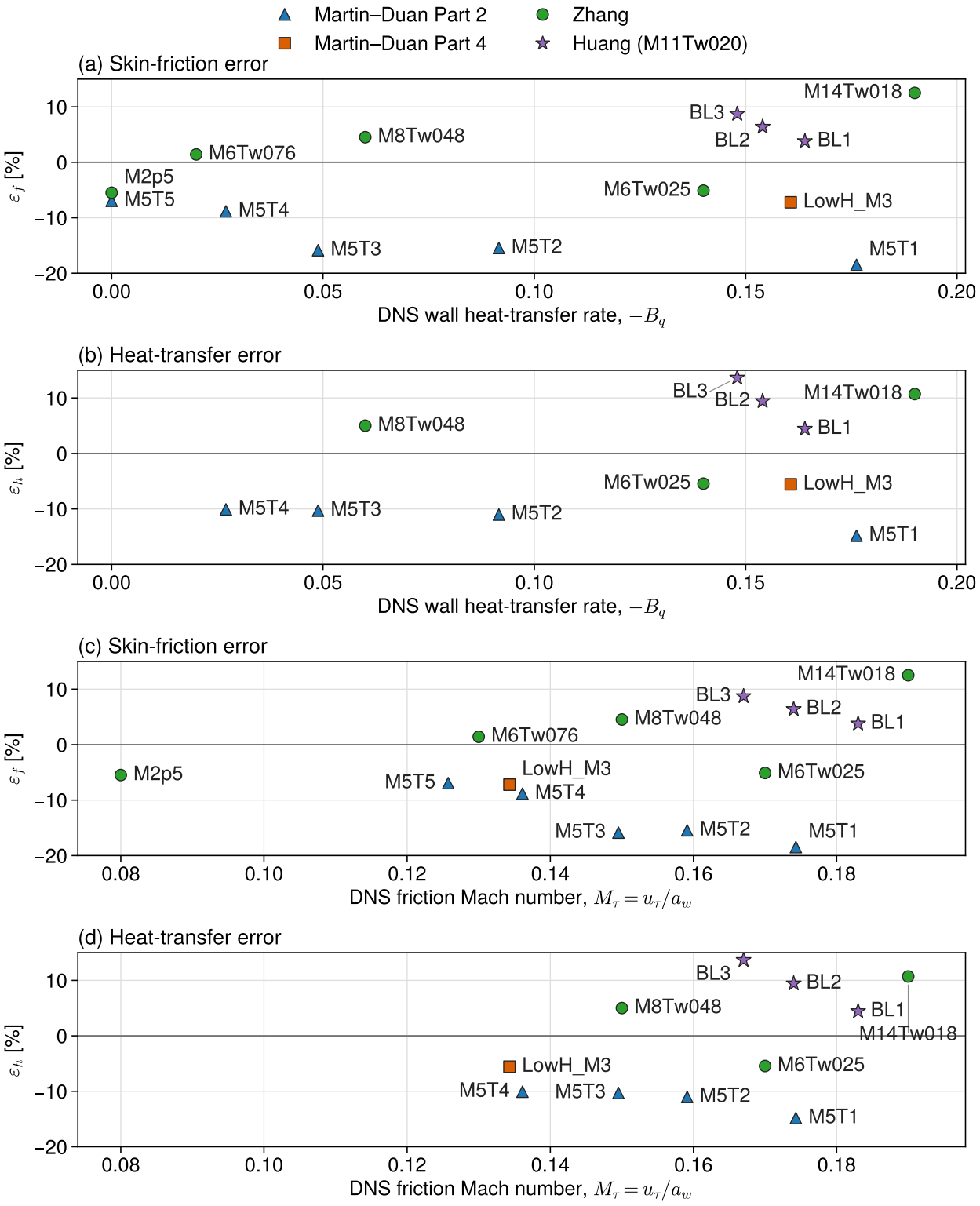



**Figure 7.3** Signed relative errors for the corrected wall-resolved Spalart–Allmaras model at $Pr_t = 0.9$, using the same references and definitions as Figure 7.2. Panels (a,b): errors in $C_f$ and $C_h$, respectively, versus DNS $-B_q$. Panels (c,d): the corresponding errors versus DNS $M_\tau$. Positive values indicate overprediction of the reference coefficient. Heat-transfer errors are omitted for the adiabatic references and M6Tw076.

### 7.2.2 Boundary layer profiles and temperature–velocity relations

Wall coefficients alone do not show whether the flow supplied to a wall function has the correct internal structure. The detailed profile comparisons therefore examine both velocity and temperature, with particular emphasis on the mean temperature–velocity relation $T(U)$ introduced in Section 2.3.3. This relation is directly relevant to the energy closures of the investigated analytical wall functions. A discrepancy in $T(U)$ implies that a wall closure fitted to DNS and the resolved RANS field need not describe the same relation between thermal and momentum transport, even when their wall heat-fluxes are close.

Figures 7.4 and 7.5 illustrate the contrast using M6Tw025 and LowH_M3. Corresponding comparisons for the other ten benchmark cases are provided in Appendix C.

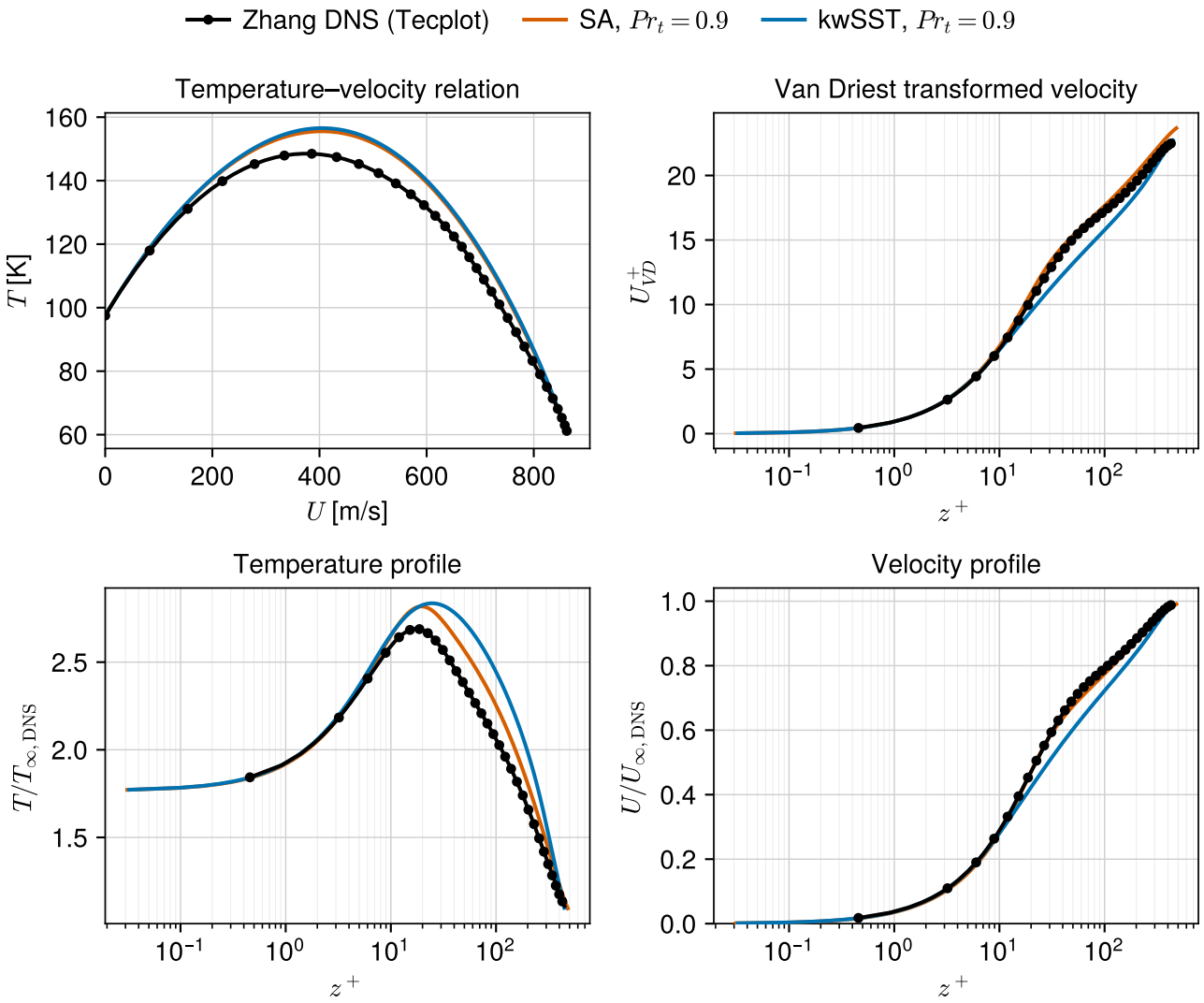



**Figure 7.4** Wall-resolved Spalart–Allmaras and $k$–$\omega$ SST profiles for M6Tw025 at $Pr_t = 0.9$, matched at $Re_{\delta 2} = 1135$ and compared with Zhang et al.'s digital DNS profiles [5]. Top: dimensional temperature–velocity relation $T(U)$ (left) and van Driest–transformed velocity $U^+_{\mathrm{VD}}$ versus wall distance $z^+$ (right). Bottom: normalized temperature $T/T_{\infty,\mathrm{DNS}}$ (left) and velocity $U/U_{\infty,\mathrm{DNS}}$ (right) versus $z^+$. Profiles are truncated at their $99\%$-velocity boundary layer edge. The thermal maximum is overpredicted by both turbulence models despite differences between their velocity profiles.

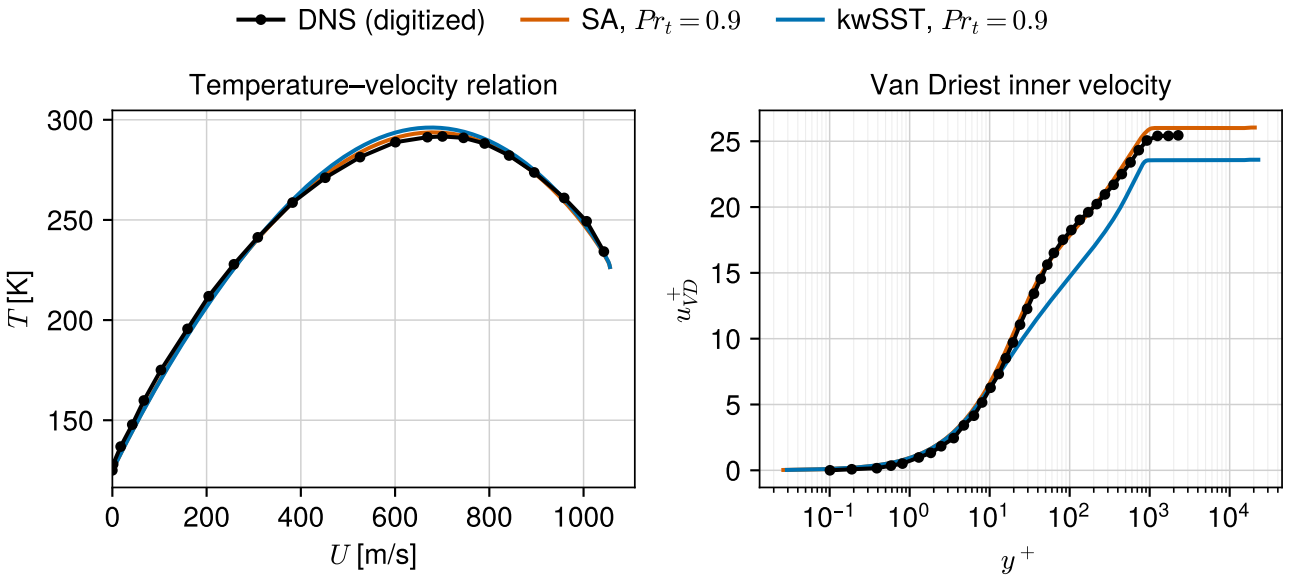



**Figure 7.5** Wall-resolved Spalart–Allmaras and $k$–$\omega$ SST profiles for Martin–Duan Part 4 LowH_M3 at $Re_{\delta 2} = 1554.5$ and $Pr_t = 0.9$, compared with digitized DNS [39]. Left: dimensional temperature–velocity relation $T(U)$. Right: van Driest–transformed velocity $U^+_{\mathrm{VD}}$ versus wall distance $y^+$. The DNS wall-normal reconstruction retains the provisional boundary layer-edge endpoint assumption indicated in the plot. Both models closely reproduce the displayed $T(U)$ relation, while their transformed velocity profiles differ.

For M6Tw025, both Spalart–Allmaras and $k$–$\omega$ SST overpredict the temperature maximum and the recovering part of the thermal profile (see Figure 7.4). Their $T(U)$ curves are very similar, although the transformed velocity profiles and the locations of the thermal maximum in wall units differ. SST's close wall-coefficient agreement (see Figure 7.2) consequently coexists with a visibly incorrect thermal profile. SA, meanwhile, follows the displayed velocity profiles comparatively closely despite its thermal discrepancy. Section 7.3.3 tests the consequence for wall-function heating by applying the same Li closure to the DNS and resolved SA profiles.

The LowH_M3 case presents the complementary behaviour: both models closely reproduce the displayed $T(U)$ relation (see Figure 7.5), yet their wall coefficients retain appreciable errors, especially for SST (see Figure 7.2 and Figure 7.3). Among the selected cold-wall cases, LowH_M3 provides the only qualitatively close agreement in the thermal $T(U)$ relation.

This distinction follows from the derivative character of the wall quantities. As described in Section 3.4, resolved wall heat-flux and shear stress depend on the temperature and velocity gradients at the wall. A profile can agree over much of the boundary layer while retaining a different wall gradient and conversely, a similar wall gradient does not establish agreement throughout the profile. Visual comparison of $T(U)$ therefore complements the coefficient errors and cannot replace them. The same reasoning will apply to the wall-function closures: a visually close temperature–velocity relation is insufficient to establish the accuracy of the wall quantities recovered from it.

### 7.2.3 Evidence concerning the turbulent Prandtl number

The similarity of the SA and SST $T(U)$ discrepancies throughout all but one of the investigated hypersonic cold-wall ZPG FP cases motivates examination of their common thermal closure. Both extend the eddy viscosity to thermal transport through $\kappa_t = c_p \mu_t / Pr_t$. Fixing $Pr_t$ fixes the local ratio of modelled turbulent momentum and thermal diffusivity, even though their physical relationship may vary across a cold-wall boundary layer.

Figure 7.6 compares the DNS turbulent-Prandtl-number distributions across the boundary layer with the adopted constant value of $Pr_t = 0.9$. It is clearly visible, that much of the outer boundary layer has values

of order unity - arguing for a constant $Pr_t$ - but the near-wall regions show pronounced variation, overshoots and sign changes. This also uncovers a notable difference in the LowH_M3 and M6Tw025 case, which may explain their difference in the ability of the turbulence model to reprduce their thermal boundary layer profile (see Figure 7.5 vs. 7.2). The LowH_M3 case recovers toward values near unity after its sharp near-wall excursion, whereas the M6Tw025 distribution falls below unity over a broader part of the inner boundary layer. So while the constant $Pr_t = 0.9$ is still a reasonable assumption for for hypersonic adiabatic wall cases and in the outer region of hypersonic cold-wall cases, it is not in the near-wall region for the majority of the hypersonic cold-wall cases. From this analysis, it is not possible to conclude that the turbulent Prandtl number is the sole cause of the observed discrepancies, but it is a plausible contributor. As discussed in later Section 7.2.5 this work can also not find an immediate explanation for the difference in the turbulent Prandtl number distributions between the two cases.

The sharply varying DNS values require particular care. The diagnostic $Pr_t$ is obtained from turbulent fluxes and mean gradients, rather than being an independently measured transport constant. Duan and Martin associate its overshoots and sign changes with the crossover locations of the turbulent heat-flux and mean temperature gradient [39, Section 5]. Near these locations the inferred local ratio can become poorly conditioned, and a temperature maximum alone does not imply a divergence of $Pr_t$. Such behaviour also exposes limitations of describing the turbulent thermal flux everywhere by a positive scalar diffusivity and a local temperature gradient. Directly substituting the displayed singular or negative values into the present RANS closure would therefore require a different modelling argument.

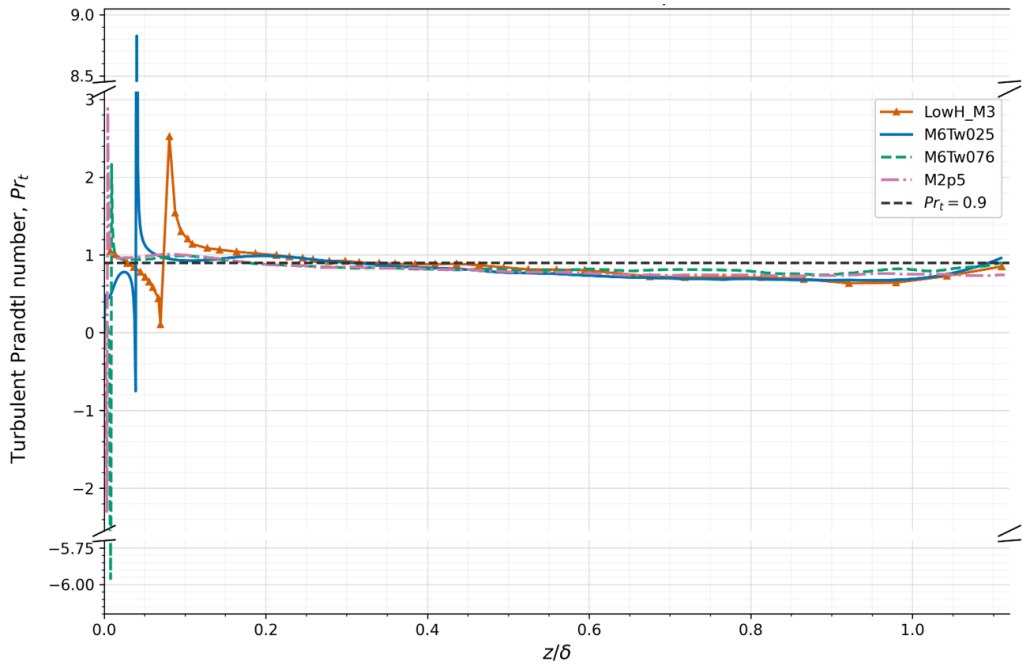



**Figure 7.6** Turbulent Prandtl number across the whole boundary layer height, with wall-normal distance $z$ normalized by the respective boundary layer thickness $\delta$. M2p5, M6Tw025 and M6Tw076 use Zhang et al.'s DNS data [5]; LowH_M3 uses the digitized Martin–Duan Part 4 distribution [39]. The dashed horizontal line marks the RANS value $Pr_t = 0.9$. Breaks in the vertical axis retain the large excursions near the wall while exposing the order-unity variation over the rest of the profile. The M2p5 and M6Tw076 are adiabatic cases, whereas LowH_M3 and M6Tw025 are strongly cold-wall cases (see Section 6.1).

The RANS $Pr_t$sensitivity calculations represented in Figure 7.7 provides further evidence that the thermal closure materially affects the $T(U)$ profile. Reducing the constant $Pr_t$ from $1.0$ to $0.6$ lowers the predicted temperature maximum in both models and brings it closer to DNS. No displayed constant reproduces the entire relation: the curves retain differences on the upper, recovering region. This aligns with the previous observation that the M6Tw025 case has a near-wall $Pr_t$ distribution that remains below unity over a broader part of the inner boundary layer.

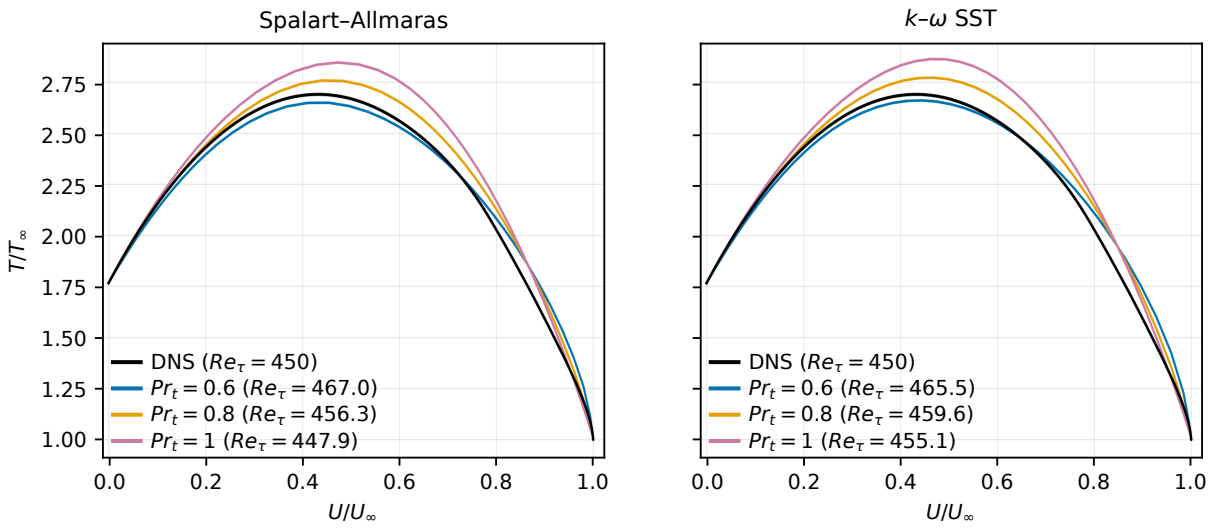



**Figure 7.7** Sensitivity of the M6Tw025 mean temperature–velocity relation to constant $Pr_t = 0.6$, $0.8$ and $1.0$ in wall-resolved SA (left) and $k$–$\omega$ SST (right). Temperature and velocity are normalized by the freestream values. The DNS reference has $Re_\tau = 450$; the RANS stations have approximately $448 \leq Re_\tau \leq 467$, as listed in the legends. These stations differ from the $Re_{\delta 2}$-matched baseline protocol. Lower $Pr_t$ reduces the thermal maximum but does not remove the discrepancy over the complete relation.

Taken together, the DNS distributions, the similar thermal-profile errors of two different momentum closures, and the constant-$Pr_t$ sensitivity identify the thermal closure as a plausible contributor to the discrepancies. They do not isolate it from errors in the eddy viscosity or from other modelling assumptions. Related hypersonic calculations by Huang et al. found improved temperature predictions with an algebraic turbulent-energy-flux closure compared with constant $Pr_t$, using an algebraic momentum model [82]. That result supports further investigation of the thermal closure, while its different momentum model prevents direct attribution of the present SA and SST errors to $Pr_t$ alone.

### 7.2.4 Assessment of results and comparison to existing publications

Several published RANS assessments use the Zhang and Huang benchmarks included in Table 6.1. They provide context for the wall-load errors and thermal-profile discrepancies found here, but common case and model names do not ensure equivalent calculations. The following comparison identifies differences in model formulation, mean-flow coupling, inflow conditions and sampling location that affect interpretation of Table 7.2.

#### Huang et al. (2019)

Huang et al. [87] assess SA and SST for M8Tw048 and report wall-coefficient errors within $5\%$, broadly consistent with the small errors obtained here for this case. Their calculations use $Pr_t = 0.85$ and $Re_\tau$ matching, whereas the present baseline uses $Pr_t = 0.9$ and $Re_{\delta 2}$ matching (Section 6.1.3). Moreover, reconstructing the DNS heat-flux from the published, rounded $-B_q$ introduces a $\pm 8.33\%$ rounding bound (Table 6.10). This limits comparisons based on that reconstruction and it is not an uncertainty estimate for Huang et al.'s underlying DNS data.

#### Aiken et al. (2022)

Aiken et al. [89] assess SA, $k$–$\omega$ BSL and SST against six Zhang and Huang benchmarks at $Pr_t = 0.9$. They report thermal-profile discrepancies similar to those in Section 7.2.2 and identify the thermal closure as a possible contributor. Their tested freestream-turbulence variations have little downstream influence for

M8Tw048. However, their SA model includes the Catris–Aupoix correction, distinct from the implementation correction in Section 3.5.3, and their original SST formulation differs from the updated version described in Section 2.2. They match $Re_\tau$, while reporting similar errors with $Re_\theta$ matching. Their Table 1 also lists total enthalpies inconsistent with the stated flow conditions, and Table 3 contains inconsistent $Re_\theta/\theta$ values for M8Tw048. These reporting issues limit reproducibility without establishing that the imposed flow states were incorrect. Together with the formulation differences, they prevent treating agreement or disagreement as a direct code-verification result, consistent with the reporting concerns of Viti et al. [97].

**Danis (2023)**

Danis [98, Figure 4.10] compares original SST skin-friction predictions with DNS for M6Tw025, M14Tw018 and M11Tw020 as functions of $Re_\tau$, without a corresponding SST wall-heat-flux comparison. The calculations use VULCAN-CFD and cite Menter's 1994 SST model, but the turbulent Prandtl number and detailed model settings are not specified in that assessment. Differences from the present skin-friction results therefore cannot be attributed to a particular modelling or implementation choice.

**Vaughan et al. (2025)**

Vaughan et al. [99] and Vaughan's master's thesis [100] examine overlapping Zhang and Huang cases. The thesis likewise finds cold-wall thermal-profile discrepancies at $Pr_t = 0.9$, but uses SA-neg and a vorticity-based SST limiter, rather than the present model variants. It selects sampling locations by matching the initial DNS boundary layer thickness and then advancing by the reported downstream distance, rather than matching $Re_\tau$. This is the shifted-station approach discussed in Section 6.1.3.

For SST, Vaughan's documented mean-flow equations retain the isotropic turbulent stress, turbulent kinetic energy in total energy, and its diffusive transport, whereas the present HiSA formulation omits these contributions. Zhang et al. [101] found sensitivity to these terms in shock interactions but little influence on skin-friction and velocity profiles for their single Mach-5 flat-plate. That result does not establish negligible heating sensitivity for the present cases.

Vaughan also interprets the DNS freestream states as post-shock values and supplies modified pre-shock inflows [100, Tables 2.2–2.3]. The reconstruction procedure is unspecified and a discrepancy arises for M14Tw018, where the tabulated pre-shock mean-flow total enthalpy $c_pT + U^2/2$ exceeds the DNS value by $4.56\%$. Table 7.3 tests these inflows in the present SST implementation for four Zhang cases, retaining each run's own $Re_{\delta 2}$-matched station. This demonstrates sensitivity to the inflow specification, but neither establishes which inflow better reproduces DNS nor reproduces Vaughan's complete setup.

**Table 7.3** Inflow sensitivity of the present wall-resolved $k$–$\omega$ SST simulations against Zhang's DNS [5]. Original inflows use the DNS freestream conditions in Table 6.3; modified inflows use Vaughan's pre-shock values [100, Table 2.2]. Each run is evaluated at its own $Re_{\delta 2}$-matched station using its respective inlet density and velocity. Parentheses give $100(\mathrm{RANS}-\mathrm{DNS})/\mathrm{DNS}$. Heat-flux is positive into the wall; shear stress is its magnitude. The M6Tw076 heat-flux reference is nominal and retained only for this sensitivity comparison, subject to the rounding and wall-temperature limitations in Section 6.1.

| Case | Reference / inflow | $q_w$ [$\mathrm{kW\,m^{-2}}$] | $\tau_w$ [Pa] |
|---|---|---|---|
| M6Tw025 | **Zhang DNS** | **11.4888** | **28.3210** |
| M6Tw025 | Original | 11.8290 ($+2.96\%$) | 29.2326 ($+3.22\%$) |
| M6Tw025 | Vaughan | 11.4180 ($-0.62\%$) | 28.2175 ($-0.37\%$) |
| M6Tw076 | **Zhang DNS** | **2.1720** | **16.2533** |
| M6Tw076 | Original | 1.8925 ($-12.87\%$) | 15.3712 ($-5.43\%$) |
| M6Tw076 | Vaughan | 1.8585 ($-14.43\%$) | 15.1801 ($-6.60\%$) |
| M8Tw048 | **Zhang DNS** | **4.5495** | **13.2915** |
| M8Tw048 | Original | 4.6231 ($+1.62\%$) | 13.3289 ($+0.28\%$) |
| M8Tw048 | Vaughan | 4.5881 ($+0.85\%$) | 13.2067 ($-0.64\%$) |
| M14Tw018 | **Zhang DNS** | **10.3028** | **12.1640** |
| M14Tw018 | Original | 11.7582 ($+14.13\%$) | 14.0633 ($+15.61\%$) |
| M14Tw018 | Vaughan* | 12.3612 ($+19.98\%$) | 14.3311 ($+17.82\%$) |

* Vaughan's tabulated M14Tw018 pre-shock mean-flow total enthalpy $c_pT+U^2/2$ is $4.56\%$ higher than that of the DNS freestream state.

#### Cheng et al. (2026)

Cheng et al. [90] use OpenFOAM-v2312 and the HiSA solver framework to investigate energy closures with the Durbin–Yin $k$–$\omega_0$ model, including additional SST–$\omega_0$ tests. Their baseline therefore differs from the present SST model even before the proposed energy-closure corrections. Their stated energy equation includes turbulent-kinetic-energy transport [90, Eq. 2.7], which is absent from the present HiSA formulation. The paper does not specifically state having performed the corresponding solver modifications outside of the stated RANS equations. Shared solver ancestry consequently does not establish equivalent mean-flow coupling. Station selection adds another difference, although the present tests in Tables 6.7 and 6.8 give changes below $1\%$ in $-B_q$ and $2\%$ in $C_h$ between the shifted and $Re_{\delta 2}$-matched stations. Their Table 5 also normalizes $B_q$ with DNS wall quantities, whereas Table 7.2 uses each solution's own wall state. These values require consistent normalization before comparison.

#### Implications for the present assessment

The literature supports treating the present discrepancies as part of a broader difficulty in predicting cold-wall boundary layers with conventional RANS closures. It does not provide an equivalent setup for direct code verification or isolate the cause of differences in wall loads. In particular, the thermal-profile deficiencies reported by Aiken and Vaughan reinforce the evidence in Sections 7.2.2 and 7.2.3, without establishing constant $Pr_t$ as their sole cause. The subsequent wall-function assessment therefore retains both DNS and the present wall-resolved SA solution as distinct references, as defined in Table 7.1.

### 7.2.5 Selection of cases for detailed wall-function assessment

M6Tw025 and LowH_M3 are retained for the detailed coupled wall-function analysis because they expose different relationships between wall-coefficient and profile accuracy. M6Tw025 supplies digital DNS profiles for a strongly cooled Mach-6 boundary layer and exhibits a clear thermal-profile discrepancy in both RANS models. It therefore tests the wall functions when their analytical thermal relation is coupled to a resolved RANS field that already differs from DNS. Its detailed reference profiles also permit an independent prescribed-profile closure assessment, subject to the heat-flux reporting limitations stated above.

LowH_M3 supplies a cold-wall supersonic comparison in which the resolved RANS $T(U)$ relation closely follows the digitized DNS. This reduces the pronounced thermal-profile mismatch present in M6Tw025 and makes the case useful for examining additional errors introduced by the wall function and the CFD mesh. The nonzero wall-coefficient errors remain part of its baseline. In the subsequent prescribed-profile calculations, the resolved SA field is used as the LowH_M3 input, so those closure errors measure consistency with SA rather than an independent reconstruction of DNS wall quantities.

The contrasting thermal-profile accuracy of LowH_M3 and M6Tw025 cannot be attributed to the models' incompressible origins alone. The corresponding DNS studies [5, 39] report approximate Morkovin-type similarity and outer-layer agreement with modified strong Reynolds analogies for both cases (ref. Section 2.1.2). This weakens an explanation based on a fundamental loss of applicability between the two flow conditions, without excluding differences in intrinsic compressibility, mean-property gradients or thermal transport. The cause of the contrast remains unresolved, with the following analysis therefore examining the shared constant turbulent Prandtl number as a possible contributor.

## 7.3 Wall-function closure assessment

The previous wall-resolved assessment established the discrepancy of the underlying RANS calculation from DNS. Adding a wall function to a RANS simulation introduces a further relationship between the sampled flow state and the resulting wall shear stress and heat-flux. To examine that relationship separately, the analytical closures are now evaluated on prescribed boundary layer profiles. This a-priori assessment asks how well a closure can recover the known wall heat-flux and shear stress, when its off-wall sampled inputs are ideal, i.e. supplied from the respectilve known reference boundary layer profile. It provides the comparison needed to distinguish a wall function's underlying closure behaviour from the additional changes that occur when the same wall function is coupled to full FVM CFD simulation.

The detailed discussion focuses on M6Tw025 and LowH_M3, which also carry the comparison into the subsequent coupled CFD assessment (as established in Section 7.2.5). The initial M6Tw025 sweeps use the digitally published DNS profiles and wall quantities of Zhang et al. [5]. LowH_M3 lacks similar high quality published DNS reference data, so it uses the wall-resolved SA profile and its corresponding wall quantities at the matched comparison station (Section 6.1.3) are used. For LowH_M3, the reported closure errors are thus relative to the wall-resolved SA wall quantities. An additional M6Tw025 comparison in Section 7.3.3 applies the same Li closure to DNS and resolved SA inputs to assess the influence of the supplied profile. Results for the additional cases are collected in Appendix D.

### 7.3.1 A-priori inverse solution

For an a-priori assessment, a virtual sampling point is moved through a known, resolved boundary layer profile. At every sampling height, the required velocity and thermodynamic state are interpolated from that given profile and supplied to the wall-function closure equations, together with the required physical wall temperature and transport properties. The wall heat-flux and shear stress are then recovered by an inverse solution of the closure equations. The reference friction velocity defines the sampling coordinate,

$$y^+_{\mathrm{ref}} = \frac{\rho_{w,\mathrm{ref}} u_{\tau,\mathrm{ref}} y}{\mu_{w,\mathrm{ref}}}, \qquad u_{\tau,\mathrm{ref}} = \sqrt{\frac{\tau_{w,\mathrm{ref}}}{\rho_{w,\mathrm{ref}}}}. \tag{7.2}$$

For the Li formulation, introduced in Section 5.3.1, the first-point velocity and enthalpy, the physical wall state, and an arbitrarily placed second reference state determine the generalized enthalpy–velocity relation. Two reference choices are examined. The two-point construction assumes two virtual cells of equal height, so their centres satisfy $y_2 = 3y_1$. Both point states are interpolated from the resolved profile. In the alternative construction, the reference state is prescribed at the boundary layer edge. Li et al propses the former, two-point construction, so that the wall function closures are independant of the boundary layer edge quantities [3], for comparison and evaluation the latter, boundary layer edge construction was selected

to be included in the analysis. In either case, the unified velocity law is inverted for $u_\tau$, and the wall outputs follow from $\tau_w = \rho_w u_\tau^2$ and the wall function's heat-flux relation.

For the Griffin formulation, introduced in Section 5.4.1, the sampled velocity and thermodynamic state are combined with the wall temperature and boundary layer-edge quantities - Griffin et al do not propose a specific approach to allow for a two-point, boundary layer edge free construction. For each sample point, the inverse-transformation model is integrated from the wall region to the sample point. The wall stress is iterated until the integrated velocity reproduces the prescribed matching point's velocity. The temperature–velocity relation then supplies the wall heat-flux. The prescribed wall stress and heat-flux of the reference profile are used to assess the recovered outputs, rather than imposed as their solution targets.

Edge states are supplied according to the available reference data. Griffin uses the supplied $\delta_{99}$ for the Zhang DNS profiles and its edge criterion for LowH_M3, whereas the prescribed-edge Li comparison for LowH_M3 uses the $U = 0.99U_\infty$ crossing.

The reported errors in this section are signed relative differences,

$$\varepsilon_X = 100\left(\frac{X_{\text{closure}}}{X_{\text{ref}}} - 1\right), \qquad X \in \{\tau_w, q_w\}. \tag{7.3}$$

The reference being either the DNS wall quantities or the wall-resolved RANS quantities, depending on the profile being sampled. The a-priori assessment is therefore a standalone evaluation of the closure's ability to recover the known wall fluxes from known inputs, and thus evaluates the wall function's closure theories in isolation.

#### Consistency with the published Griffin assessment

An independent check of the herin implemented a-priori solver on the Griffin evaluation is available from the a-priori results of Griffin et al. [4, Figure 5]. Their matching height is $y_m = 0.3\delta_{99}$. In reference wall units, this corresponds to $y_m^+ = 0.3\mathrm{Re}_\tau$. Table 7.4 compares the present inverse solutions at those heights with the errors read from the published figure for two identifiable Zhang cases.

**Table 7.4** Comparison of the present Griffin a-priori solver at $y_m = 0.3\delta_{99}$ with Griffin et al. [4, Figure 5]. Published errors are approximate values read from the figure; the comparison checks consistency of the standalone closure evaluation.

| | | $\tau_w$ error [%] | | $q_w$ error [%] | |
|---|---|---|---|---|---|
| Case | $y_m^+$ | Griffin et al. | Present solver | Griffin et al. | Present solver |
| M6Tw025 | 134.561 | ≈–13 | –11.56 | ≈–15 | –15.28 |
| M14Tw018 | 192.588 | ≈–10 | –11.32 | ≈–15 | –15.91 |

The present solutions reproduce the sign and approximate magnitude of the published errors to within the uncertainties of the published graph's resolution. Their agreement supports the use of the a-priori solver used in the following to examine matching-height sensitivity of the closure theories.

### 7.3.2 Closure errors versus sampling height

Figures 7.8–7.11 assess the closures by Li and Griffin jointly on the M6Tw025 DNS profile and the LowH_M3 wall-resolved SA profile. Both formulations recover the wall quantities with significant errors over the varying sampling height, but neither is notably more accurate than the other. But it shows, that the Li's two-reference-point method, rather than relying on boundary layer edge quantities, works without a systematic loss of accuracy (e.g. see Figure 7.8 dashed vs. solid line). And it should be note that both Li and Griffin's formulations don't show significantly higher errors in the buffer layer, highlighting that the blending between

the sublayer and law-of-the-wall for these adaptive wall functions work well and doesn't present a $y^+$ meshing region to be specifically avoided.

Across all the investigated profiles (including the cases in Appendix D) Griffin gives signed shear-stress errors of approximately $-11\%$ to $+11\%$ and heat-flux errors of $-14\%$ to $+6\%$, over the reasonable range of $1 \leq y^+_{\mathrm{ref}} \leq 100$. Over the same cases and sampling-height interval, the Li closure gives respective shear-stress errors of approximately $-7\%$ to $+13\%$ and heat-flux errors of $-15\%$ to $+18\%$. These are appreciable closure errors for prescribed, ideal, inputs and warrant ongoing research into finding better velocity transformation like by Hasan et al.[37] and improved closure formulations.

The heat-flux ranges exclude M2p5, whose adiabatic reference makes relative heat-flux errors undefined, and M6Tw076, owing to the reference-data limitations described in Section 6.1.

At the largest sampling height, Li's prescribed-edge M6Tw025 heat-flux error rises to approximately $+79\%$ at $y_1^+ = 400$, with a reported $Re_\tau = 450$ [5], this corresponds to a sampling station at about $0.89\delta_{99}$. Li's two-reference point method, is not applicable anymore at these sampling stations because the second reference point sits outside of the boundary layer, this is a limit of this method. The high error at these high sampling points, is consistent with increasing sensitivity of the enthalpy inversion as the sampled and reference velocities approach one another, especially because the first sampling point also lies outside the inner layer [3, Eq. (40)]. Griffin avoids this particular inversion sensitivity by prescribing the wall temperature–velocity slope through its Reynolds analogy. It shows significant improvement to this high $y^+$ sensitivity, but does not establish universal validity near the boundary layer edge. The published a-priori comparison uses $y_m = 0.3\delta_{99}$; the additional WMLES matching-height studies discussed in Section 5.4.3 do not extend this assessment towards the edge.

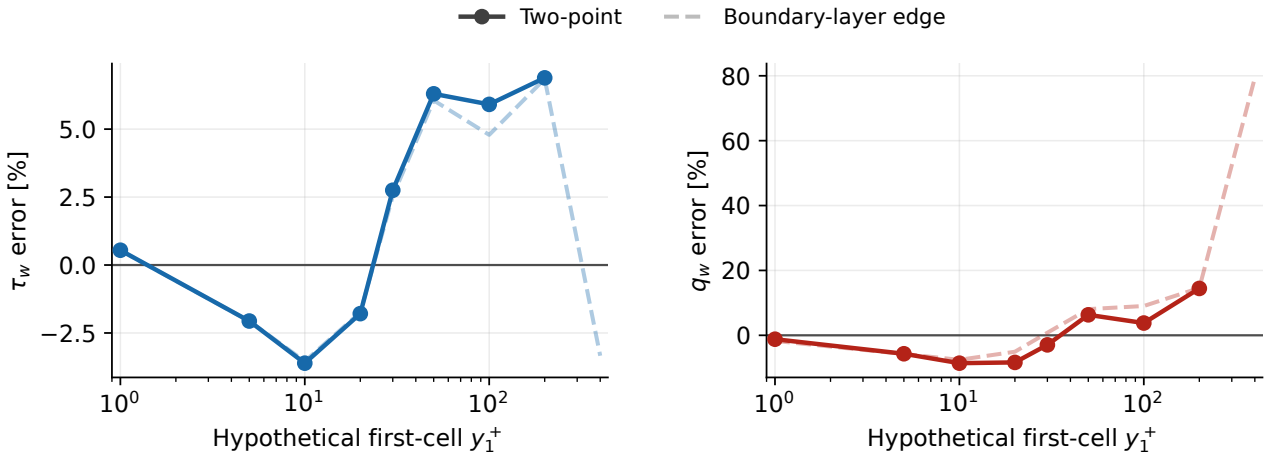



**Figure 7.8** A-priori Li wall-shear (left) and heat-flux (right) errors relative to the M6Tw025 DNS wall quantities. The solid curves use two virtual equal-height cells, with $y_2 = 3y_1$; the dashed curves use a prescribed boundary layer-edge state. The horizontal coordinate is the virtual first-cell-centre height in DNS wall units. Each point is a separate inverse closure solution on the same prescribed profile.

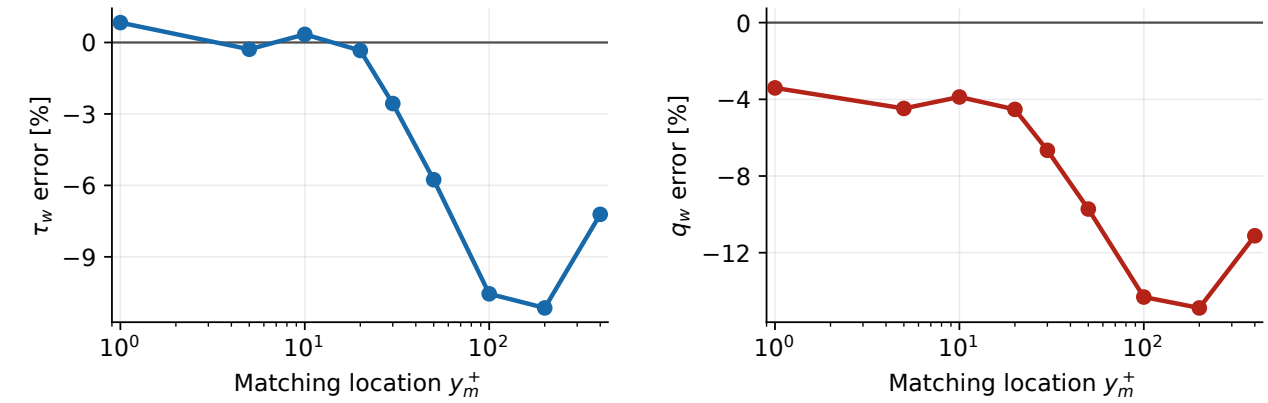



**Figure 7.9** A-priori Griffin wall-shear (left) and heat-flux (right) errors relative to the M6Tw025 DNS wall quantities. The prescribed DNS edge state remains fixed while the virtual matching height $y_m^+$ is varied. Points above $y_m = 0.3\delta_{99}$, corresponding here to $y_m^+ \simeq 134.6$, extend the assessment beyond the matching height used in the published a-priori comparison and are retained as sensitivity results.

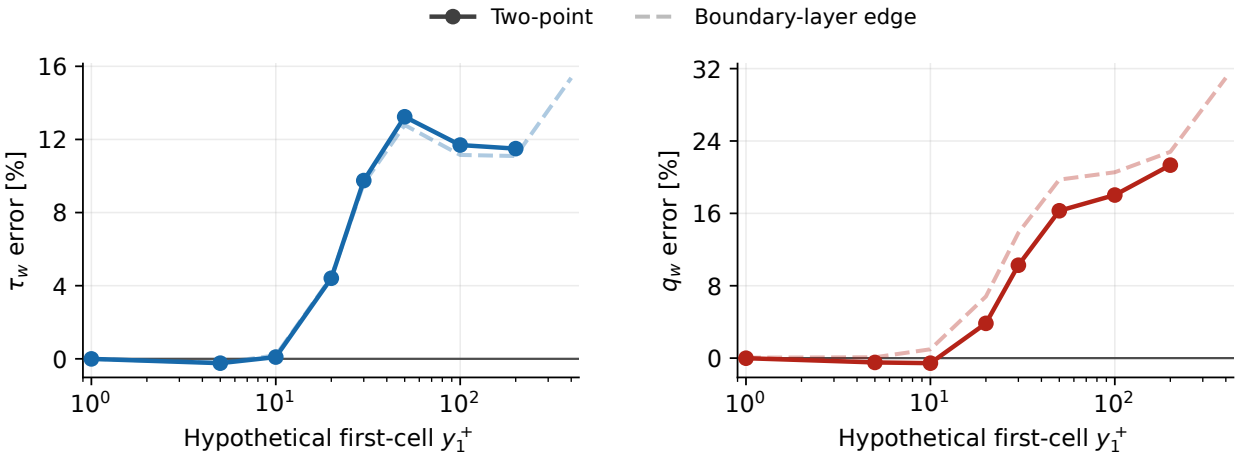



**Figure 7.10** A-priori Li wall-shear (left) and heat-flux (right) errors relative to the wall-resolved SA baseline for LowH_M3. Solid curves use the two-point formulation, with $y_2 = 3y_1$, and dashed curves use the prescribed $0.99U_\infty$ edge state. Sampling heights are expressed in wall units of the unchanged SA reference profile. These are closure errors relative to RANS, rather than errors relative to DNS.

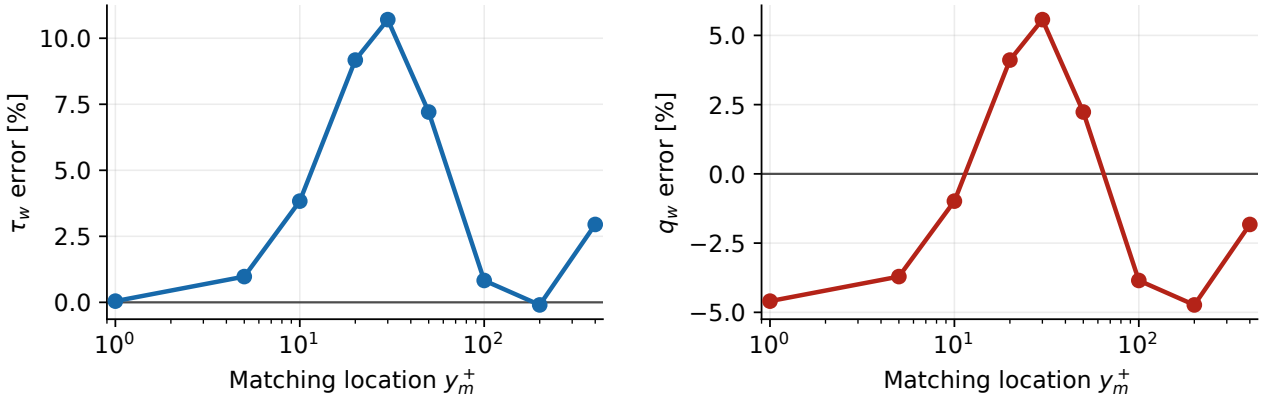



**Figure 7.11** A-priori Griffin wall-shear (left) and heat-flux (right) errors relative to the wall-resolved SA baseline for LowH_M3. The edge state is reconstructed once from the resolved profile using the Griffin boundary layer criterion, then held fixed while the virtual matching height is varied. The point at $y_m^+ = 400$ lies above $0.3\delta_{99}$ of this reconstructed boundary layer and is included as a sensitivity result.

### 7.3.3 Influence of the prescribed RANS profile

In the previous Section7.2, with the assessment of turbulence models, the wall-resolved M6Tw025 comparison in Figure 7.4 revealed a clear discrepancy in the temperature maximum and $T(U)$ relation compared to the DNS reference. To quantify the consequence of this short coming of the turbulence models to the wall function closure that rely on these profiles, the Li et al. closure is evaluated on both the DNS profile and the wall-resolved SA profile at $Pr_t = 0.9$ and subsequently compared.

Figure 7.12 expresses both a-priori solved results relative to the same DNS wall quantities. Replacing the DNS profile as input, with the wall-resolved SA profile, changes the heat-flux prediction substantially, while the shear predictions remain much closer. This aligns with the SA velocity profile matching the DNS reference (see Figure 7.4 bottom right), while the $T(U)$ profile, which feeds the wall function's energy closure, does not (see Figure 7.4 top left). At $y^+_{1,\mathrm{DNS}} = 30$, the predicted heat-flux increases by $13.4\%$ relative to the DNS-input prediction, but shear changes by only $0.1\%$. At $y^+_{1,\mathrm{DNS}} = 100$, the heat-flux error relative to DNS rises from $+3.2\%$ to $+16.0\%$. Thus, the thermal-profile discrepancy observed in the resolved calculation has a significant counterpart in the wall-function response, even where the velocity profiles appear convincing, and need to be addressed in future improvements of the turbulence models. In a fully coupled turbulence model + wall function CFD run, this error does not isolated as cleanly as in the a-priori solver here, because of the feedback of the wall function's wall quantity prediction into the overall flow field, but the underlying discrepancy still stands.

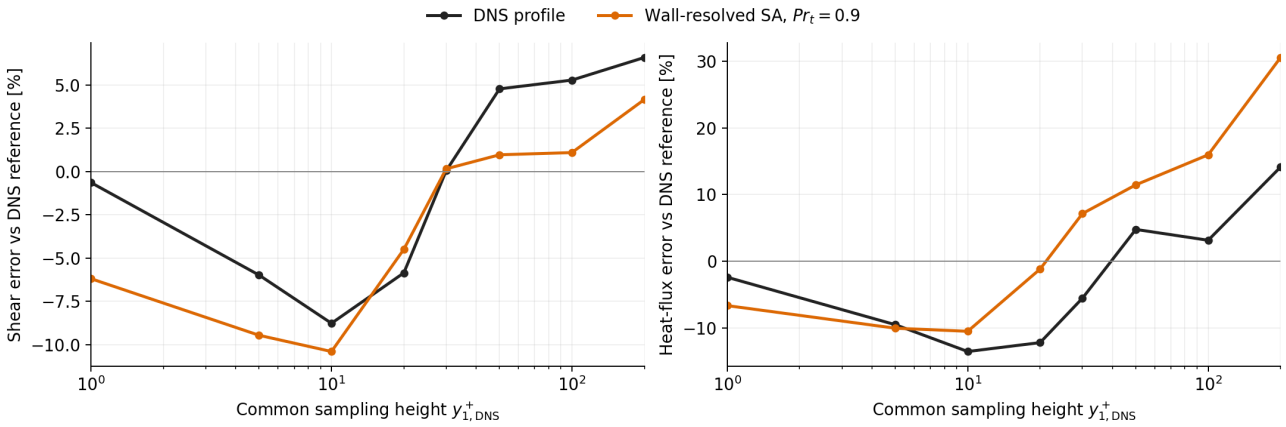



**Figure 7.12** Li closure evaluated on the M6Tw025 DNS and wall-resolved SA profiles. Left: wall-shear error; right: heat-flux error. Both curves use the DNS wall heat-flux and shear stress as their reference, hence the orange line not converging to zero error at $y^+_{1,\mathrm{DNS}} = 1$. The virtual sampling heights of $y_1$ and $y_2 = 3y_1$ are identical for both inputs; the abscissa uses the supplied DNS wall units.

The common DNS denominator in the figure must be distinguished from recovery of each profile's own wall flux. Over e.g. the three sampled heights $y^+_{1,\mathrm{DNS}} = 30$, $50$ and $100$, the mean absolute heat-flux error relative to the corresponding source wall value increases from $4.5\%$ for DNS inputs to $17.8\%$ for SA inputs. The discrepancy is therefore more than the offset between the resolved SA and DNS wall fluxes itself, hence the discrepancy in the wall-resolved boundary layer to the DNS impacts the wall function closure's accuracy.

This is also clearly seen through the immediate mechanism, which is the inferred enthalpy–velocity wall slope in the wall function's energy closure. Li's generalized recovery enthalpy is obtained from

$$h_{rg} = \frac{h_1 - h_w(1-r) - h_R r^2}{r(1-r)}, \qquad r = \frac{U_1}{U_R}, \tag{7.4}$$

where $(U_R, h_R)$ is the reference state, here the second sampling point [3, Eq. (40)].

With positive heat-transfer into the cooled wall denoted by $Q_w = |q_w|$, Li's relations give

$$S = \left.\frac{\mathrm{d}h}{\mathrm{d}U}\right|_w = \frac{h_{rg} - h_w}{U_2}, \qquad Q_w = \frac{\tau_w S}{Pr_w}, \tag{7.5}$$

where $h_{rg}$ follows from Equation (7.4) [3, Eqs. (40)–(41)]. At $y^+_{1,\mathrm{DNS}} = 30$, changing from DNS to SA inputs raises $S$ by $13.3\%$, accounting for almost all of the heat-flux increase while the recovered shear is nearly unchanged because so is the sampled velocity profile. This identifies the thermal wall slope as the immediate route through which the profile mismatch affects the predicted heating. Thus future improvements to RANS turbulence models should not only improve the heat-flux and shear stress results in isolation, but also need to tackle the challenge in hypersonic cold-wall boundary layer reproducibility.

### 7.3.4 Summary and Implications for the coupled assessment

The a-priori assessment (ref. Section 7.3.2) identifies appreciable baseline errors in both closures, even when the sampled inputs are supplied from known reference profiles. However, neither formulation shows a systematic breakdown in the buffer layer. The results therefore provide no basis for excluding a buffer-layer sampling interval because of the blending between the viscous and logarithmic relations. But the remaining closure errors do warrant investigation into better, improved velocity transformations and closure theories.

A comparison of supplying the a-priori solver with the DNS or the wall-resolved SA profile (ref. Section 7.3.3) identifies a further limitation: discrepancies in the wall-resolved RANS profile (as identified in Section 7.2.2) can substantially alter the wall-function heat-flux prediction. In the example of the Li assessment on the M6Tw025 case, it shows that a convincing velocity profile does not compensate for an inaccurate $T(U)$ relation, which changes the inferred thermal wall slope. Improving the coupled prediction therefore requires attention to both the closure's baseline accuracy and the turbulence model's reproduction of the cold-wall thermal profile. These findings provide the context for the following assessment of the additional influence of spatial discretization in Section 7.4, especially with the final isolation from the closure error by overwriting the closure theory with a fitted function in Section 7.4.5.

## 7.4 Wall-function in FVM CFD: Spatial-discretization error and sampling sensitivity

The closure assessment in Section 7.3 determines how accurately a wall function recovers wall loads from a prescribed, ideal, boundary layer state. In CFD, that state is itself part of a discrete solution. Consequently, a change in sampling height can alter both the closure response and the numerical solution of the surrounding boundary layer. This section analysis the behaviour of the wall functions on the LowH_M3 and M6Tw025 cases, by first examining their accuracy and consistency on a conventional mesh strategy as the wall functions were intended to be used e.g. by Li et al. in [3]. Subsequently, a novel boundary layer-informed mesh design is introduced to isolate the identified internal-face discretization errors. Finally the, in the previous section identified, closure discrepancies are revisited in the context of the new meshes, by presccribing an ideally fitted closure to isolate out all found error effects, but from the turbulence models.

### 7.4.1 Wall functions on conventional meshes

The conventional mesh strategy and sampling geometry are introduced in Section 5.1. The general idea how conventionally meshes for wall-functions are designed is, to increase the first-cell height until the desired $y^+_1$ is achieved. In these *conventional wall-function mesh* sweeps, increasing the physical first-cell height moves the wall-function sample outward and increases the separation of neighbouring cell centres. Thus, on *conventional wall-function mesh* the sampling location $y^+_s$ coincides with the first-cell location $y^+_1$, while the spacing of internal-face stencils, herin referred to as $h^+$, changes with it (see Figure 7.19 $h^+$ change between exemplary $y^+_1 = 10$ and $y^+_1 = 30$ meshes). One should keep in mind, that the sample

height $y_s^+$ and physical first-cell mesh resolution $y_1^+$ need not necessarily be identical, but one could also take the second off-wall cell centre as the wall function sampling point. This distinction becomes essential when the two are varied independently below.

Figure 7.13 shows a representative conventional M6Tw025 mesh with special attention to the leading edge region. The leading edge requires additional care because the local friction velocity and the boundary layer thickness differ substantially from their downstream values at which the sampling station with the desired $y_1^+$ is specified. With a constant physical first-cell height, the elevated upstream shear raises $y_1^+$ significantly above its intended value at the comparison station. For coarse meshes it was found, that $y_1^+$ at the leading edge region even spiked beyond values where the wall-function is applicable and resulted in unphyiscal behaviour which propagates downstream. Figure 7.14 illustrates this effect and its reduction by smoothly increasing the first-cell height over the initial $80\,\mathrm{mm}$ of the plate, thus giving a clearer picture of the actual wall-function behaviour.

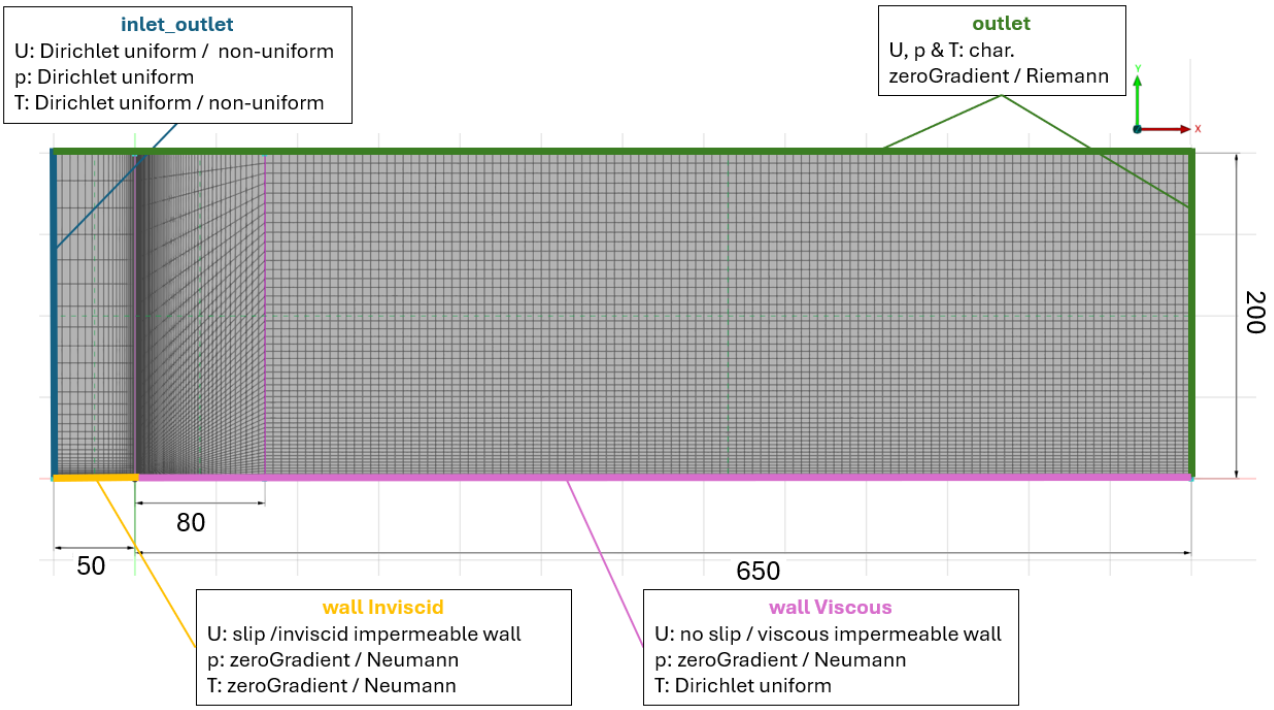



**Figure 7.13** Conventional wall-function mesh for M6Tw025 with nominal first-cell location $y_1^+ \approx 100$ at the comparison station. Streamwise refinement and a smaller first-cell height resolve the leading-edge region; the wall-normal spacing increases smoothly over the first $80\,\mathrm{mm}$.

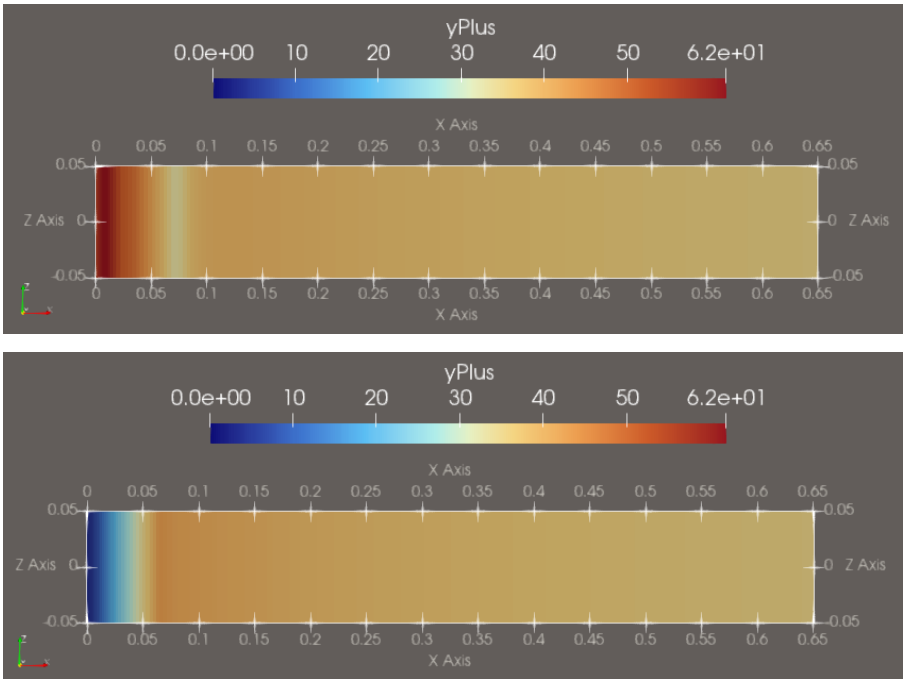



**Figure 7.14** Wall-unit first-cell location along the M6Tw025 plate for a nominal $y_1^+ \approx 40$ Li wall-function mesh. A constant physical first-cell height (top) produces elevated $y_1^+$ near the leading edge. A smooth increase of first-cell height over the first $80\,\mathrm{mm}$ (bottom) reduces this upstream excursion.

Figures 7.15–7.18 show the resulting wall heat-flux and shear-stress errors. All curves use the signed relative error to the respective DNS reference at the matched $Re_{\delta 2}$ station (see Section 6.1.3). The dashed lines show the wall-resolved SA result at the same station as presented and discussed in Section 7.2. Around the DNS reference value for th heat-flux and shear stress, the shading marks a $\pm 5\%$ error band (as selected in Section 6.3). The horizontal axis shows the actual, calculated first-cell $y_1^+$ at the matched station on these conventional meshes, so each point represents a different CFD simulation with the only change being the first-cell height in the sampling region of the used mesh.

From Figures 7.15–7.18 it is immediatly visible that the wall-function results are highly sensitive to the first-cell height. But the errors are not monotonic with increasing $y_1^+$, but instead exhibit a pronounced different behaviour in the sublayer region, the buffer and the log-layer region of the cases. The wall-function error bands are also substantially larger than the corresponding a-priori closure variations identified in Section 7.3.2. Griffin gives signed shear-stress errors of approximately $-20\%$ to $+8\%$ and heat-flux errors of $-16\%$ to $+12\%$. Over the same cases and sampling-height interval, the Li wall function gives respective shear-stress errors of approximately $-7\%$ to $+14\%$ and heat-flux errors of $-6\%$ to $+27\%$. But for an engineering context, the high sensitivity of the wall-function results to the first-cell height is arguably more concerning than the absolute error magnitude because it introduces a substantial uncertainty and not a clear bias.

Looking at the sub-layer region, the wall-function results approach the wall-resolved baseline to within the closure accuracy as expected. Looking at Li's previously found closure behaviour where at $y^+ = 1$ the wall-function closure error is basically zero (see a-priori result for M6Tw025 in Figure 7.8 and LowH_M3 in Figure 7.10), the wall-function CFD results are also very close to the wall-resolved baseline (see conventional mesh CFD result for M6Tw025 in Figure 7.17 and LowH_M3 in Figure 7.15). The same findings also apply to the Griffin wall-function results where for both cases the a-priori closure assessment in Section 7.3.2 shows a low-$y^+$ closure discrepancy of about $-5\%$ for the wall heat-flux (see a-priori result for M6Tw025 in Figure 7.9 and LowH_M3 in Figure 7.11), which is also reflected in the deviation of wall-function CFD results from the wall-resolved baseline (see conventional mesh CFD result for M6Tw025 in Figure 7.18 and LowH_M3 in Figure 7.16).

Beyond the sublyer, reaching into the buffer layer, the wall-function CFD results do not follow this behaviour where the whole error can be explained through the closure error alone. Rather it is clear that the wall-function CFD results are not only affected by the closure discrepancy, but also by additional error sources. The cause for this behaviour beyond the sublayer is analysed and examined in the next step, where the internal-face fluxues of momentum and energy where compared to the wall-resolved results, and found in the spatial discretization.

On the even higher $y^+$ conventional meshes, reaching the log-layer and beyond, a sharp reduction in the wall-function CFD error is observed. This reduction does not necessarily indicate a better wall-function performance, but rather indicates a different error sources, that happens to cancel some of the closure discrepancy. In Figure 7.18 for example the relative error becomes highly negative, instead of cancelling out to zero. And this behvior in the log-layer and beyond also, as in the buffer layer, do not follow the a-priori seen closure discrepancy. But it is a nice indication why Li et al found a good agreement of their wall-function CFD results with the DNS reference, because for their validation cases they used a conventional mesh with a first-cell height of $y_1^+ = 4, 8$ and then for their cases comfortly in the log-layer $y_1^+ = 104, 133, 156, 200$ [3, Figures 17–20 and 23–24]. Li et al. claim "[...] and it is also essential to avoid them [first grid points] falling into the bufferlayer when using the wall function method." [3, p. 61]. So in their analysis they avoided the regions with high sensitivity and large errors, but attributed it to the wrong cause. Griffin et al. use $y_m = 0.3\delta$ for their principal wall-flux comparisons, but also reduce the matching height during WMLES mesh refinement, reaching the buffer region in the lower-Reynolds-number channel case [4, Section 4.1]. Their study therefore does not support a blanket exclusion of buffer-layer sampling. As discussed in Section 5.4.3, those matching-height and LES-resolution studies differ from the present conventional RANS mesh sweeps.

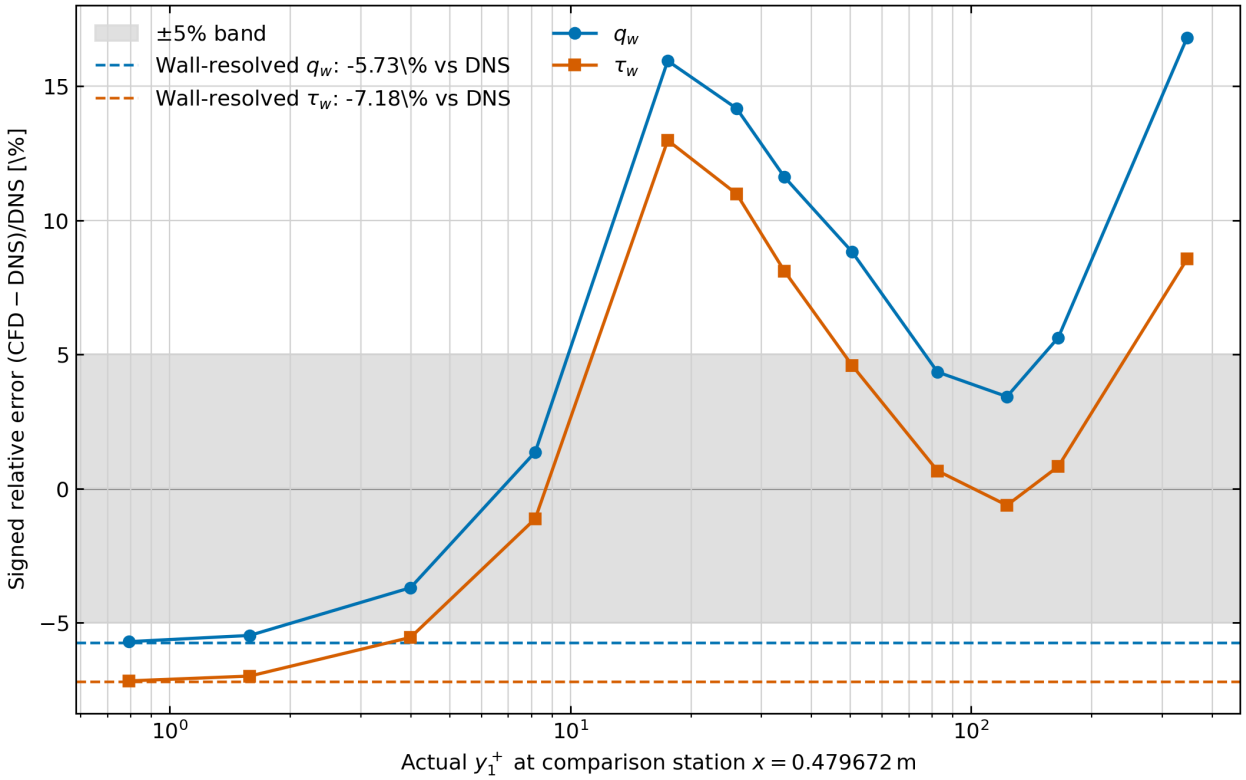



**Figure 7.15** Li wall-function CFD with SA on conventional meshes for LowH_M3: signed errors in wall heat-flux $q_w$ and shear stress $\tau_w$ relative to DNS, evaluated at $x = 0.479672\,\mathrm{m}$ against the actual first-cell $y_1^+$. Dashed lines denote the wall-resolved SA errors; shading marks $\pm 5\%$ relative to DNS.

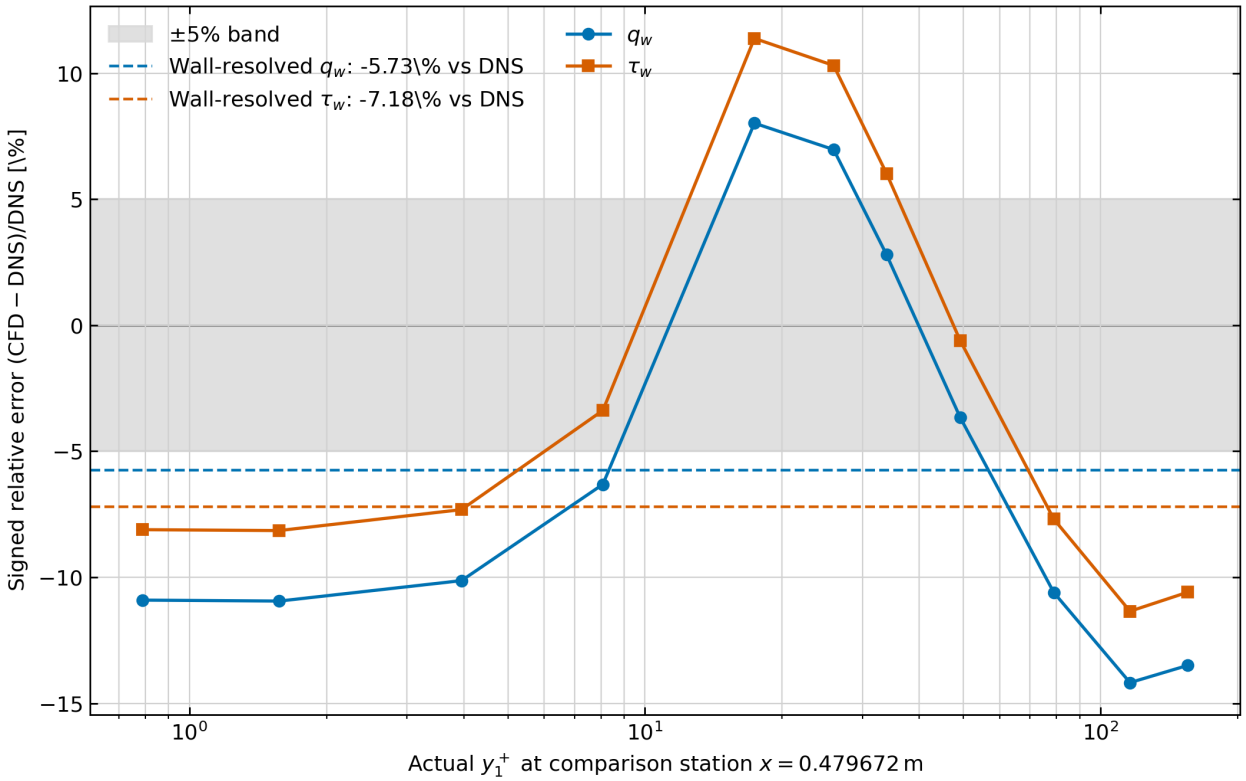



**Figure 7.16** Griffin wall-function CFD with SA on conventional meshes for LowH_M3. Wall heat-flux and shear-stress errors use the same DNS normalization and comparison station as Figure 7.15. Dashed lines denote the wall-resolved baseline; shading marks $\pm 5\%$ relative to DNS.

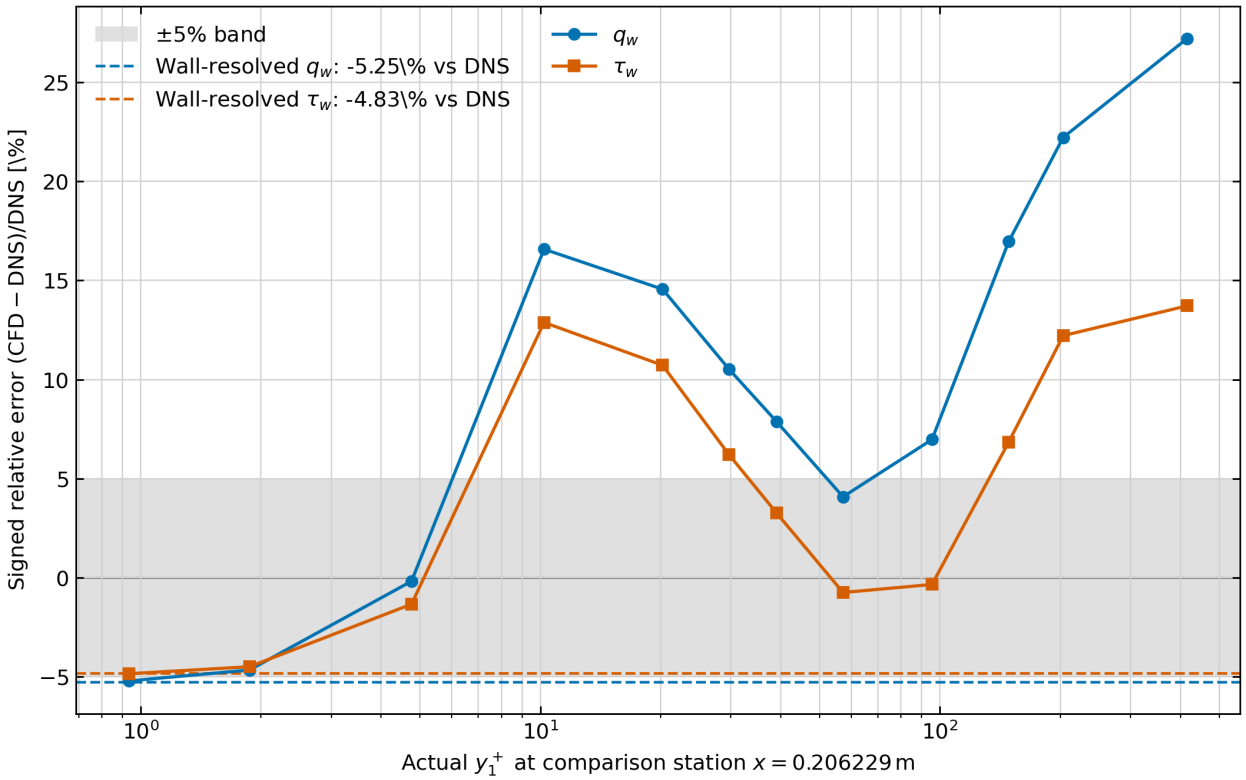



**Figure 7.17** Li wall-function CFD with SA on conventional meshes for M6Tw025: signed wall heat-flux and shear-stress errors relative to DNS at $x = 0.206229\,\mathrm{m}$ against the actual first-cell $y_1^+$. Dashed lines denote the wall-resolved SA errors; shading marks $\pm 5\%$ relative to DNS.

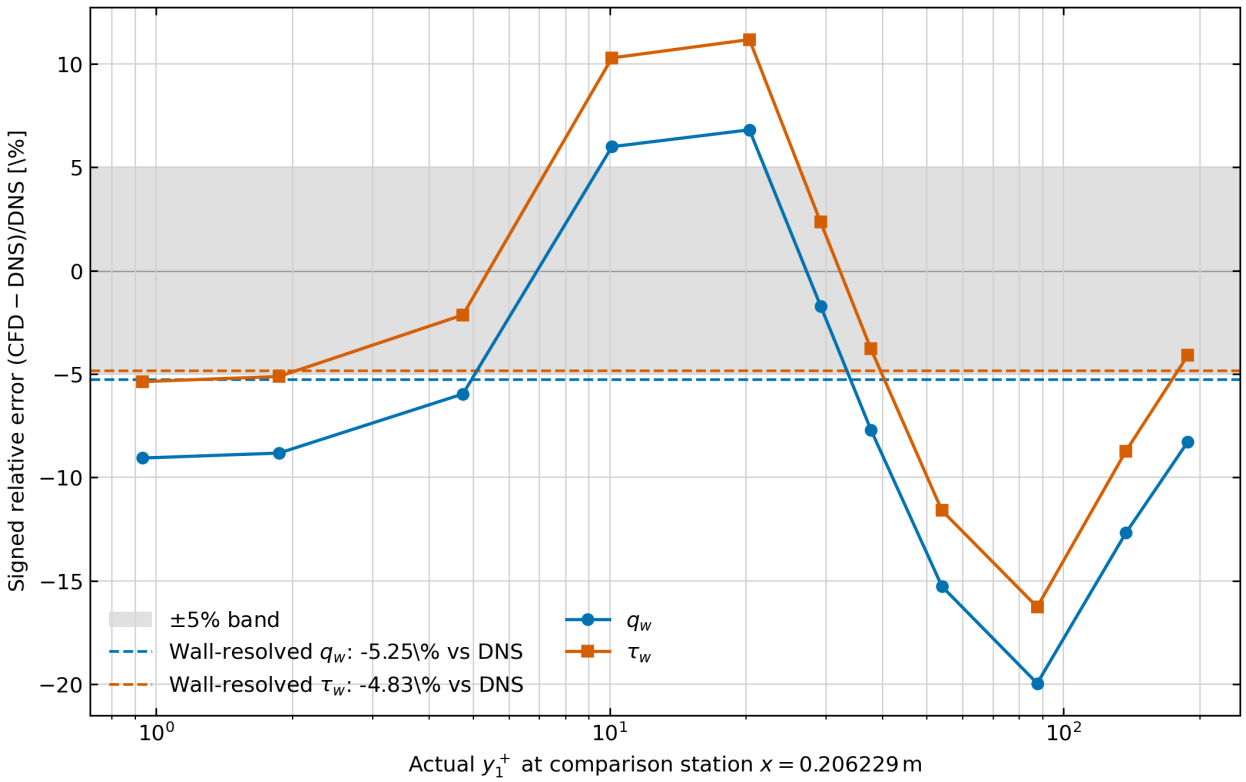



**Figure 7.18** Griffin wall-function CFD with SA on conventional meshes for M6Tw025, using the DNS normalization and station of Figure 7.17. Dashed lines denote the wall-resolved baseline; shading marks $\pm 5\%$ relative to DNS.

### 7.4.2 Internal-face errors and a wall-normal spacing criterion

For the FVM CFD, the numerical framework, is described in Section 3.3 and distinguishes convective reconstruction from viscous flux evaluation. The direct wall-flux delivery of the wall-functions, as described in Section 5.2.3, replaces the gradient-based evaluation at the wall, but gradients and transport coefficients at internal faces must still be approximated from the cell-centre values. For a predominantly streamwise boundary layer flow, neglecting $\partial V/\partial x$ relative to $\partial U/\partial y$ gives

$$\tau_{xy} \simeq \mu_{\mathrm{eff}} \frac{\partial U}{\partial y}, \qquad q_y = -\kappa_{\mathrm{eff}} \frac{\partial T}{\partial y}. \tag{7.6}$$

Here $\tau_{xy}$ is the effective shear stress and $q_y$ is the effective heat-flux in the positive wall-normal direction. At internal faces parallel to the wall, these fluxes contribute to the discrete momentum and energy balances. As Equation 7.6 shows, their evaluation relies both on wall-normal gradient approximations as well as interpolation of the effective viscosity $\mu_{\mathrm{eff}}$ and thermal conductivity $\kappa_{\mathrm{eff}}$ at the face location. Errors in either can therefore alter the flow field solution, which in turn feeds into the wall-function.

Figure 7.19 visualizes the geometry of these operations on an exemplary conventional coarse wall-function mesh with a first-cell height of $y_1^+ = 10$ on the left and $y_1^+ = 30$ on the right. For equal adjacent cell thicknesses, the first centre at $y_1^+ = 10$ and the next at $y_2^+ = 30$ surround the internal face at $y_f^+ = 20$. The full centre separation, which the gradient approximation and coefficient interpolation needs to gap, is thus $h^+ = 20$. Increasing the first-cell height to $y_1^+ = 30$ also increases the stencil separation to $y_f^+ = h^+ = 60$. Thus conventional coarsening changes both the wall-function sampling height and the distance over which internal-face transport is evaluated. The former already isolately evaluated in Section 7.3 and the latter is evaluated in this section.

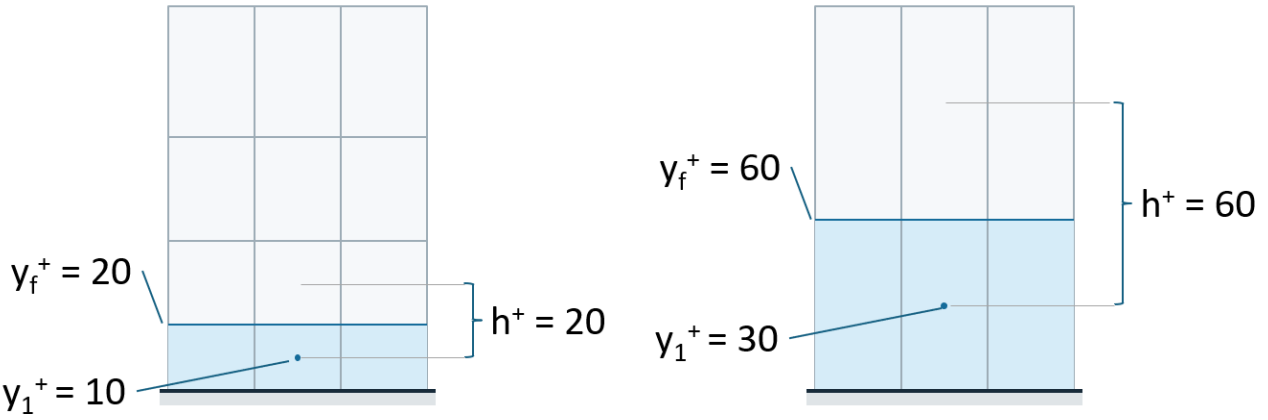



**Figure 7.19** Illustrative conventional near-wall meshes. For equal cell thicknesses adjacent to the first internal face, $y_1^+ = 10$ gives $y_f^+ = h^+ = 20$ (left), while $y_1^+ = 30$ gives $y_f^+ = h^+ = 60$ (right). Here $h^+$ is the full centre separation. The lower cell extends from the wall to the face, establishing an upper geometric link between the centre spacing $h^+$ and cell height of $h^+ = y_f^+$.

Figure 7.20 uses the M6Tw025 DNS temperature profile to illustrate a potential limitation of second-order accurate internal-face gradient reconstruction in a hypersonic cold-wall boundary layer. For the symmetric two-centre stencil considered here, the second-order accurate face-gradient approximation is the slope of a linear secant through the neighbouring cell-centre values. In this example, the secant connects the temperatures at $y^+ = 10$ and $30$, while the required gradient is the tangent slope at $y_f^+ = 20$. The pronounced cold-wall temperature peak causes substantial variation in curvature across this coarse stencil, and the two slopes differ visibly. Thus even a symmetric, second-order accurate approximation can poorly represent the local thermal gradient. This example illustrates the mechanism, while the spacing analysis below quantifies it over the profile. Here second-order accuracy refers to the leading error scaling with the square of the centre separation, not to a quadratic interpolating curve.

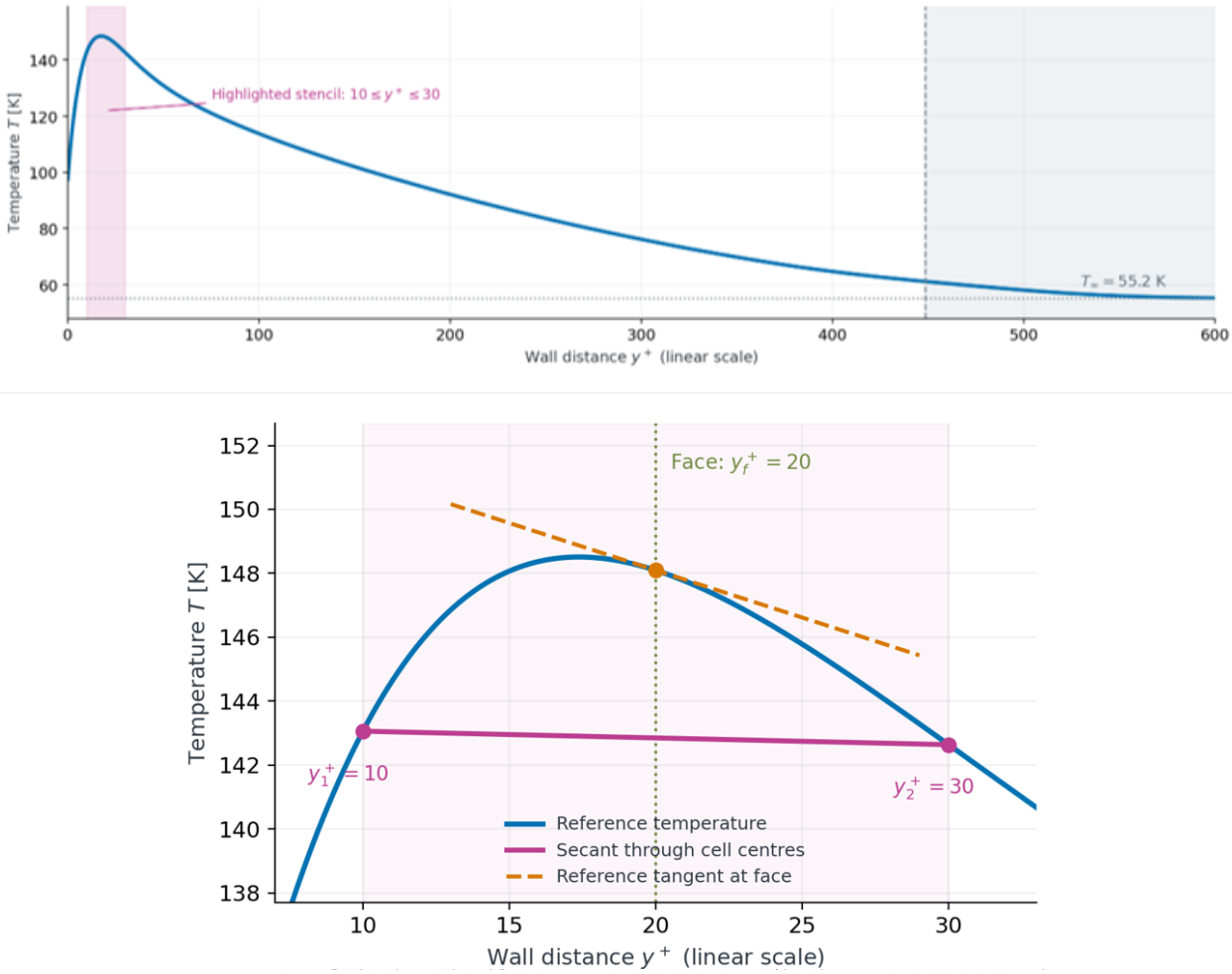



**Figure 7.20** M6Tw025 DNS temperature profile (top) and detail near its maximum (bottom). For the illustrative stencil in Figure 7.19, the purple secant between $y^+ = 10$ and $30$ differs from the local tangent at the internal face $y_f^+ = 20$ (orange dashed line). The slopes illustrate the gradient contribution to thermal-diffusion error; the corresponding face flux also depends on the interpolated effective conductivity.

The two-centre face-gradient approximation of a second-order accurate gradient reconstruction scheme is the slope of the linear secant through the neighbouring profile values:

$$D_f\phi = \frac{\phi(y_R) - \phi(y_L)}{y_R - y_L} = \frac{\phi(y_f + b) - \phi(y_f - a)}{a + b}, \qquad \phi \in \{T, U\}, \tag{7.7}$$

where $a = y_f - y_L$, $b = y_R - y_f$ and $h = a + b$. Unequal spacing introduces a truncation-error contribution $(b - a)\phi''(y_f)/2$. For the symmetric stencil, $a = b = h/2$, however this contribution vanishes and the secant slope approximates the face gradient with second-order accuracy. Taylor expansion expresses the secant slope as the exact tangent slope plus its error:

$$\frac{\phi(y_f + h/2) - \phi(y_f - h/2)}{h} = \phi'(y_f) + \frac{h^2}{24}\phi'''(y_f) + O(h^4), \qquad \phi \in \{T, U\}. \tag{7.8}$$

For this symmetric stencil, the leading gradient error therefore depends on the change in curvature, $\phi'''$, rather than curvature alone. Dividing its magnitude by $|\phi'(y_f)|$ and prescribing a relative accuracy of the face gradient $\epsilon$ gives the spacing estimation $h_{\text{grad}}$ for the second order accurate gradient reconstruction on a symmetric stencil as

$$h_{\text{grad}} \simeq \sqrt{24\epsilon \left| \frac{\phi'}{\phi'''} \right|}. \tag{7.9}$$

Similarly, for symmetric linear interpolation of a coefficient $\Gamma \in \{\mu_{\text{eff}}, \kappa_{\text{eff}}\}$, the face value is approximated by the mean of the two cell-centre values. Its leading error is $h^2\Gamma''(y_f)/8$, giving

$$h_{\text{interp}} \simeq \sqrt{8\epsilon \left|\frac{\Gamma}{\Gamma''}\right|}. \tag{7.10}$$

Equations (7.8)–(7.10) identify the profile derivatives controlling the two operations. The envelopes below evaluate the corresponding finite-stencil errors directly on cubic-interpolation reconstructions of the digital reference profiles. The resulting spacing limits are the maximum centre separation that satisfies a prescribed relative error $\epsilon$ for each operation.

Figure 7.21 shows exemplary the four profiles of the M6Tw025 case passed through this spacing analysis. DNS supplies M6Tw025 temperature and velocity, the matched wall-resolved SA solution supplies effective viscosity and conductivity. The effective transport properties need to be taken from the wall-resolved SA reference because not published with the DNS profiles. For the subsequent LowH_M3 mesh design, all profiles come from the wall-resolved SA reference, because of the lack of published high-quality digital DNS data. In the Figure 7.21 profiles it is immediatly apparant that the thermal profile present significantly more change in curvature than the velocity profile because of the sharp temperature peak near the wall. As identified in Equation (7.10), the transport coefficients rely on the curvature of the repsetive profiles, rather than the change in curvature, which for the M6Tw025 case has three pronounced regions of high curvature, directly at the wall because of the zero gradient Neumann boundary conditions there, at $y \approx 0.0025\text{m}$ and $y \approx 0.005\text{m}$ (see Figure 7.21 bottom plots).

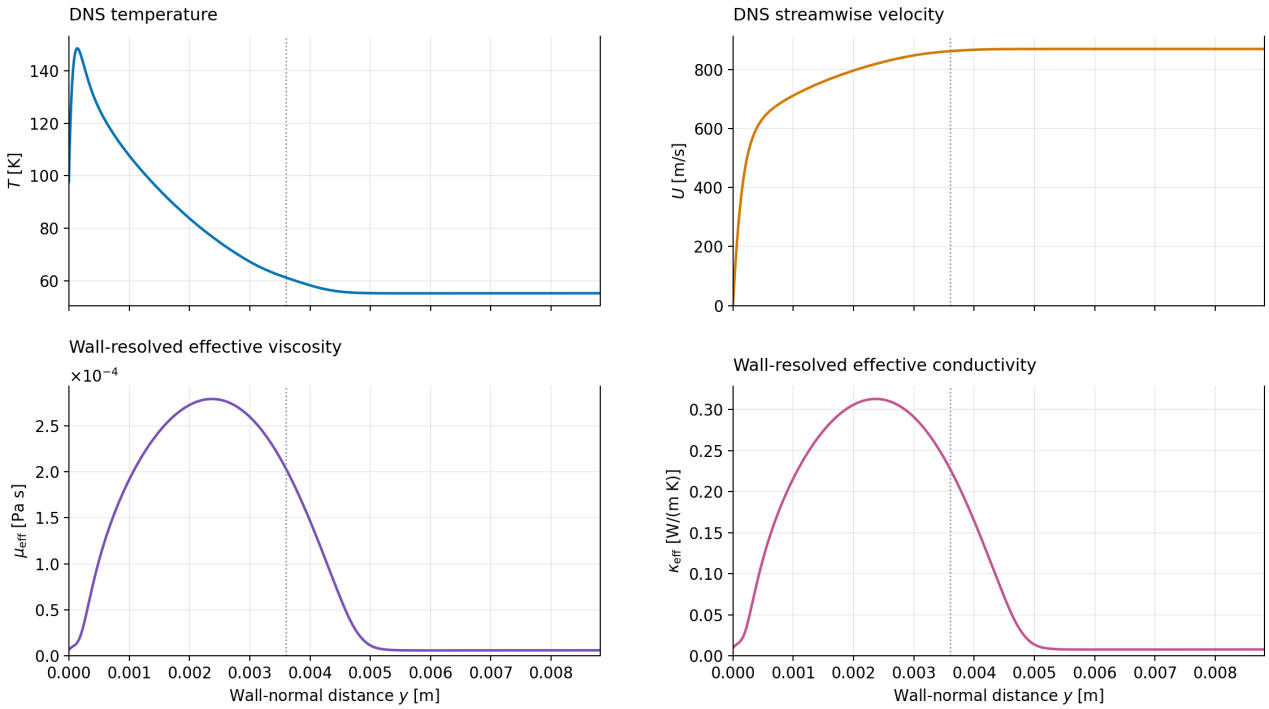



**Figure 7.21** Profiles underlying the M6Tw025 spacing assessment, plotted against physical wall-normal distance. DNS temperature and velocity are shown above; effective viscosity and thermal conductivity from the matched wall-resolved SA solution are shown below. The temperature maximum and the variation of effective transport motivate separate gradient and coefficient-interpolation criteria.

Using the found Equations (7.9) and (7.10) that link the cell-centre spacing to the respective gradient reconstruction and interpolation errors, the M6Tw025 profiles of Figure 7.21 are analysed. This results in the four individual spacing limits as shown in Figure 7.22. The resulting envelopes show, the allowable cell spacing $h$, for a given cell face position $y_f^+$ so that the relative errors for the gradient reconstruction or interpolation stay below a given tolerance of e.g. $\epsilon_{max} = 3\%$.

The cell spacing must also adhere to the upper geometric limit of $h^+ \leq y_f^+$. At this limit two equal hight cells around a given face have centres at $y_f^+/2$ and $3y_f^+/2$, thus resulting in the lower cell extending all the way from the face to the wall. Thus the full centre separation equals the face-to-wall distance. This is the conventional coarse wall-function mesh geometry illustrated above in Figure 7.19. A limit below this line requires a smaller centre separation at the same face position. a limit reaching it permits that conventional stencil under the selected local criterion. The cap is an adopted mesh constraint, not an error estimate.

Figure 7.22 shows the individual limits for $\epsilon = 0.03$. The temperature-gradient limit has a sharp minimum near the temperature maximum as expected and hypthesized above. The coefficient interpolation and velocity-gradient reconstruction somewhat restrict other parts of the inner layer. At the exact Temperature peak, $T' = 0$ thus the relative gradient error of Equations (7.9) appraoches a asymptote - for the M6Tw025 case this is at around $y_f^+ \approx 18$, in-line with the plotted temperature profile in Figure 7.20. The sharp minimum reflects this conditioning as good as the cubic-interpolation of the given profile allows.

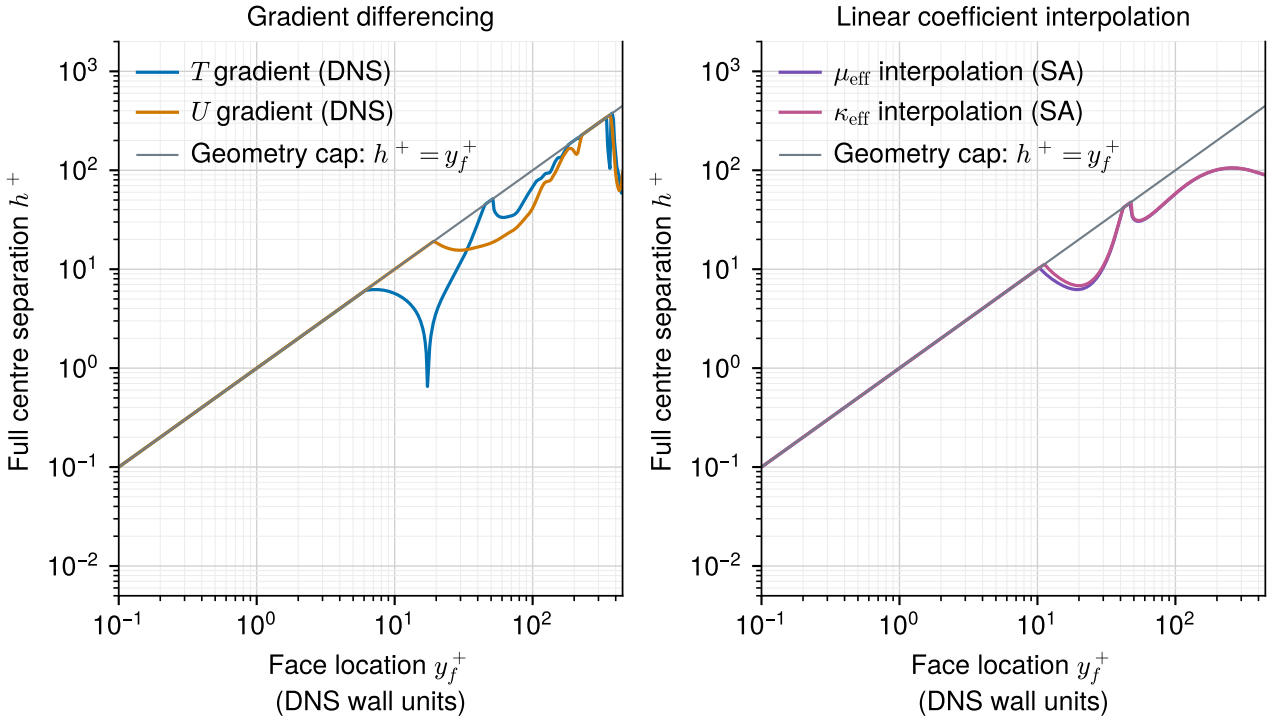



**Figure 7.22** M6Tw025 spacing limits at a $3\%$ local component-error tolerance. The allowable full centre separation $h^+$ is plotted against internal-face position $y_f^+$ for centred differencing of DNS temperature and velocity (left) and linear interpolation of wall-resolved SA effective transport (right). Each limit is the first tolerance crossing evaluated on cubic profile reconstructions, capped by the adopted geometric condition $h^+ \leq y_f^+$. Both coordinates use DNS wall units.

To satisfy all four component criteria of Figure 7.22 simultaneously, the allowable spacing is their minimum at each face position, further limited by profile support and the geometric cap. Figure 7.23 is the resulting $3\%$ envelope. It converts the separate operator restrictions into a single wall-normal spacing requirement. Near the cold-wall temperature maximum, its large separation from $h^+ = y_f^+$ quantifies why the conventional coarse stencil is inadequate and how much smaller the centre separation must be to avoid excessive errors. It must be emphasized that this is a local operator criterion, not a $3\%$ error bound on the wall heat-flux or shear stress. Additionally these four component errors can add constructively or destructively, so the combined envelope is not a direct error estimate, but it limits their individual contributions.

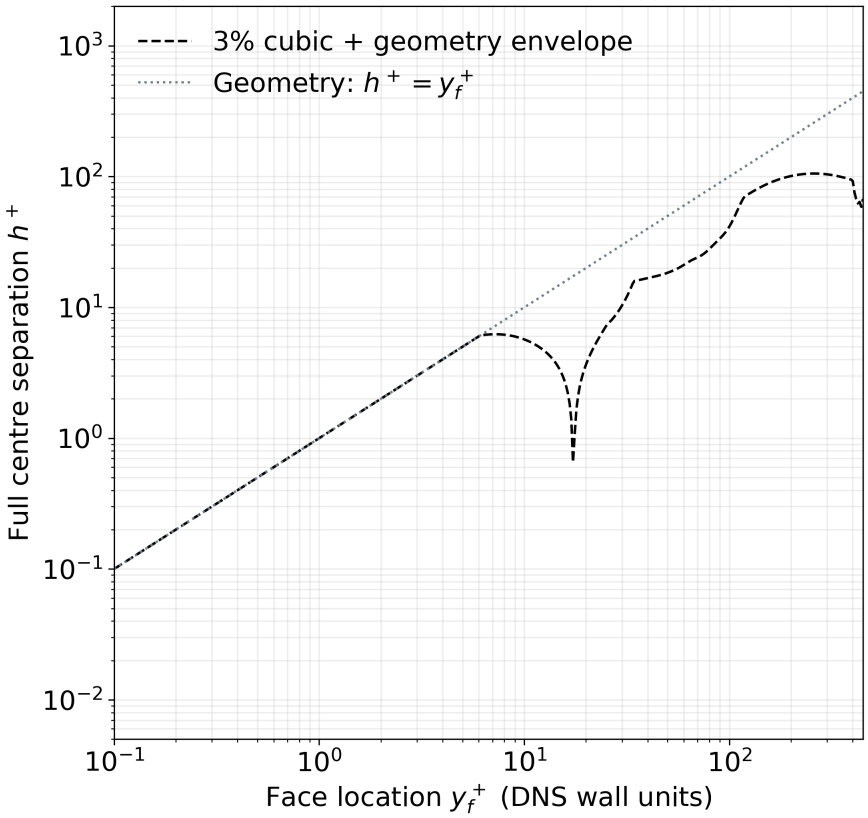



**Figure 7.23** Combined M6Tw025 spacing envelope for a $3\%$ component-error tolerance: the minimum of the four cubic-reference limits in Figure 7.22, profile support and the geometric cap $h^+ = y_f^+$. The restrictive region near the temperature maximum requires substantially smaller centre separation than a conventional coarse stencil.

Repeating the same analysis with $\epsilon = 0.10$ gives Figure 7.24. The restrictive minimum persists in the buffer region, whereas substantially larger spacings are admitted farther into the logarithmic region. Around $y_f^+ \approx 150$, the envelope reaches the geometric cap. Meaning a conventional symmetric coarse wall-function mesh stencil can locally satisfy the $10\%$ component tolerance and supports the finding discussed in Section 7.4.1, that these log-layer regions are the ones used in the papers of Li et al. and Griffin et al. for validation purposes.

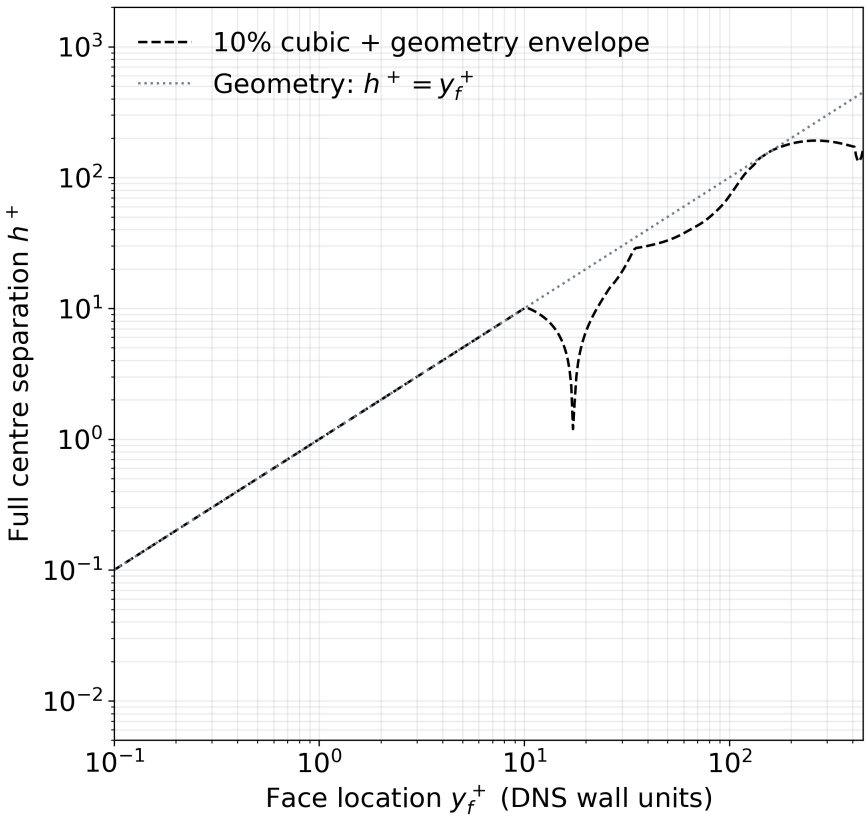



**Figure 7.24** Combined M6Tw025 spacing envelope with the local component-error tolerance relaxed to $10\%$. The restrictive minimum near the temperature maximum persists, while larger spacings are permitted farther outward. Around $y_f^+ \approx 150$, the envelope reaches the geometric cap and locally admits the conventional symmetric first-face stencil of Figure 7.19. This is a local operator criterion, rather than a $10\%$ bound on CFD wall quantities.

The physical reason on why this meshing restriction only arises now in hypersonic cold-wall boundary layers, and not in the subsonic or adiabatic-wall hypersonic cases can be seen in the comparative Figures 7.25 and 7.26. There the M6Tw025 case is compared to the near-adiabatic M6Tw076 case and a transonic cold-wall case *C09* of [6]. Neither of the latter two cases show a pronounced temperature peak, and therefore the curvature variation is much less.

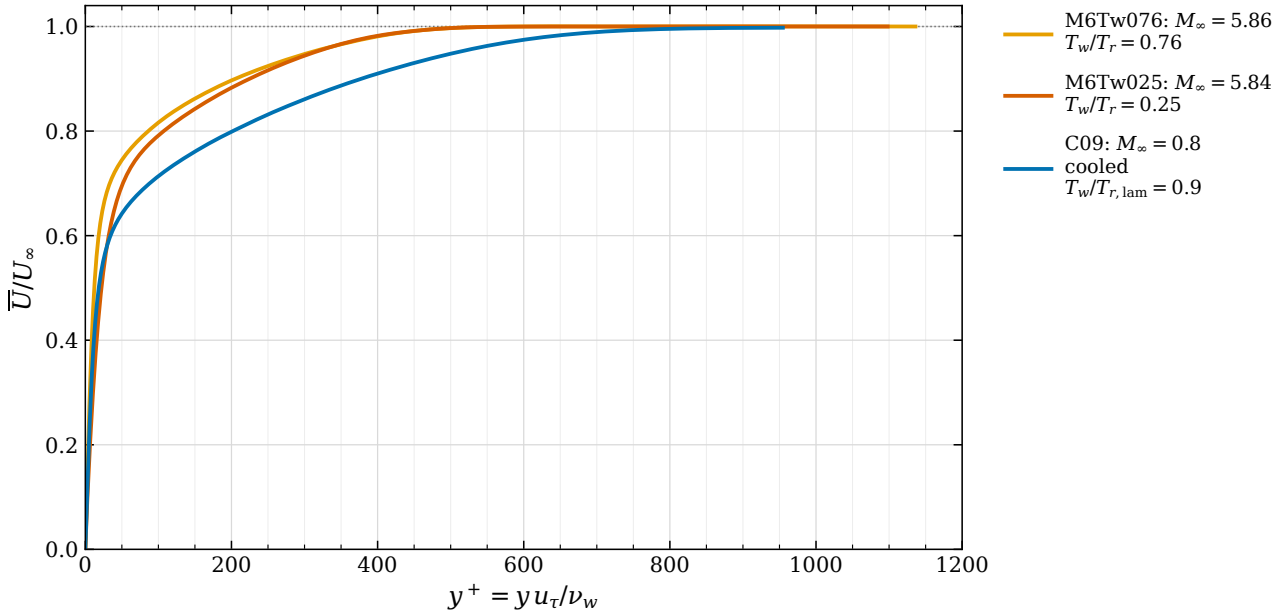



**Figure 7.25** Freestream-normalized DNS velocity profiles for M6Tw076, M6Tw025 and C09, using the same data and normalization as Figure 2.1, now plotted against linear $y^+$. The native wall points at $y^+ = 0$ are included.

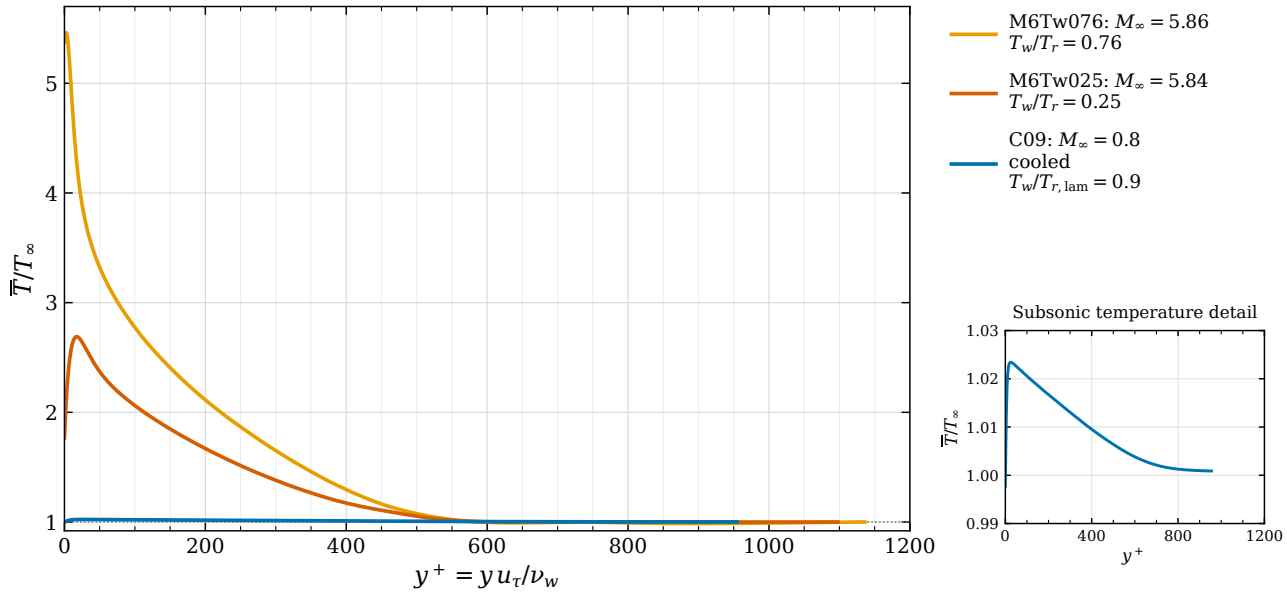



**Figure 7.26** DNS temperature profiles from Figure 2.2, shown on a linear $y^+$ axis to better visualize the pronounced change in curvature associated with the hypersonic cold-wall M6Tw025 temperature peak. Since $y^+$ is proportional to physical $y$ within each case, this view connects the thermal-profile structure to finite stencil spacing and the reconstruction error illustrated by the secant–tangent comparison in Figure 7.20. The C09 inset uses a separate vertical scale.

All in all, the above analysis shows that the conventional coarse wall-function mesh stencil, paired with second-order accurate schemes, is not sufficient to resolve the internal-face transport in hypersonic cold-wall boundary layers. The pronounced temperature peak causes a large change in curvature, which in turn requires a much smaller centre separation than the conventional coarse wall-function mesh stencil provides. This novel mesh requirement is thus only dependant on the boundary layer profile and numerical schemes, in turn this requirement also applies to wall-resolved runs. In other words, the wall-functions can be understood as just a different heat-flux and shear stress relation than the convential linear correlation

between the first off-wall cell cetner to the wall, as described for wall-resolved RANS in Equation (3.2). But with the very low first-cell requirement of these wall-resolved runs on hypersonic cold-walls (i.e. the first-cell acoustic Reynolds number must be below 1 as discussed in Section 6.2), only very extreme growth rates of the boundary layer mesh may run into any appreciable internal-face transport error, thus it remained undiscussed so far.

The following section will show how proof this hypothesis by isolating these discretization errors through constructing boundary layer-informed meshes that satisfy the above discussed $3\%$ criterion.

### 7.4.3 Wall functions on boundary layer-informed meshes

To test the above discussed error due to spatial discretization in CFD, families of boundary layer-informed meshes are constructed for prescribed wall-function sampling positions. This allows the wall-function sample to be swept over the whole $y^+$ range similar to the wall-closures a-priori analysis in Section 7.3.1 and conventional mesh CFD runs in Figures 7.15–7.18, while the internal-face spacing is controlled by the above discussed $3\%$ criterion. This arrangement reduces the coupling between moving the wall-function sample and the boundary layer spatial discretization.

Figures 7.27 and 7.28 show the resulting families for the M6Tw025 and LowH_M3 case respectivly. The dotted and dashed lines are taken and constructed as described in the boundary layer analysis Section 7.4.2 above - specifically for the M6Two025 case, the Figure 7.28 dashed and dotted lines are the same as in the Figure 7.23. For the construction of the mesh families (see Figures 7.27 and 7.28 right plots), a desired wall-function sampling height is prescribed and the internal-face spacing around it is then grown up until the allowable discretization error of e.g. $3\%$ against the known profile is reached. This is continued until the whole boundary layer is covered. The resulting meshes are not a bound on the total CFD error or a proof of a globally optimal meshing, but aim to provide an isolation from this error source. The generated meshes are subsequently re-checked against the known boundary layer profiles, using their actual, generally unsymmetric, centre-to-face distances to ensure the desired spacing criterion is really satisfied.

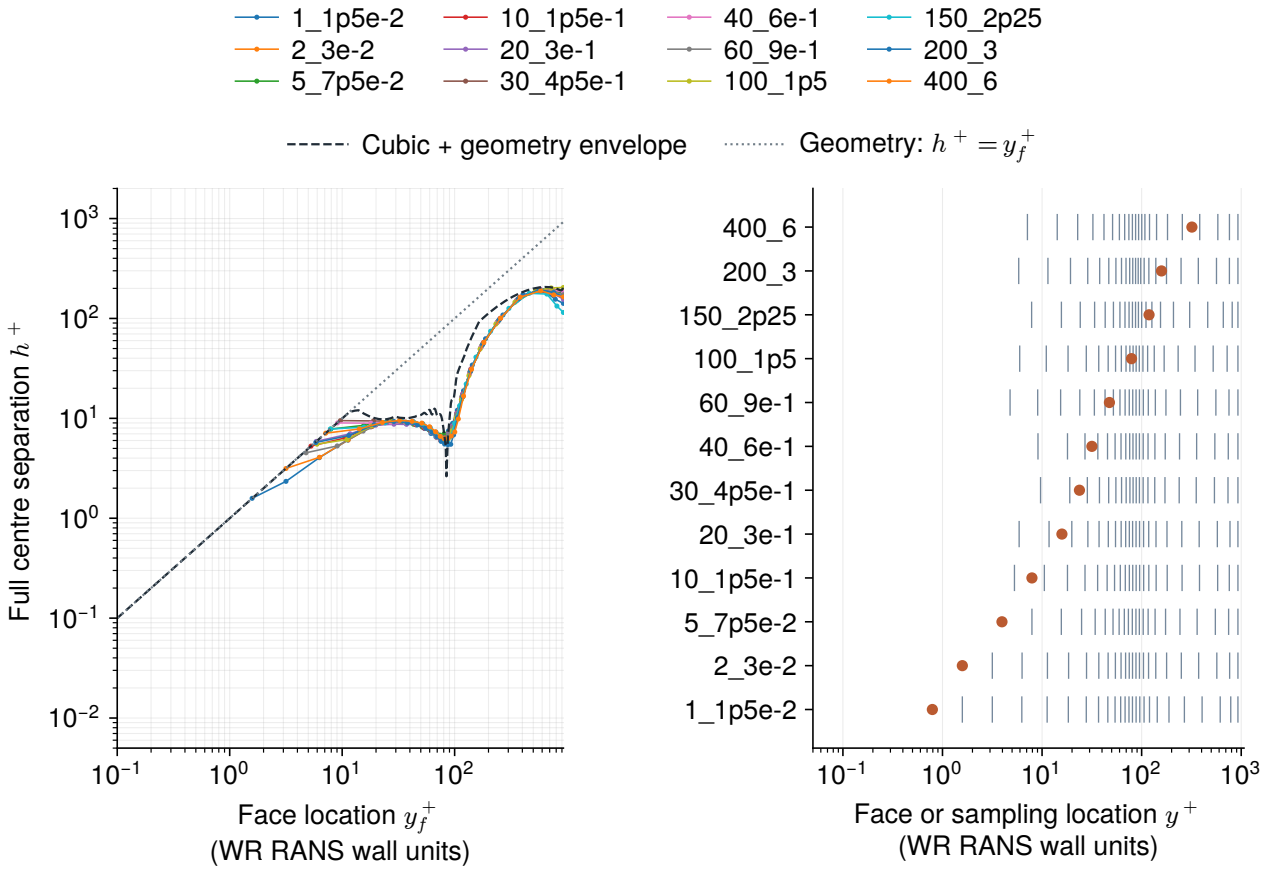



**Figure 7.27** Boundary layer-informed $3\%$ spatial discretization error mesh family for LowH_M3. The naming of the mesh families follows the structure *desired wall-function sampling point in* $y^+$*_desired wall-function sampling point in millimetres*. Left: Black dotted and dashed lines represent the envelope stemming from the boundary layer analysis as described in Section 7.4.2. The colourfull lines show the new boundary layer-informed meshes' actual spatial discretization error against the known boundary layer profiles, with all of them consistently staying below the $3\%$ envelope. Right: The actual mesh design (blue lines represent the position of the cell faces) and wall-function sampling centres (orange) are shown, with the pronounced clustering around the $y^+ \approx 85$ where the temperature peak sits for this case.

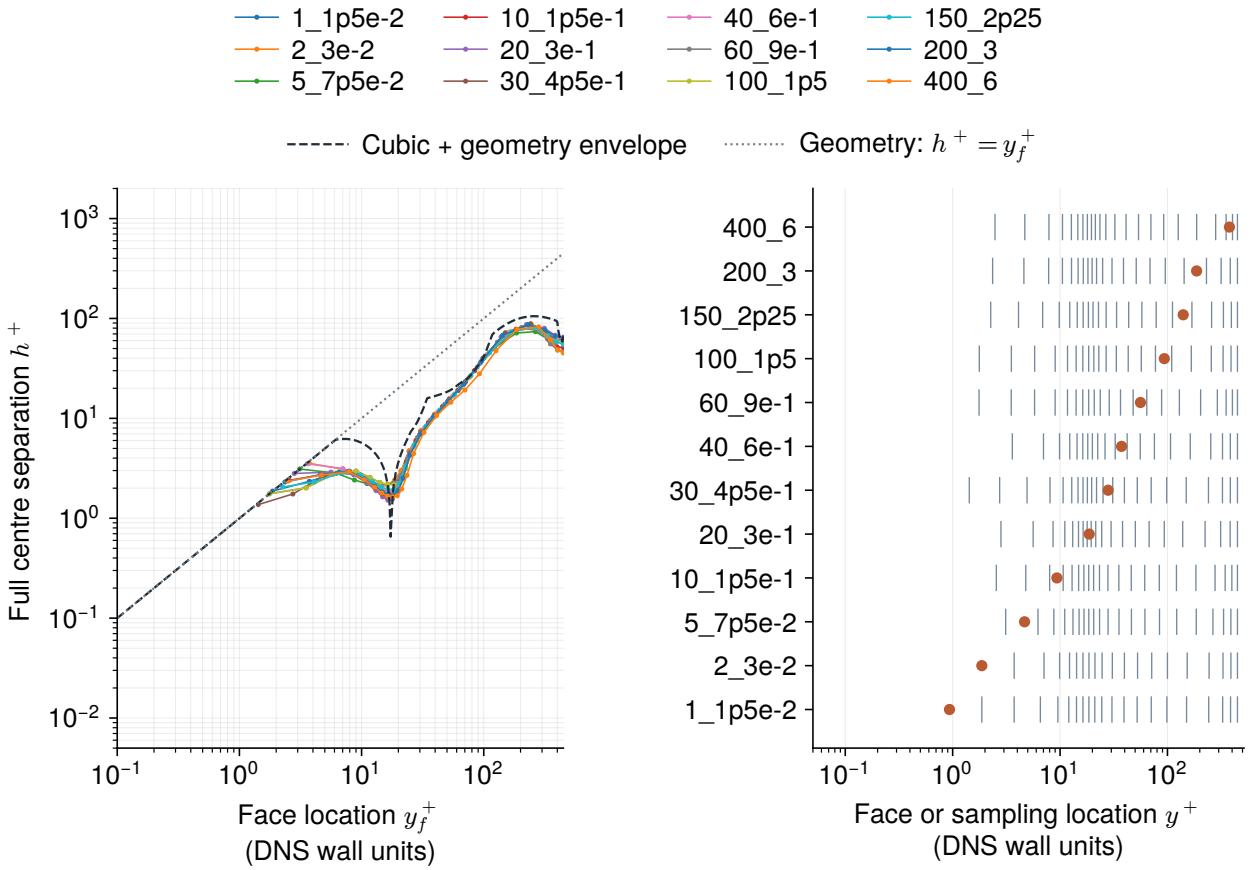



**Figure 7.28** Boundary layer-informed $3\%$ spatial discretization error mesh family for M6Tw025. The naming of the mesh families follows the structure *desired wall-function sampling point in $y^+$_desired wall-function sampling point in millimetres*. Left: Black dotted and dashed lines represent the envelope stemming from the boundary layer analysis as described in Section 7.4.2. The colourfull lines show the new boundary layer-informed meshes' actual spatial discretization error against the known boundary layer profiles, with all of them consistently staying below the $3\%$ envelope. Right: The actual mesh design (blue lines represent the position of the cell faces) and wall-function sampling centres (orange) are shown, with the pronounced clustering around the $y^+ \approx 18$ where the temperature peak sits for this case.

Figure 7.29 presents the CFD sweeps on these meshes. The abscissa is now the wall-function's sample location $y_s^+$, rather than necessarily the physical first-cell $y_1^+$. The reader is encouraged to look at the a-priori analysis, conventional coarase wall-function mesh CFD and herein presented boundary layer-informed mesh CFD results side by side for the four investigated combionations of Li wall function and Griffin wall function on the LowH_M3 and M6Tw025 case (e.g. for the Li wall function on the LowH_M3 case, see Figures 7.10, 7.15 and 7.29 top).

It is immediatly apparant that the boundary layer-informed meshes follow the general error trend of the a-priori found closure error, significantly better than the conventional coarse wall-function meshes. The pronounced overshoot in the buffer layer that the conventional mesh exhibits in Figure 7.15 is removed and the boundary layer-informed meshes in Figure 7.29 (top) show significantly closer trend to the a-priori results of Figure 7.10. Also indicative is, that Figure 7.15 from $y^+ \approx 2$ onwards deviated from the wall-resolved baseline, whereas the boundary layer-informed meshes in Figure 7.29 (top) follow the a-priori in Figure 7.10, of effectly zero error, up to the same $y^+ \approx 10$.

Because of error addition and cancellation, the wall-function CFD results on the boundary layer-informed meshes do not exactly match the quantiataive errors of the a-priori analysis. Arguably an even finer boundary layer-informed mesh with $\epsilon_max \leq 0.1\%$ could be designed to further isolate the discretization errors, this highlights how big of an impact even these small discretization errors can have. For comparing

the M6Tw025 cases, one should also keep in mind, that an additonal uncertainty arises in the discovered fact, that the turbulence model is not capable to reproduce the DNS temperature peak, as discussed in Section 7.2.2 and investigated implications in Section 7.3.3.

The highest sampling positions require particular caution. The LowH_M3 Li curve rises markedly at its outermost sample, where inner-layer and approximately constant-stress assumptions become less suitable as discussed in the a-prori analysis in Section 7.4.4. This point does not establish a general acceptable upper $y_s^+$, since the relevant boundary layer region and the closure's usable range are case dependent.

From the analysis there is a strong conclusion from the CFD results of the herin used boundary layer-informed meshes vs. the previous conventional meshes, that the large error in the buffer layer and lower log-layer is driven by the spatial discretization error discussed as hypthesized.

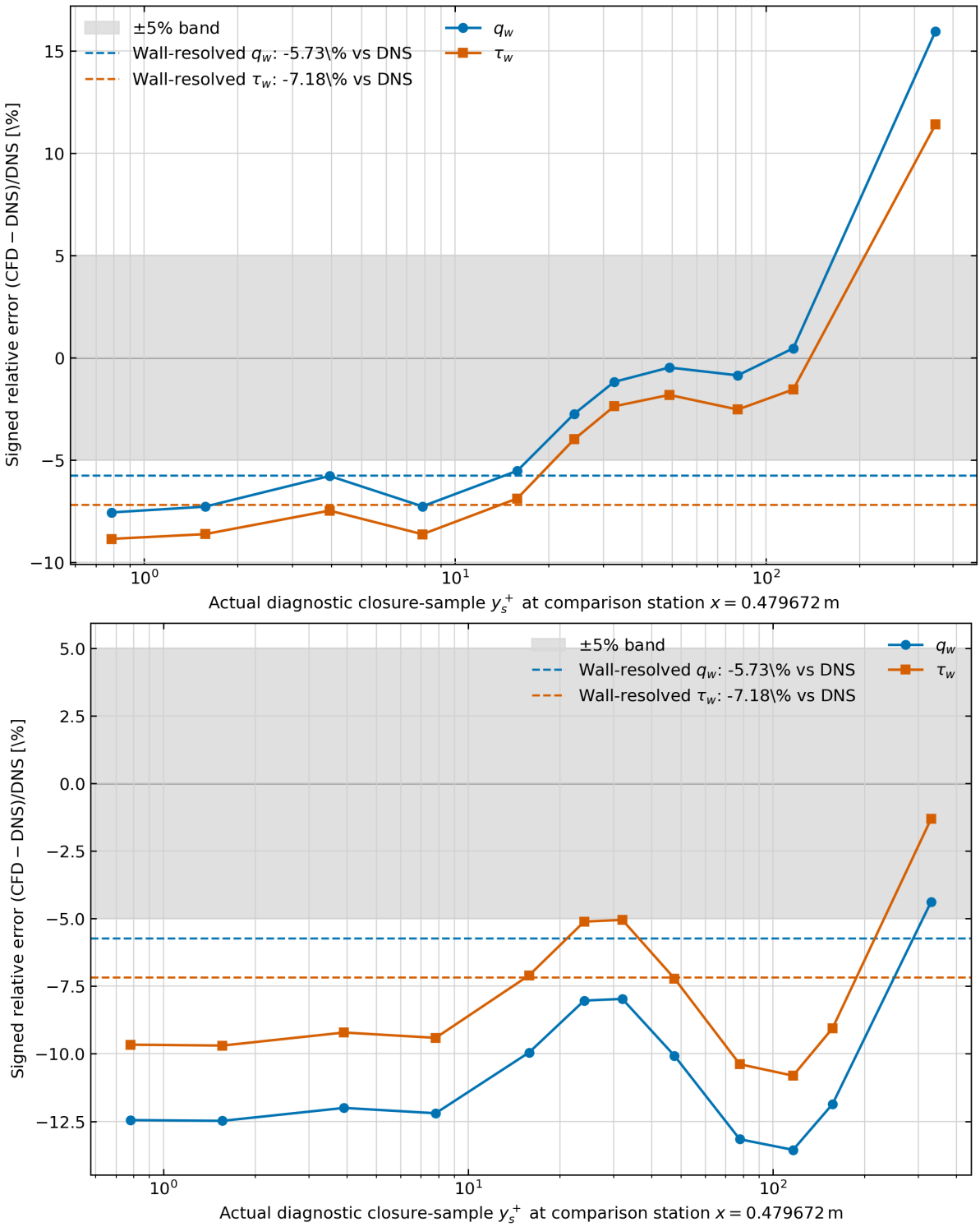



**Figure 7.29** Wall-function CFD on boundary layer-informed meshes designed with the $3\%$ cubic-reference component criterion. LowH_M3 results are shown for Li (top) and Griffin (bottom). The abscissa is the actual diagnostic sampling location $y_s^+$ for the wall-function, not the first cell height. Dashed lines denote the wall-resolved baseline; shading marks $\pm 5\%$ relative to DNS.

q

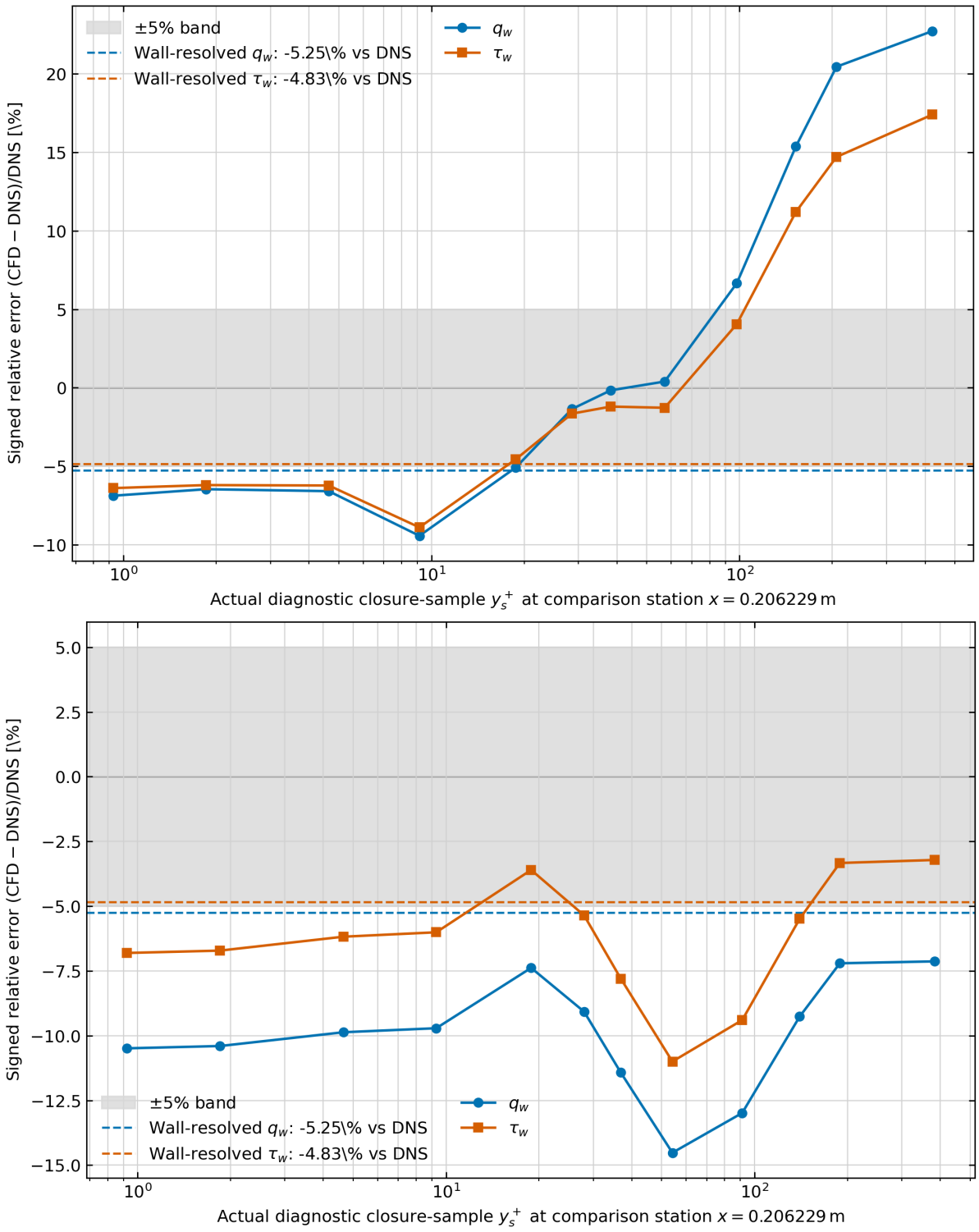



**Figure 7.29** M6Tw025 results for Li (top) and Griffin (bottom), with SA and DNS normalization at $x = 0.206229\,\mathrm{m}$. The mesh criterion and sampling-height definition are the same as in the preceding panels.

### 7.4.4 Conditional a-priori estimation of the mesh requirement

The above analysis and the novel proposed boundary layer informed meshing criterion, has one practical shortcoming: it requires knowledge of the resolved boundary layer profile. For an engineering approach, it is desirable to have a mesh design tool that does not require the resolved profile, but can still provide a-priori guidance on the required mesh spacing. The closures of the analytical Li et al. wall-function [3]

have this knowledge about hypersonic boundary layer backed in by design. Given wall shear and heat-flux, wall thermodynamic and transport properties, and an edge state, the wall-function closures can reconstruct the underlying profiles from which then the same finite-stencil limits can be evaluated. In the constant-$c_p$ form used here, the enthalpy relation supplies temperature from velocity, and the velocity law supplies the wall-normal coordinate. Under the additional constant-total-shear assumption, effective viscosity follows from $\mu_{\text{eff}} = \tau_w/(\partial U/\partial y)$. Prescribed molecular and turbulent Prandtl numbers then give an effective conductivity estimate.

Figure 7.30 compares this estimate with the M6Tw025 profile-based envelope at a $10\%$ error tolerance. The reconstruction uses scalar DNS metadata for the wall loads and edge/wall states, without using the resolved profile values. It is therefore an assessment conditional on known loads, rather than an independent prediction of those loads. The broad inner-layer spacing trend is recovered, but the most restrictive minimum is shifted because the Li reconstruction places the temperature maximum closer to the wall. Near that maximum, even a modest position error can produce a large difference in the relative-gradient criterion. Differences toward the boundary layer edge also reflect the limited outer-layer representation of the reconstructed inner law.

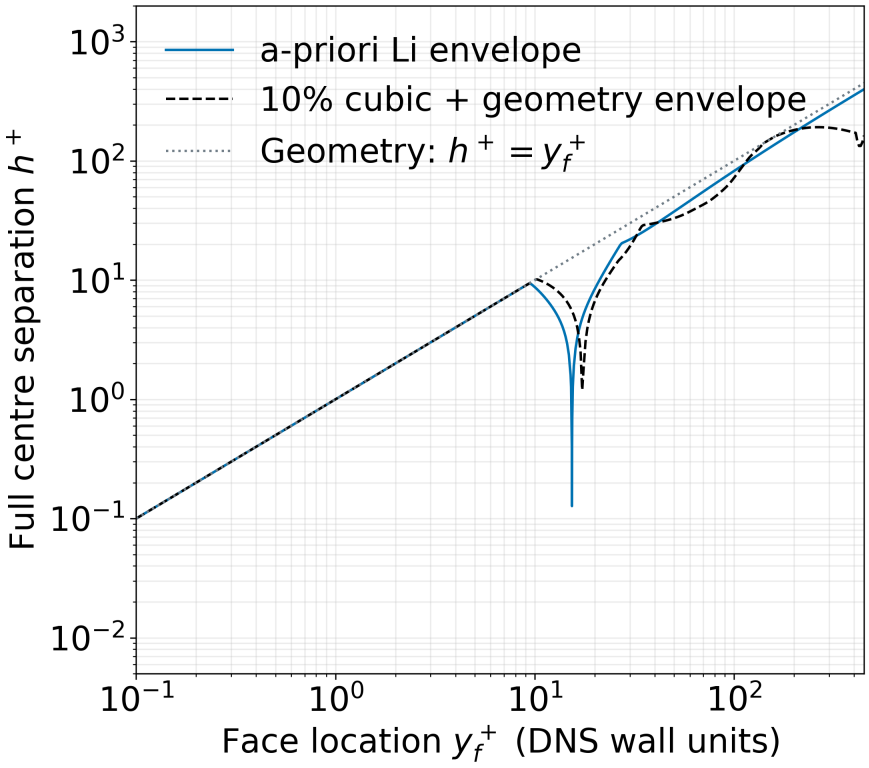



**Figure 7.30** Conditional a-priori estimate of the M6Tw025 $10\%$ spacing envelope from the Li relations (blue), compared with the profile-based cubic-reference envelope (black dashed). The reconstruction uses scalar wall loads, wall/edge states and transport properties, without the resolved boundary layer profile. The geometric cap is dotted. The displaced restrictive minimum near the temperature maximum limits the estimate's use as an accuracy guarantee.

The estimate can consequently guide an initial mesh design or identify a likely region of refinement. Practical use also requires suitable estimates of wall loads and edge quantities. For the present zero-pressure-gradient flat-plates these inputs are comparatively well defined. The boundary layer edge states can be approximated to the freestream conditions, and for the wall heat-flux and shear stress on zero-pressure-gradient flat-plates analytical approximations like the theory of van Driest are available. Their approximation and the validity of the equilibrium assumptions are however more challenging for complex flows.

### 7.4.5 Recovery of the wall-resolved baseline with fitted closures

The remaining question, to close the error isolation loop, is whether the wall-function formulation can recover the wall-resolved solution when both its closure relations and spatial discretization errors are removed. LowH_M3 provides a suitable diagnostic case because its wall-resolved SA thermal-profile shape agrees comparatively well with DNS, thus also eliviating the implications of the turbulence model. The Li formulation is retained, but its velocity and enthalpy relations are fitted to the wall-resolved SA field of the LowH_M3 case.

Lookup tables for the wall function's momentum and enthalpy closures are constructed from the wall-resolved SA boundary layer profiles, using the reference wall shear stress, wall heat-flux and local boundary layer edge velocity and temperature. The tables store the fitted velocity relation and a correction to the Li enthalpy relation. During CFD, the fitted closure relations determine wall shear stress and wall heat-flux from the wall-function sample station's velocity and temperature, while the fitted velocity-law derivative supplies the near-wall constraint on the SA variable $\tilde{\nu}$ (Section 5.3.2). The construction therefore requires reference profiles at multiple streamwise stations and the singular published DNS profiles are insufficient.

Figure 7.31 first checks the fitted closure relations using the same a-priori approach as in Section 7.3.1. The recovered heat-flux and shear stress differ from the wall-resolved targets by order $10^{-7}\%$. This near-exact recovery verifies the reference-conditioned inverse construction at the sampled states.

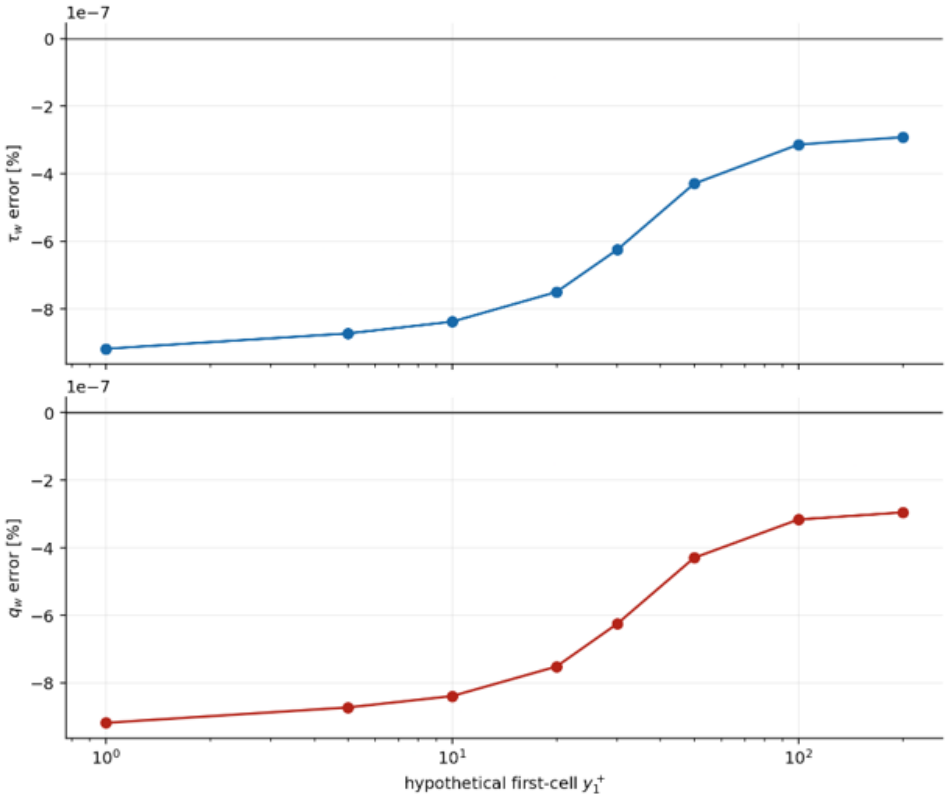



**Figure 7.31** A-priori inverse recovery using the fitted LowH_M3 Li relations and samples from the wall-resolved SA reference. Recovered shear stress (top) and heat-flux (bottom) are compared with that same reference. Errors are of order $10^{-7}\%$, as indicated by the axis multipliers; this is consistency of the reference-conditioned inverse model, rather than independent DNS validation.

The corresponding CFD result is shown in Figure 7.32. With the fitted relations and boundary layer-informed meshes, heat-flux and shear stress remain within approximately $\pm 2\%$ of the wall-resolved values over the completed sampling sweep. Thus successfully isolating the errors caused by spatial discretization and wall-function closure theories. The last persistent error to the DNS reference, is the baseline error of the underlying wall-resolved RANS as discovered in the first Section 7.2.1. So it can be concluded, that with a

boundary layer-informed mesh, improved closure relations and improved RANS turbulence modelling, the wall-function CFD could recover perfect wall heat-flux and shear stress.

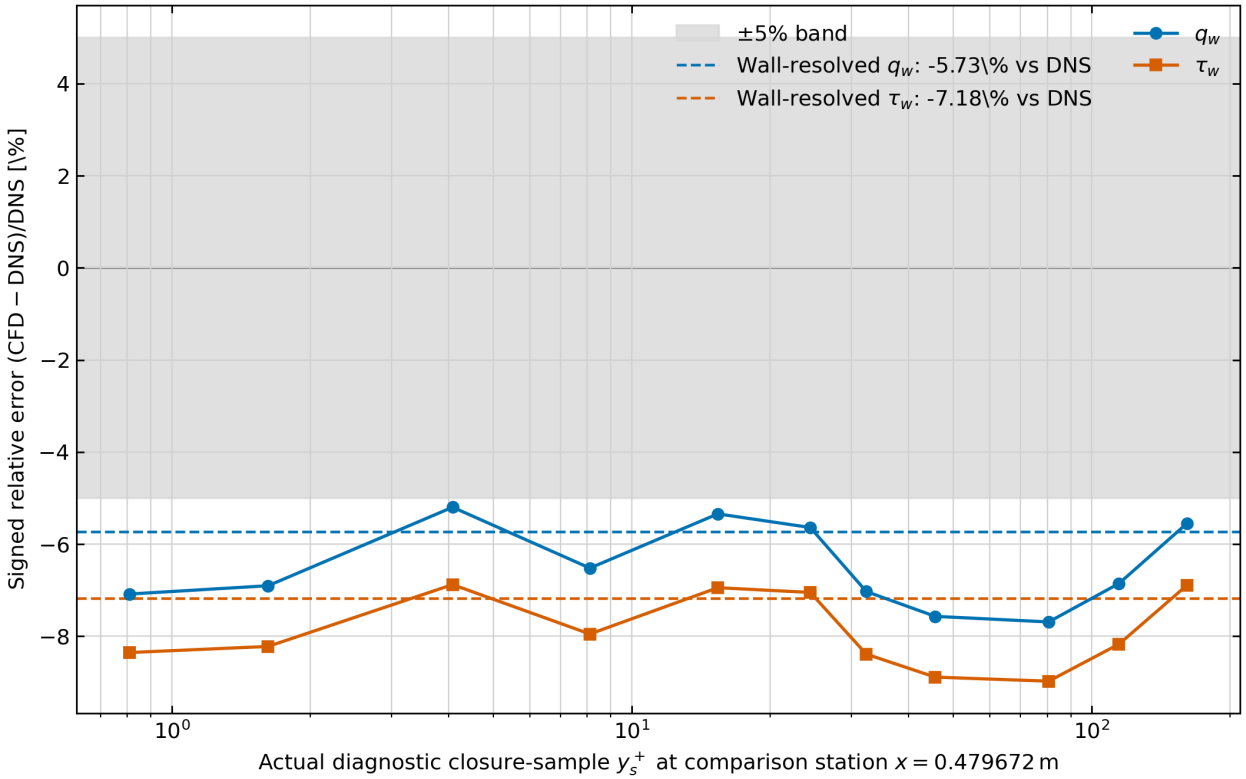



**Figure 7.32** LowH_M3 CFD with fitted Li relations, SA and boundary layer-informed $3\%$ meshes. Signed errors in heat-flux and shear stress relative to DNS are plotted against the actual sampling location at $x = 0.479672\,\mathrm{m}$. The curves remain close to the wall-resolved baseline (dashed), with deviations of approximately $\pm 2\%$ when normalized by the corresponding wall-resolved quantities. Shading marks $\pm 5\%$ relative to DNS.

# 8 Conclusion and Outlook

Through a successive isolation of errors from the turbulence model (ref. Section 7.2), the underlying wall-function closure theory (ref. Section 7.3) and the spatial discretization (ref. Section 7.4), novel findings and strong assessments to the accuracy of the aerothermal prediction of RANS with and without wall-functions was formed.

Figure 8.1 shows the isolation of errors, from conventional coarse wall function mesh (top), to novel boundary layer informed mesh (middle) to additional posterior fitting of the closure (bottom), which then all in all results in the wall function CFD to be within the wall resolved result to within +-2% - i.e. all the errors have been fully isolated and investigated (within error cancellation still being a thing also).

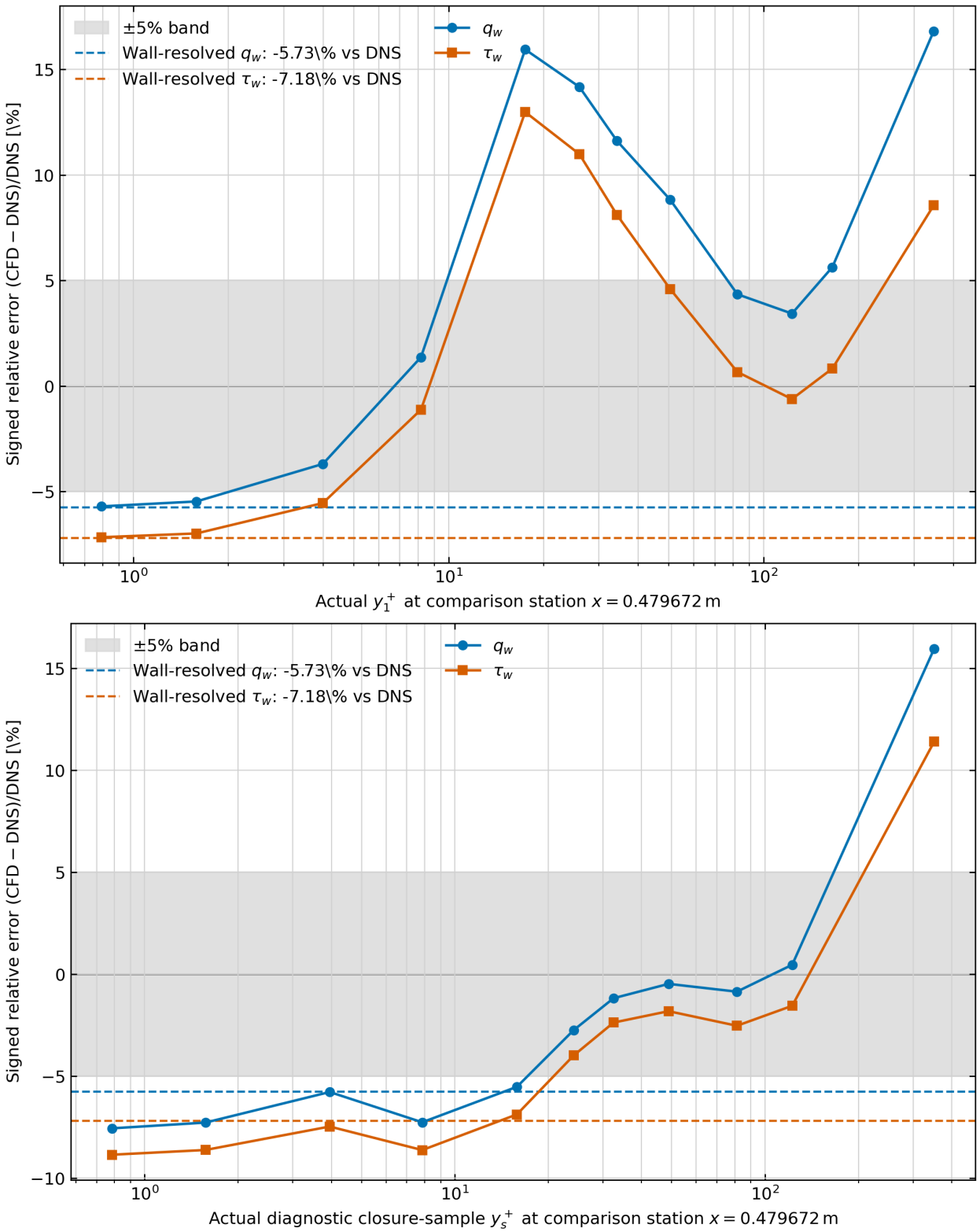



**Figure 8.1** Error-isolation sequence for LowH_M3 with Li wall functions and SA: conventional meshes (top), boundary layer-informed meshes (bottom), and fitted closures on informed meshes (continued). Signed wall heat-flux and shear-stress errors are relative to DNS; dashed lines show the wall-resolved SA baseline and shading marks the $\pm 5\%$ DNS band. The abscissa is the first-cell location $y_1^+$ on conventional meshes and the sampling location $y_s^+$ on informed meshes.

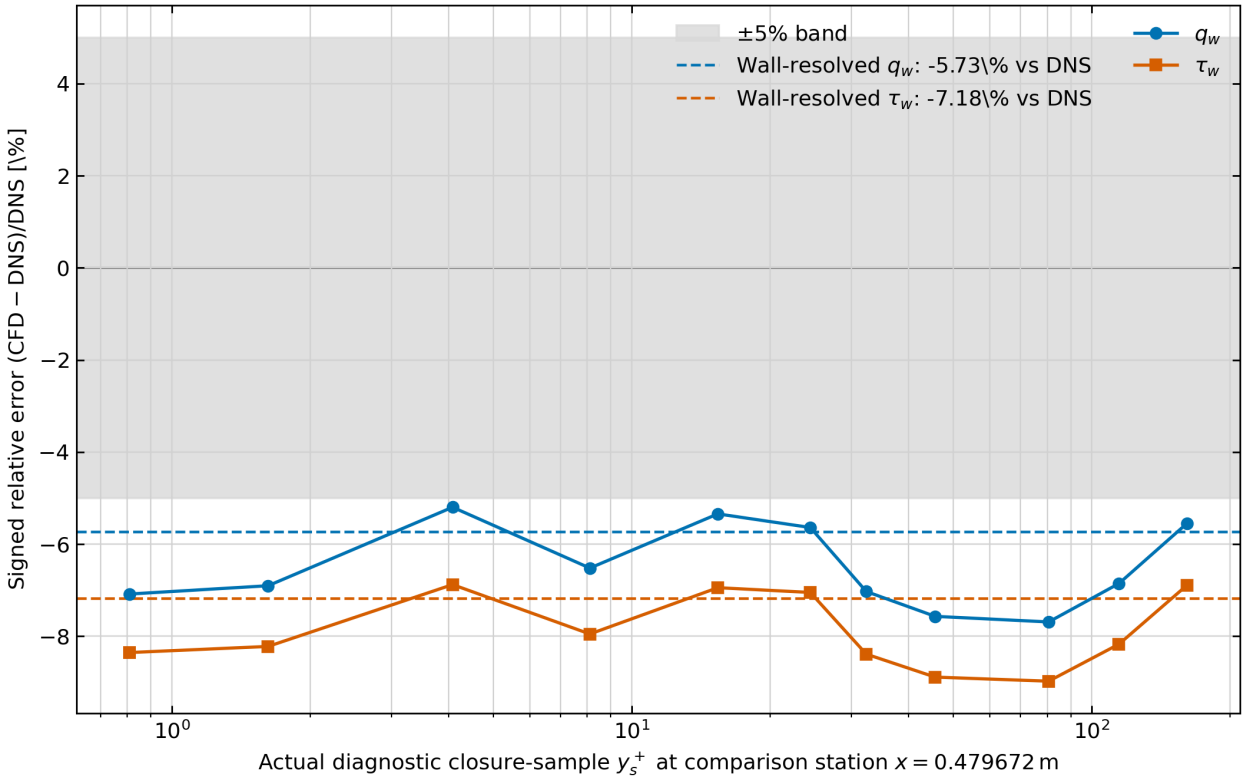



**Figure 8.1** (continued) Closures fitted to the wall-resolved SA reference recover its wall heat-flux and shear stress to within approximately $\pm 2\%$ on the informed meshes. The baseline discrepancy from DNS remains.

## 8.1 Summary

The main contribution of this work is a novel boundary layer-informed mesh criterion, which is necessary for wall-function simulations of hypersonic cold-wall flat plates. It relates cell spacing to the local profiles, controlling internal-face gradient and transport-coefficient interpolation errors (Sections 7.4.2 and 7.4.3). This criterion also provides the basis for a proposed a-priori mesh-design approach, conditional on estimates of wall loads and wall/edge states (Section 7.4.4).

A-priori assessment, independently of coupled CFD, of the Li and Griffin wall-function closures was performed. Across the investigated adiabatic and cold-wall flat plates, relative shear-stress and heat-flux errors are of order $\pm 10\%$ over the assessed inner-layer sampling range, with some larger deviations in some cases (Section 7.3.2). This demonstrates promising closure accuracy, although reliance on reconstructed boundary layer-edge quantities introduces an additional sensitivity in CFD (Section 5.4.2).

The wall-resolved assessment of the Spalart–Allmaras and $k$–$\omega$ SST turbulence models across twelve zero-pressure-gradient flat-plate cases reinforces existing evidence of their limitations in hypersonic cold-wall flows. Wall heat-flux and shear-stress predictions show substantial, case-dependent discrepancies from DNS, reaching approximately $\pm 20\%$ (Section 7.2.1).

Consistent with previous RANS assessments, both turbulence models were also found to struggle to reproduce the thermal profiles of hypersonic cold-wall boundary layers (Section 7.2.2). These profile discrepancies affect the wall functions that sample the RANS field (Section 7.3.3). The evidence points to the shared constant turbulent Prandtl number $Pr_t$ assumption as a plausible cause, although its contribution to wall heat-flux and shear-stress errors has not been isolated (Section 7.2.3).

The wall-resolved mesh studies also revisit an existing acoustic-Reynolds-number criterion. Using wall properties and the wall-to-first-cell-centre distance, $Re_{\mathrm{cell1}} \leq 1$ provides a more reliable practical resolu-

tion guideline than $y_1^+ \leq 1$ alone for the investigated hypersonic cold-wall cases, particularly with SST (Section 6.2).

## 8.2 Outlook

#### Thermal closure and turbulence modelling

The similar SA and SST thermal-profile errors and their sensitivity to $Pr_t$ in Section 7.2.3 make the thermal closure a priority for further research. Variable turbulent Prandtl numbers have long been studied [102], alongside more general turbulent heat-flux closures [103]. Huang et al. [82], for example, improve cold-wall temperature profiles using an algebraic energy-flux model. Their use of an algebraic momentum model, however, leaves the energy-flux closure's performance with the present SA and SST formulations to be assessed.

Recent results also show why a variable $Pr_t$ alone may be insufficient. Cheng et al. [90] attribute most of the reduction in the excessive temperature maximum to revised turbulent kinetic-energy contributions in the governing equations, with their variable-$Pr_t$ relation provides a smaller additional improvement, but no isolated variable-$Pr_t$ relation assessment was performed. On the other hand, Hasan et al. [104] find little effect from replacing $Pr_t = 0.9$ with DNS-derived values in corrected inner-layer calculations with prescribed wall loads. Their inner-layer thermal-profile improvements combine turbulence and energy-source corrections, where wall-load predictions are improved while leaving temperature profiles largely unchanged.

Compressibility corrections provide another possible direction, but their benefits are also conditional. Allmaras, Johnson and Spalart [23, Section 2.2] reaffirm the same standard SA equation for incompressible and compressible flows, without prescribing an additional compressibility correction. This does not establish its accuracy throughout the hypersonic cold-wall regime. Xue et al. [88] show that additional SA corrections can improve some predictions while worsening others as compressibility increases. Recent work distinguishing variable-property effects from intrinsic compressibility provides further guidance [8, 104, 9], without establishing a universal correction for the present discrepancies.

Future work may give interesting insight by varying the thermal closure, momentum-model corrections and energy-equation modelling separately before assessing their combinations. Boundary layer profiles, wall heat-flux and wall shear stress should be evaluated under otherwise consistent conditions. This would test how much of the present discrepancy is attributable to the constant-$Pr_t$ assumption and which additional changes are required. While at the same time not only the exact turbulence model formulation, but also its coupling to the RANS field needs assessing (ref. Section 7.2.4).

#### DNS reference data

The uncertainties and reporting inconsistencies identified with the DNS references of the investigated ZPG flat plates in Section 6.1 limit conclusions drawn from these widely used benchmarks. Recomputing selected cases and publishing complete flow-field data, including inlet profiles and downstream development, would strengthen their value for future model assessment. Zhang's digitally available profiles illustrate the benefit of accessible reference data. Extending this coverage could also support more controlled thermal-closure tests. Additional cases could also broaden the coverage of wall cooling and compressibility shown in Figure 6.1.

#### Comparison across RANS implementations

A coordinated comparison and collective summary of the turbulence models variants and their respective code implementations discussed in Section 7.2.4 could help explain why similar thermal-profile discrepancies coexist with markedly different wall-load predictions. The recurring temperature overshoot in studies using

$Pr_t = 0.9$ is a useful starting point. Common meshes, physical models, inflow conditions and matching procedures would help distinguish implementation differences from modelling limitations.

#### Boundary layer-informed meshing for engineering applications

The meshes in Section 7.4.3 demonstrate the spacing criterion, but do not optimize the cell count. An engineering algorithm should seek efficient meshes while controlling the relevant internal-face errors, while additionally retaining sufficient leading-edge resolution (Section 7.4.1). It should permit conventional spacing wherever the local profiles satisfy the criterion, including less restrictive adiabatic or lower-speed conditions. Extension beyond the investigated ZPG flat plates remains necessary. The adaptive wall-functions applicability over the whole boundary layer height (Section 7.3.2) may facilitate local wall-resolved refinement in regions of complex flow interactions, but this does not automatically establish the wall-functions validity in complex flows.

#### A-priori mesh design

The conditional estimate in Section 7.4.4 demonstrates how the Li wall-function relations can guide mesh design without a resolved reference profile. Future work should revisit this approach with improved closures and assess its sensitivity to estimated wall loads and wall/edge states, particularly near the temperature maximum. This is necessary before treating the estimate as a reliable engineering meshing tool.

#### Higher-order discretization

Higher-order viscous-gradient and transport-coefficient approximations could relax the identified spacing requirement. The compression-corner calculations of Tu et al. [105] show that discretization order can substantially affect separation and wall heating. Extending such comparisons to wall-function RANS could establish whether higher-order treatment can improve accuracy or reduce the required cell count.

#### Complex flows and the Wang–Craft wall function

The Wang–Craft formulation merits further investigation because of its reported performance in shock-wave/boundary-layer interactions, including separated flows [79]. The implementation difficulties documented in Section 5.5 should first be resolved, ideally through collaboration with the authors and access to their implementation. Subsequent assessment could thus extend the present error-isolation approach to pressure gradients, separation and complex geometries.

# A Reynolds-number inconsistencies in Martin–Duan Part 2

This appendix examines the consistency of the Reynolds numbers reported for the Part 2 database [81], using the published temperature data.

## A.1 Reynolds-number inconsistency and matching ambiguity

The definitions in Part 2, Section 2.1, use the same edge quantities and momentum thickness:

$$Re_\theta = \frac{\rho_\delta u_\delta \theta}{\mu_\delta}, \qquad Re_{\delta 2} = \frac{\rho_\delta u_\delta \theta}{\mu_w}, \qquad \boxed{\frac{Re_\theta}{Re_{\delta 2}} = \frac{\mu_w}{\mu_\delta}}. \tag{A.1}$$

For M5T1, the printed $T_w = T_\delta$ therefore requires $Re_\theta = Re_{\delta 2}$ for any temperature-dependent viscosity law. Instead, Table 1 gives $1279.1/1538.4 = 0.83145 \neq 1$, a $16.86\%$ deficit relative to unity. This is an inconsistency of the printed row, independent of density, velocity, momentum thickness, or the chosen viscosity exponent.

The other cases also fail a common power-law test. Part 2 inherits the power-law viscosity model stated in Part 1 [85, Section 2]. Writing $\mu = CT^\omega$, Equation (A.1) requires $Re_\theta/Re_{\delta 2} = (T_w/T_\delta)^\omega$. The implied exponents $\omega = \ln(Re_\theta/Re_{\delta 2})/\ln(T_w/T_\delta)$ in Table A.1 have no common intersection even after allowing for rounding of the printed entries.

**Table A.1** Part 2 Table 1 ratio test. Intervals allow half a unit in each last printed digit; endpoints are rounded outward.

| Case | $T_w/T_\delta$ | $Re_\theta$ | $Re_{\delta 2}$ | Implied $\omega$ interval |
|---|---|---|---|---|
| M5T2 | 1.90 | 2300.2 | 1521.5 | [0.6412, 0.6467] |
| M5T3 | 2.89 | 3011.6 | 1524.7 | [0.6402, 0.6425] |
| M5T4 | 3.74 | 3819.3 | 1526.4 | [0.6945, 0.6961] |
| M5T5 | 5.40 | 4840.5 | 1536.8 | [0.6799, 0.6808] |

A separate check illustrates the dimensional ambiguity. Part 1's M5 [85, Table 2] and Part 2's M5T5 [81, Table 1] have effectively identical printed edge states, while $\theta$ changes from $0.657$ to $0.655\,\mathrm{mm}$. With the same edge viscosity, $Re_\theta \propto \theta$, giving $6225(0.655/0.657) = 6206.1$, whereas Part 2 prints $4840.5$, approximately $22.0\%$ lower. Temporal evolution alone cannot reconcile these columns while preserving the reported edge state and viscosity model.

Accordingly, this thesis retains the published $Re_{\delta 2} \approx 1530$ values as the authors' intended matching targets [81, Section 2.1]. They are nominal reference labels, not independently verified dimensional reconstructions. The tables do not determine unique corrected Reynolds numbers or establish whether the mismatch reflects transport conventions, post-processing, or sampling. For M5T1, the accepted $T_w = T_\delta$ requires equality of the two Reynolds numbers; additionally retaining $Re_{\delta 2} = 1538.4$ would give $Re_\theta = 1538.4$. This is conditional on that matching target, rather than an independent reconstruction of both Reynolds numbers.

# B Wall-resolved mesh convergence

The following wall-normal mesh studies supplement Section 6.2 for all twelve ZPG cases, including Huang's three stations. Coefficients are evaluated at each mesh's $Re_{\delta 2}$-matched station. Dashed lines show Eça–Hoekstra extrapolations from the four finest meshes; shaded bands mark $\pm 1\%$ of each extrapolated value and do not measure agreement with DNS.

## B.1 Martin–Duan Parts 2 and 4

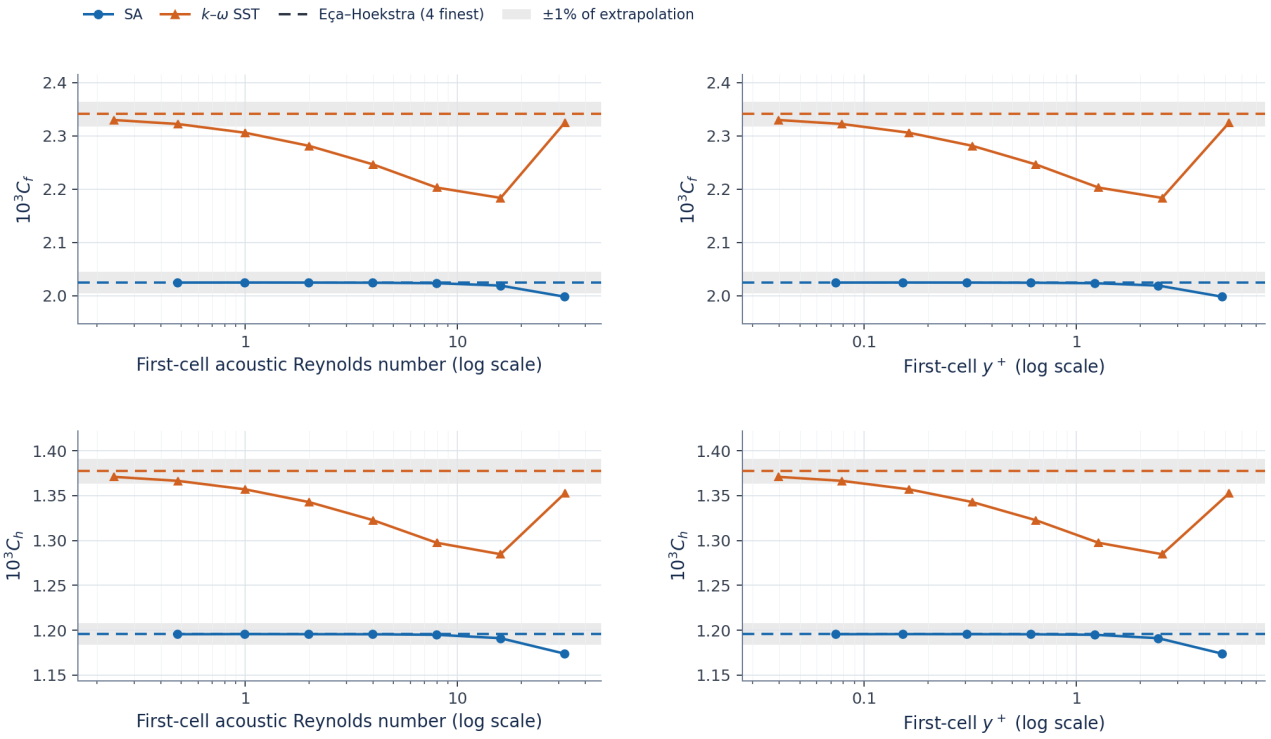



**Figure B.1** Wall-normal mesh sensitivity for Martin–Duan Part 2 M5T1, comparing Spalart–Allmaras and $k$–$\omega$ SST. Skin-friction coefficient $C_f$ (top) and Stanton number $C_h$ (bottom) versus first-cell acoustic Reynolds number (left) and first-cell $y^+$ (right) at the matched $Re_{\delta 2}$ = 1538.4 station. Dashed line represents Eça–Hoekstra extrapolations from the four finest mesh results; shaded bands mark $\pm 1\%$ of each extrapolated value.

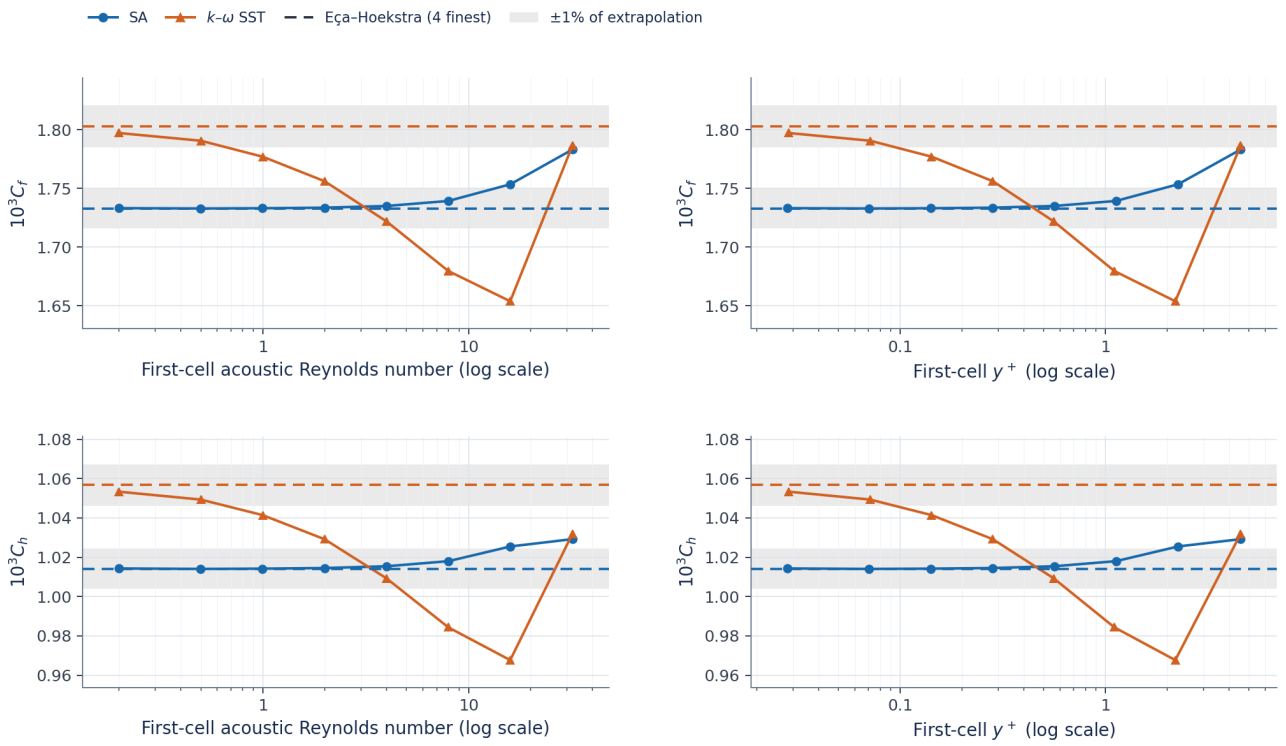



**Figure B.2** Wall-normal mesh sensitivity for Martin–Duan Part 2 M5T2, comparing Spalart–Allmaras and $k$–$\omega$ SST. Skin-friction coefficient $C_f$ (top) and Stanton number $C_h$ (bottom) versus first-cell acoustic Reynolds number (left) and first-cell $y^+$ (right) at the matched $Re_{\delta 2}$ = 1521.5 station. Dashed line represents Eça–Hoekstra extrapolations from the four finest mesh results; shaded bands mark $\pm 1\%$ of each extrapolated value.

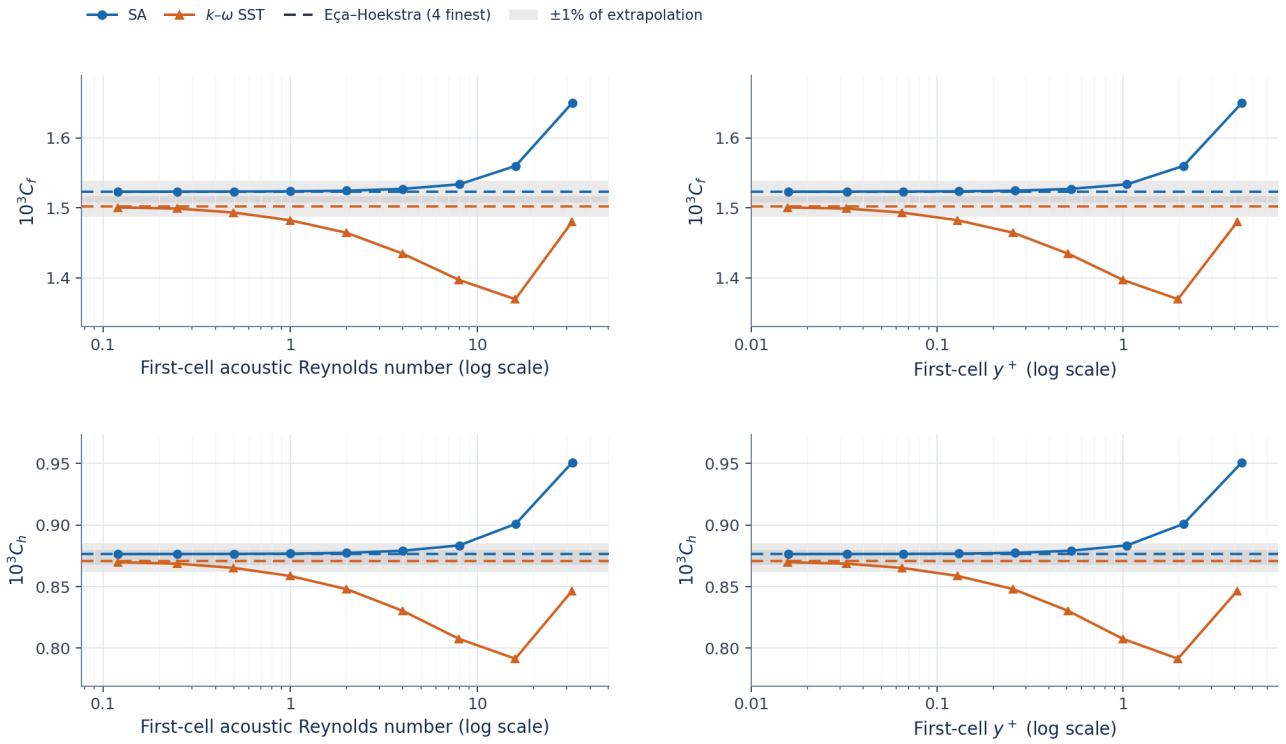



**Figure B.3** Wall-normal mesh sensitivity for Martin–Duan Part 2 M5T3, comparing Spalart–Allmaras and $k$–$\omega$ SST. Skin-friction coefficient $C_f$ (top) and Stanton number $C_h$ (bottom) versus first-cell acoustic Reynolds number (left) and first-cell $y^+$ (right) at the matched $Re_{\delta 2}$ = 1524.7 station. Dashed line represents Eça–Hoekstra extrapolations from the four finest mesh results; shaded bands mark $\pm 1\%$ of each extrapolated value.

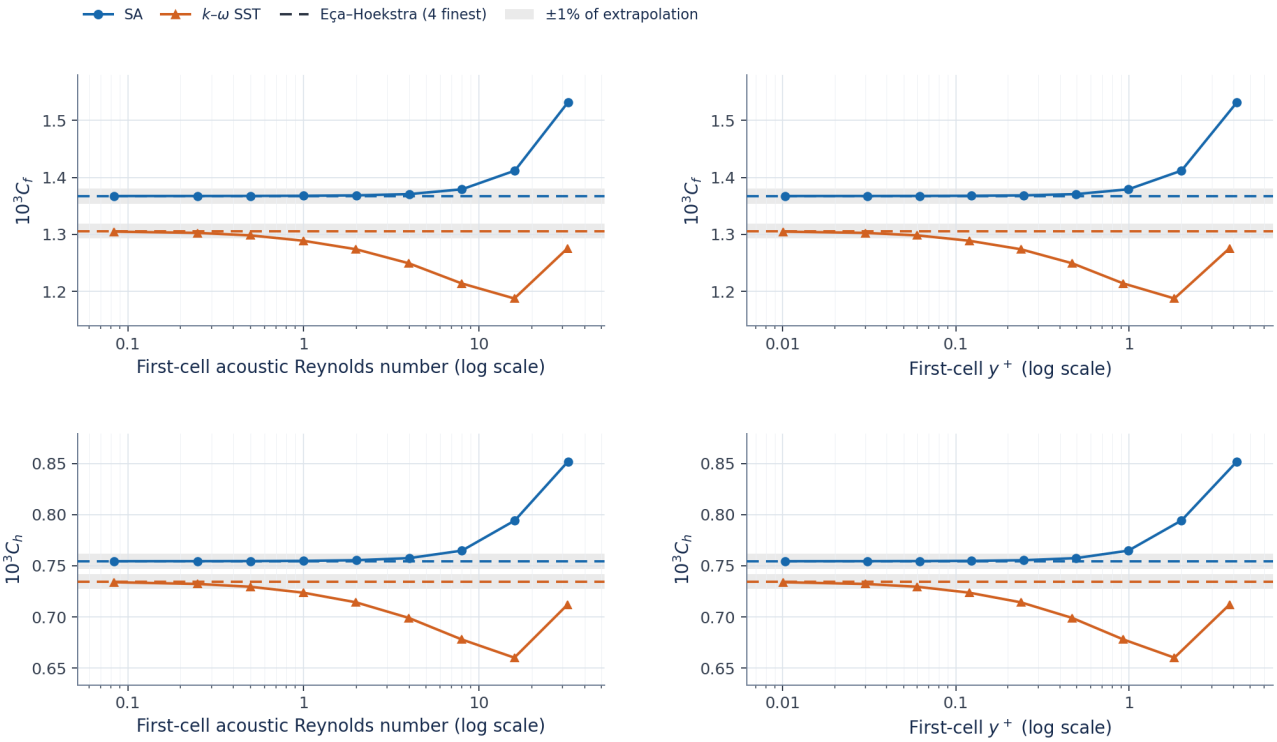



**Figure B.4** Wall-normal mesh sensitivity for Martin–Duan Part 2 M5T4, comparing Spalart–Allmaras and $k$–$\omega$ SST. Skin-friction coefficient $C_f$ (top) and Stanton number $C_h$ (bottom) versus first-cell acoustic Reynolds number (left) and first-cell $y^+$ (right) at the matched $Re_{\delta 2}$ = 1526.4 station. Dashed line represents Eça–Hoekstra extrapolations from the four finest mesh results; shaded bands mark $\pm 1\%$ of each extrapolated value.

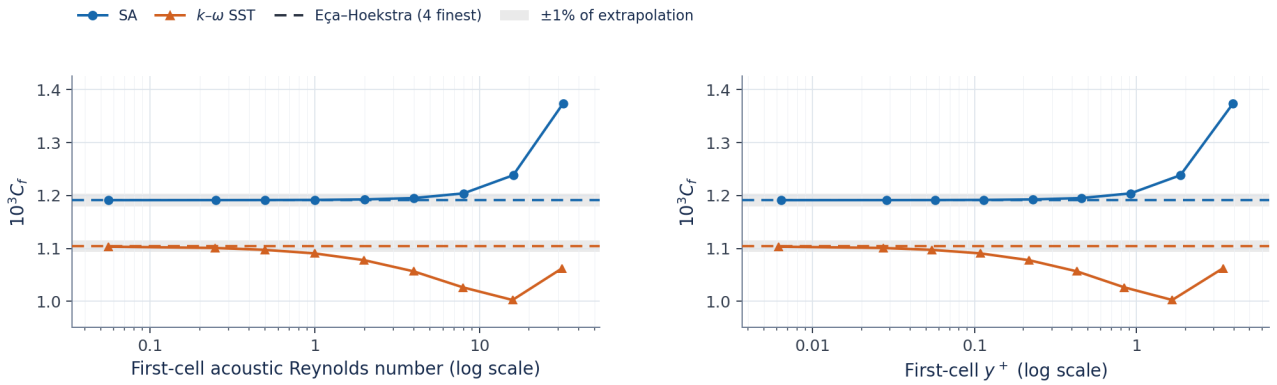



**Figure B.5** Wall-normal mesh sensitivity for Martin–Duan Part 2 M5T5, comparing Spalart–Allmaras and $k$–$\omega$ SST. Skin-friction coefficient $C_f$ versus first-cell acoustic Reynolds number (left) and first-cell $y^+$ (right) at the matched $Re_{\delta 2}$ = 1536.8 station. Dashed line represents Eça–Hoekstra extrapolations from the four finest mesh results; shaded bands mark $\pm 1\%$ of each extrapolated value. Heat-transfer is not assessed for this adiabatic case.

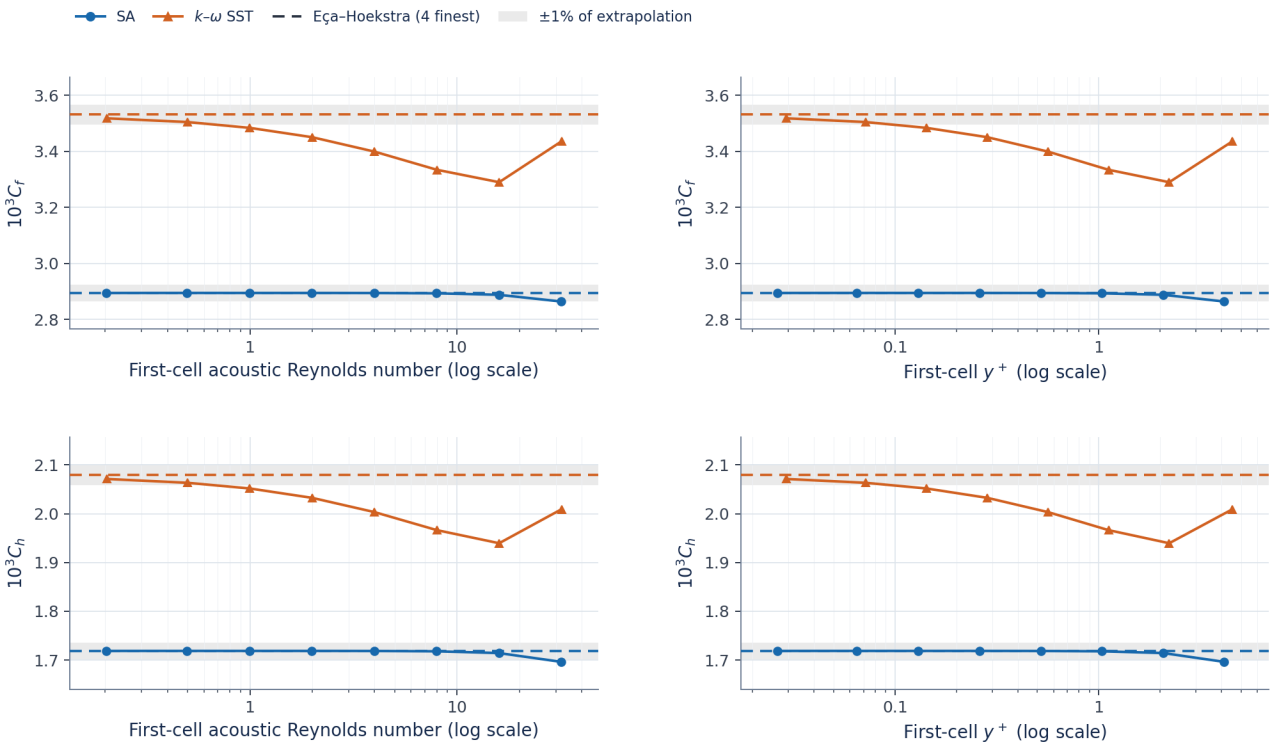



**Figure B.6** Wall-normal mesh sensitivity for Martin–Duan Part 4 LowH_M3, comparing Spalart–Allmaras and $k$–$\omega$ SST. Skin-friction coefficient $C_f$ (top) and Stanton number $C_h$ (bottom) versus first-cell acoustic Reynolds number (left) and first-cell $y^+$ (right) at the matched $Re_{\delta 2}$ = 1554.5 station. Dashed line represents Eça–Hoekstra extrapolations from the four finest mesh results; shaded bands mark $\pm 1\%$ of each extrapolated value.

## B.2 Zhang cases

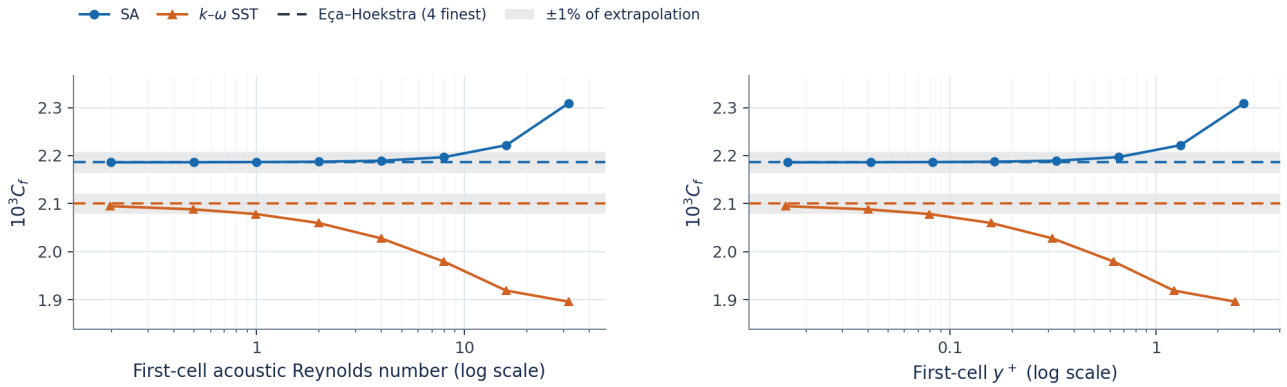



**Figure B.7** Wall-normal mesh sensitivity for Zhang M2p5, comparing Spalart–Allmaras and $k$–$\omega$ SST. Skin-friction coefficient $C_f$ versus first-cell acoustic Reynolds number (left) and first-cell $y^+$ (right) at the matched $Re_{\delta 2}$ = 1657 station. Dashed line represents Eça–Hoekstra extrapolations from the four finest mesh results; shaded bands mark $\pm 1\%$ of each extrapolated value. Heat-transfer is not assessed for this adiabatic case.

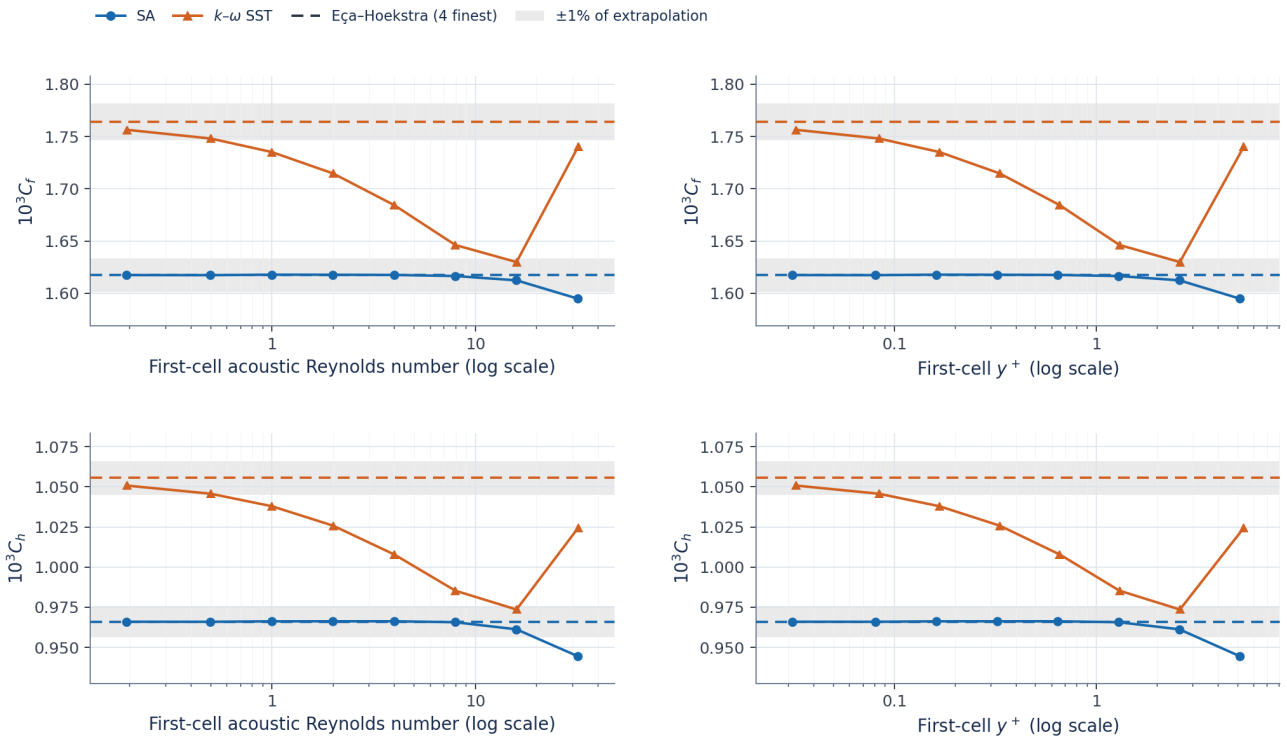



**Figure B.8** Wall-normal mesh sensitivity for Zhang M6Tw025, comparing Spalart–Allmaras and $k$–$\omega$ SST. Skin-friction coefficient $C_f$ (top) and Stanton number $C_h$ (bottom) versus first-cell acoustic Reynolds number (left) and first-cell $y^+$ (right) at the matched $Re_{\delta 2}$ = 1135 station. Dashed line represents Eça–Hoekstra extrapolations from the four finest mesh results; shaded bands mark $\pm 1\%$ of each extrapolated value.

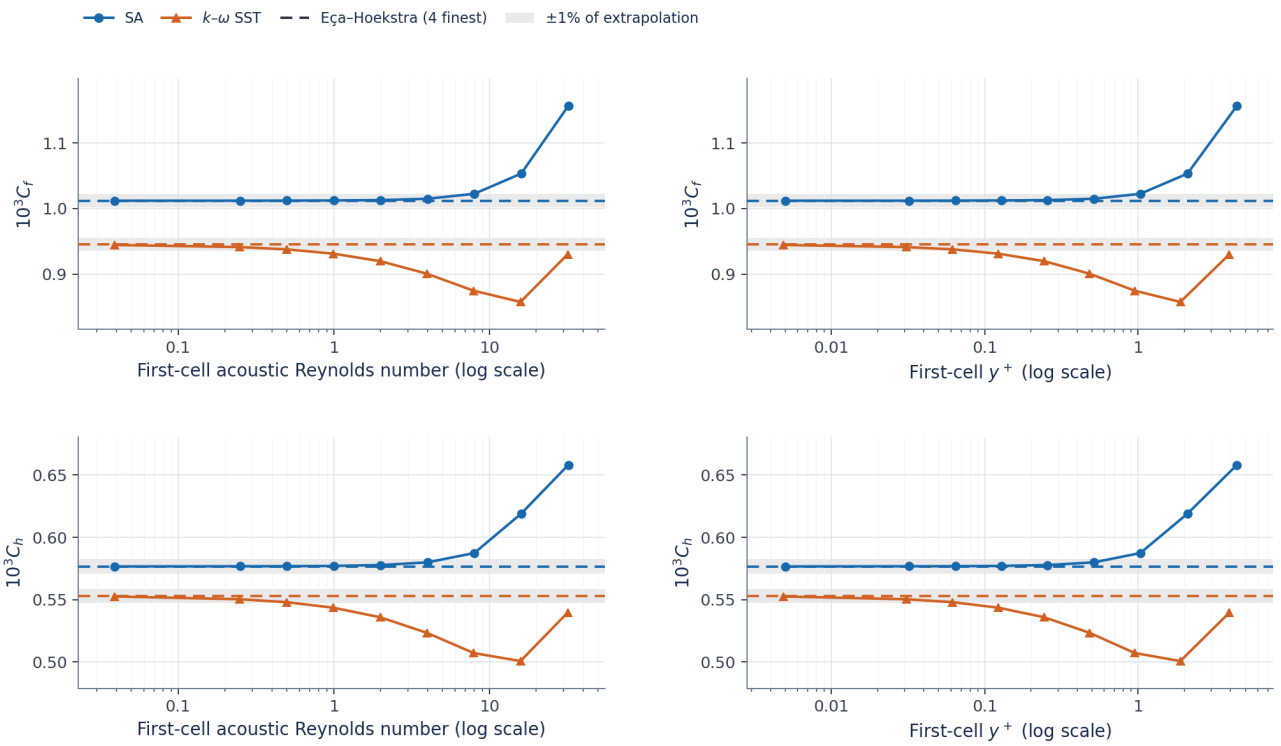



**Figure B.9** Wall-normal mesh sensitivity for Zhang M6Tw076, comparing Spalart–Allmaras and $k$–$\omega$ SST. Skin-friction coefficient $C_f$ (top) and Stanton number $C_h$ (bottom) versus first-cell acoustic Reynolds number (left) and first-cell $y^+$ (right) at the matched $Re_{\delta 2}$ = 1746 station. Dashed line represents Eça–Hoekstra extrapolations from the four finest mesh results; shaded bands mark $\pm 1\%$ of each extrapolated value. The $C_h$ curves document mesh sensitivity only; the reference-data limitations in Section 6.1 preclude their use for quantitative heat-flux validation against DNS.

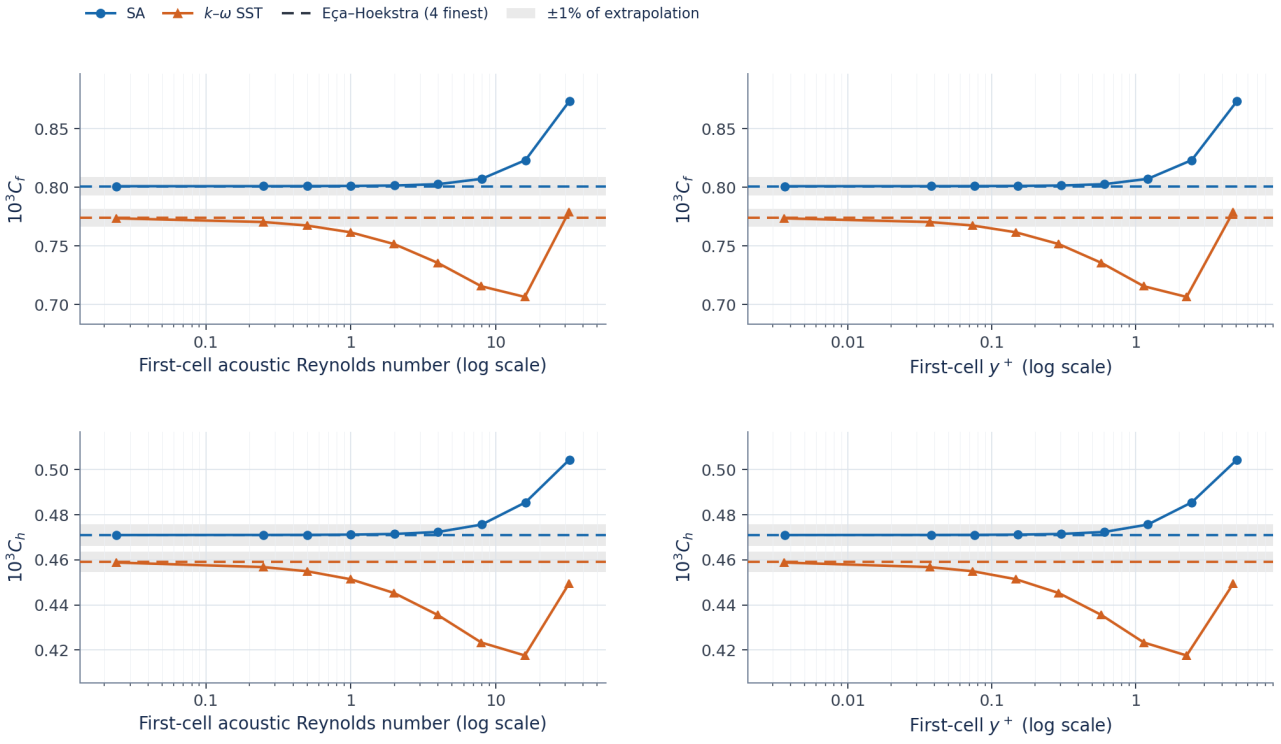



**Figure B.10** Wall-normal mesh sensitivity for Zhang M8Tw048, comparing Spalart–Allmaras and $k$–$\omega$ SST. Skin-friction coefficient $C_f$ (top) and Stanton number $C_h$ (bottom) versus first-cell acoustic Reynolds number (left) and first-cell $y^+$ (right) at the matched $Re_{\delta 2}$ = 1990 station. Dashed line represents Eça–Hoekstra extrapolations from the four finest mesh results; shaded bands mark $\pm 1\%$ of each extrapolated value.

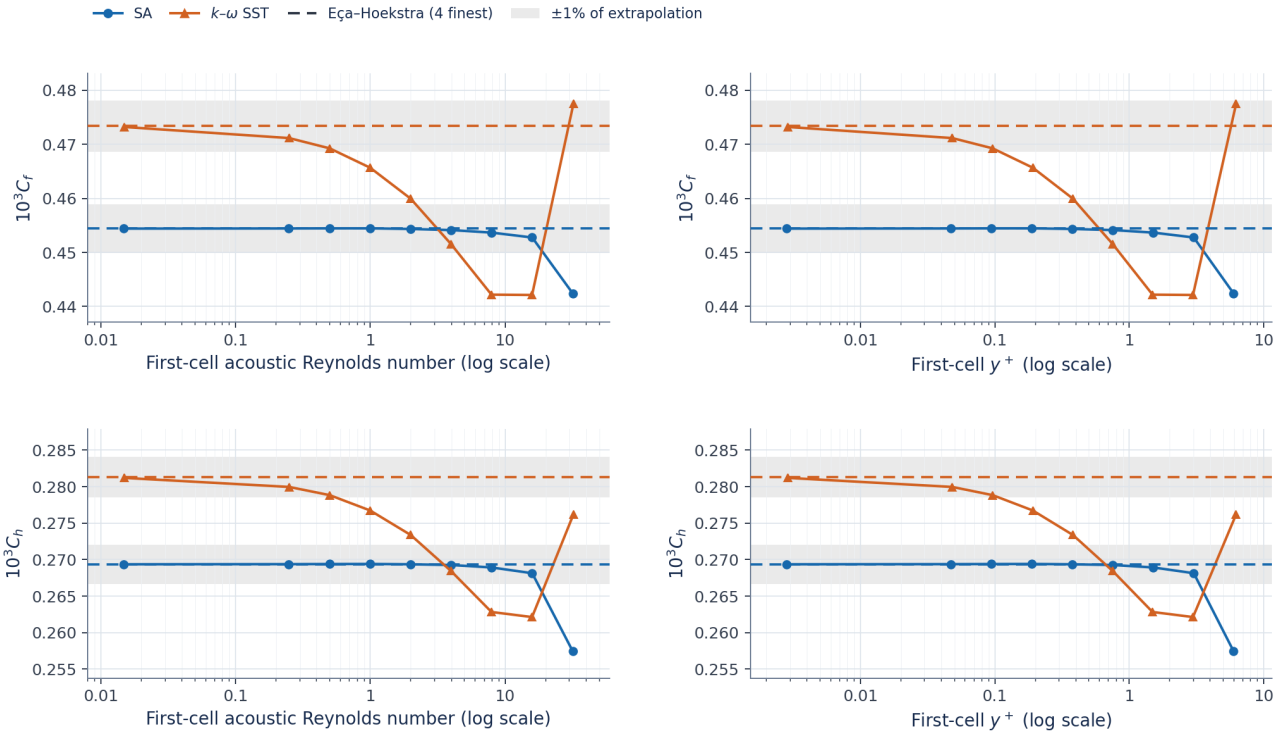



**Figure B.11** Wall-normal mesh sensitivity for Zhang M14Tw018, comparing Spalart–Allmaras and $k$–$\omega$ SST. Skin-friction coefficient $C_f$ (top) and Stanton number $C_h$ (bottom) versus first-cell acoustic Reynolds number (left) and first-cell $y^+$ (right) at the matched $Re_{\delta 2}$ = 2354 station. Dashed line represents Eça–Hoekstra extrapolations from the four finest mesh results; shaded bands mark $\pm 1\%$ of each extrapolated value.

## B.3 Huang case at stations BL1–BL3

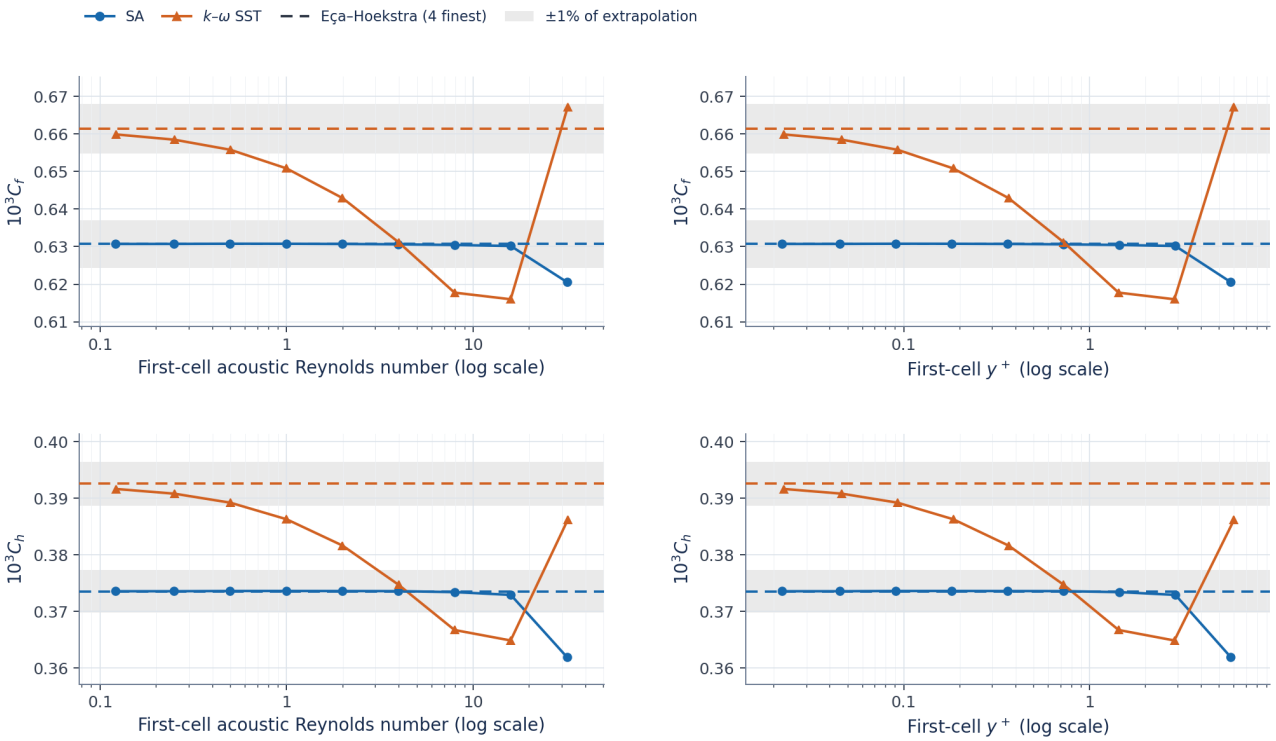



**Figure B.12** Wall-normal mesh sensitivity for Huang M11Tw020 at station BL1, comparing Spalart–Allmaras and $k$–$\omega$ SST. Skin-friction coefficient $C_f$ (top) and Stanton number $C_h$ (bottom) versus first-cell acoustic Reynolds number (left) and first-cell $y^+$ (right) at the matched $Re_{\delta 2}$ = 2204 station. Dashed line represents Eça–Hoekstra extrapolations from the four finest mesh results; shaded bands mark $\pm 1\%$ of each extrapolated value.

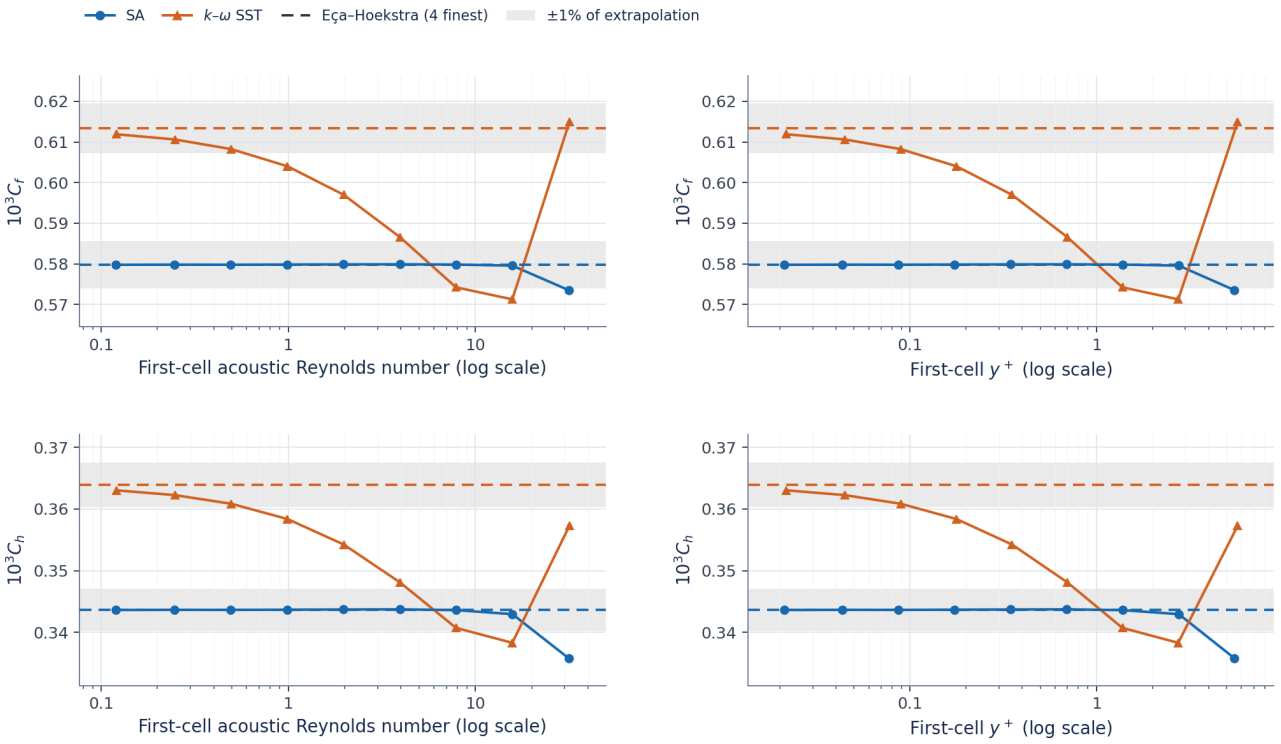



**Figure B.13** Wall-normal mesh sensitivity for Huang M11Tw020 at station BL2, comparing Spalart–Allmaras and $k$–$\omega$ SST. Skin-friction coefficient $C_f$ (top) and Stanton number $C_h$ (bottom) versus first-cell acoustic Reynolds number (left) and first-cell $y^+$ (right) at the matched $Re_{\delta 2}$ = 3419 station. Dashed line represents Eça–Hoekstra extrapolations from the four finest mesh results; shaded bands mark $\pm 1\%$ of each extrapolated value.

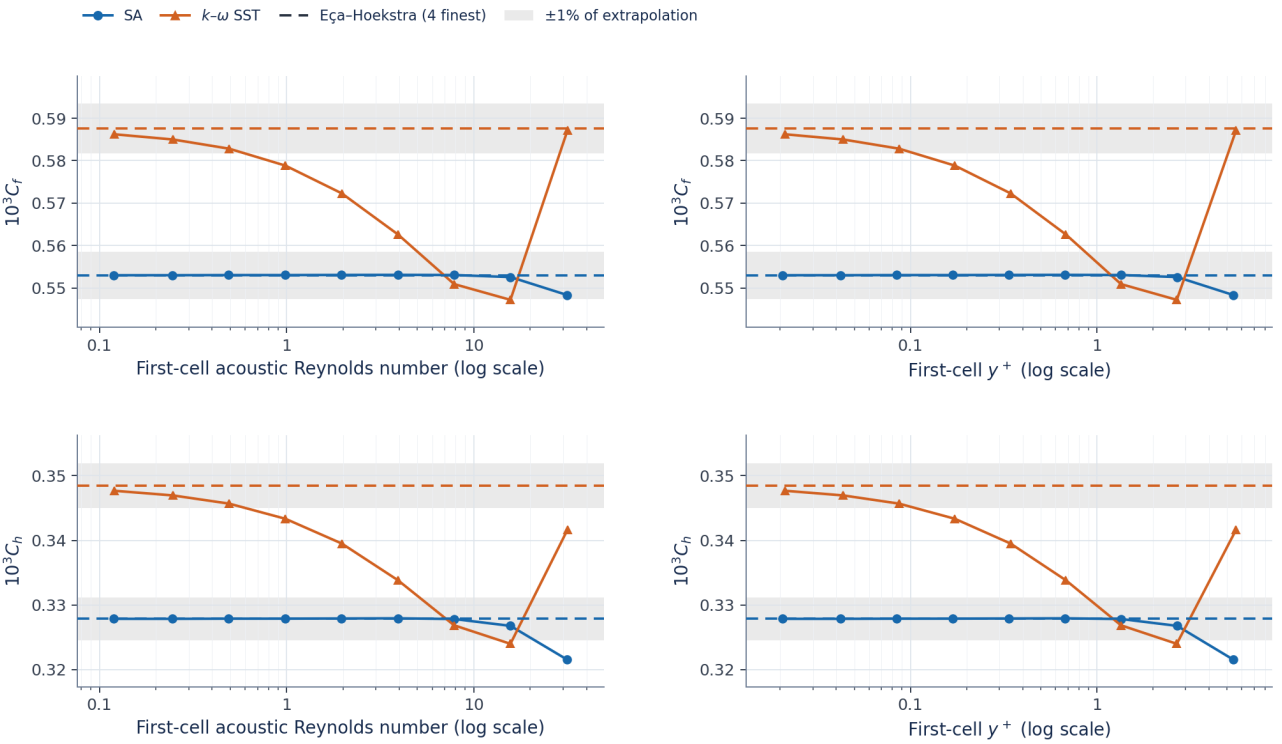



**Figure B.14** Wall-normal mesh sensitivity for Huang M11Tw020 at station BL3, comparing Spalart–Allmaras and $k$–$\omega$ SST. Skin-friction coefficient $C_f$ (top) and Stanton number $C_h$ (bottom) versus first-cell acoustic Reynolds number (left) and first-cell $y^+$ (right) at the matched $Re_{\delta 2}$ = 4390 station. Dashed line represents Eça–Hoekstra extrapolations from the four finest mesh results; shaded bands mark $\pm 1\%$ of each extrapolated value.

# C Additional wall-resolved boundary layer profiles

This appendix supplements Section 7.2 with the ten case comparisons accompanying the M6Tw025 and LowH_M3 profiles shown in the main text. The plots compare wall-resolved Spalart–Allmaras and $k$–$\omega$ SST predictions at $Pr_t = 0.9$ with the DNS references. Matching conventions and reference-data limitations are described in Section 6.1.

## C.1 Martin–Duan Part 2

The digitized profiles use the inferred temperature normalization described in Section 6.1. The published $Re_{\delta 2}$ values remain nominal matching targets, subject to the inconsistency discussed in Appendix A.1.

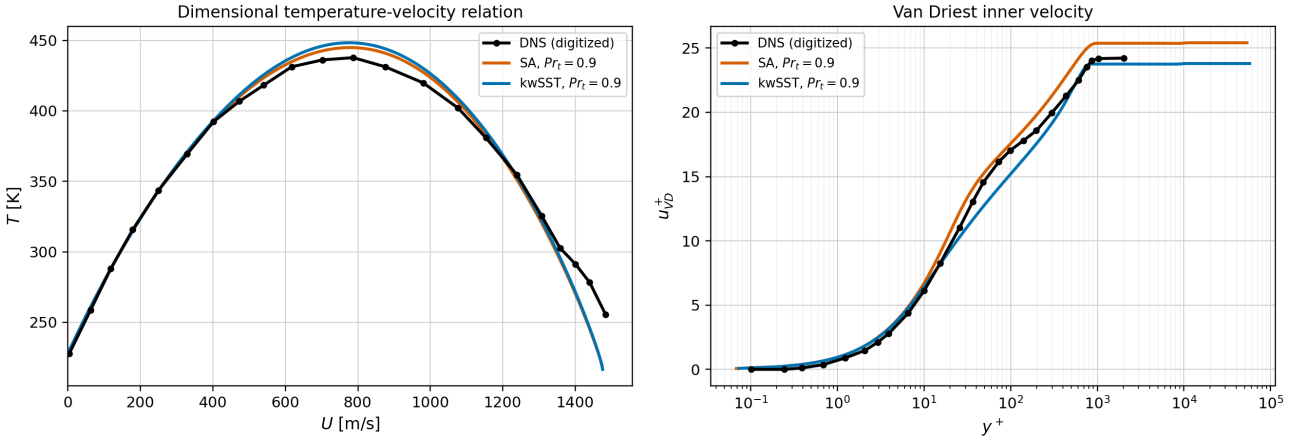



**Figure C.1** Wall-resolved Spalart–Allmaras and $k$–$\omega$ SST profiles for M5T1 at $Pr_t = 0.9$, compared with digitized Martin–Duan Part 2 DNS data [81]. Left: dimensional temperature–velocity relation $T(U)$. Right: van Driest–transformed velocity $U^+_{\mathrm{VD}}$ versus wall distance $y^+$ at the matched $Re_{\delta 2}$ = 1538.4 station.

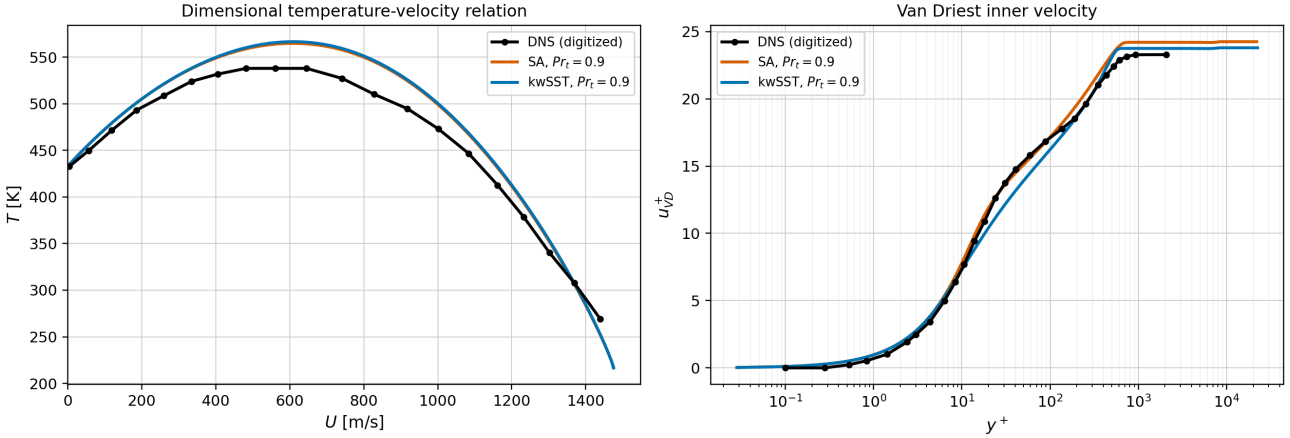



**Figure C.2** Wall-resolved Spalart–Allmaras and $k$–$\omega$ SST profiles for M5T2 at $Pr_t = 0.9$, compared with digitized Martin–Duan Part 2 DNS data [81]. Left: dimensional temperature–velocity relation $T(U)$. Right: van Driest–transformed velocity $U^+_{\mathrm{VD}}$ versus wall distance $y^+$ at the matched $Re_{\delta 2}$ = 1521.5 station.

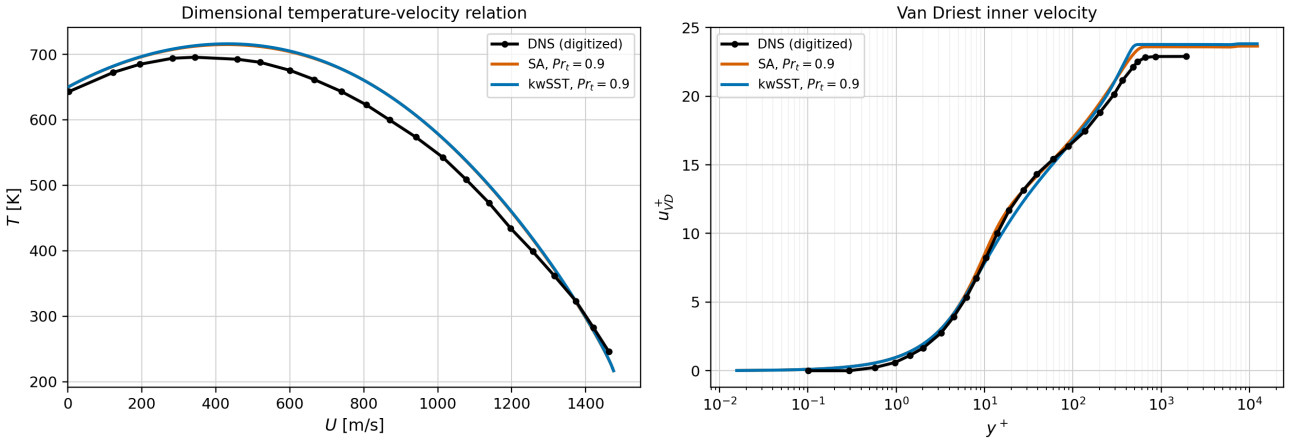



**Figure C.3** Wall-resolved Spalart–Allmaras and $k$–$\omega$ SST profiles for M5T3 at $Pr_t = 0.9$, compared with digitized Martin–Duan Part 2 DNS data [81]. Left: dimensional temperature–velocity relation $T(U)$. Right: van Driest–transformed velocity $U^+_{\mathrm{VD}}$ versus wall distance $y^+$ at the matched $Re_{\delta 2}$ = 1524.7 station.

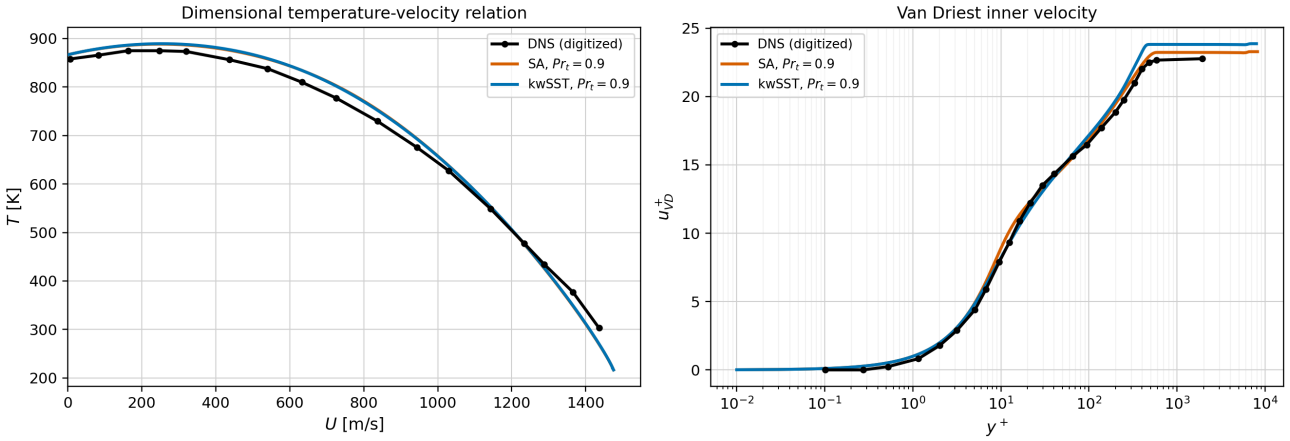



**Figure C.4** Wall-resolved Spalart–Allmaras and $k$–$\omega$ SST profiles for M5T4 at $Pr_t = 0.9$, compared with digitized Martin–Duan Part 2 DNS data [81]. Left: dimensional temperature–velocity relation $T(U)$. Right: van Driest–transformed velocity $U^+_{\mathrm{VD}}$ versus wall distance $y^+$ at the matched $Re_{\delta 2}$ = 1526.4 station.

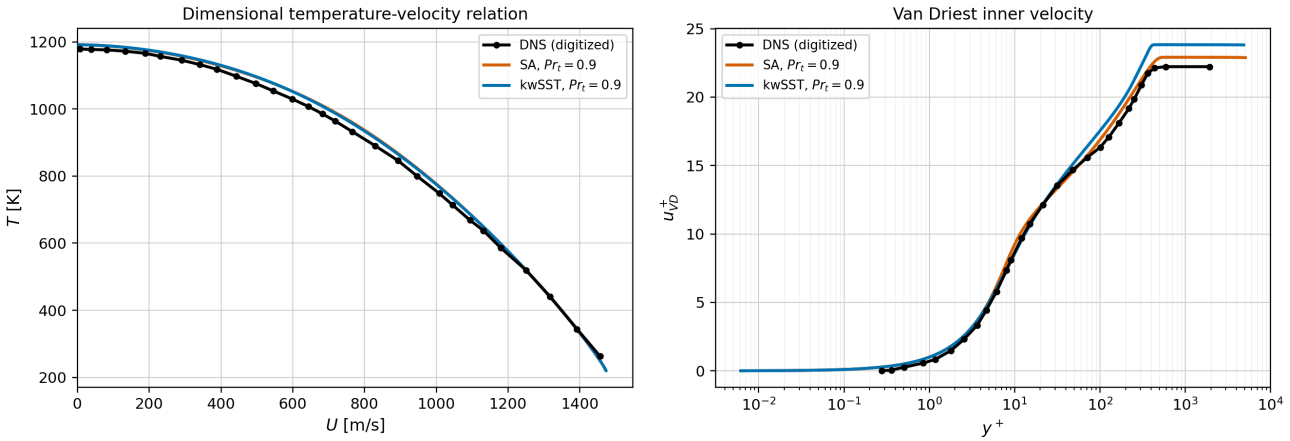



**Figure C.5** Wall-resolved Spalart–Allmaras and $k$–$\omega$ SST profiles for M5T5 at $Pr_t = 0.9$, compared with digitized Martin–Duan Part 2 DNS data [81]. Left: dimensional temperature–velocity relation $T(U)$. Right: van Driest–transformed velocity $U^+_{\mathrm{VD}}$ versus wall distance $y^+$ at the matched $Re_{\delta 2}$ = 1536.8 station.

## C.2 Zhang cases

The following comparisons use the digital DNS profiles of Zhang et al. [5]. The M6Tw076 profiles are retained as qualitative comparisons; the wall-temperature inconsistency and limitations on quantitative heat-flux assessment remain those described in Section 6.1.

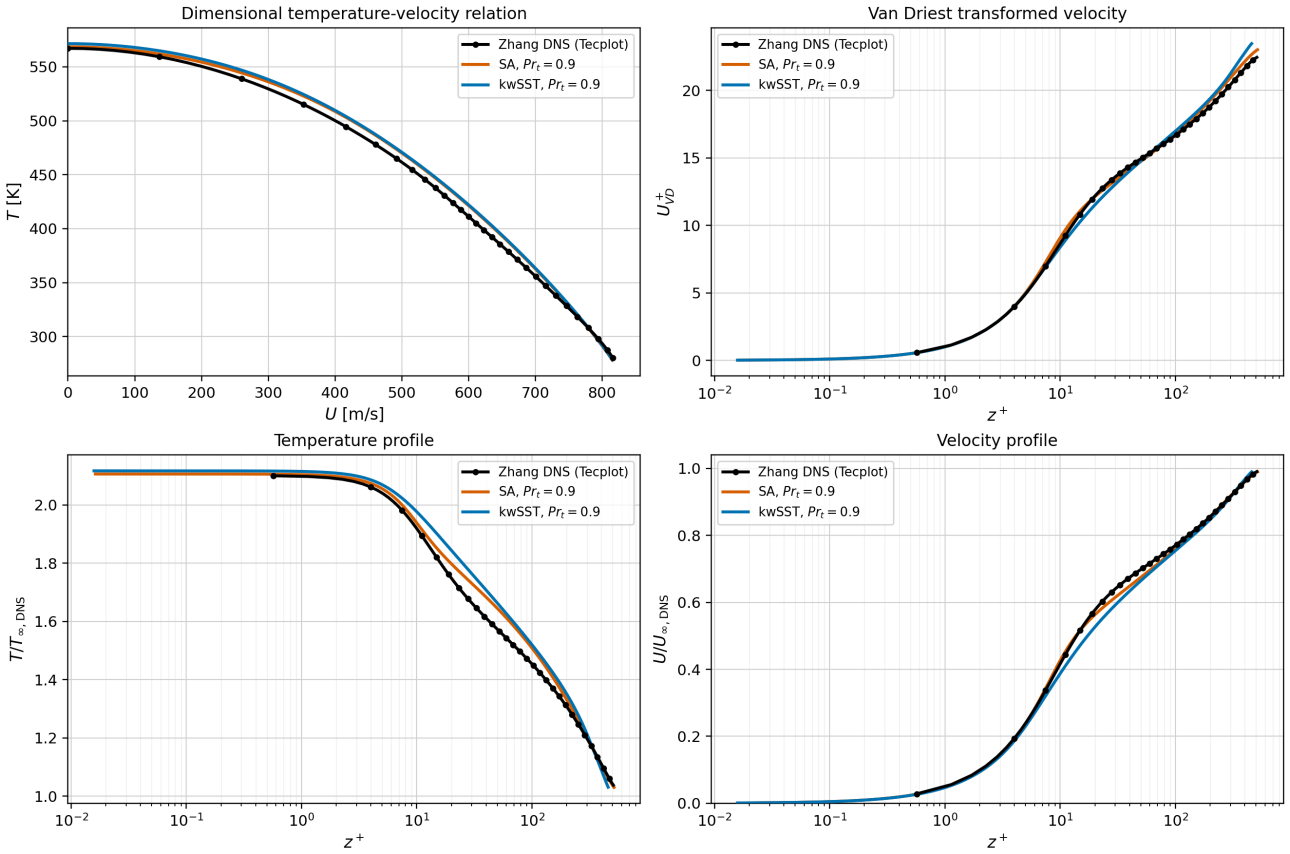



**Figure C.6** Wall-resolved Spalart–Allmaras and $k$–$\omega$ SST profiles for M2p5 at $Pr_t = 0.9$, compared with Zhang et al.'s DNS [5]. Top: dimensional $T(U)$ (left) and van Driest–transformed velocity $U^+_{\mathrm{VD}}$ versus $z^+$ (right). Bottom: $T/T_{\infty,\mathrm{DNS}}$ (left) and $U/U_{\infty,\mathrm{DNS}}$ (right), both versus $z^+$ at the matched $Re_{\delta 2}$ = 1657 station.

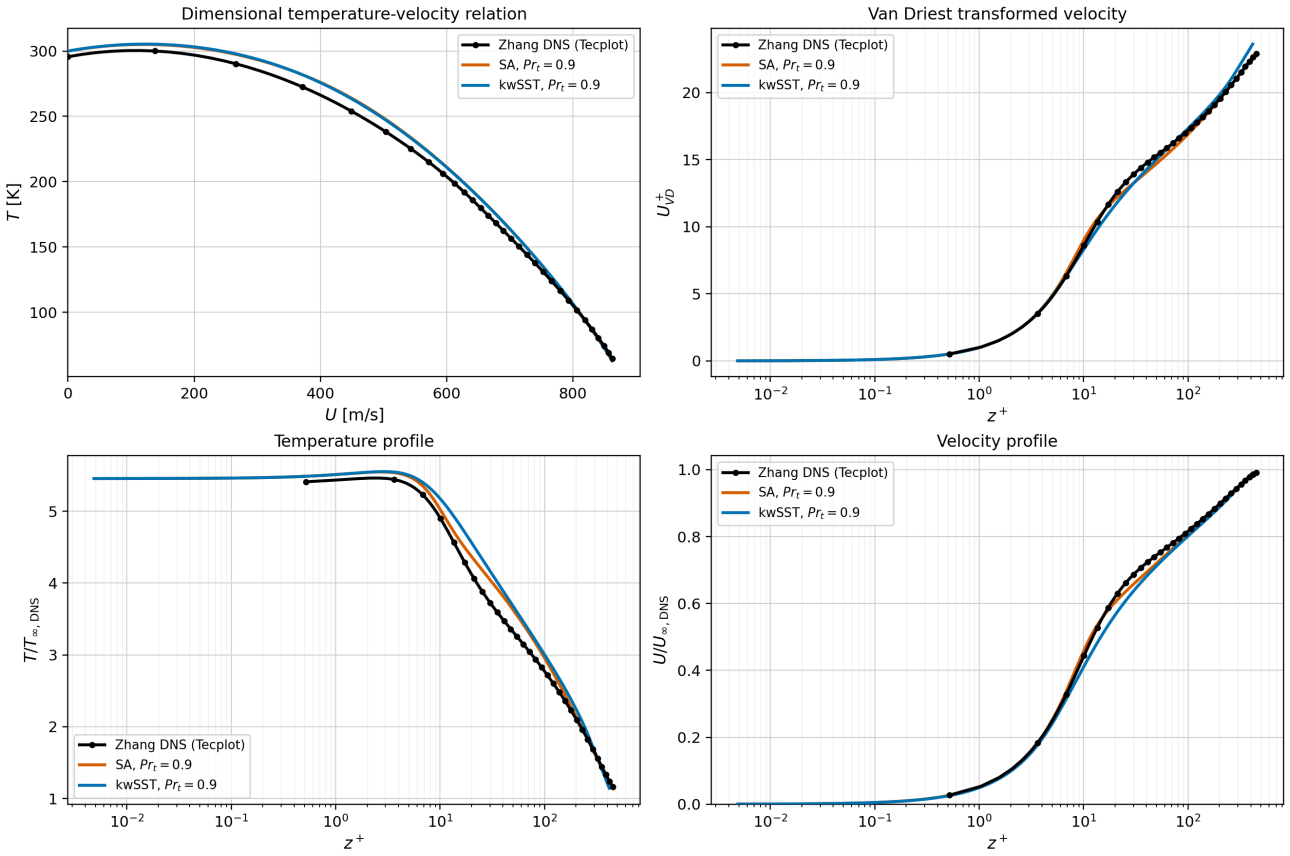



**Figure C.7** Wall-resolved Spalart–Allmaras and $k$–$\omega$ SST profiles for M6Tw076 at $Pr_t = 0.9$, compared with Zhang et al.'s DNS [5]. Top: dimensional $T(U)$ (left) and van Driest–transformed velocity $U^+_{\mathrm{VD}}$ versus $z^+$ (right). Bottom: $T/T_{\infty,\mathrm{DNS}}$ (left) and $U/U_{\infty,\mathrm{DNS}}$ (right), both versus $z^+$ at the matched $Re_{\delta 2}$ = 1746 station.

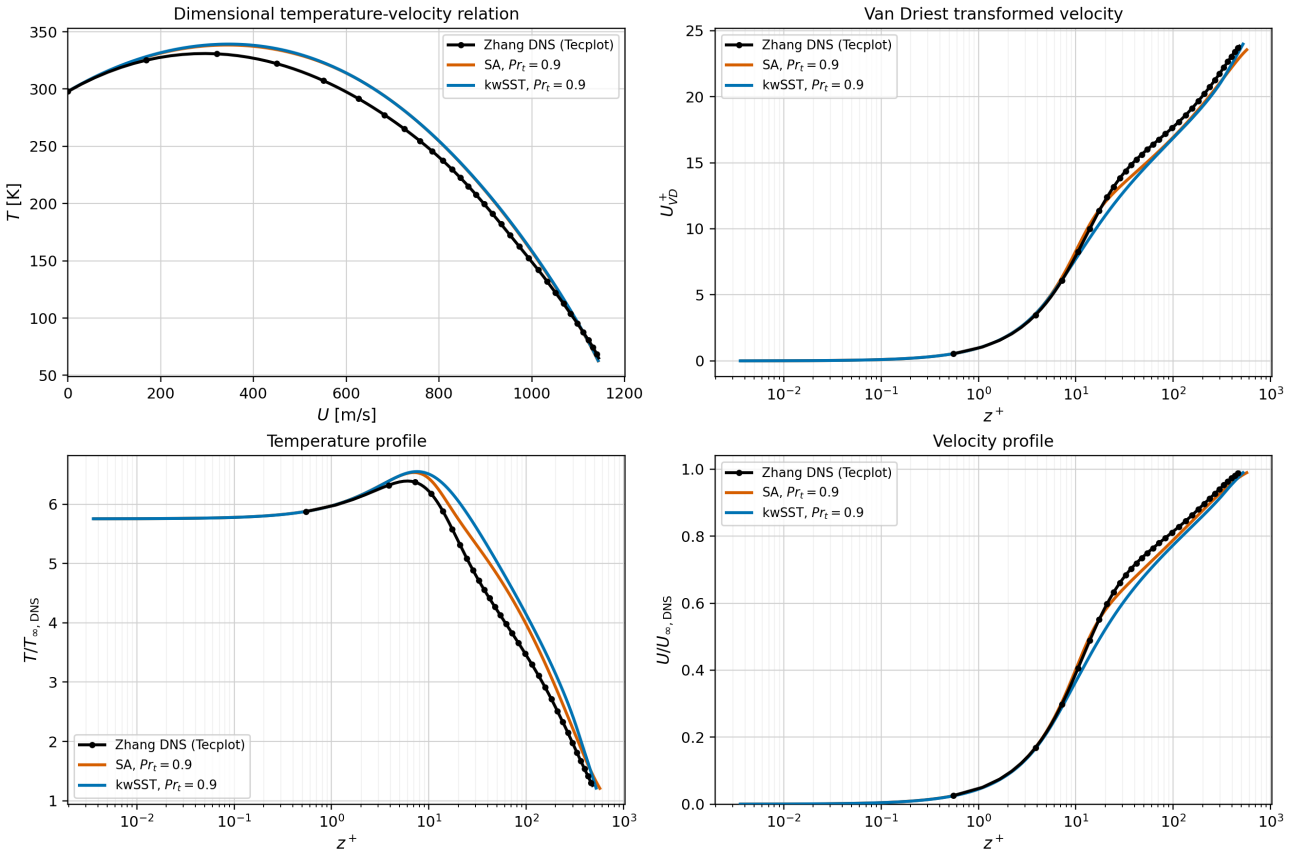



**Figure C.8** Wall-resolved Spalart–Allmaras and $k$–$\omega$ SST profiles for M8Tw048 at $Pr_t = 0.9$, compared with Zhang et al.'s DNS [5]. Top: dimensional $T(U)$ (left) and van Driest–transformed velocity $U^+_{\mathrm{VD}}$ versus $z^+$ (right). Bottom: $T/T_{\infty,\mathrm{DNS}}$ (left) and $U/U_{\infty,\mathrm{DNS}}$ (right), both versus $z^+$ at the matched $Re_{\delta 2}$ = 1990 station.

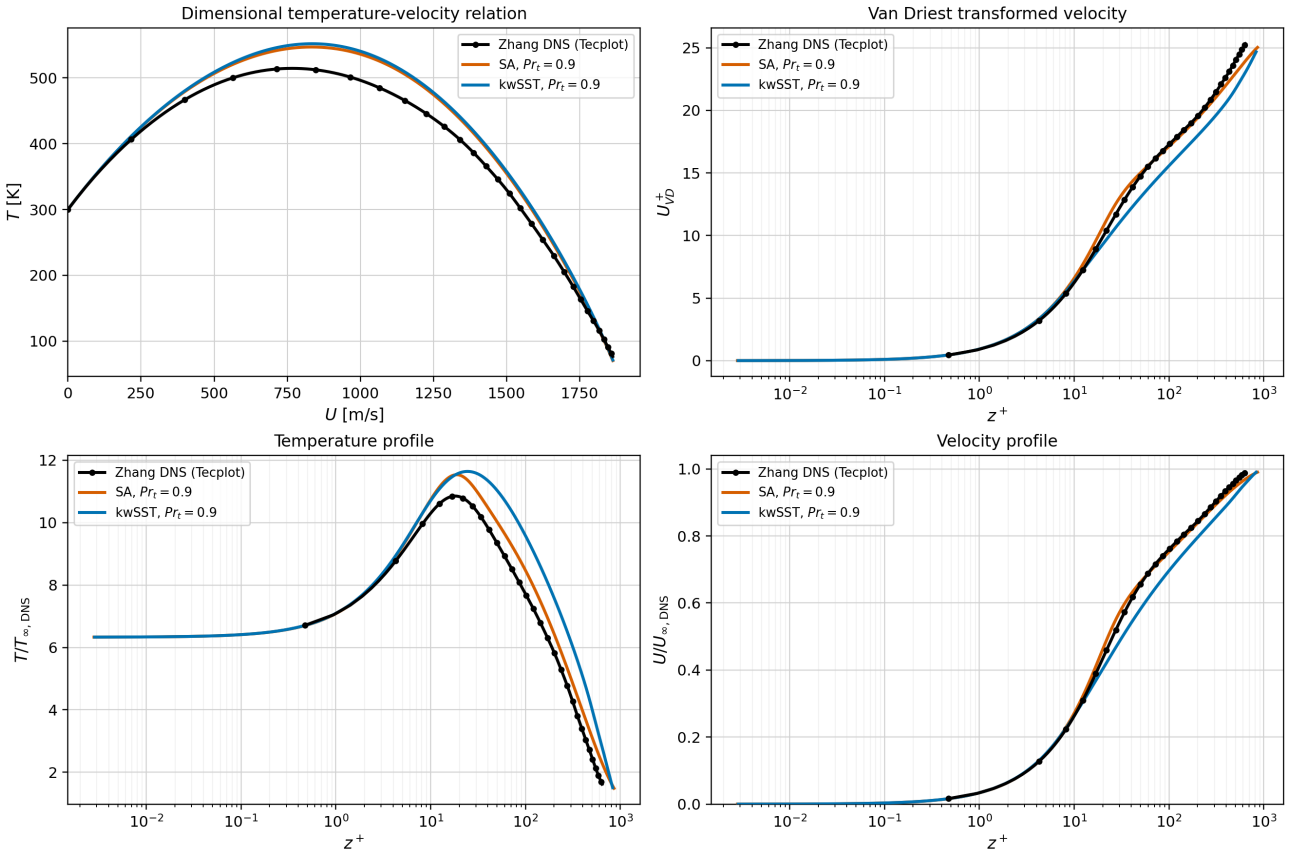



**Figure C.9** Wall-resolved Spalart–Allmaras and $k$–$\omega$ SST profiles for M14Tw018 at $Pr_t = 0.9$, compared with Zhang et al.'s DNS [5]. Top: dimensional $T(U)$ (left) and van Driest–transformed velocity $U^+_{\mathrm{VD}}$ versus $z^+$ (right). Bottom: $T/T_{\infty,\mathrm{DNS}}$ (left) and $U/U_{\infty,\mathrm{DNS}}$ (right), both versus $z^+$ at the matched $Re_{\delta 2}$ = 2354 station.

## C.3 Huang case at station BL2

The Huang M11Tw020 case is represented by station BL2, this is the only of the three station for which the reference velocity and temperature profiles are available.

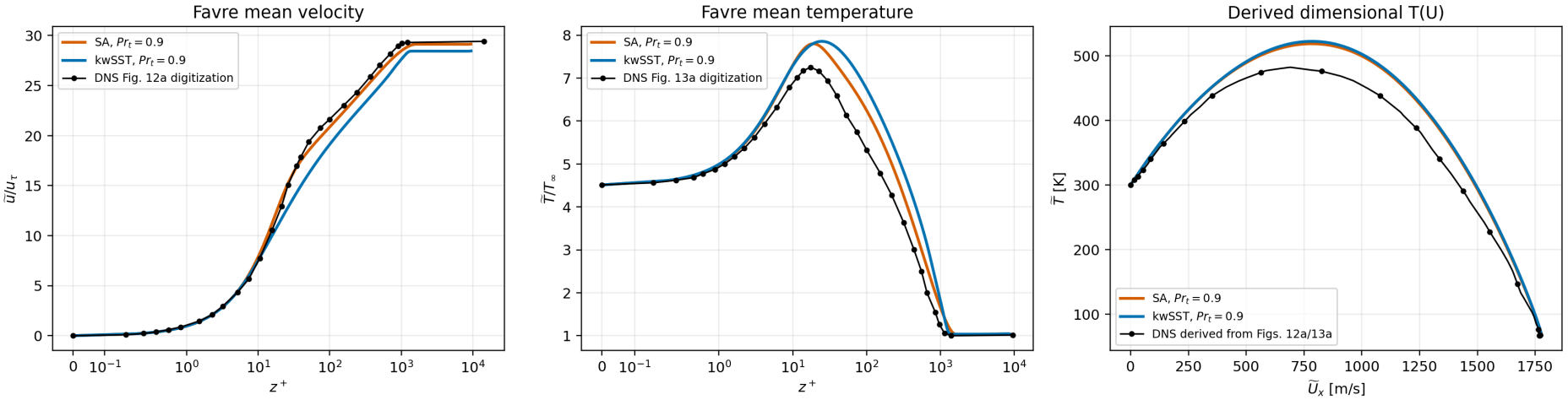



**Figure C.10** Wall-resolved Spalart–Allmaras and $k$–$\omega$ SST profiles for M11Tw020 at station BL2 and $Pr_t = 0.9$, compared with DNS data digitized from Huang et al. [82, Figures 12(a) and 13(a)]. Left: Favre-averaged velocity $\widetilde{u}/u_\tau$ versus $z^+$. Middle: temperature $\widetilde{T}/T_\infty$ versus $z^+$ at the matched $Re_{\delta 2}$ = 3419 station. Right: the corresponding dimensional temperature–velocity relation.

# D A-priori wall-function assessments

This appendix supplements Section 7.3 with Li and Griffin assessments for the additional Zhang cases M2p5, M6Tw076, M8Tw048 and M14Tw018. The prescribed-profile procedure is unchanged. For adiabatic M2p5, relative heat-flux errors are undefined. The M6Tw076 heat-flux curves are retained for completeness, subject to the reference-data limitations in Section 6.1.

## D.1 Zhang cases

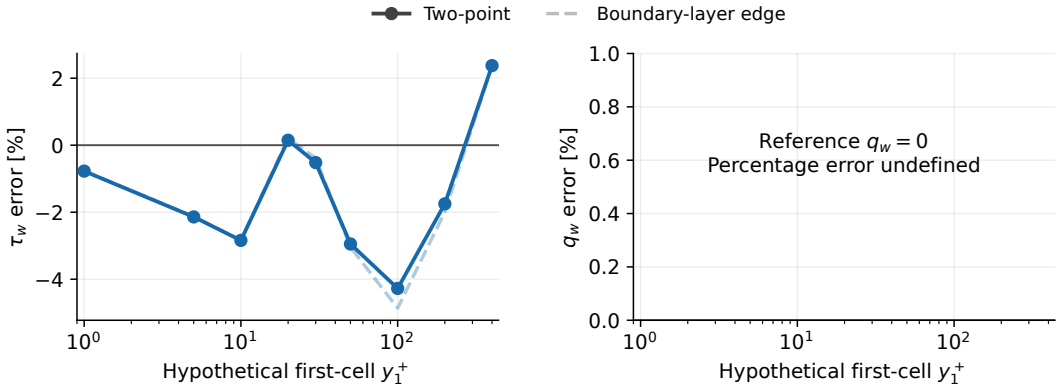



**Figure D.1** A-priori Li assessment for M2p5; $Re_\tau = 510$ [5]. Signed relative errors in wall shear stress $\tau_w$ (left) and heat-flux $q_w$ (right) versus $y_1^+$. Solid: equal-height two-cell formulation; dashed: boundary layer-edge reference. The heat-flux percentage error is undefined for this adiabatic reference.

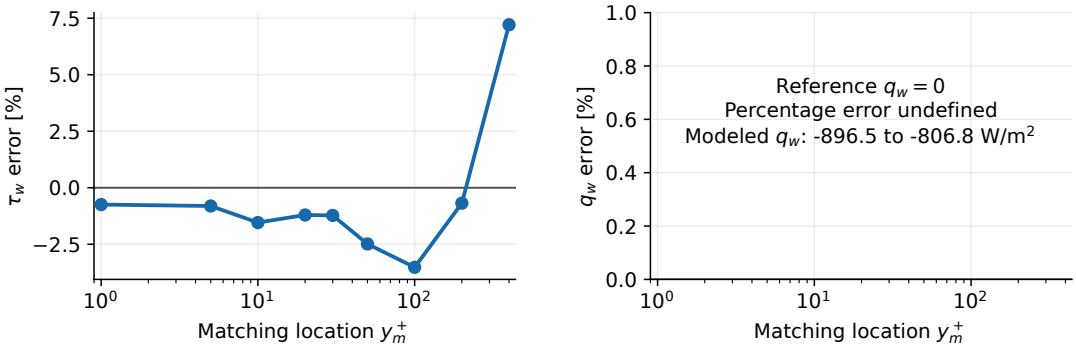



**Figure D.2** A-priori assessment of the Griffin wall function for M2p5; $Re_\tau = 510$ [5]. Signed relative errors in wall shear stress $\tau_w$ (left) and wall heat-flux $q_w$ (right) versus matching height $y_m^+$. The relative heat-flux error is undefined because the reference heat-flux is zero.

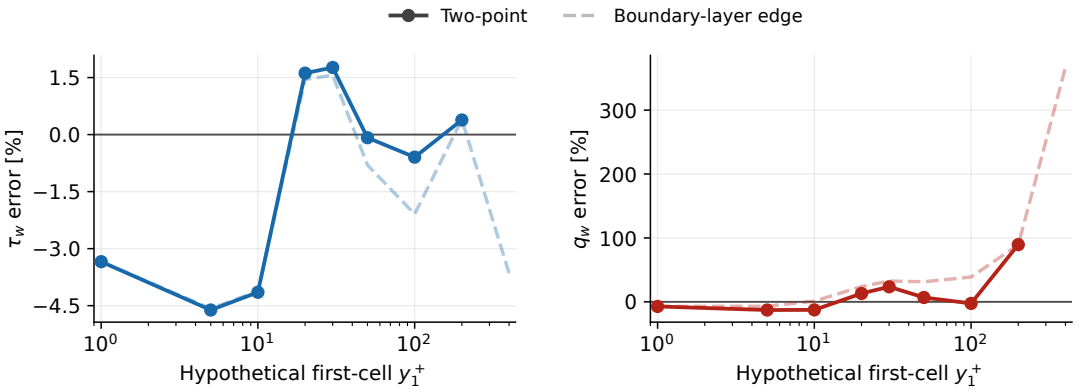



**Figure D.3** A-priori assessment of the Li wall function for M6Tw076; $Re_\tau = 453$ [5]. Signed relative errors in wall shear stress $\tau_w$ (left) and wall heat-flux $q_w$ (right) versus virtual first-cell height $y_1^+$. Solid curves use the equal-height two-cell formulation; dashed curves use the boundary layer-edge reference. The heat-flux curves are shown for completeness and excluded from quantitative assessment against DNS.

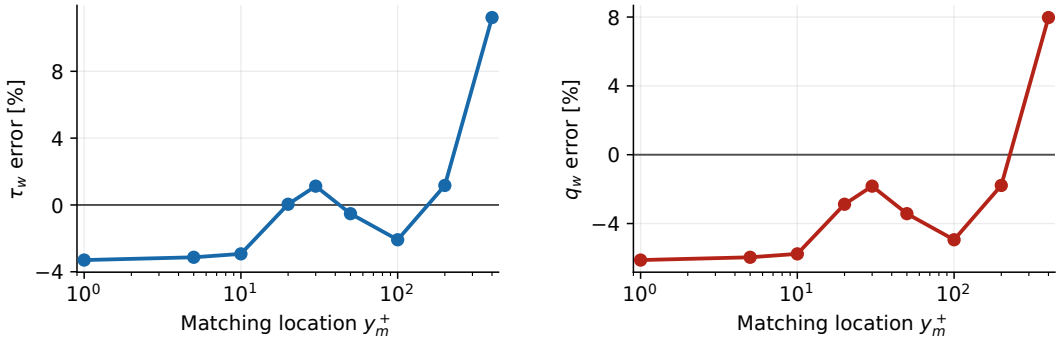



**Figure D.4** A-priori assessment of the Griffin wall function for M6Tw076; $Re_\tau = 453$ [5]. Signed relative errors in wall shear stress $\tau_w$ (left) and wall heat-flux $q_w$ (right) versus matching height $y_m^+$. The heat-flux curves are shown for completeness and excluded from quantitative assessment against DNS.

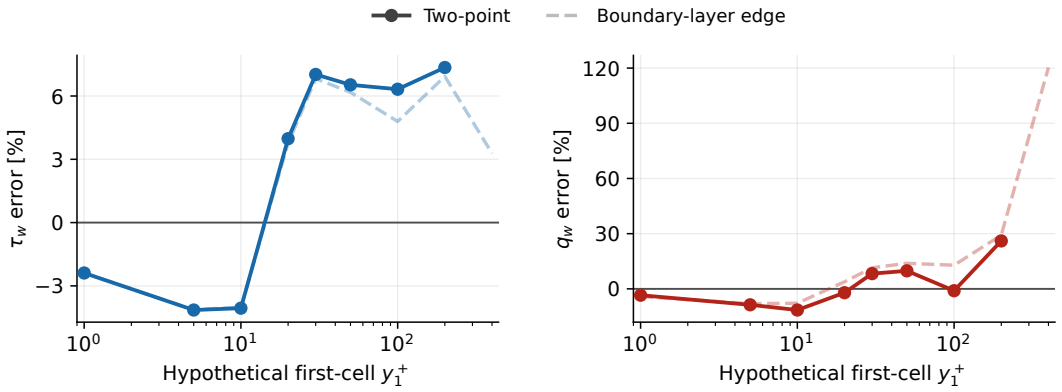



**Figure D.5** A-priori assessment of the Li wall function for M8Tw048; $Re_\tau = 480$ [5]. Signed relative errors in wall shear stress $\tau_w$ (left) and wall heat-flux $q_w$ (right) versus virtual first-cell height $y_1^+$. Solid curves use the equal-height two-cell formulation; dashed curves use the boundary layer-edge reference.

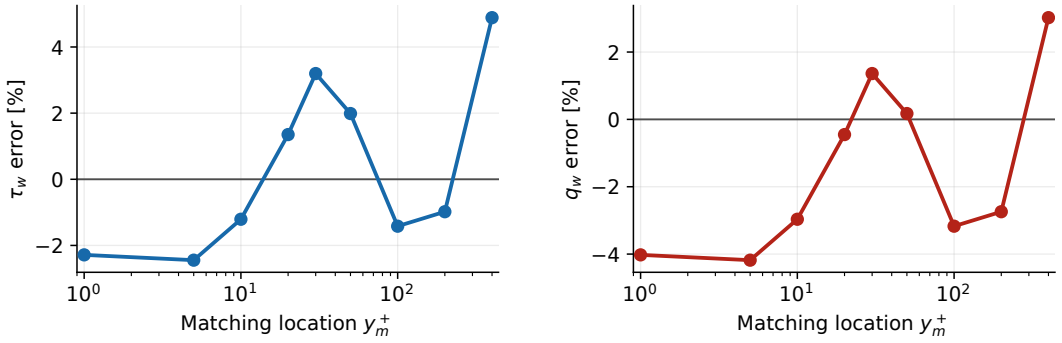



**Figure D.6** A-priori assessment of the Griffin wall function for M8Tw048; $Re_\tau = 480$ [5]. Signed relative errors in wall shear stress $\tau_w$ (left) and wall heat-flux $q_w$ (right) versus matching height $y_m^+$.

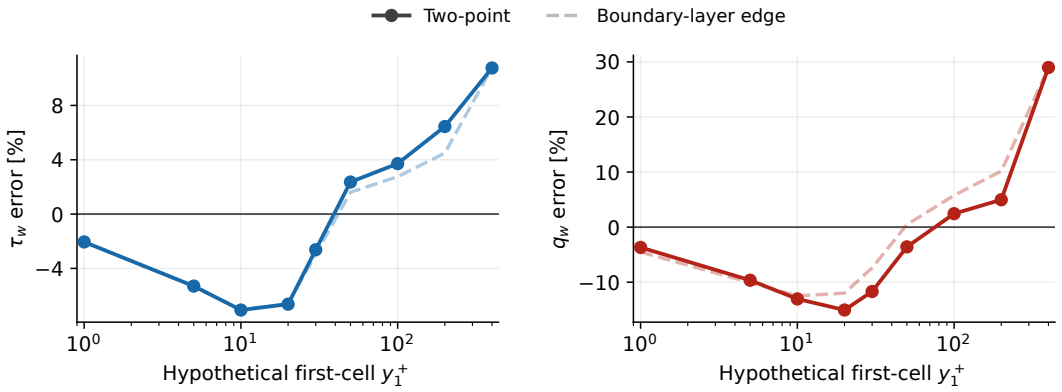



**Figure D.7** A-priori assessment of the Li wall function for M14Tw018; $Re_\tau = 646$ [5]. Signed relative errors in wall shear stress $\tau_w$ (left) and wall heat-flux $q_w$ (right) versus virtual first-cell height $y_1^+$. Solid curves use the equal-height two-cell formulation; dashed curves use the boundary layer-edge reference.

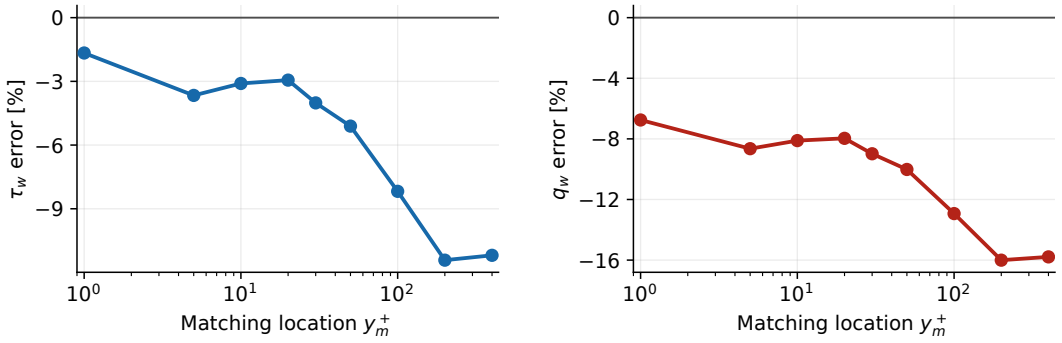



**Figure D.8** A-priori assessment of the Griffin wall function for M14Tw018; $Re_\tau = 646$ [5]. Signed relative errors in wall shear stress $\tau_w$ (left) and wall heat-flux $q_w$ (right) versus matching height $y_m^+$.

# E Sutherland-law coefficient in Malik and Anderson (1991)

## E.1 Summary

This appendix concerns M. R. Malik and E. C. Anderson, "Real gas effects on hypersonic boundary-layer stability," *Physics of Fluids A* **3**, 803–821 (1991), doi:10.1063/1.858012 [2]. Their Eq. (56) states the Sutherland law

$$\mu(T) = d_1 \frac{\sqrt{T}}{1 + d_2/T}, \qquad d_2 = 110\ \mathrm{K},$$

with $d_{1,\mathrm{print}} = 1.0869 \times 10^{-6}$. When $T$ is measured in kelvin and $\mu$ in SI units, dimensional consistency requires

$$[d_1] = \mathrm{kg\,m^{-1}\,s^{-1}\,K^{-1/2}}.$$

The unit printed with Eq. (56) omits the factor $\mathrm{K}^{-1/2}$; this dimensional typo is separate from the numerical-coefficient inconsistency investigated here.

The Mach-15 calorically imperfect mean-flow profiles in Fig. 3 cannot be reproduced with the printed coefficient under the stated Sutherland, Keyes, and Eggers closures. The tested candidate $d_{1,\mathrm{cand}} = 1.5869 \times 10^{-6}\,\mathrm{kg\,m^{-1}\,s^{-1}\,K^{-1/2}}$ gives substantially closer agreement with the digitized Fig. 3 profiles. Analytically, the distinction is consistent: the Sutherland constant $d_1$ cancels from the Reynolds-normalized perfect-gas similarity problem, whereas it remains in the calorically imperfect energy equation through the thermal-diffusion coefficient $K \propto d_1^{-1}$.

- **Strong indication of a typo:** the printed coefficient in Eq. (56) is inconsistent with the calorically imperfect mean-flow profiles in Fig. 3, whereas $d_1 = 1.5869 \times 10^{-6}$ gives substantially closer agreement. This strongly indicates a typographical or implementation inconsistency, although the exact replacement value is not uniquely established by the digitized profiles.
- **Perfect-gas cases are unaffected:** although they use the same Sutherland law, $d_1$ cancels from the Reynolds-normalized perfect-gas similarity equations and from the plotted wall coordinate. This appendix demonstrates the cancellation analytically and confirms it with the profile reproductions.
- **Real-gas cases are unaffected by Eq. (56):** the real-gas calculations, which form the main contribution of the paper, use a Wilke mixture-viscosity model instead of the disputed Sutherland coefficient. They are therefore outside the direct scope of this inconsistency.
- **Figure 13 remains unresolved:** Fig. 13 includes a calorically imperfect stability curve and may therefore be affected. Neither its perfect-gas nor its imperfect-gas curve was reproduced here, so no conclusion is drawn about the magnitude or direction of a possible effect.

### Outline

- Section E.3 shows that the discrepancy does not originate in the length scale. Reconstructing $y$ from the Levy–Lees similarity coordinate $\eta$ and then normalizing it as $Y = y/L_r$ introduces no explicit dependence on $d_1$: the factor cancels.
- Section E.4 shows that the Reynolds-normalized perfect-gas profiles are independent of $d_1$. The momentum and energy similarity equations contain no explicit $d_1$.

- Section E.5 shows why the calorically imperfect profiles do depend on $d_1$. The thermal-diffusion coefficient satisfies $K \propto 1/d_1$, so $d_1$ affects the temperature profile directly and the velocity profile through the coupled similarity equations.
- Section E.6.1 compares the perfect- and calorically imperfect-gas solutions with the digitized profiles from Fig. 3 of Malik and Anderson. For the Mach-15 calorically imperfect case, the candidate $d_{1,\mathrm{cand}} = 1.5869 \times 10^{-6}$ is substantially more consistent with the published profiles than the printed value. A non-converged printed-coefficient marching attempt is also shown as a visual diagnostic; neither comparison is a unique fit of the exact replacement value.

The detailed mathematical derivations are given in Sections E.7–E.9 and referenced throughout this appendix.

## E.2 Malik–Anderson property setup

The paper gives the Sutherland viscosity law as Eq. (56) [2]:

$$\mu(T) = d_1 \frac{\sqrt{T}}{1 + d_2/T} = d_1 \frac{T^{3/2}}{T + d_2}, \qquad d_2 = 110\ \mathrm{K}. \tag{E.1}$$

The printed and candidate values compared in this appendix are

$$d_{1,\mathrm{print}} = 1.0869 \times 10^{-6}\ \mathrm{kg\,m^{-1}\,s^{-1}\,K^{-1/2}}, \tag{E.2}$$

$$d_{1,\mathrm{cand}} = 1.5869 \times 10^{-6}\ \mathrm{kg\,m^{-1}\,s^{-1}\,K^{-1/2}}. \tag{E.3}$$

The second value is treated as a tested candidate, not as a uniquely fitted correction.

**Malik–Anderson Mach 10 / 15 "perfect"-gas model**

$$p = \rho R T, \qquad c_p = \mathrm{const.}, \qquad Pr_0 = 0.7, \qquad k = \frac{\mu c_p}{Pr_0}. \tag{E.4}$$

**Malik–Anderson Mach 15 "imperfect"-gas model**

$$p = \rho R T, \qquad R = \mathrm{const.}, \tag{E.5}$$

$$c_p = c_p(T) \quad \text{from the Eggers vibrational formula (Malik–Anderson Eq. 59),} \tag{E.6}$$

$$k = k(T) \quad \text{from the Keyes formula (Malik–Anderson Eq. 58),} \tag{E.7}$$

$$\mu = \mu(T) \quad \text{from Sutherland's law (Malik–Anderson Eq. 56).} \tag{E.8}$$

The important distinction is that, in the perfect gas, $k$ is proportional to $\mu$. In the imperfect gas, $k(T)$ is specified independently of $\mu(T)$.

### E.2.1 Cases, boundary conditions, and held parameters

The profile comparisons use the published edge states and unit Reynolds numbers, with the numerical station and wall conditions listed below. The same values are held fixed when the two candidate coefficients are compared.

| Case | $M_e$ | $T_e$ [K] | $Re_{\mathrm{unit}}$ [m$^{-1}$] | $x$ [m] | Wall condition |
|---|---|---|---|---|---|
| Fig. 1, perfect | 10 | 350 | $6.6 \times 10^6$ | 0.1 | adiabatic |
| Fig. 3, perfect | 15 | 700 | $3.4 \times 10^6$ | 0.1 | $T_w/T_e = 1.75$ |
| Fig. 3, imperfect | 15 | 700 | $3.4 \times 10^6$ | 0.1 | $T_w/T_e = 1.75$ |

For every similarity solve,

$$f(0) = 0, \qquad f'(0) = 0, \qquad f'(\infty) = 1, \tag{E.9}$$

and the thermal boundary conditions are

$$\left(\frac{T}{T_e}\right)'_w = 0 \quad \text{(adiabatic)}, \qquad \frac{T_w}{T_e} = 1.75 \quad \text{(isothermal)}, \qquad \frac{T(\infty)}{T_e} = 1. \tag{E.10}$$

Also held fixed are $R = 287\ \mathrm{J\,kg^{-1}\,K^{-1}}$, $\gamma = 1.4$ for the perfect-gas reference, $d_2 = 110\ \mathrm{K}$, the Keyes and Eggers constants where applicable, and the definitions of $Y$ and $Re_x$. At fixed $M_e$, $T_e$, and unit Reynolds number, changing $d_1$ changes $\mu_e$; the edge density and pressure are consequently recomputed to preserve the prescribed unit Reynolds number. The comparison is therefore a fixed nondimensional-flow-condition test, not a fixed-edge-pressure test.

## E.3 Wall-coordinate normalization: explicit cancellation of $d_1$

This section shows that the Malik–Anderson paper's plotted wall-normal coordinate $Y$ contains no explicit dependence on $d_1$; it therefore cannot introduce or remove any $d_1$-dependence in the similarity equations. The similarity equations are solved as functions of the similarity variable $\eta$:

$$\eta \mapsto \left(Y(\eta), \frac{u}{u_e}(\eta)\right), \qquad \eta \mapsto \left(Y(\eta), \frac{T}{T_e}(\eta)\right).$$

The physical wall-normal distance $y$ is then reconstructed from $\eta$ using the Levy–Lees transformation, and the plotted coordinate is obtained by normalizing this distance as $Y = y/L_r$. Therefore, the full coordinate mapping

$$\eta \mapsto y(\eta) \mapsto Y(\eta)$$

must be checked for explicit $d_1$-dependence.

### E.3.1 Levy–Lees transformation $\eta \mapsto y$

Schlichting and Gersten give the Levy–Lees transformation in integral form as their Eq. (10.68) [1]:

$$\eta = \frac{u_e}{\sqrt{2\xi}} \int_0^y \rho \,\mathrm{d}y, \qquad \xi = \int_0^x \rho_e \mu_e u_e \,\mathrm{d}x. \tag{E.11}$$

At a fixed streamwise station $x$, the quantities $\xi$ and $u_e$ are constants with respect to $y$. Differentiating the first relation with respect to $y$ therefore gives

$$\mathrm{d}\eta = \frac{u_e}{\sqrt{2\xi}} \rho \,\mathrm{d}y, \qquad \text{or} \qquad \mathrm{d}y = \frac{\sqrt{2\xi}}{\rho u_e} \,\mathrm{d}\eta. \tag{E.12}$$

For the Malik–Anderson flat-plate zero-pressure-gradient case, the boundary layer edge quantities are constant in $x$, so

$$\xi = \int_0^x \rho_e \mu_e u_e \,\mathrm{d}x = \rho_e \mu_e u_e x. \tag{E.13}$$

Using $Re_x = \rho_e u_e x / \mu_e$, this can be rewritten as

$$\xi = Re_x \, \mu_e^2. \tag{E.14}$$

Inserting this into Eq. (E.12) gives

$$\mathrm{d}y = \frac{\sqrt{2 Re_x \, \mu_e^2}}{\rho u_e} \,\mathrm{d}\eta, \tag{E.15}$$

with an apparent explicit dependence on $\mu_e$ and thus $d_1$. However, the subsequent normalization step cancels this explicit dependence, as shown in the next section.

### E.3.2 Wall-normal coordinate normalization $y \mapsto Y$

Malik–Anderson normalize the physical wall-normal distance $y$ by the length scale $L_r = \sqrt{\nu_e x/u_e} = x/\sqrt{Re_x}$. Thus the plotted wall-normal coordinate is

$$Y = \frac{y}{L_r} = y\sqrt{\frac{\rho_e u_e}{\mu_e x}} = \frac{y\sqrt{Re_x}}{x}, \tag{E.16}$$

or in differential form,

$$\mathrm{d}Y = \frac{\sqrt{Re_x}}{x}\,\mathrm{d}y. \tag{E.17}$$

This is the Malik–Anderson normalization $y \mapsto Y$. Written alone, it appears to contain $\mu_e^{-1/2}$ through $\sqrt{Re_x}$.

### E.3.3 Full coordinate mapping $\eta \mapsto y \mapsto Y$ and explicit cancellation of $d_1$

Substitution of Eq. (E.15) into Eq. (E.17) gives

$$\begin{aligned}\mathrm{d}Y &= \frac{\sqrt{Re_x}}{x}\,\mathrm{d}y \\ &= \frac{\sqrt{Re_x}}{x}\frac{\sqrt{2Re_x\,\mu_e^2}}{\rho u_e}\,\mathrm{d}\eta \\ &= \frac{\sqrt{2}\,Re_x\,\mu_e}{x\rho u_e}\,\mathrm{d}\eta.\end{aligned} \tag{E.18}$$

Now insert $Re_x = \rho_e u_e x/\mu_e$:

$$\mathrm{d}Y = \frac{\sqrt{2}}{x\rho u_e}\left(\frac{\rho_e u_e x}{\mu_e}\right)\mu_e\,\mathrm{d}\eta = \sqrt{2}\,\frac{\rho_e}{\rho}\,\mathrm{d}\eta \tag{E.19}$$

and, with the thermally perfect equation of state and $p = p_e$ across a zero-pressure-gradient boundary layer, so that $\rho_e/\rho = T/T_e$, the normalized coordinate satisfies

$$\mathrm{d}Y = \sqrt{2}\,\frac{T}{T_e}\,\mathrm{d}\eta, \qquad Y(\eta) = \sqrt{2}\int_0^\eta \frac{T(\tilde{\eta})}{T_e}\,\mathrm{d}\tilde{\eta}. \tag{E.20}$$

- The edge-viscosity $\mu_e$, and therefore $d_1$, cancels during the transformation-normalization mapping $\eta \mapsto y(\eta) \mapsto Y(\eta)$.
- The remaining dependence of $Y(\eta)$ is only through the solved temperature profile $T(\eta)/T_e$.
- For the perfect-gas case, $T(\eta)/T_e$ is independent of $d_1$, so $Y(\eta)$ is also independent of $d_1$ (see Section E.4).
- For the calorically imperfect case, $T(\eta)/T_e$ changes through the $d_1$-dependent energy equation (see Section E.5). $Y(\eta)$ changes indirectly, but does not cancel out any $d_1$ dependence.

## E.4 Analysis of the perfect gas momentum and energy similarity equations

In this section, the perfect-gas boundary layer similarity equations are introduced and examined with respect to their dependence on $d_1$.
The perfect-gas similarity problem is governed by the streamwise momentum equation (derivation in Section E.7.1)

$$(Cf'')' + ff'' = 0, \qquad C = \frac{\rho\mu}{\rho_e\mu_e}, \tag{E.21}$$

The coefficient $C$ is the Chapman–Rubesin viscosity-density coefficient; local quantities $\rho$ and $\mu$ are evaluated at the local static temperature $T(\eta)$, while subscript $e$ denotes boundary layer edge values.
Section E.7.2.2 derives the corresponding static-temperature energy equation:

$$\left[\frac{C}{Pr_0}\left(\frac{T}{T_e}\right)'\right]' + f\left(\frac{T}{T_e}\right)' + CQ\left(f''\right)^2 = 0, \qquad Q = \frac{u_e^2}{c_p T_e}. \tag{E.22}$$

In this equation, $T/T_e$ is the normalized static-temperature profile, $Pr_0$ is the prescribed constant perfect-gas Prandtl number, and $Q$ is the nondimensional kinetic-energy scale that comes from writing total enthalpy as $H = h + u^2/2$. For the perfect gas, $c_p$ is constant.
Thus, the normalized perfect-gas profiles depend only on $C(T/T_e)$, $Q$, and the prescribed value $Pr_0 = 0.7$.

$$\begin{aligned} C\left(\frac{T}{T_e}\right) &\text{ is independent of } d_1 \text{ (see Section E.8)},\\ Pr_0 = 0.7 &\text{ is prescribed (Section E.2)},\\ Q = \frac{u_e^2}{c_p T_e} &\text{ is independent of } d_1. \end{aligned} \tag{E.23}$$

For fixed boundary conditions and the perfect-gas closure stated in Section E.2, the similarity equations contain no explicit $d_1$-dependence. The normalized velocity and temperature profiles are therefore insensitive to $d_1$. The Mach-10 and Mach-15 perfect-gas reproductions illustrate this cancellation, but cannot diagnose the printed-coefficient inconsistency (see Figures E.1 and E.2).

## E.5 Analysis of the calorically imperfect gas momentum and energy similarity equations

In this section, the calorically imperfect-gas boundary layer similarity equations are introduced and examined with respect to their dependence on $d_1$. Calorically imperfect herein is as defined in the Malik–Anderson paper, where $R$ is constant but $c_p(T)$ and $k(T)$ vary and are prescribed independently of $\mu(T)$ (see Section E.2).
The imperfect-gas similarity problem is governed by the streamwise momentum equation (derivation in Section E.7.1)

$$\left(Cf''\right)' + ff'' = 0, \qquad C = \frac{\rho\mu}{\rho_e\mu_e}, \tag{E.24}$$

The coefficient $C$ is the Chapman–Rubesin viscosity-density coefficient. The momentum equation therefore has the same form as for the perfect gas (see Eq. (E.21)); the difference between the gas models only enters through the temperature field used to evaluate the local properties.
Section E.7.2.1 derives the corresponding static-temperature energy equation:

$$\left[K\left(\frac{T}{T_e}\right)'\right]' + \frac{c_p(T)}{c_{p,e}} f\left(\frac{T}{T_e}\right)' + CQ\left(f''\right)^2 = 0, \tag{E.25}$$

with

$$K = \frac{\rho k(T)}{\rho_e\mu_e c_{p,e}}, \qquad C = \frac{\rho\mu}{\rho_e\mu_e}, \qquad Q = \frac{u_e^2}{c_{p,e}\,T_e}. \tag{E.26}$$

Here $c_{p,e} = c_p(T_e)$ is the edge heat capacity, and $K$ is the normalized thermal-diffusion coefficient multiplying the static-temperature gradient. The conductivity $k(T)$ is the Keyes thermal conductivity law used by Malik–Anderson (see Section E.2). As in the perfect-gas equation, $Q$ is the nondimensional kinetic-energy scale in the energy equation.

$$C\left(\frac{T}{T_e}\right) \text{ contains no explicit } d_1 \text{ at fixed } T/T_e \text{ (see Section E.8)},$$
$$Q = \frac{u_e^2}{c_{p,e}\,T_e} \text{ is independent of } d_1,$$
$$\frac{c_p(T)}{c_{p,e}} \text{ is prescribed by the Eggers law (see Section E.2).} \tag{E.27}$$
$$K = \frac{\rho k(T)}{\rho_e \mu_e c_{p,e}} \propto \frac{1}{d_1} \quad \text{(see Section E.9).}$$

The calorically imperfect energy equation retains explicit $d_1$-dependence through $K$. The temperature profile therefore changes directly, and the velocity profile changes indirectly because the momentum equation is coupled to the changed temperature field (see Figure E.3).

## E.6 Diagnostic result and scope

**Perfect gas with constant $Pr$.**

$d_1$ cancels from all normalized profile equations and from the plotted $Y$ coordinate.

Thus no $d_1$-dependence is expected in these plotted mean-flow profiles.

**Calorically imperfect gas with Keyes $k(T)$ and Eggers $c_p(T)$.**

$d_1$ remains in $K$, so the energy equation changes.

Temperature changes directly; velocity changes through coupling to the changed temperature profile. While the transformation-normalization mapping $Y(\eta)$ contains no explicit $d_1$-dependence, there is no cancellation of $d_1$ from the energy equation. The $d_1$-dependence of the energy equation therefore remains in the plotted profiles (see Figure E.3).

A perfect-gas replication provides a consistency check on the solver and normalization, but it cannot distinguish

$$d_1 = 1.0869 \times 10^{-6} \qquad \text{from} \qquad d_1 = 1.5869 \times 10^{-6}.$$

The Mach-15 calorically imperfect mean-flow profiles in Fig. 3 are therefore the discriminating case examined here.

These results do not prove that the exact replacement is $d_1 = 1.5869 \times 10^{-6}$, nor do they distinguish a typographical error in Eq. (56) from an undocumented difference in the original implementation. They establish the narrower result that the calorically imperfect Fig. 3 mean-flow profiles are inconsistent with the printed value under the published closures and are much better matched by the tested value near $1.59 \times 10^{-6}$.

Figure 13 of Malik–Anderson also includes a calorically imperfect stability result, so it is potentially within the scope of the Eq. (56) inconsistency. Neither the perfect-gas nor the imperfect-gas stability curve in Fig. 13 was reproduced successfully in this work. Figure 13 is therefore outside the quantitative conclusions of this appendix: no claim is made about whether it changes, or about the magnitude or direction of any change.

### E.6.1 Profile validation plots

The self-similar boundary-value solutions provide the coefficient-sensitivity comparison. For the calorically imperfect case, the solver retains the temperature-dependent $c_p(T)$, $\mu(T)$, and $k(T)$ closures stated by Malik–Anderson.

A separate parabolic streamwise-marching implementation uses a Crank–Nicolson-type implicit finite-difference discretization based on the method class documented by Anderson and Lewis [106], which Malik–Anderson cite for their mean-flow calculation. The present marching code is an independent implementation of the method described by Anderson and Lewis. Because Malik and Anderson do not provide all numerical implementation details, it should not be regarded as an exact reproduction of their original solver. Both marching tests are initialized from the converged self-similar $d_1 = 1.5869 \times 10^{-6}$ profile. Retaining that coefficient during the march gives a result that agrees closely with the self-similar solution and therefore provides a cross-check of the numerical formulation and discretization.

For comparison, a second march changes the Sutherland constant in the transport closure to the printed $d_1 = 1.0869 \times 10^{-6}$ while retaining the same initial profile; no converged state can be reached. The resulting strongly distorted velocity and temperature profiles are included in Figure E.3 because they make the incompatibility of the printed coefficient with this marching reconstruction visually immediate. They remain an explicitly non-converged diagnostic, however, and are not used as quantitative coefficient evidence. In particular, their failure cannot establish how the original Malik–Anderson implementation behaved or uniquely identify the corrected value.

For the two self-similar Mach-15 calorically imperfect profiles plotted in Figure E.3, the unweighted errors relative to the digitized Fig. 3 points are:

| Tested $d_1$ | RMSE in $u/u_e$ | RMSE in $T/T_e$ |
|---|---|---|
| 1.0869 × $10^{-6}$ | 0.0425 | 1.343 |
| 1.5869 × $10^{-6}$ | 0.00393 | 0.165 |

These descriptive errors quantify the improvement between the two sampled coefficients; they are not a continuous fit and do not include digitization uncertainty.

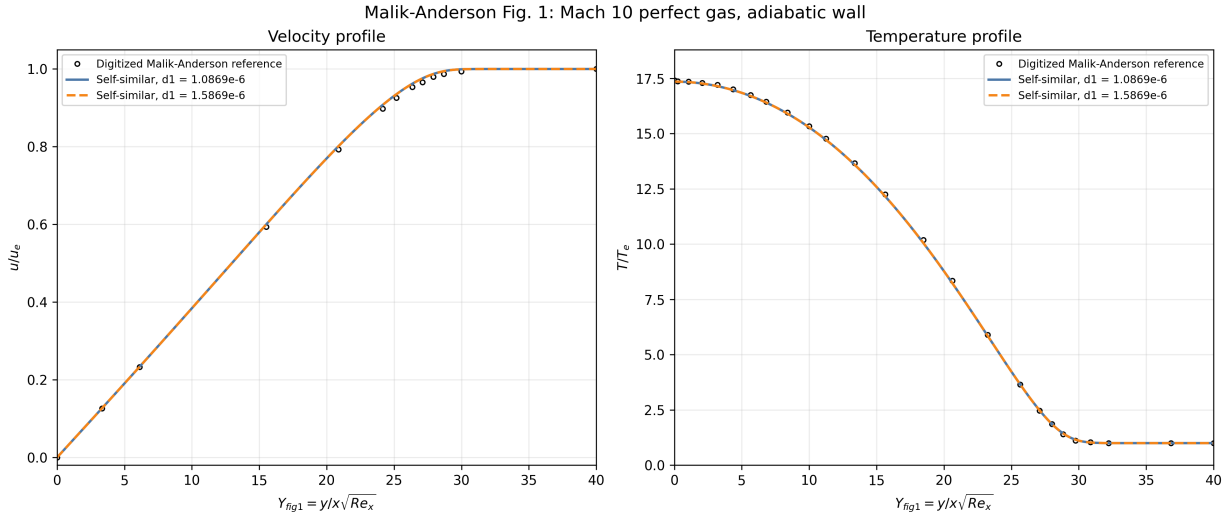



**Figure E.1** Mach-10 perfect-gas comparison with Malik–Anderson Fig. 1. The self-similar profiles for $d_1 = 1.0869 \times 10^{-6}$ and $d_1 = 1.5869 \times 10^{-6}$ are indistinguishable on the normalized axes.

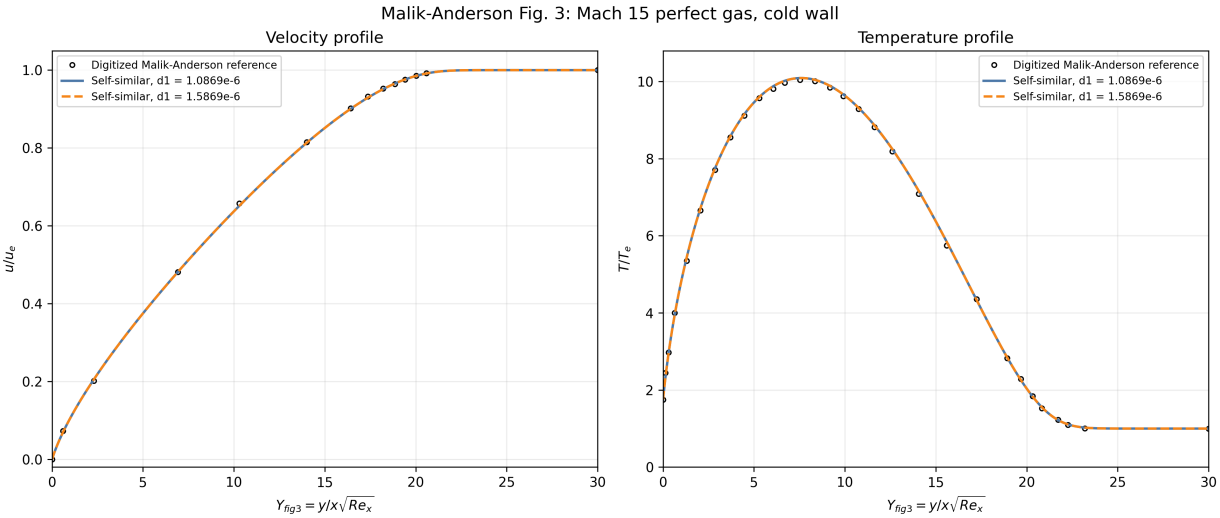



**Figure E.2** Mach-15 perfect-gas comparison with Malik–Anderson Fig. 3. As in the Mach-10 case, the self-similar perfect-gas profiles are insensitive to the multiplicative Sutherland coefficient.

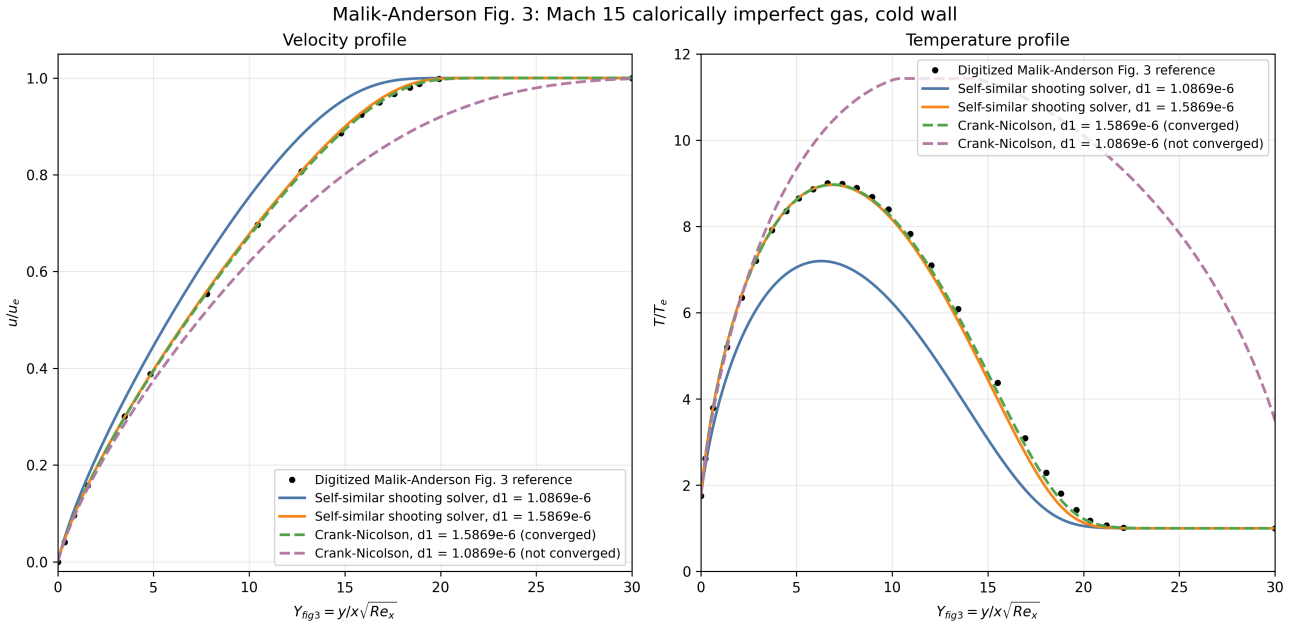



**Figure E.3** Mach-15 calorically imperfect-gas comparison with Malik–Anderson Fig. 3. The self-similar solutions reveal the difference between the two tested $d_1$ values. The converged Crank–Nicolson-type result at $d_1 = 1.5869 \times 10^{-6}$ agrees closely with the corresponding self-similar solution and serves as a numerical-formulation check. The printed-coefficient marching attempt (purple dashed) reaches the Picard iteration cap and produces strongly distorted profiles; it is included as an explicitly visual diagnostic.

## E.7 Levy–Lees reduction of the Malik–Anderson boundary layer equations

This section starts from the boundary layer equations of Malik–Anderson (Eqs. (38)–(41b)) [2] and uses the Levy–Lees transformation given by Schlichting and Gersten (Eq. (10.68)) [1]. The transformed equations are first reduced under the assumptions used for the flat-plate profiles and are then rewritten in static-temperature form. This identifies the coefficients $C$, $Q$, and $K$ without relying on a particular numerical discretization.

Schlichting and Gersten define

$$\xi = \int_0^x \rho_e \mu_e u_e \,\mathrm{d}x, \qquad \eta = \frac{u_e}{\sqrt{2\xi}} \int_0^y \rho \,\mathrm{d}y, \qquad \frac{u}{u_e} = f'(\xi,\eta), \qquad g(\xi,\eta) = \frac{H - H_e}{H_e}, \tag{E.28}$$

where $H = h + u^2/2$ is dimensional total enthalpy, $H_e = h_{t,e}$, and a prime denotes partial differentiation with respect to $\eta$. The normalized total enthalpy used by Malik–Anderson approaches one at the edge, whereas $g$ approaches zero; the two definitions differ only by an additive constant, so their derivatives are identical after the edge scale is restored.

### E.7.1 Common momentum similarity equation and the coefficient $C$

The transformed momentum equation of Schlichting and Gersten (Eqs. (10.70) and (10.72)) is

$$(Cf'')' + ff'' + \beta\left(\frac{\rho_e}{\rho} - f'^2\right) = 2\xi\left(f'\frac{\partial f'}{\partial \xi} - f''\frac{\partial f}{\partial \xi}\right), \qquad \beta = \frac{2\xi}{u_e}\frac{\mathrm{d}u_e}{\mathrm{d}\xi}, \qquad C = \frac{\rho\mu}{\rho_e\mu_e}. \tag{E.29}$$

For the planar Malik–Anderson flat-plate cases, the geometry exponent in Eqs. (38)–(40a) is $j = 0$. The edge state is uniform and the pressure gradient is zero, so $u_e$ is constant and $\beta = 0$. The present calculation seeks the similar solution

$$f(\xi,\eta) = f(\eta), \qquad g(\xi,\eta) = g(\eta), \tag{E.30}$$

so all partial derivatives with respect to $\xi$ on the right-hand side of Eq. (E.29) vanish. The constant edge state and streamwise-invariant wall condition (constant $T_w/T_e$, or an adiabatic wall) ensure that the remaining nondimensional property ratios are functions of $\eta$ only. Hence

$$(Cf'')' + ff'' = 0, \qquad C = \frac{\rho\mu}{\rho_e\mu_e}. \tag{E.31}$$

This is the momentum equation used in the remainder of the analysis for both gas models.

### E.7.2 Energy similarity equation and static-temperature reductions

The energy equation is where the perfect-gas and calorically imperfect closures differ. Schlichting and Gersten write the transformed total-enthalpy equation as their Eq. (10.71):

$$\left[\frac{C}{Pr}g' + C\frac{u_e^2}{H_e}\left(1 - \frac{1}{Pr}\right)f'f''\right]' + fg' = 2\xi\left(f'\frac{\partial g}{\partial \xi} - g'\frac{\partial f}{\partial \xi}\right). \tag{E.32}$$

Under Eq. (E.30), the right-hand side vanishes, giving

$$\left(\frac{C}{Pr}g'\right)' + fg' + \left[C\frac{u_e^2}{H_e}\left(1 - \frac{1}{Pr}\right)f'f''\right]' = 0. \tag{E.33}$$

Schlichting and Gersten derive Eq. (10.71) for an ideal gas with constant $c_p$ and constant $Pr$. It is used here as the standard Levy–Lees structure and notation, not as the constitutive derivation for the calorically imperfect model. For that model, applying Eq. (E.11) directly to the planar ($j = 0$) Malik–Anderson Eq. (40a), after restoring the edge scales implicit in $f$ and $g$, gives the same reduced flux structure with the local $Pr(T) = \mu(T)c_p(T)/k(T)$. The reduction below is therefore based on Malik–Anderson Eq. (40a): first scale the total-enthalpy equation with $c_{p,e}T_e$, then rewrite the total-enthalpy derivative in terms of static temperature, and finally use the momentum equation to collapse the kinetic-energy terms. The perfect-gas equation follows as the constant-$c_p$, constant-$Pr$ special case.

#### E.7.2.1 Imperfect-gas static-temperature energy equation

Define the edge heat capacity and introduce the nondimensional kinetic-energy scale $Q$ as

$$c_{p,e} \equiv c_p(T_e), \qquad Q = \frac{u_e^2}{c_{p,e}\,T_e}. \tag{E.34}$$

Multiplying Eq. (E.33) by $H_e/(c_{p,e}\,T_e)$ gives

$$\left[\frac{C}{Pr}\frac{H_e}{c_{p,e}\,T_e}g'\right]' + f\frac{H_e}{c_{p,e}\,T_e}g' + \left[C\frac{u_e^2}{c_{p,e}\,T_e}\left(1-\frac{1}{Pr}\right)f'f''\right]' = 0. \tag{E.35}$$

Since $H_e$ is an edge constant, the scaled total-enthalpy derivative can be rewritten as

$$g' = \left(\frac{H}{H_e}\right)' = \frac{1}{H_e}H', \tag{E.36}$$

Because $H_e$, $c_{p,e}$, and $T_e$ are edge constants at the $x$-station, Eq. (E.36) rewrites the terms $H_e/(c_{p,e}T_e)\,g'$ in Eq. (E.35) as

$$\frac{H_e}{c_{p,e}\,T_e}g' = \left(\frac{H}{c_{p,e}\,T_e}\right)'. \tag{E.37}$$

Substituting Eqs. (E.37) and $Q$ from (E.34) into Eq. (E.35) gives

$$\left[\frac{C}{Pr}\left(\frac{H}{c_{p,e}\,T_e}\right)'\right]' + f\left(\frac{H}{c_{p,e}\,T_e}\right)' + \left[CQ\left(1-\frac{1}{Pr}\right)f'f''\right]' = 0. \tag{E.38}$$

For the calorically imperfect gas,

$$H = h(T) + \frac{u^2}{2}, \qquad \frac{\mathrm{d}h}{\mathrm{d}T} = c_p(T), \qquad \frac{u}{u_e} = f'. \tag{E.39}$$

The $\eta$-derivative of $H$ is obtained by applying the chain rule:

$$\begin{aligned} H' &= \frac{\mathrm{d}}{\mathrm{d}\eta}\left[h(T(\eta)) + \frac{u(\eta)^2}{2}\right] \\ &= \frac{\mathrm{d}h}{\mathrm{d}T}\frac{\mathrm{d}T}{\mathrm{d}\eta} + u\frac{\mathrm{d}u}{\mathrm{d}\eta} \\ &= c_p(T)T' + u\frac{\mathrm{d}u}{\mathrm{d}\eta}. \end{aligned} \tag{E.40}$$

Therefore, using Eq. (E.40), the scaled total-enthalpy derivative is

$$\begin{aligned} \left(\frac{H}{c_{p,e}\,T_e}\right)' &= \frac{H'}{c_{p,e}\,T_e} \\ &= \frac{1}{c_{p,e}\,T_e}\left[c_p(T)T' + u\frac{\mathrm{d}u}{\mathrm{d}\eta}\right] \\ &= \frac{c_p(T)}{c_{p,e}}\left(\frac{T}{T_e}\right)' + \frac{u_e^2}{c_{p,e}\,T_e}\frac{u}{u_e}\left(\frac{u}{u_e}\right)' \\ &= \frac{c_p(T)}{c_{p,e}}\left(\frac{T}{T_e}\right)' + Qf'f''. \end{aligned} \tag{E.41}$$

Substituting Eq. (E.41) into Eq. (E.38) gives

$$\begin{aligned} &\left\{\frac{C}{Pr}\left[\frac{c_p(T)}{c_{p,e}}\left(\frac{T}{T_e}\right)' + Qf'f''\right]\right\}' + f\left[\frac{c_p(T)}{c_{p,e}}\left(\frac{T}{T_e}\right)' + Qf'f''\right] \\ &\quad + \left[CQ\left(1-\frac{1}{Pr}\right)f'f''\right]' = 0. \end{aligned} \tag{E.42}$$

Now split Eq. (E.42) into the static-temperature-gradient $T/T_e$ terms and the terms proportional to $Q$:

$$\left[\frac{C}{Pr}\frac{c_p(T)}{c_{p,e}}\left(\frac{T}{T_e}\right)'\right]' + f\frac{c_p(T)}{c_{p,e}}\left(\frac{T}{T_e}\right)' + Q\left\{\left[\frac{C}{Pr}f'f''\right]' + ff'f'' + \left[C\left(1-\frac{1}{Pr}\right)f'f''\right]'\right\} = 0. \tag{E.43}$$

Using $Pr(T) = \mu(T)c_p(T)/k(T)$ and the definition of the Chapman–Rubesin coefficient, the first coefficient multiplying the diffusive static-temperature-gradient term is

$$\begin{aligned}\frac{C}{Pr}\frac{c_p(T)}{c_{p,e}} &= \frac{\rho\mu(T)}{\rho_e\mu_e}\frac{k(T)}{\mu(T)c_p(T)}\frac{c_p(T)}{c_{p,e}} \\ &= \frac{\rho k(T)}{\rho_e\mu_e c_{p,e}} \equiv K.\end{aligned} \tag{E.44}$$

Therefore the static-temperature-gradient group in Eq. (E.43) becomes

$$\left[K\left(\frac{T}{T_e}\right)'\right]' + \frac{c_p(T)}{c_{p,e}}f\left(\frac{T}{T_e}\right)'. \tag{E.45}$$

The $Q$-group in Eq. (E.43) reduces by combining the two derivative terms and then using the momentum equation:

$$\begin{aligned}Q\left\{\left[\frac{C}{Pr}f'f''\right]' + ff'f'' + \left[C\left(1-\frac{1}{Pr}\right)f'f''\right]'\right\} &= Q\left\{[Cf'f'']' + ff'f''\right\} \\ &= Q\left\{f'\left[(Cf'')' + ff''\right] + C(f'')^2\right\} \\ &= CQ(f'')^2,\end{aligned} \tag{E.46}$$

where the last step uses the reduced momentum equation $(Cf'')' + ff'' = 0$ (see Eq. (E.31)). Combining Eqs. (E.45) and (E.46) yields the calorically imperfect static-temperature similarity equation

$$\left[K\left(\frac{T}{T_e}\right)'\right]' + \frac{c_p(T)}{c_{p,e}}f\left(\frac{T}{T_e}\right)' + CQ\left(f''\right)^2 = 0, \tag{E.47}$$

with

$$K = \frac{\rho k(T)}{\rho_e\mu_e c_{p,e}}, \qquad C = \frac{\rho\mu}{\rho_e\mu_e}, \qquad Q = \frac{u_e^2}{c_{p,e}T_e}. \tag{E.48}$$

This is the calorically imperfect static-temperature energy equation used in the remainder of the analysis.

#### E.7.2.2 Perfect-gas static-temperature energy equation as a special case

The perfect-gas static-temperature equation follows directly as a special case. For a perfect gas,

$$c_p(T) = c_p(T_e) = c_p, \qquad \frac{c_p(T)}{c_{p,e}} = 1, \qquad Q = \frac{u_e^2}{c_pT_e}, \qquad k = \frac{\mu c_p}{Pr_0}. \tag{E.49}$$

Therefore,

$$K = \frac{\rho}{\rho_e\mu_e c_p}\frac{\mu c_p}{Pr_0} = \frac{C}{Pr_0}. \tag{E.50}$$

Substituting $K = C/Pr_0$ and $c_p(T)/c_{p,e} = 1$ into Eq. (E.47) gives

$$\left[\frac{C}{Pr_0}\left(\frac{T}{T_e}\right)'\right]' + f\left(\frac{T}{T_e}\right)' + CQ\left(f''\right)^2 = 0, \qquad Q = \frac{u_e^2}{c_pT_e}. \tag{E.51}$$

This is the perfect-gas static-temperature energy equation used in the remainder of the analysis.

## E.8 Cancellation of $d_1$ in the Chapman–Rubesin coefficient $C(T/T_e)$

The Chapman–Rubesin coefficient used in the transformed boundary layer equations is the viscosity-density coefficient

$$C = \frac{\rho\mu}{\rho_e\mu_e}. \tag{E.52}$$

Here $\rho$ and $\mu$ are local boundary layer values evaluated at $T(\eta)$, while $\rho_e$ and $\mu_e$ are boundary layer edge values. This coefficient appears in the transformed momentum equation (Eq. (E.31)) and in both static-temperature energy equations (Eqs. (E.47) and (E.51)).
For both gas models considered here, the gas is thermally perfect:

$$p = \rho RT. \tag{E.53}$$

Because $p = p_e$ across a zero-pressure-gradient boundary layer,

$$\frac{\rho}{\rho_e} = \frac{T_e}{T}. \tag{E.54}$$

For the viscosity ratio, using a shortened notation for Sutherland's law:

$$\mu(T) = d_1\phi(T), \qquad \mu_e = \mu(T_e) = d_1\phi(T_e), \qquad \phi(T) = \frac{\sqrt{T}}{1 + d_2/T}, \tag{E.55}$$

Then

$$\frac{\mu}{\mu_e} = \frac{d_1\phi(T)}{d_1\phi(T_e)} = \frac{\phi(T)}{\phi(T_e)}. \tag{E.56}$$

Therefore,

$$C = \frac{\rho\mu}{\rho_e\mu_e} = \frac{T_e}{T}\frac{\phi(T)}{\phi(T_e)}. \tag{E.57}$$

At fixed $T/T_e$, $C$ contains no explicit $d_1$ for either gas model, as long as both use the thermally perfect equation of state $p = \rho RT$ and the same Sutherland form for $\mu(T)$. The coefficient can still depend on $T/T_e$, $T_e$, and $d_2$. In the calorically imperfect problem, $C(\eta)$ can consequently change *indirectly* when the $d_1$-dependent energy equation changes $T(\eta)$.

## E.9 Dependence of the thermal-diffusion coefficient $K$ on $d_1$

For the calorically imperfect gas model, the static-temperature energy equation contains the thermal-diffusion coefficient $K$ multiplying the static-temperature-gradient term (see Eq. (E.47)). The coefficient is defined as

$$K = \frac{\rho k(T)}{\rho_e\mu_e c_{p,e}}, \qquad c_{p,e} = c_p(T_e). \tag{E.58}$$

The question is whether this coefficient contains the Sutherland constant $d_1$. Write Sutherland's law locally as

$$\mu(T) = d_1\phi(T), \qquad \mu_e = \mu(T_e) = d_1\phi(T_e), \qquad \phi(T) = \frac{\sqrt{T}}{1 + d_2/T}. \tag{E.59}$$

In contrast, Malik–Anderson define the thermal conductivity independently by the Keyes law (their Eq. 58),

$$k(T) = d_3\frac{\sqrt{T}}{1 + (d_4/T)\,10^{-d_5/T}}, \qquad d_3 = 2.646 \times 10^{-3}, \quad d_4 = 245.4\ \mathrm{K}, \quad d_5 = 12\ \mathrm{K}. \tag{E.60}$$

This conductivity closure contains the constants $d_3, d_4, d_5$, but no Sutherland constant $d_1$. Using the thermally perfect equation of state and $p = p_e$ across the boundary layer gives $\rho/\rho_e = T_e/T$. Therefore

$$K = \frac{\rho k(T)}{\rho_e \mu_e c_{p,e}} \tag{E.61}$$

$$= \frac{T_e}{T} \frac{k(T)}{d_1 \phi(T_e) c_{p,e}} \propto \frac{1}{d_1}. \tag{E.62}$$

$K$ is proportional to $1/d_1$. Therefore $d_1$ remains in the calorically imperfect static-temperature energy equation through the thermal-diffusion coefficient $K$.

# F Consistency of the density treatment in the compressible Spalart–Allmaras model of OpenFOAM v2412

## F.1 Summary and principal finding

This appendix audits the single-phase compressible Spalart–Allmaras (SA) transport equation assembled by OpenFOAM v2412 (which is unchanged to the, as of writing, latest v2606). It compares the native finite-volume equation with the material-form equation cited by OpenFOAM from Spalart and Allmaras (1994) and with its equivalent density-conservative form given by Allmaras, Johnson, and Spalart (2012) [22, 23].

The issue is confined to the diffusion term. Define

$$D_{\widetilde{\nu}} = \frac{\nu + \widetilde{\nu}}{\sigma}.$$

Native OpenFOAM uses the density-weighted diffusive flux

$$\boldsymbol{\nabla} \cdot \left( \rho D_{\widetilde{\nu}} \boldsymbol{\nabla} \widetilde{\nu} \right),$$

whereas multiplying the cited material equation by density requires

$$\rho \boldsymbol{\nabla} \cdot \left( D_{\widetilde{\nu}} \boldsymbol{\nabla} \widetilde{\nu} \right).$$

These expressions are not equal for variable density. The product rule gives

$$\boldsymbol{\nabla} \cdot \left( \rho D_{\widetilde{\nu}} \boldsymbol{\nabla} \widetilde{\nu} \right) = \rho \boldsymbol{\nabla} \cdot \left( D_{\widetilde{\nu}} \boldsymbol{\nabla} \widetilde{\nu} \right) + D_{\widetilde{\nu}} \boldsymbol{\nabla} \rho \cdot \boldsymbol{\nabla} \widetilde{\nu}.$$

The 2012 conservative equation cancels the final cross-gradient contribution explicitly; native OpenFOAM does not. Consequently, expanding the native equation back into material form produces the additional term

$$\boxed{\frac{\nu + \widetilde{\nu}}{\sigma} \boldsymbol{\nabla} \ln \rho \cdot \boldsymbol{\nabla} \widetilde{\nu}} \tag{F.1}$$

relative to the cited 1994 equation. Equivalently, relative to the 2012 conservative equation, the native physical right-hand side lacks

$$\boxed{-\frac{\nu + \widetilde{\nu}}{\sigma} \boldsymbol{\nabla} \rho \cdot \boldsymbol{\nabla} \widetilde{\nu}}. \tag{F.2}$$

- **Single identified mismatch.** Native OpenFOAM places $\rho$ inside the SA diffusive flux without the cross-gradient cancellation required to recover the cited material equation.
- **Two equivalent checks.** The native material equation has the additional positive term in Eq. (F.1); equivalently, its conservative physical right-hand side lacks the negative term in Eq. (F.2). These are two descriptions of the same continuum mismatch, not two independent inconsistencies.
- **Corrected implementation.** The custom `SpalartAllmarasDensityCorrected` model adds the required term with the coefficient and matrix sign needed to cancel the native density-product term. It matches the 2012 diffusion coefficient when `sigmaNut` is identified with the paper's $\sigma$.
- **Scope.** The mismatch vanishes for constant density or when $\boldsymbol{\nabla}\rho \cdot \boldsymbol{\nabla}\widetilde{\nu} = 0$. The derivation concerns the continuum transport/diffusion sector; it does not establish complete closure identity, exact discrete equivalence, or improved predictions.
- **Independent NASA implementation warning.** Note 4 of the NASA Turbulence Modeling Resource warns that some SA implementations "incorrectly introduce the density inside the derivatives of the SA equation" and states that this changes predictions for supersonic flows; it explicitly distinguishes the separately formulated Catris–Aupoix proposal as having a valid basis [107]. The NASA note does not identify OpenFOAM by name. Its relevance here follows from the independent reconstruction of Eq. (F.14), which establishes that the native implementation has precisely the warned-about derivative structure.

### Outline

- Section F.2 identifies the literature equations, citation basis, and audit scope.
- Section F.3 follows the actual OpenFOAM operators from code to the native OpenFOAMimplemented conservative continuum equation.
- Section F.4 directly compares the Allmaras 2012 and native OpenFOAM conservative equations and identifies the missing term.
- Section F.5 summarizes the established algebraic result and interprets the native equation as a modified SA model.
- Section F.6 audits the custom correction, including the `fvm::Su` sign and a dimensional check.
- Section F.7 retains the complete line-by-line derivation from the material equation to its density-conservative form.

## F.2 Reference equations and audit scope

The SA working variable is denoted by $\widetilde{\nu}$, the molecular kinematic viscosity by $\nu$, density by $\rho$, velocity by $\boldsymbol{u}$, and the model diffusion constant by $\sigma$. The production, destruction, and trip contributions are grouped as $P$, $D$, and $T$, respectively. The diffusion coefficient is left explicitly as $(\nu + \widetilde{\nu})/\sigma$ so that every factor remains visible.

### F.2.1 The cited 1994 material formulation

The audited OpenFOAM v2412 header cites the 1994 journal article by Spalart and Allmaras as the standard-model reference. That article explicitly defines $\mathrm{D}/\mathrm{D}t$ as the Lagrangian or material derivative and gives the full Version-Ia working-variable equation in Appendix Eq. (A2), printed on p. 20 (PDF page 16):

$$\begin{aligned}\frac{\mathrm{D}}{\mathrm{D}t}\widetilde{\nu} = c_{b1}(1-f_{t2})\widetilde{S}\widetilde{\nu} \\ + \frac{1}{\sigma}\left\{\boldsymbol{\nabla}\cdot[(\nu+\widetilde{\nu})\boldsymbol{\nabla}\widetilde{\nu}] + c_{b2}\left|\boldsymbol{\nabla}\widetilde{\nu}\right|^2\right\} \\ - \left(c_{w1}f_w - \frac{c_{b1}}{\kappa^2}f_{t2}\right)\left(\frac{\widetilde{\nu}}{d}\right)^2 + f_{t1}(\Delta U)^2.\end{aligned} \tag{F.3}$$

Production, destruction, and tripping differ in detail from the implementation audited below, but the material transport and diffusion structure is the one relevant to the density comparison.

The final paragraph of the 1994 appendix states that Eqs. (A1)–(A6) had been applied directly in compressible flow and reports only a weak influence of density variations in the slightly supersonic boundary layers then examined. Thus the cited journal article supports a density-free material equation for compressible use. It does not give a density-conservative finite-volume recasting or prescribe where $\rho$ should be placed within such a recasting. The Turbulence Modeling Resource identifies this journal article as the official publication arising from the 1992 conference paper and states that the journal version takes precedence [22, 107].

### F.2.2 The Allmaras 2012 standard compressible form

This resource explicitly states a conservative form of the SA model, which can be directly compared to the OpenFOAM implementation. For this Allmaras et al. (2012) first write their transport equation Eq. (2), printed on p. 2, as

$$\frac{\mathrm{D}}{\mathrm{D}t}\widetilde{\nu} = P - D + T + \frac{1}{\sigma}\left\{\boldsymbol{\nabla}\cdot[(\nu+\widetilde{\nu})\boldsymbol{\nabla}\widetilde{\nu}] + c_{b2}\left|\boldsymbol{\nabla}\widetilde{\nu}\right|^2\right\}. \tag{F.4}$$

where production, wall destruction and trip terms are

$$P = c_{b1}(1-f_{t2})\widetilde{S}\widetilde{\nu}, \qquad D = \left(c_{w1}f_w - \frac{c_{b1}}{\kappa^2}f_{t2}\right)\left[\frac{\widetilde{\nu}}{d}\right]^2, \qquad T = f_{t1}(\Delta u)^2.$$

This formulation is equivalent to the original 1994 model equation (see F.3)

Their Section 2.2 explicitly states that this equation applies to both incompressible and compressible flow and should be treated as the standard compressible form. They then give the equivalent conservation form as their Eq. (9), printed on p. 3:

**density conservative SA model equation**

$$\begin{aligned}0 = \frac{\partial(\rho\widetilde{\nu})}{\partial t} + \boldsymbol{\nabla}\cdot(\rho\boldsymbol{u}\widetilde{\nu}) - \rho(P-D+T) \\ - \frac{1}{\sigma}\boldsymbol{\nabla}\cdot[\rho(\nu+\widetilde{\nu})\boldsymbol{\nabla}\widetilde{\nu}] - \frac{c_{b2}}{\sigma}\rho\left|\boldsymbol{\nabla}\widetilde{\nu}\right|^2 \\ + \frac{\nu+\widetilde{\nu}}{\sigma}\boldsymbol{\nabla}\rho\cdot\boldsymbol{\nabla}\widetilde{\nu}.\end{aligned} \tag{F.5}$$

The final term is positive because Eq. (F.5) is an all-terms-left residual. It is negative when moved to the physical right-hand side. Allmaras 2012 Eq. (10), their shock-layer integral, independently repeats the negative right-hand-side sign [23].

Section F.7 contains the full mathematical derivation from the material equation Eq. (F.3) to the density conservative form Eq. (F.5) and proofs that no density external correction or similar is being applied, but that it is a clean mathematical derivation. Thus Eq. (F.5) should be understood as the faithful density conservative model equation of the original SA model.

### F.2.3 The 1992 Outlook on possible compressibility additions

The earlier conference paper is retained here only for its historical Outlook. That discussion mentions possible compressibility additions involving $\boldsymbol{\nabla}\rho \cdot \boldsymbol{\nabla}\widetilde{\nu}$, considers writing the transport equation for $\widetilde{\mu}$ instead of $\widetilde{\nu}$, and notes freedom to place density inside or outside derivatives. It does not prescribe a final coefficient, sign, or unique conservative equation, and therefore does not define the density-weighted flux used by native OpenFOAM or establish its equivalence to the standard material equation [108].

## F.3 Reconstruction of the native OpenFOAM equation

The governing equation is assembled in the shared SA base class. The relevant OpenFOAM v2412 code is shown first and subsequently the continuum equation is translated from it term-by-term to the underlying mathematical equation. Which can then be compared to the Allmaras 2012 published density conservative SA model equation.

**Listing F.1** Native SA equation assembly in SpalartAllmarasBase.C.

```
468 fvm::ddt(alpha, rho, nuTilda_)
469 + fvm::div(alphaRhoPhi, nuTilda_)
470 - fvm::laplacian(alpha*rho*DnuTildaEff(), nuTilda_)
471 - Cb2_/sigmaNut_*alpha()*rho()*magSqr(fvc::grad(nuTilda_)()())
472 ==
473   Cb1_*alpha()*rho()*Stilda()*nuTilda_()*(1 - ft2())
474 - fvm::Sp
475   (
476     (Cw1_*fw(Stilda, dTilda) - Cb1_/sqr(kappa_)*ft2())
477    *alpha()*rho()*nuTilda_()/sqr(dTilda()),
478     nuTilda_
479   )
480 + fvOptions(alpha, rho, nuTilda_)
```

This code is at `SpalartAllmarasBase.C:466--481`. The coefficient used by the Laplacian is defined at lines 383–389 as

$$\texttt{DnuTildaEff()} = \frac{\nu + \widetilde{\nu}}{\sigma}. \tag{F.6}$$

Here and below, the code coefficient `sigmaNut` is denoted by $\sigma$. Its v2412 default is the configurable decimal `0.66666`; exact numerical agreement with the paper value $\sigma = 2/3$ requires configuring the same value. The custom correction reuses `DnuTildaEff` and therefore cancels the native density-product term for either setting.

The thin RAS wrapper does not replace this equation; it supplies wall distance and viscosity-related updates and inherits the base-class transport equation.

The reconstruction below is restricted to the bulk/interior transport equation assembled in Listing F.1. It precedes and does not include the subsequent `relax()`, `fvOptions.constrain()`, `boundaryManipulate()` (including any boundary-row edits made by constrained patch fields), `solve()`, the post-solve `bound()`, or `nuTilda_.correctBoundaryConditions()` calls. Those

operations may modify matrix coefficients, sources, boundary-adjacent rows, or the accepted field, but in an audit it was not found that these operations correct the identified model deviation.

### F.3.1 Translate each code term into continuum notation

For the registered single-phase compressible model, $\alpha = 1$, $\rho$ is the density field, and `alphaRhoPhi` is the face mass flux corresponding to $\rho \boldsymbol{u} \cdot \boldsymbol{S}_f$. The mappings below are for this single-phase registration and interpret the displayed matrix as an all-terms-left residual before reporting a physical right-hand-side sign.

#### F.3.1.1 Transient and convection operators

With `fvm::ddt` and `fvm::div` read as their canonical unbounded continuum operators, the first two code terms,

$$\texttt{fvm::ddt(alpha, rho, nuTilda_)} + \texttt{fvm::div(alphaRhoPhi, nuTilda_)},$$

represent

$$\frac{\partial(\rho\widetilde{\nu})}{\partial t} + \boldsymbol{\nabla}\cdot(\rho\boldsymbol{u}\widetilde{\nu}), \tag{F.7}$$

This is initially a conservative transport operator, not directly the material derivative. For differentiable fields, applying the product rule gives

$$\begin{aligned}&\frac{\partial(\rho\widetilde{\nu})}{\partial t} + \boldsymbol{\nabla}\cdot(\rho\boldsymbol{u}\widetilde{\nu}) \\ &\quad = \rho\frac{\mathrm{D}}{\mathrm{D}t}\widetilde{\nu} + \widetilde{\nu}\left[\frac{\partial\rho}{\partial t} + \boldsymbol{\nabla}\cdot(\rho\boldsymbol{u})\right].\end{aligned} \tag{F.8}$$

Mass continuity, $\partial_t\rho + \boldsymbol{\nabla}\cdot(\rho\boldsymbol{u}) = 0$, removes the bracket. Consequently,

$$\texttt{ddt + div} \quad\longrightarrow\quad \rho\frac{\mathrm{D}}{\mathrm{D}t}\widetilde{\nu},$$

and it becomes the unweighted material derivative only after the complete equation is divided by $\rho > 0$.

This is a continuum/operator mapping, not a scheme-independent statement about the assembled matrix. The selected `fvSchemes` may wrap the transient or convection operator in a bounded form. In particular, the bounded three-field time scheme subtracts

$$\texttt{fvm::Sp(fvc::ddt(alpha,rho),vf)},$$

whereas the bounded convection scheme subtracts

$$\texttt{fvm::Sp(fvc::surfaceIntegrate(faceFlux),vf)}.$$

These terms change how a discrete mass-continuity residual is apportioned between transport and source; they do not contain the diffusion cross-gradient $(\nu + \widetilde{\nu})\boldsymbol{\nabla}\rho\cdot\boldsymbol{\nabla}\widetilde{\nu}/\sigma$.

#### F.3.1.2 Laplacian operator

Using Eq. (F.6), the matrix-left term

$$-\texttt{fvm::laplacian}\left(\rho\frac{\nu+\widetilde{\nu}}{\sigma},\widetilde{\nu}\right)$$

represents

$$-\frac{1}{\sigma}\boldsymbol{\nabla}\cdot[\rho(\nu+\widetilde{\nu})\boldsymbol{\nabla}\widetilde{\nu}]\,. \tag{F.9}$$

Density is inside the coefficient passed to the Laplacian. The code therefore discretizes a density-weighted conservative diffusive flux; it does not discretize $\rho$ multiplied by a density-free Laplacian.

#### F.3.1.3 Explicit $c_{b2}$ term

The code term

```
−Cb2_/sigmaNut_*rho()*magSqr(fvc::grad(nuTilda_)())
```

maps to

$$-\frac{c_{b2}}{\sigma}\rho\,|\boldsymbol{\nabla}\widetilde{\nu}|^2\,. \tag{F.10}$$

The $c_{b2}$ term is part of the modelled nonlinear diffusion in both papers. It is distinct from the density product-rule term and must not be interpreted as the missing cancellation.

#### F.3.1.4 Production and destruction terms

The first term on the code right-hand side represents the paper's production term:

```
Cb1_*rho()*Stilda()*nuTilda_()*(1-ft2())
```
$$\longrightarrow \quad \rho P,$$

where

$$P = c_{b1}(1-f_{t2})\widetilde{S}\widetilde{\nu}. \tag{F.11}$$

The implicit `fvm::Sp` term represents destruction. Because it is preceded by a minus sign on the code right-hand side, its continuum meaning is $-\rho D$, where

$$D = \left(c_{w1}f_w - \frac{c_{b1}}{\kappa^2}f_{t2}\right)\left(\frac{\widetilde{\nu}}{d}\right)^2. \tag{F.12}$$

Thus the modelled right-hand side in the native equation is $\rho(P-D)$. Native OpenFOAM has no trip term $T$; comparison with the full paper equation therefore uses $T=0$. The final `fvOptions(alpha, rho, nuTilda_)` entry permits additional case-defined sources. It is not part of the SA paper equation and is omitted from the reconstructed native equation.

For the density audit, $P$ and $D$ are held symbolically fixed so that the transport and diffusion sectors can be compared directly. This does not assert complete closure identity: native OpenFOAM uses the $\widetilde{S}$ limiter $\max[\Omega + f_{v2}\widetilde{\nu}/(\kappa d)^2, C_s\Omega]$, which is not identical to the Allmaras 2012 baseline and proposed negative-$\widetilde{S}$ continuation.

### F.3.2 Assemble the mathematical equation

Combining the preceding mappings gives the all-terms-left continuum meaning of the native finite-volume assembly:

$$\begin{aligned} 0 = {} & \frac{\partial(\rho\widetilde{\nu})}{\partial t} + \boldsymbol{\nabla}\cdot(\rho\boldsymbol{u}\widetilde{\nu}) - \rho(P-D) \\ & - \frac{1}{\sigma}\boldsymbol{\nabla}\cdot[\rho(\nu+\widetilde{\nu})\boldsymbol{\nabla}\widetilde{\nu}] - \frac{c_{b2}}{\sigma}\rho\,|\boldsymbol{\nabla}\widetilde{\nu}|^2\,. \end{aligned} \tag{F.13}$$

Equivalently, moving the modelled and diffusion terms to the physical right-hand side gives:

**Native OpenFOAM continuum equation**

$$\frac{\partial(\rho\widetilde{\nu})}{\partial t} + \boldsymbol{\nabla}\cdot(\rho\boldsymbol{u}\widetilde{\nu}) = \rho(P-D) + \frac{1}{\sigma}\boldsymbol{\nabla}\cdot[\rho(\nu+\widetilde{\nu})\boldsymbol{\nabla}\widetilde{\nu}] + \frac{c_{b2}}{\sigma}\rho\left|\boldsymbol{\nabla}\widetilde{\nu}\right|^2. \tag{F.14}$$

Equation (F.14) is not copied from either paper; it is the continuum equation represented by the audited native finite-volume assembly, expressed in the papers' notation with $\alpha = 1$, `fvOptions` suppressed, and $T = 0$. For the incompressible registration, both $\alpha$ and $\rho$ are compile-time one fields, so the density issue vanishes exactly.

## F.4 Direct comparison and identified error

Equation (F.5) is the Allmaras 2012 conservative equation, and Eq. (F.14) is the equation reconstructed term by term from native OpenFOAM. Set $T = 0$, as required by the native implementation, and hold $P$, $D$, and $\sigma$ fixed. After moving the modelled and diffusion terms of Eq. (F.5) to the physical right-hand side, the complete comparison is

$$\mathrm{RHS}|_{\text{Allmaras Eq. (F.5)},\,T=0} = \mathrm{RHS}|_{\text{OpenFOAM Eq. (F.14)}} - \frac{\nu+\widetilde{\nu}}{\sigma}\boldsymbol{\nabla}\rho\cdot\boldsymbol{\nabla}\widetilde{\nu}. \tag{F.15}$$

Every other displayed transport, production, destruction, and diffusion term is identical under this comparison. Equation (F.14) therefore lacks the final negative density-gradient term.

That term is the cancellation required by the product rule

$$\boldsymbol{\nabla}\cdot[\rho(\nu+\widetilde{\nu})\boldsymbol{\nabla}\widetilde{\nu}] = \rho\boldsymbol{\nabla}\cdot[(\nu+\widetilde{\nu})\boldsymbol{\nabla}\widetilde{\nu}] + (\nu+\widetilde{\nu})\boldsymbol{\nabla}\rho\cdot\boldsymbol{\nabla}\widetilde{\nu}.$$

Without the cancellation, rewriting the native equation in material form produces the corresponding positive cross-gradient term.

**Identified error in the OpenFOAMimplemented equation.**
Relative to Eq. (F.5), the physical right-hand side of Eq. (F.14) lacks

$$-\frac{\nu+\widetilde{\nu}}{\sigma}\boldsymbol{\nabla}\rho\cdot\boldsymbol{\nabla}\widetilde{\nu}.$$

Equivalently, the native material equation contains the additional term

$$+\frac{\nu+\widetilde{\nu}}{\sigma}\boldsymbol{\nabla}\ln\rho\cdot\boldsymbol{\nabla}\widetilde{\nu}.$$

The equations agree when density is constant, $\boldsymbol{\nabla}\widetilde{\nu} = 0$, or the density and working-variable gradients are orthogonal. A nonzero density gradient by itself is not sufficient; the cross-gradient alignment determines the contribution.

## F.5 Conclusion

Native single-phase compressible OpenFOAM v2412 uses the density-weighted diffusive flux

$$\frac{1}{\sigma}\boldsymbol{\nabla}\cdot[\rho(\nu+\widetilde{\nu})\boldsymbol{\nabla}\widetilde{\nu}]$$

without the explicit conservative cancellation needed to recover

$$\frac{\rho}{\sigma}\boldsymbol{\nabla}\cdot[(\nu+\widetilde{\nu})\boldsymbol{\nabla}\widetilde{\nu}]\,.$$

Its material form therefore contains the additional positive term in Eq. (F.1). The custom `SpalartAllmarasDensityCorrected` source adds the corresponding positive term to the matrix residual, which is negative on the physical right-hand side and cancels the native product-rule contribution. With the same $\sigma$, the resulting diffusion structure agrees with Allmaras 2012 Eq. (9). This cancellation is exact at the continuum level under the assumptions stated during the reconstruction in Section F.3; it is not a claim that all nondiffusive closure choices are identical.

Equation (F.14) nevertheless defines a mathematically consistent conservative scalar-transport model if the diffusive flux

$$\boldsymbol{J}_{\widetilde{\nu}} = -\rho D_{\widetilde{\nu}}\boldsymbol{\nabla}\widetilde{\nu}, \qquad D_{\widetilde{\nu}} \equiv \frac{\nu+\widetilde{\nu}}{\sigma}, \tag{F.16}$$

is chosen intentionally. That choice, however, generates the additional density-gradient term in the corresponding material equation. Without the cancellation prescribed by Allmaras et al. (2012), the native equation is therefore a modified SA model rather than the intended compressible SA formulation.

The defensible conclusion is thus a *confirmed transport/diffusion-sector deviation from both the 1994 material formulation cited by OpenFOAM and the Allmaras 2012 conservative standard*. This audit cannot determine whether the deviation was intentional or arose from an unintended derivation error. The 1992 Outlook shows that alternative compressibility additions and density placements were contemplated, but it does not define the native density-weighted flux or establish that flux as the intended SA model [22, 108, 23]. The native OpenFOAM equation and the cited standard therefore represent different models in compressible flow, with the latter and herein corrected, representing a version in-line with the understanding of Spalart–Allmaras and stated as the standard form for compressible flow [23].

## F.6 Implementation of the corrected model

The custom model first assembles the native equation and then adds the source shown below. The relevant code is in `SpalartAllmarasDensityCorrected.C:54--66,136--175`.

**Listing F.2** Density-gradient source in SpalartAllmarasDensityCorrected.

```
54 const tmp<volVectorField> tgradRho = fvc::grad(rho);
55
56 // The stock conservative diffusion contributes
57 //     div(rho*DnuTildaEff*grad(nuTilda))
58 // on the physical RHS.  Adding this explicit source to the matrix LHS
59 // subtracts DnuTildaEff*grad(rho).grad(nuTilda) from that RHS, giving
60 // exactly rho*div(DnuTildaEff*grad(nuTilda)).
61 nuTildaEqn += fvm::Su
62 (
63     DnuTildaEff()
64    *(tgradRho()() & gradNuTilda()),
65     this->nuTilda_
66 );
```

The `fvm::Su` convention resolves the apparent sign ambiguity. The explicit calls to the field-internal accessors make the expression in Listing F.2 a temporary `DimensionedField`. The overload in `fvmSup.C:92--103` forwards that temporary to the field implementation at lines 72–89, where the supplied field is stored as `matrix.source() -= V*source`. Adding the source therefore adds

$$+\frac{\nu+\widetilde{\nu}}{\sigma}\boldsymbol{\nabla}\rho\cdot\boldsymbol{\nabla}\widetilde{\nu} \tag{F.17}$$

to the all-terms-left matrix residual, or the negative of this expression to the physical right-hand side. It has the form and sign of the final term in Eq. (F.5), is exactly the product-rule cancellation identified in Section F.4 for the native `DnuTildaEff`, and has the Allmaras 2012 numerical coefficient when `sigmaNut` is set to $2/3$.

The dimensions are consistent:

$$\left[\frac{\nu+\widetilde{\nu}}{\sigma}\right]=L^2T^{-1},\qquad[\boldsymbol{\nabla}\rho]=ML^{-4},\qquad[\boldsymbol{\nabla}\widetilde{\nu}]=LT^{-1}, \tag{F.18}$$

and hence

$$\left[\frac{\nu+\widetilde{\nu}}{\sigma}\boldsymbol{\nabla}\rho\cdot\boldsymbol{\nabla}\widetilde{\nu}\right]=ML^{-1}T^{-2}=\left[\frac{\partial(\rho\widetilde{\nu})}{\partial t}\right]. \tag{F.19}$$

The custom header provides a no-op overload for `geometricOneField` density, so the incompressible registration remains the native SA equation. The library is registered only in the single-phase compressible and incompressible turbulence tables. Its omission of a phase-fraction-gradient term is consistent with that scope; applying the same source to a variable-$\alpha$ multiphase equation would require a new derivation.

Finally, the modelled term

$$\frac{c_{b2}}{\sigma}\,|\boldsymbol{\nabla}\widetilde{\nu}|^2 \tag{F.20}$$

is not the density correction. Expanding $\boldsymbol{\nabla}\cdot[(\nu+\widetilde{\nu})\boldsymbol{\nabla}\widetilde{\nu}]$ already retains the derivatives of $\nu+\widetilde{\nu}$, while $c_{b2}$ is a separate model coefficient. The custom cancellation neither removes nor duplicates it.

## F.7 Allmaras 2012 material form to conservative form

This section retains the complete conversion from Allmaras 2012 Eq. (2) to Eq. (9). It makes explicit which terms are supplied by mass conservation and which term is created by rewriting the diffusion operator.

The Allmaras 2012 material equation is Eq. (F.4):

$$\frac{\mathrm{D}}{\mathrm{D}t}\widetilde{\nu} = P - D + T + \frac{1}{\sigma}\left[\boldsymbol{\nabla}\cdot((\nu+\widetilde{\nu})\boldsymbol{\nabla}\widetilde{\nu}) + c_{b2}\left|\boldsymbol{\nabla}\widetilde{\nu}\right|^2\right]. \tag{F.21}$$

### F.7.1 Put the material equation in zero-residual form

Move every right-hand-side term to the left and expand the material derivative:

$$\begin{aligned}\{\mathrm{SA}\} = \frac{\partial\widetilde{\nu}}{\partial t} + \boldsymbol{u}\cdot\boldsymbol{\nabla}\widetilde{\nu} - (P - D + T)\\ - \frac{1}{\sigma}\boldsymbol{\nabla}\cdot[(\nu+\widetilde{\nu})\boldsymbol{\nabla}\widetilde{\nu}] - \frac{c_{b2}}{\sigma}\left|\boldsymbol{\nabla}\widetilde{\nu}\right|^2 = 0.\end{aligned} \tag{F.22}$$

The braces label the complete residual; they introduce no new model quantity.

### F.7.2 Add the zero mass residual

Mass conservation in zero-residual form is

$$\{\mathrm{mass}\} = \frac{\partial\rho}{\partial t} + \boldsymbol{\nabla}\cdot(\rho\boldsymbol{u}) = 0. \tag{F.23}$$

Allmaras et al. (2012) combine the two residuals as

$$0 = \rho\{\mathrm{SA}\} + \widetilde{\nu}\{\mathrm{mass}\}. \tag{F.24}$$

Because both residuals are zero, multiplying and adding them does not change the solution.

### F.7.3 Multiply out the residuals

Multiplying Eq. (F.22) by $\rho$ gives

$$\begin{aligned}\rho\{\mathrm{SA}\} = \rho\frac{\partial\widetilde{\nu}}{\partial t} + \rho\boldsymbol{u}\cdot\boldsymbol{\nabla}\widetilde{\nu} - \rho(P - D + T)\\ - \frac{\rho}{\sigma}\boldsymbol{\nabla}\cdot[(\nu+\widetilde{\nu})\boldsymbol{\nabla}\widetilde{\nu}] - \frac{c_{b2}}{\sigma}\rho\left|\boldsymbol{\nabla}\widetilde{\nu}\right|^2.\end{aligned} \tag{F.25}$$

Multiplying Eq. (F.23) by $\widetilde{\nu}$ gives

$$\widetilde{\nu}\{\mathrm{mass}\} = \widetilde{\nu}\frac{\partial\rho}{\partial t} + \widetilde{\nu}\boldsymbol{\nabla}\cdot(\rho\boldsymbol{u}). \tag{F.26}$$

Adding Eqs. (F.25) and (F.26) yields

$$\begin{aligned}0 = \rho\frac{\partial\widetilde{\nu}}{\partial t} + \widetilde{\nu}\frac{\partial\rho}{\partial t}\\ + \rho\boldsymbol{u}\cdot\boldsymbol{\nabla}\widetilde{\nu} + \widetilde{\nu}\boldsymbol{\nabla}\cdot(\rho\boldsymbol{u}) - \rho(P - D + T)\\ - \frac{\rho}{\sigma}\boldsymbol{\nabla}\cdot[(\nu+\widetilde{\nu})\boldsymbol{\nabla}\widetilde{\nu}] - \frac{c_{b2}}{\sigma}\rho\left|\boldsymbol{\nabla}\widetilde{\nu}\right|^2.\end{aligned} \tag{F.27}$$

### F.7.4 Combine transient and convection products

For the transient terms,

$$\frac{\partial(\rho\widetilde{\nu})}{\partial t} = \rho\frac{\partial\widetilde{\nu}}{\partial t} + \widetilde{\nu}\frac{\partial\rho}{\partial t}. \tag{F.28}$$

For a scalar $\widetilde{\nu}$ and vector $\rho\boldsymbol{u}$,

$$\boldsymbol{\nabla}\cdot(\rho\boldsymbol{u}\widetilde{\nu}) = \rho\boldsymbol{u}\cdot\boldsymbol{\nabla}\widetilde{\nu} + \widetilde{\nu}\boldsymbol{\nabla}\cdot(\rho\boldsymbol{u}). \tag{F.29}$$

Substitution into Eq. (F.27) gives

$$\begin{aligned} 0 = &\frac{\partial(\rho\widetilde{\nu})}{\partial t} + \boldsymbol{\nabla}\cdot(\rho\boldsymbol{u}\widetilde{\nu}) - \rho(P - D + T) \\ &- \frac{\rho}{\sigma}\boldsymbol{\nabla}\cdot[(\nu+\widetilde{\nu})\boldsymbol{\nabla}\widetilde{\nu}] - \frac{c_{b2}}{\sigma}\rho\left|\boldsymbol{\nabla}\widetilde{\nu}\right|^2. \end{aligned} \tag{F.30}$$

Only the diffusion term remains to be rewritten.

### F.7.5 Rewrite the diffusion term

Apply the vector product rule:

$$\begin{aligned} &\boldsymbol{\nabla}\cdot[\rho(\nu+\widetilde{\nu})\boldsymbol{\nabla}\widetilde{\nu}] \\ &\quad= \rho\boldsymbol{\nabla}\cdot[(\nu+\widetilde{\nu})\boldsymbol{\nabla}\widetilde{\nu}] + (\nu+\widetilde{\nu})\boldsymbol{\nabla}\rho\cdot\boldsymbol{\nabla}\widetilde{\nu}. \end{aligned} \tag{F.31}$$

Solve for the density-multiplied diffusion that appears in Eq. (F.30):

$$\begin{aligned} &\rho\boldsymbol{\nabla}\cdot[(\nu+\widetilde{\nu})\boldsymbol{\nabla}\widetilde{\nu}] \\ &\quad= \boldsymbol{\nabla}\cdot[\rho(\nu+\widetilde{\nu})\boldsymbol{\nabla}\widetilde{\nu}] - (\nu+\widetilde{\nu})\boldsymbol{\nabla}\rho\cdot\boldsymbol{\nabla}\widetilde{\nu}. \end{aligned} \tag{F.32}$$

The term in Eq. (F.30) is negative, so

$$\begin{aligned} &-\frac{\rho}{\sigma}\boldsymbol{\nabla}\cdot[(\nu+\widetilde{\nu})\boldsymbol{\nabla}\widetilde{\nu}] \\ &\quad= -\frac{1}{\sigma}\boldsymbol{\nabla}\cdot[\rho(\nu+\widetilde{\nu})\boldsymbol{\nabla}\widetilde{\nu}] + \frac{\nu+\widetilde{\nu}}{\sigma}\boldsymbol{\nabla}\rho\cdot\boldsymbol{\nabla}\widetilde{\nu}. \end{aligned} \tag{F.33}$$

The positive final sign follows from two negative signs: the one already in front of the residual diffusion and the one introduced by rearranging the product rule.

### F.7.6 Recover Allmaras 2012 Eq. (9)

Substituting Eq. (F.33) into Eq. (F.30) gives

$$\begin{aligned} 0 = &\frac{\partial(\rho\widetilde{\nu})}{\partial t} + \boldsymbol{\nabla}\cdot(\rho\boldsymbol{u}\widetilde{\nu}) - \rho(P - D + T) \\ &- \frac{1}{\sigma}\boldsymbol{\nabla}\cdot[\rho(\nu+\widetilde{\nu})\boldsymbol{\nabla}\widetilde{\nu}] - \frac{c_{b2}}{\sigma}\rho\left|\boldsymbol{\nabla}\widetilde{\nu}\right|^2 \\ &+ \frac{\nu+\widetilde{\nu}}{\sigma}\boldsymbol{\nabla}\rho\cdot\boldsymbol{\nabla}\widetilde{\nu}. \end{aligned} \tag{F.34}$$

This is Allmaras 2012 Eq. (9). The mass residual creates conservative storage and convection. The diffusion product rule, not continuity, creates the final density-gradient term.

# Bibliography


[1] Hermann Schlichting and Klaus Gersten. *Boundary-Layer Theory*. Springer, Berlin, Heidelberg, ninth edition, 2017. doi: 10.1007/978-3-662-52919-5.

[2] M. R. Malik and E. C. Anderson. Real gas effects on hypersonic boundary-layer stability. *Physics of Fluids A*, 3(5):803–821, 1991. doi: 10.1063/1.858012.

[3] Bingliang Li, Zhenxun Gao, Fan Mo, and Chongwen Jiang. Improved wall function method for hypersonic reacting turbulent boundary layers. *AIAA Journal*, 63(1):55–71, 2025. doi: 10.2514/1.J064306.

[4] Kevin P Griffin, Lin Fu, and Parviz Moin. Near-wall model for compressible turbulent boundary layers based on an inverse velocity transformation. *Journal of Fluid Mechanics*, 970:A36, 2023. doi: 10.1017/jfm.2023.627.

[5] Chao Zhang, Lian Duan, and Meelan M Choudhari. Direct numerical simulation database for supersonic and hypersonic turbulent boundary layers. *AIAA journal*, 56(11):4297–4311, 2018. doi: 10.2514/1.J057296.

[6] Yuta Iwatani, Hiroyuki Asada, and Soshi Kawai. Direct numerical simulation of subharmonic transition in pseudo-adiabatic, heated and cooled boundary layers at Mach number 0.8. *International Journal of Heat and Fluid Flow*, 116:109896, 2025. doi: 10.1016/j.ijheatfluidflow.2025.109896. Digital profiles available from the Kawai Lab DNS/LES database, `https://www.klab.mech.tohoku.ac.jp/database/index.html`.

[7] Alexander J Smits and Jean-Paul Dussauge. *Turbulent shear layers in supersonic flow*. Springer, second edition, 2006. doi: 10.1007/b137383.

[8] Asif Manzoor Hasan, Pedro Costa, Johan Larsson, Sergio Pirozzoli, and Rene Pecnik. Intrinsic compressibility effects in near-wall turbulence. *Journal of Fluid Mechanics*, 1006:A14, 2025. doi: 10.1017/jfm.2024.1238.

[9] Rene Pecnik and Asif Manzoor Hasan. Turbulence modeling: Compressible wall-bounded flows. In *Turbulence and Transition in Supersonic and Hypersonic Flows*, pages 453–500. Elsevier, 2026. doi: 10.1016/b978-0-44-318785-8.00016-x.

[10] Mark V Morkovin. Effects of compressibility on turbulent flows. In A. Favre, editor, *Mécanique de la Turbulence*, pages 367–380. CNRS, Paris, 1962.

[11] L Fu, M Karp, S. T. Bose, P Moin, and J Urzay. Turbulence statistics in a high Mach number boundary layer downstream of an incident shock wave. *Center for Turbulence Research Annual Research Briefs*, 2019:41–54, 2019.

[12] Zhangfeng Huang, Heng Zhou, and Jisheng Luo. Direct numerical simulation of a supersonic turbulent boundary layer on a flat plate and its analysis. *Science in China Series G: Physics, Mechanics and Astronomy*, 48(5):626–640, 2005. doi: 10.1360/142005-184.

[13] David C. Wilcox. *Turbulence modeling for CFD*. DCW Industries, third edition, 2006.

[14] Florian R. Menter and Thomas Esch. Elements of industrial heat transfer predictions. In *16th Brazilian Congress of Mechanical Engineering (COBEM), Uberlandia, Brazil, 2001*, 2001.

[15] Florian R. Menter, Martin Kuntz, and Robin Langtry. Ten years of industrial experience with the SST turbulence model. In K. Hanjalić, Y. Nagano, and M. Tummers, editors, *Turbulence, Heat and Mass Transfer 4*, pages 625–632. Begell House, 2003.

[16] Florian R Menter. Two-equation eddy-viscosity turbulence models for engineering applications. *AIAA journal*, 32(8):1598–1605, 1994. doi: 10.2514/3.12149.

[17] Sang Dug Kim and Dong Joo Song. Modified shear-stress transport turbulence model for supersonic flows. *Journal of aircraft*, 42(5):1118–1125, 2005. doi: 10.2514/1.10223.

[18] Yu Zeng, Hongbo Wang, Dapeng Xiong, Yixin Yang, Mingbo Sun, and Weidong Liu. A multi-physics correction of the shear-stress transport turbulence model for supersonic flows. *Physics of Fluids*, 36 (5):055139, 2024. doi: 10.1063/5.0205391.

[19] Chao Wang, Mingbo Sun, Yixin Yang, Hongbo Wang, Xu Liu, Dapeng Xiong, and Yanan Wang. Improved SST turbulence model for supersonic flows with APG/separation. *Computers & Fluids*, 274: 106237, 2024. doi: 10.1016/j.compfluid.2024.106237.

[20] Jimmy-John O. E. Hoste, Nicholas N. Gibbons, Tobias Ecker, Chiara Amato, Doyle Knight, Artemii Sattarov, Olivier Thiry, Jean-Pierre Hickey, Fahri Erinç Hizir, Tolga Köktürk, Neil Castelino, Valerio Viti, Megan A. Roldan, Steven Qiang, James G. Coder, Robert A. Baurle, and Jeffery A. White. A review of Reynolds-averaged Navier–Stokes modeling for hypersonic large cone–flares. *Physics of Fluids*, 37(2):025193, 2025. doi: 10.1063/5.0246552.

[21] Holger Frahnert and Uwe Ch Dallmann. Examination of the eddy-viscosity concept regarding its physical justification. In *New Results in Numerical and Experimental Fluid Mechanics III: Contributions to the 12th STAB/DGLR Symposium Stuttgart, Germany 2000*, pages 255–262. Springer, 2002. doi: 10.1007/978-3-540-45466-3_31.

[22] P. R. Spalart and S. R. Allmaras. A one-equation turbulence model for aerodynamic flows. *La Recherche Aérospatiale*, (1):5–21, 1994.

[23] S. R. Allmaras, F. T. Johnson, and P. R. Spalart. Modifications and clarifications for the implementation of the Spalart–Allmaras turbulence model. In *Seventh International Conference on Computational Fluid Dynamics*, 2012. ICCFD7-1902.

[24] J. E. Bardina, P. G. Huang, and T. J. Coakley. Turbulence modeling validation. In *28th Fluid dynamics conference*, 1997. doi: 10.2514/6.1997-2121. AIAA Paper 1997-2121.

[25] Christopher J Roy and Frederick G Blottner. Assessment of one- and two-equation turbulence models for hypersonic transitional flows. *Journal of Spacecraft and Rockets*, 38(5):699–710, 2001. doi: 10.2514/2.3755.

[26] Christopher J Roy and Frederick G Blottner. Review and assessment of turbulence models for hypersonic flows. *Progress in aerospace sciences*, 42(7-8):469–530, 2006. doi: 10.1016/j.paerosci.2006.12.002.

[27] E. Reginald van Driest. The problem of aerodynamic heating. *Aeronautical Engineering Review*, 15 (10):26–41, 1956.

[28] Edward J Hopkins and Mamoru Inouye. An evaluation of theories for predicting turbulent skin friction and heat transfer on flat plates at supersonic and hypersonic Mach numbers. *AIAA Journal*, 9(6): 993–1003, 1971. doi: 10.2514/3.6323.

[29] Zhiye Zhao and Lin Fu. Revisiting the theory of van Driest: a general scaling law for the skin-friction coefficient of high-speed turbulent boundary layers. *Journal of Fluid Mechanics*, 1012:R3, 2025. doi: 10.1017/jfm.2025.10209.

[30] Kemal Hanjalić and Brian Launder. *Modelling turbulence in engineering and the environment: rational alternative routes to closure*. Cambridge University Press, second edition, 2022. doi: 10.1017/9781108875400.

[31] Dudley B Spalding. A single formula for the law of the wall. *J. Appl. Mech*, 28(3):455–458, 1961. doi: 10.1115/1.3641728.

[32] Donald Coles. The law of the wake in the turbulent boundary layer. *Journal of Fluid Mechanics*, 1(2): 191–226, 1956. doi: 10.1017/s0022112056000135.

[33] Luís Eça, Gonçalo Saraiva, Guilherme Vaz, and Hugo Abreu. The pros and cons of wall functions. In *Proceedings of the ASME 2015 34th International Conference on Ocean, Offshore and Arctic Engineering: CFD and VIV*, volume 2, page V002T08A012. American Society of Mechanical Engineers, 2015. doi: 10.1115/omae2015-41518.

[34] Edward R Van Driest. Turbulent boundary layer in compressible fluids. *Journal of the Aeronautical Sciences*, 18(3):145–160, 1951. doi: 10.2514/8.1895.

[35] Andrew Trettel and Johan Larsson. Mean velocity scaling for compressible wall turbulence with heat transfer. *Physics of Fluids*, 28(2):026102, 2016. doi: 10.1063/1.4942022.

[36] Kevin Patrick Griffin, Lin Fu, and Parviz Moin. Velocity transformation for compressible wall-bounded turbulent flows with and without heat transfer. *Proceedings of the National Academy of Sciences*, 118 (34):e2111144118, 2021. doi: 10.1073/pnas.2111144118.

[37] Asif Manzoor Hasan, Johan Larsson, Sergio Pirozzoli, and Rene Pecnik. Incorporating intrinsic compressibility effects in velocity transformations for wall-bounded turbulent flows. *Physical Review Fluids*, 8(11):L112601, 2023. doi: 10.1103/PhysRevFluids.8.L112601.

[38] Alfred Walz. Compressible turbulent boundary layers with heat transfer and pressure gradient in flow direction. *Journal of Research of the National Bureau of Standards, Section B: Mathematics and Mathematical Physics*, 63B(1):53–70, 1959. doi: 10.6028/jres.063B.008.

[39] Lian Duan and M. P. Martín. Direct numerical simulation of hypersonic turbulent boundary layers. part 4. effect of high enthalpy. *Journal of Fluid Mechanics*, 684:25–59, 2011. doi: 10.1017/jfm.2011.252.

[40] You-Sheng Zhang, Wei-Tao Bi, Fazle Hussain, and Zhen-Su She. A generalized Reynolds analogy for compressible wall-bounded turbulent flows. *Journal of Fluid Mechanics*, 739:392–420, 2014. doi: 10.1017/jfm.2013.620.

[41] Florian R Menter and Alexey Matyushenko. Generalized k- $\omega$ (GEKO) two-equation turbulence model. *AIAA Journal*, 63(11):4590–4606, 2025. doi: 10.2514/1.J065393.

[42] Johan A Heyns, Oliver F Oxtoby, and Adriaan Steenkamp. Modelling high-speed flow using a matrix-free coupled solver. In *Proceedings of the 9th OpenFOAM Workshop, Zagreb, Croatia*, 2014. URL `https://hisa.gitlab.io/doc.html`. 23–26 June 2014.

[43] Youcef Saad and Martin H Schultz. GMRES: A generalized minimal residual algorithm for solving nonsymmetric linear systems. *SIAM Journal on scientific and statistical computing*, 7(3):856–869, 1986. doi: 10.1137/0907058.

[44] Seokkwan Yoon and Antony Jameson. Lower-upper symmetric-Gauss-Seidel method for the Euler and Navier-Stokes equations. *AIAA journal*, 26(9):1025–1026, 1988. doi: 10.2514/3.10007.

[45] Ll Macia, R Castilla, Pedro Javier Gamez-Montero, Sergi Camacho, and E Codina. Numerical simulation of a supersonic ejector for vacuum generation with explicit and implicit solver in openfoam. *Energies*, 12(18):3553, 2019. doi: 10.3390/en12183553.

[46] Daeyoung Jun, Hansol Nam, Hanphil Yoo, Bok Jik Lee, and Hyoungjin Kim. A data-driven body vortex modeling for engineering-level missile aerodynamic prediction. *International Journal of Aeronautical and Space Sciences*, 27(3):2070–2084, 2026. doi: 10.1007/s42405-025-01082-7.

[47] Zifei Yin. Compressibility correction to the k- $\omega$ turbulence model that considers the wall-cooling effect. *AIAA Journal*, 64(3):1309–1319, 2026. doi: 10.2514/1.J065875.

[48] Muhammed Osama Balkhi and Bayram Çelik. Numerical investigation of heating effect in 3D shock-wave/boundary layer interactions near protrusions in hypersonic flow. In *Advances in Computational Heat and Mass Transfer II: Proceedings of the 15th International Conference on Computational Heat and Mass Transfer, ICCHMT 2025*, Lecture Notes in Mechanical Engineering, pages 3–12. Springer, 2026. doi: 10.1007/978-3-032-16135-2_1.

[49] J. R. Edwards. Reflections on the early development of the "AUSM family" of Riemann solvers. *Shock Waves*, 29(5):601–609, 2019. doi: 10.1007/s00193-018-0863-8.

[50] Johan Heyns and Oliver Oxtoby. *High Speed Aerodynamic Solver: User Guide*. CSIR, July 2021. URL `https://hisa.gitlab.io/downloads/userManual.pdf`.

[51] Oliver Oxtoby. Inquiry regarding the characteristicWallPressure of HiSA solver. Personal communication by email to Michael Gnadlinger, 11 August 2026, 2026.

[52] S. Phongthanapanich and A. Matthujak. An improvement of the AUSMDV$^+$ scheme on unstructured grids. *Shock Waves*, 31(8):901–927, 2021. doi: 10.1007/s00193-021-01060-4.

[53] L. Eça and M. Hoekstra. A procedure for the estimation of the numerical uncertainty of CFD calculations based on grid refinement studies. *Journal of Computational Physics*, 262:104–130, 2014. doi: 10.1016/j.jcp.2014.01.006.

[54] P. J. Roache. Perspective: A method for uniform reporting of grid refinement studies. *Journal of Fluids Engineering*, 116(3):405–413, 1994. doi: 10.1115/1.2910291.

[55] Ismail B. Celik, Urmila Ghia, Patrick J. Roache, Christopher J. Freitas, Hugh Coleman, and Peter E. Raad. Procedure for estimation and reporting of uncertainty due to discretization in CFD applications. *Journal of Fluids Engineering*, 130(7):078001, 2008. doi: 10.1115/1.2960953.

[56] Yuyan Tian, Zhenxun Gao, Chongwen Jiang, and Chun-Hian Lee. A correction for Reynolds-averaged-Navier–Stokes turbulence model under the effect of shock waves in hypersonic flows. *International Journal for Numerical Methods in Fluids*, 95(2):313–333, 2023. doi: 10.1002/fld.5150.

[57] Chul-Soo Kim. Experimental studies of supersonic flow past a circular cylinder. *Journal of the Physical Society of Japan*, 11(4):439–445, 1956. doi: 10.1143/JPSJ.11.439.

[58] H. G. Hornung and G. H. Smith. The influence of relaxation on shock detachment. *Journal of Fluid Mechanics*, 93(2):225–239, 1979. doi: 10.1017/S0022112079001865.

[59] Paul A Durbin. On the $k$–$\varepsilon$ stagnation point anomaly. *International journal of heat and fluid flow*, 17 (1):89–90, 1996. doi: 10.1016/0142-727X(95)00073-Y.

[60] Fumio Tamaki. A divergent shock tube for obtaining supersonic flows. *Journal of the Physical Society of Japan*, 11(4):434–439, 1956. doi: 10.1143/JPSJ.11.434.

[61] Frank M. White. *Viscous fluid flow*. McGraw-Hill, third edition, 2006.

[62] W Frels, D. R. Smith, and T Ashworth. Vapour pressure of nitrogen below the triple point. *Cryogenics*, 14(1):3–7, 1974. doi: 10.1016/0011-2275(74)90035-6.

[63] J.-Ch. Robinet, J. Gressier, G. Casalis, and J.-M. Moschetta. Shock wave instability and the carbuncle phenomenon: same intrinsic origin? *Journal of Fluid Mechanics*, 417:237–263, 2000. doi: 10.1017/S0022112000001129.

[64] Michael Dumbser, Jean-Marc Moschetta, and Jérémie Gressier. A matrix stability analysis of the carbuncle phenomenon. *Journal of Computational Physics*, 197(2):647–670, 2004. doi: 10.1016/j.jcp.2003.12.013.

[65] Volker Elling. The carbuncle phenomenon is incurable. *Acta Mathematica Scientia*, 29(6):1647–1656, 2009. doi: 10.1016/S0252-9602(10)60007-0.

[66] Sutthisak Phongthanapanich. Healing of the carbuncle phenomenon for AUSMDV scheme on triangular grids. *International Journal of Nonlinear Sciences and Numerical Simulation*, 17(1):15–28, 2016. doi: 10.1515/ijnsns-2015-0008.

[67] Olaf Marxen, Thierry Magin, Gianluca Iaccarino, and Eric S. G. Shaqfeh. A high-order numerical method to study hypersonic boundary-layer instability including high-temperature gas effects. *Physics of Fluids*, 23(8):084108, 2011. doi: 10.1063/1.3614526.

[68] Mitchel H. Bertram. Hypersonic laminar viscous interaction effects on the aerodynamics of two-dimensional wedge and triangular planform wings. NASA Technical Note D-3523, National Aeronautics and Space Administration, August 1966.

[69] John D. Anderson, Jr. *Hypersonic and High-Temperature Gas Dynamics*. American Institute of Aeronautics and Astronautics, second edition, 2006. doi: 10.2514/4.861956.

[70] Shibin Cao, Jiaao Hao, Igor Klioutchnikov, Chih-Yung Wen, Herbert Olivier, and Karl Alexander Heufer. Transition to turbulence in hypersonic flow over a compression ramp due to intrinsic instability. *Journal of Fluid Mechanics*, 941:A8, 2022. doi: 10.1017/jfm.2022.277.

[71] Shibin Cao, Jiaao Hao, Igor Klioutchnikov, Herbert Olivier, Karl Alexander Heufer, and Chih-Yung Wen. Leading-edge bluntness effects on hypersonic three-dimensional flows over a compression ramp. *Journal of Fluid Mechanics*, 923:A27, 2021. doi: 10.1017/jfm.2021.552.

[72] Giuseppe Chiapparino and Christian Stemmer. Numerical investigation of a Mach 6 laminar shock-wave/boundary-layer interaction on a two-dimensional ramp with 3D controlled surface roughness. *International Journal of Heat and Fluid Flow*, 103:109193, 2023. doi: 10.1016/j.ijheatfluidflow.2023.109193.

[73] Shibin Cao, Jiaao Hao, Igor Klioutchnikov, Herbert Olivier, and Chih-Yung Wen. Unsteady effects in a hypersonic compression ramp flow with laminar separation. *Journal of Fluid Mechanics*, 912:A3, 2021. doi: 10.1017/jfm.2020.1093.

[74] Vance Dippold and Keven Lenahan. Mach 4.5 turbulent flow over a flat plate. NASA Glenn Research Center, NPARC Alliance Validation Archive, 2012. URL `https://www.grc.nasa.gov/WWW/wind/valid/fpmach45/fpmach45.html`. Updated 27 January 2012; accessed 22 September 2026.

[75] Donald Earl Coles. *Measurements in the Boundary Layer on a Smooth Flat Plate in Supersonic Flow*. PhD thesis, California Institute of Technology, 1953.

[76] Kevin P Griffin, Lin Fu, and Parviz Moin. Near-wall model for compressible turbulent boundary layers based on an inverse velocity transformation. *Center for Turbulence Research Annual Research Briefs*, pages 3–12, 2022.

[77] Kevin Patrick Griffin, Lin Fu, and Parviz Moin. General method for determining the boundary layer thickness in nonequilibrium flows. *Physical Review Fluids*, 6(2):024608, 2021. doi: 10.1103/PhysRevFluids.6.024608.

[78] Min Gao and Yu-Xin Ren. Adaptive wall-modeling strategy for large-eddy simulation of nonequilibrium turbulent flows. *AIAA Journal*, 64(10):5604–5621, 2026. doi: 10.2514/1.J066429.

[79] Xinguang Wang, Jianqiang Chen, Timothy Craft, and Hector Iacovides. Extension of analytical wall functions to supersonic and hypersonic flows. *Flow, Turbulence and Combustion*, 111(1):1–34, 2023. doi: 10.1007/s10494-023-00412-7.

[80] Xinguang Wang. *Advanced RANS and Near-Wall Turbulence Modelling for High-Speed Flow*. PhD thesis, The University of Manchester, 2019.

[81] Lian Duan, Izaak Beekman, and M. P. Martín. Direct numerical simulation of hypersonic turbulent boundary layers. part 2. effect of wall temperature. *Journal of Fluid Mechanics*, 655:419–445, 2010. doi: 10.1017/S0022112010000959.

[82] Junji Huang, Gary L Nicholson, Lian Duan, Meelan M Choudhari, and Rodney D Bowersox. Simulation and modeling of cold-wall hypersonic turbulent boundary layers on flat plate. In *AIAA Scitech 2020 Forum*, page 0571, 2020. doi: 10.2514/6.2020-0571.

[83] Thomas Reimer, Giuseppe Di Martino, Lucas Dauth, Luis Baier, Ali Gülhan, Florian Klingenberg, and Dorian Hargarten. STORT hypersonic flight experiment forebody temperature results. In *AIAA SCITECH 2025 Forum*, page 1342, 2025. doi: 10.2514/6.2025-1342.

[84] Ali Gülhan, Dorian Amadeus Hargarten, Maximilian Zurkaulen, Florian Klingenberg, Frank Siebe, Thomas Reimer, and Giuseppe Di Martino. Objectives and achievements of the hypersonic flight experiment STORT. In *73rd International Astronautical Congress, IAC 2022*, 2022.

[85] M Pino Martin. Direct numerical simulation of hypersonic turbulent boundary layers. part 1. initialization and comparison with experiments. *Journal of Fluid Mechanics*, 570:347–364, 2007. doi: 10.1017/S0022112006003107.

[86] Kyle B. Thompson, Christopher O. Johnston, Bil Kleb, Alireza Mazaheri, Thomas K. West, and William A. Wood. LAURA users manual: 5.7. Technical Memorandum NASA/TM-20240010688, NASA Langley Research Center, Hampton, Virginia, September 2024. URL `https://ntrs.nasa.gov/citations/20240010688`.

[87] Junji Huang, Jorge-Valentino Bretzke, and Lian Duan. Assessment of turbulence models in a hypersonic cold-wall turbulent boundary layer. *Fluids*, 4(1):37, 2019. doi: 10.3390/fluids4010037.

[88] Yunlong Xue, Yongliang Feng, and Xiaojing Zheng. Assessment of compressibility corrections on Spalart–Allmaras turbulence model for high-Mach-number flows. *AIAA Journal*, 62(1):92–107, 2024. doi: 10.2514/1.J063274.

[89] Timothy T Aiken, Iain D Boyd, Lian Duan, and Junji Huang. Assessment of Reynolds averaged Navier-Stokes models for a hypersonic cold-wall turbulent boundary layer. In *AIAA Scitech 2022 Forum*, page 0586, 2022. doi: 10.2514/6.2022-0586.

[90] Yuxiao Cheng, Sijie Wang, Yitong Fan, Paul A Durbin, and Zifei Yin. Reynolds-averaged closure modelling of the energy equation for compressible wall turbulence based on the $k$–$\omega$ equations. *Journal of Fluid Mechanics*, 1043:A28, 2026. doi: 10.1017/jfm.2026.12029.

[91] Hans-Hermann Fernholz and P. J. Finley. A critical commentary on mean flow data for two-dimensional compressible turbulent boundary layers. AGARDograph AGARD-AG-253, Advisory Group for Aerospace Research and Development, 1980.

[92] Clive A. J. Fletcher. *Computational techniques for fluid dynamics 2: Specific techniques for different flow categories*. Springer Science & Business Media, second edition, 2012. doi: 10.1007/978-3-642-58239-4. Second edition originally published in 1991; eBook edition published in 2012.

[93] Arthur D. Dilley. Evaluation of CFD turbulent heating prediction techniques and comparison with hypersonic experimental data. Contractor Report NASA/CR-2001-210837, NASA Langley Research Center, March 2001.

[94] Periklis Papadopoulos, Ethiraj Venkatapathy, Dinesh Prabhu, Mark P Loomis, and Dave Olynick. Current grid-generation strategies and future requirements in hypersonic vehicle design, analysis and testing. *Applied Mathematical Modelling*, 23(9):705–735, 1999. doi: 10.1016/S0307-904X(99)00007-4.

[95] Mark P. Loomis, Ethiraj Venkatapathy, Periklis Papadopoulos, Carol B. Davies, Scott Berry, Tom Horvath, and Charles Campbell. Aeroheating and aerodynamic CFD validation and prediction for the X-38 program. In *32nd Thermophysics Conference*, page 2478, 1997. doi: 10.2514/6.1997-2478.

[96] Asif Manzoor Hasan, Johan Larsson, Sergio Pirozzoli, and Rene Pecnik. Estimating mean profiles and fluxes in high-speed turbulent boundary layers using inner/outer-layer scalings. *AIAA Journal*, 62 (2):848–853, 2024. doi: 10.2514/1.J063335.

[97] Valerio Viti, George Huang, and Peter Bradshaw. Numerical study of stress-transport turbulence models: Implementation and validation issues. *Computers & Fluids*, 36(8):1373–1383, 2007. doi: 10.1016/j.compfluid.2006.11.006.

[98] Mustafa Engin Danis. *Direct discontinuous Galerkin methods and hybrid turbulence modeling for high-speed compressible flows*. PhD thesis, Iowa State University, 2023.

[99] Joshua C Vaughan, Christopher J Roy, and William A Jordan. Assessment of turbulence models for hypersonic turbulent flat plate boundary layers. In *AIAA SciTech 2025 Forum*, 2025. doi: 10.2514/6.2025-1689. AIAA Paper 2025-1689.

[100] Joshua Clinton Vaughan. Assessment of RANS turbulence models for hypersonic flat plates. Master's thesis, Virginia Tech, 2025.

[101] H Zhang, T. J. Craft, and H Iacovides. The formulation of the RANS equations for supersonic and hypersonic turbulent flows. *The Aeronautical Journal*, 125(1285):525–555, 2021. doi: 10.1017/aer.2020.93.

[102] William M Kays. Turbulent Prandtl number—where are we? *Journal of Heat Transfer*, 116(2):284–295, 1994. doi: 10.1115/1.2911398.

[103] Rodney D. W. Bowersox. Extension of equilibrium turbulent heat flux models to high-speed shear flows. *Journal of fluid mechanics*, 633:61–70, 2009. doi: 10.1017/S0022112009007691.

[104] Asif Manzoor Hasan, Alex José Elias, Florian Menter, and Rene Pecnik. Variable-property and intrinsic compressibility corrections for turbulence models using near-wall scaling theories. *Journal of Fluid Mechanics*, 1019:A8, 2025. doi: 10.1017/jfm.2025.10572.

[105] Guohua Tu, Xiaogang Deng, and Meiliang Mao. Assessment of two turbulence models and some compressibility corrections for hypersonic compression corners by high-order difference schemes. *Chinese Journal of Aeronautics*, 25(1):25–32, 2012. doi: 10.1016/S1000-9361(11)60358-0.

[106] E. C. Anderson and C. H. Lewis. Laminar or turbulent boundary-layer flows of perfect gases or reacting gas mixtures in chemical equilibrium. Technical Report NASA CR-1893, National Aeronautics and Space Administration, Washington, DC, 1971.

[107] NASA Langley Research Center. The Spalart–Allmaras turbulence model. Turbulence Modeling Resource. Accessed September 1, 2026.

[108] P. R. Spalart and S. R. Allmaras. A one-equation turbulence model for aerodynamic flows. In *30th Aerospace Sciences Meeting and Exhibit*, 1992. doi: $10.2514/6.1992\text{-}439$. AIAA Paper 92-0439.

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